Lecture Notes in Physics, Volumes I–IV

Oscillators · Variational Mechanics · Spacetime and Fields · Quantum Mechanics

Technical reference for lecture slides

August 2, 2026

This volume binds the four standalone documents of the series. Each Part reproduces its source verbatim, with its own internal numbering, so the plain-text cross-references between Parts — written as [Osc. §n], [Mech. §n], [Fld. §n] — read exactly as in the standalone documents. Conventions are stated per Part and occasionally break at Part boundaries by design (notably: the summation convention is introduced only in Part III). Future courses (quantum field theory, general relativity) are intended to join as further Parts.

Contents

Part I

Oscillators: From One Spring to the Wave Equation

EVERYTHING You Need to Know About Oscillators

A technical reference on linear oscillations: from one mass on a spring to the wave equation

This document is a self-contained technical reference on the classical harmonic oscillator, intended as source material for teaching slides. Every result is derived in full, with all intermediate steps displayed. The topics, in order: the simple harmonic oscillator (where the complex-exponential method is introduced and justified), the damped oscillator (with a complete classification of the three damping regimes), the driven oscillator (treated separately in the undamped and damped cases), resonance (amplitude response, peak frequency, width, and quality factor), coupled oscillators (matrix form, normal modes, decoupling by a change of coordinates), and the forced coupled oscillator (mode-by-mode response and the antiresonance phenomenon). The treatment then extends to the N-mass chain — built up explicitly from the three-, four-, and five-mass cases — whose normal modes and dispersion relation follow from a spatial exponential ansatz, and closes with the continuum limit, in which the chain becomes a stretched rope and the equations of motion become the wave equation. The general solution of the wave equation leads naturally to the Fourier expansion of initial data, and a closing outlook introduces the wave equation in two and three dimensions. The treatment is entirely Newtonian.

1 Conventions and Notation

Throughout, $x(t)$ denotes a real displacement from equilibrium, dots denote time derivatives, and i is the imaginary unit. Whether a symbol is real or complex is visible from its mark:

Mark	Meaning	Examples
plain italic	real scalar	$x(t)$, A, ϕ, ω_0
tilde $\tilde{}$	complex-valued *function* of t (stand-in for a real one)	$\tilde{z}(t)$, $\tilde{F}(t)$
hat $\hat{}$	complex *constant* (amplitude/phasor)	$\hat{z}$, $\hat{c}_1$, $\hat{D}(\omega)$
arrow $\vec{}$	real column vector	$\vec{x}$, $\vec{v}$
sans-serif	matrix	A

Complex conjugation is written with a superscript star, $\hat{c}^*$ (never a bar, to avoid stacking accents). A hatted quantity may depend on the driving frequency ω — $\hat{D}(\omega)$ is still a "constant" in the relevant sense: constant *in time*. The physical quantity behind any tilde or hat is always its *real part*.

Whenever a parameter is defined for notational convenience, the definition is displayed and its dimensions stated. The recurring definitions are collected here for reference; each is re-introduced where it first appears.

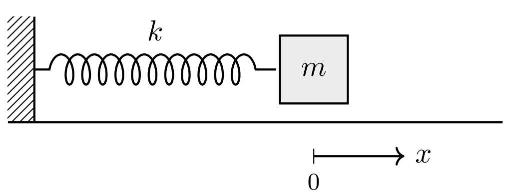
Symbol	Definition	Dimensions	Meaning
ω_0	$\omega_0^2 = k/m$	$[\text{time}]^{-1}$	natural (angular) frequency
β	$2\beta = b/m$	$[\text{time}]^{-1}$	damping constant
ω_d	$\omega_d = \sqrt{\omega_0^2 - \beta^2}$	$[\text{time}]^{-1}$	damped oscillation frequency
f_0	$f_0 = F_0/m$	$[\text{length}][\text{time}]^{-2}$	driving force per unit mass
ω_c	$\omega_c^2 = k_c/m$	$[\text{time}]^{-1}$	coupling frequency (§6)
Q	$Q = \omega_0/(2\beta)$	dimensionless	quality factor
a	lattice spacing	[length]	equilibrium mass separation (§8)
c	$c^2 = T/\mu$	$[\text{length}][\text{time}]^{-1}$	wave speed, μ = mass/length (§9)

Convention 1.1 (Complexification)**.** To solve a linear differential equation with a real sinusoidal inhomogeneity $f_0 \cos \omega t$, we will replace it by the complex inhomogeneity $f_0\, \mathrm{e}^{\mathrm{i}\omega t}$, solve for a complex $\tilde{z}(t)$, and recover the physical solution as $x(t) = \operatorname{Re} \tilde{z}(t)$. The justification is Lemma 2.3 below, proved once and used throughout.

2 The Simple Harmonic Oscillator

2.1 Physical setup and equation of motion

A block of mass m slides on a frictionless horizontal surface, attached to a rigid wall by an ideal massless spring of stiffness k. Let x denote the displacement of the block from the position at which the spring is relaxed (the *equilibrium position*); $x > 0$ means the spring is stretched.



By Hooke's law the spring exerts a restoring force $F_{\text{spring}} = -kx$: for $x > 0$ (stretched) the force points in the $-x$ direction, for $x < 0$ (compressed) in the $+x$ direction. Newton's second law, $m\ddot{x} = \sum F$, gives

$$m\ddot{x} = -kx. \tag{1}$$

Convention 2.1 (Natural frequency)**.** Define

$$\boxed{\omega_0^2 \equiv \frac{k}{m}} \qquad [\omega_0] = \sqrt{\frac{[\text{force}]/[\text{length}]}{[\text{mass}]}} = \frac{1}{[\text{time}]}. \tag{2}$$

ω_0 is the only parameter of the system with the dimensions of an inverse time; it will turn out to be the angular frequency of the free oscillation, hence the name *natural frequency*.

Dividing (1) by m puts the equation in standard form:

$$\boxed{\ddot{x} + \omega_0^2\, x = 0} \tag{3}$$

Remark 2.2 (Why this equation is everywhere)*.* The block and spring stand in for something far more general. Let a particle of mass m move in *any* smooth potential $V(x)$ with a stable equilibrium at x_0: $V'(x_0) = 0$ and $V''(x_0) > 0$. Taylor expansion about the minimum gives

$$V(x) = V(x_0) + \tfrac{1}{2}\, V''(x_0)\,(x - x_0)^2 + O\big((x - x_0)^3\big), \tag{4}$$

the linear term vanishing precisely because x_0 is an equilibrium, so for small displacements Newton's law is Eq. (3) with the effective stiffness $k = V''(x_0)$. Every small oscillation about a stable equilibrium is this equation with its own k — the pendulum, for instance: $V = mg\ell(1 - \cos\theta)$ gives, in the angle coordinate, $\omega_0^2 = g/\ell$. The rest of this document therefore applies wherever equilibria are stable and amplitudes are small, which is why one equation deserves a document.

This is a linear, homogeneous, second-order ordinary differential equation with constant real coefficients. Its solution space is a two-dimensional real vector space: a solution is uniquely determined by the two initial data $x(0)$ and $\dot{x}(0)$,[1] and any linear combination of solutions is again a solution (superposition).

2.2 First solutions: sine and cosine

Equation (3) asks for a function whose second derivative is a negative multiple of itself. Two functions with exactly this property are known from calculus: try $x(t) = \cos\omega_0 t$,

$$\dot{x} = -\omega_0 \sin\omega_0 t, \qquad \ddot{x} = -\omega_0^2 \cos\omega_0 t = -\omega_0^2\, x \quad \checkmark \tag{5}$$

and $x(t) = \sin\omega_0 t$,

$$\dot{x} = \omega_0 \cos\omega_0 t, \qquad \ddot{x} = -\omega_0^2 \sin\omega_0 t = -\omega_0^2\, x \quad \checkmark \tag{6}$$

Both solve (3). (The frequency must be ω_0: trying $\cos\alpha t$ gives $\ddot{x} = -\alpha^2 x$, which matches only for $\alpha^2 = \omega_0^2$.)

[1] Uniqueness is the one fact this document imports from the general theory of differential equations, and for the equations used here it has a three-line self-contained proof. Let x_1, x_2 solve $\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$ (any $\beta \geq 0$, including the undamped case $\beta = 0$; any driving f) with the same initial data, and set $\delta = x_1 - x_2$. Then δ solves the homogeneous equation with $\delta(0) = \dot{\delta}(0) = 0$, and $E_\delta \equiv \frac{1}{2}\dot{\delta}^2 + \frac{1}{2}\omega_0^2\delta^2$ satisfies $\dot{E}_\delta = \dot{\delta}\,(\ddot{\delta} + \omega_0^2\delta) = -2\beta\,\dot{\delta}^{\,2} \leq 0$. Since $E_\delta(0) = 0$ and $E_\delta \geq 0$, E_δ vanishes identically, so $\delta \equiv 0$. The same argument, run with the potential-energy quadratic form of §6, covers the coupled systems. Existence is exhibited constructively throughout, by writing the solutions down.

Linear combinations also work. Because the equation is linear and homogeneous, any combination $x = B_1 \cos\omega_0 t + B_2 \sin\omega_0 t$ with constants B_1, B_2 is again a solution — verify directly:

$$\ddot{x} + \omega_0^2 x = B_1 \big(\underbrace{-\omega_0^2 \cos\omega_0 t + \omega_0^2 \cos\omega_0 t}_{=0}\big) + B_2 \big(\underbrace{-\omega_0^2 \sin\omega_0 t + \omega_0^2 \sin\omega_0 t}_{=0}\big) = 0. \tag{7}$$

Moreover this two-parameter family is *all* the solutions: the solution space is two-dimensional (a solution is pinned down by the two numbers $x(0)$, $\dot{x}(0)$), and $\cos\omega_0 t$, $\sin\omega_0 t$ are linearly independent, so they span it. The general real solution is

$$\boxed{x(t) = B_1 \cos\omega_0 t + B_2 \sin\omega_0 t} \tag{8}$$

Initial-value problem. Given $x(0) = x_0$ and $\dot{x}(0) = v_0$:

$$x(0) = B_1 = x_0, \tag{9}$$

$$\dot{x}(t) = -B_1\omega_0 \sin\omega_0 t + B_2\omega_0 \cos\omega_0 t \implies \dot{x}(0) = B_2\,\omega_0 = v_0 \implies B_2 = \frac{v_0}{\omega_0}. \tag{10}$$

Hence

$$\boxed{x(t) = x_0 \cos\omega_0 t + \frac{v_0}{\omega_0} \sin\omega_0 t} \tag{11}$$

2.3 Amplitude–phase form

The two-term form (8) hides the physics: the motion is in fact a *single* sinusoid. To see it, try to write

$$B_1 \cos\omega_0 t + B_2 \sin\omega_0 t \stackrel{?}{=} A\cos(\omega_0 t - \phi) \tag{12}$$

for some amplitude $A \geq 0$ and phase ϕ. Expand the right side with the cosine addition formula:

$$A\cos(\omega_0 t - \phi) = A\cos\phi\,\cos\omega_0 t + A\sin\phi\,\sin\omega_0 t. \tag{13}$$

Since $\cos\omega_0 t$ and $\sin\omega_0 t$ are linearly independent, the two sides agree for all t if and only if the coefficients match:

$$B_1 = A\cos\phi, \qquad B_2 = A\sin\phi. \tag{14}$$

These are precisely polar coordinates for the point (B_1, B_2) in the plane: they can always be solved, with

$$\boxed{A = \sqrt{B_1^2 + B_2^2}, \qquad \tan\phi = \frac{B_2}{B_1}} \tag{15}$$

(quadrant of ϕ chosen so that $\cos\phi$ has the sign of B_1 and $\sin\phi$ the sign of B_2). The general solution is therefore equally

$$\boxed{x(t) = A\cos(\omega_0 t - \phi)} \tag{16}$$

a single oscillation of angular frequency ω_0, amplitude A, phase ϕ; the phase records *when* the maxima occur (at $\omega_0 t = \phi + 2\pi n$), the amplitude *how large* they are. In terms of initial data, from (11):

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega_0^2}}, \qquad \tan\phi = \frac{v_0}{\omega_0 x_0}. \tag{17}$$

Everything above was elementary: one good guess plus trigonometry. It worked because for *this* equation the correct guess is common knowledge. Add a damping term $2\beta\dot{x}$ (as we will in

§3) and pure sines and cosines no longer solve the equation — the guess would have to be revised to decaying oscillations, with the decay rate and frequency both unknown, and the trigonometric bookkeeping grows accordingly. Before facing that, we set up a method that does not depend on guessing correctly.

2.4 The complex-exponential method

The ansatz. Rather than guessing oscillations, guess the one function family that differentiation cannot change: exponentials,

$$x(t) = \mathrm{e}^{\lambda t} \quad \Longrightarrow \quad \dot{x} = \lambda\, \mathrm{e}^{\lambda t}, \qquad \ddot{x} = \lambda^2 \mathrm{e}^{\lambda t}, \tag{18}$$

with the constant λ to be determined *by the equation*, not by us. Substituting into (3):

$$\lambda^2 \mathrm{e}^{\lambda t} + \omega_0^2\, \mathrm{e}^{\lambda t} = \big(\lambda^2 + \omega_0^2\big)\mathrm{e}^{\lambda t} = 0. \tag{19}$$

Since $\mathrm{e}^{\lambda t} \neq 0$ for all t, the ansatz solves the equation if and only if λ satisfies the *characteristic equation*

$$\lambda^2 + \omega_0^2 = 0. \tag{20}$$

This has *no real solutions* — consistent with §2.2, where the real solutions turned out to be oscillations, not real exponentials. Here is the decision point: **if we allow the solution to take complex values**, the characteristic equation is solvable,

$$\lambda = \pm\, \mathrm{i}\, \omega_0, \tag{21}$$

and we obtain two complex solutions $\mathrm{e}^{\mathrm{i}\omega_0 t}$ and $\mathrm{e}^{-\mathrm{i}\omega_0 t}$. At the end we will extract the physical (real) part. The next two ingredients make this procedure precise.

Euler's formula. For real θ,

$$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta. \tag{22}$$

(One way to see this: the power series of $\mathrm{e}^{\mathrm{i}\theta}$, split into even and odd powers of θ, reproduces the series of $\cos\theta$ and $\mathrm{i}\sin\theta$ respectively.) In particular $\mathrm{e}^{\mathrm{i}\theta}$ traces the unit circle in the complex plane with angular speed 1 in θ, and

$$\cos\theta = \frac{\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta}}{2}, \qquad \sin\theta = \frac{\mathrm{e}^{\mathrm{i}\theta} - \mathrm{e}^{-\mathrm{i}\theta}}{2\mathrm{i}}. \tag{23}$$

So the complex solutions just found are familiar: $\mathrm{e}^{\pm\mathrm{i}\omega_0 t} = \cos\omega_0 t \pm \mathrm{i}\sin\omega_0 t$ — the sine and cosine of §2.2, packaged in complex pairs.

Why taking the real part is legitimate.

Lemma 2.3 (Real and imaginary parts of complex solutions)**.** *Let L be a linear differential operator with* real *coefficients, e.g. $L = \frac{\mathrm{d}^2}{\mathrm{d}t^2} + 2\beta\frac{\mathrm{d}}{\mathrm{d}t} + \omega_0^2$ with $\beta, \omega_0 \in \mathbb{R}$.*

(i) If a complex-valued function $\tilde{z}(t)$ satisfies $L[\tilde{z}] = 0$, then $\operatorname{Re}\tilde{z}$ and $\operatorname{Im}\tilde{z}$ are real solutions of $L[x] = 0$.

(ii) If $\tilde{z}(t)$ satisfies $L[\tilde{z}] = \tilde{g}(t)$ for a complex inhomogeneity $\tilde{g}$, then $\operatorname{Re}\tilde{z}$ satisfies $L[x] = \operatorname{Re}\tilde{g}$.

Proof. Write $\tilde{z} = u + \mathrm{i}v$ with u, v real. Because differentiation acts on real and imaginary parts separately and the coefficients of L are real,

$$L[\tilde{z}] = L[u + \mathrm{i}v] = L[u] + \mathrm{i}\, L[v], \tag{24}$$

where $L[u]$ and $L[v]$ are real-valued. (i) If $L[\tilde{z}] = 0$, then both the real part $L[u]$ and the imaginary part $L[v]$ vanish separately, so $u = \operatorname{Re}\tilde{z}$ and $v = \operatorname{Im}\tilde{z}$ solve the homogeneous equation. (ii) If $L[\tilde{z}] = \tilde{g}$, then matching real parts of $L[u] + \mathrm{i}L[v] = \operatorname{Re}\tilde{g} + \mathrm{i}\operatorname{Im}\tilde{g}$ gives $L[u] = \operatorname{Re}\tilde{g}$. □

Equivalently: since the coefficients are real, taking the complex conjugate of $L[\tilde{z}] = \tilde{g}$ gives $L[\tilde{z}^*] = \tilde{g}^*$, so $\tilde{z}^*$ is a solution of the conjugated problem, and $\operatorname{Re}\tilde{z} = \frac{1}{2}(\tilde{z} + \tilde{z}^*)$ is a solution of the problem with inhomogeneity $\frac{1}{2}(\tilde{g} + \tilde{g}^*) = \operatorname{Re}\tilde{g}$ by linearity. We will invoke Lemma 2.3 every time we complexify.

As a first check, apply part (i) to $\tilde{z} = \mathrm{e}^{\mathrm{i}\omega_0 t}$: $\operatorname{Re}\tilde{z} = \cos\omega_0 t$ and $\operatorname{Im}\tilde{z} = \sin\omega_0 t$ must be real solutions — and they are exactly the two solutions of §2.2. The complex method has *re-derived* the guess.

General solution and the reality condition. The general complex solution of (3) is the two-parameter family

$$\tilde{z}(t) = \hat{c}_1\, \mathrm{e}^{\mathrm{i}\omega_0 t} + \hat{c}_2\, \mathrm{e}^{-\mathrm{i}\omega_0 t}, \qquad \hat{c}_1, \hat{c}_2 \in \mathbb{C}. \tag{25}$$

We want the real solutions among these. Imposing $\tilde{z}(t) = \tilde{z}(t)^*$ for all t:

$$\hat{c}_1 \mathrm{e}^{\mathrm{i}\omega_0 t} + \hat{c}_2 \mathrm{e}^{-\mathrm{i}\omega_0 t} = \hat{c}_1^* \mathrm{e}^{-\mathrm{i}\omega_0 t} + \hat{c}_2^* \mathrm{e}^{\mathrm{i}\omega_0 t} \quad \Longleftrightarrow \quad (\hat{c}_1 - \hat{c}_2^*)\, \mathrm{e}^{\mathrm{i}\omega_0 t} = (\hat{c}_1^* - \hat{c}_2)\, \mathrm{e}^{-\mathrm{i}\omega_0 t}. \tag{26}$$

Since $\mathrm{e}^{\mathrm{i}\omega_0 t}$ and $\mathrm{e}^{-\mathrm{i}\omega_0 t}$ are linearly independent functions, both coefficients must vanish, i.e.

$$\boxed{\hat{c}_2 = \hat{c}_1^*} \qquad \text{(reality condition).} \tag{27}$$

Write the one remaining free complex constant in polar form, $\hat{c}_1 = \frac{A}{2}\, \mathrm{e}^{-\mathrm{i}\phi}$ with $A \geq 0$ and $\phi \in \mathbb{R}$. Then

$$\begin{aligned} x(t) &= \frac{A}{2}\, \mathrm{e}^{-\mathrm{i}\phi} \mathrm{e}^{\mathrm{i}\omega_0 t} + \frac{A}{2}\, \mathrm{e}^{+\mathrm{i}\phi} \mathrm{e}^{-\mathrm{i}\omega_0 t} = \frac{A}{2}\Big(\mathrm{e}^{\mathrm{i}(\omega_0 t - \phi)} + \mathrm{e}^{-\mathrm{i}(\omega_0 t - \phi)}\Big) \\ &= A\cos(\omega_0 t - \phi), \end{aligned} \tag{28}$$

using (23): the amplitude–phase form (16) of §2.3 drops out *directly* — no trigonometric identity was needed, because the polar decomposition of the complex constant $\hat{c}_1$ *is* the amplitude–phase decomposition. Equivalently, one may skip the reality condition and simply take

$$x(t) = \operatorname{Re}\big[\, \hat{C}\, \mathrm{e}^{\mathrm{i}\omega_0 t} \,\big], \qquad \hat{C} = A\, \mathrm{e}^{-\mathrm{i}\phi} \in \mathbb{C}, \tag{29}$$

which is legitimate by Lemma 2.3(i) and contains the same two real parameters packaged into one complex number.

Three equivalent forms. Collecting the results of §2.2–§2.4, the general solution now has three parametrizations,

$$x(t) = \underbrace{B_1 \cos\omega_0 t + B_2 \sin\omega_0 t}_{\text{(I), §2.2}} = \underbrace{A\cos(\omega_0 t - \phi)}_{\text{(II), §2.3}} = \underbrace{\operatorname{Re}\big[\hat{C}\mathrm{e}^{\mathrm{i}\omega_0 t}\big]}_{\text{(III)}}, \tag{30}$$

with conversion dictionary (the $B \leftrightarrow (A,\phi)$ part is (14)):

$$B_1 = A\cos\phi, \quad B_2 = A\sin\phi; \qquad A = \sqrt{B_1^2 + B_2^2}, \quad \tan\phi = \frac{B_2}{B_1}; \qquad \hat{C} = B_1 - \mathrm{i}B_2 = A\mathrm{e}^{-\mathrm{i}\phi}. \tag{31}$$

(To verify the last identity: $\mathrm{Re}[(B_1 - \mathrm{i}B_2)(\cos\omega_0 t + \mathrm{i}\sin\omega_0 t)] = B_1\cos\omega_0 t + B_2\sin\omega_0 t$.) Form (I) is best for initial conditions, (II) for reading off the physics, (III) for calculation — one complex number carries amplitude as its modulus and phase as (minus) its argument. This packaging is the essential convenience of the complex method, and from the next section on it is the form we compute in.

2.5 Energy

The total mechanical energy is kinetic plus spring potential:

$$E(t) = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2. \tag{32}$$

Differentiate along a solution and use (1):

$$\frac{\mathrm{d}E}{\mathrm{d}t} = m\dot{x}\,\ddot{x} + kx\,\dot{x} = \dot{x}\left(m\ddot{x} + kx\right) = \dot{x}\cdot 0 = 0. \tag{33}$$

Energy is conserved. Its value in terms of the amplitude: insert $x = A\cos(\omega_0 t - \phi)$, $\dot{x} = -A\omega_0\sin(\omega_0 t - \phi)$,

$$\begin{aligned} E &= \frac{1}{2}mA^2\omega_0^2\sin^2(\omega_0 t - \phi) + \frac{1}{2}kA^2\cos^2(\omega_0 t - \phi) \\ &= \frac{1}{2}kA^2\left[\sin^2(\omega_0 t - \phi) + \cos^2(\omega_0 t - \phi)\right] = \frac{1}{2}kA^2, \end{aligned} \tag{34}$$

where $m\omega_0^2 = k$ was used to combine the two terms. The energy is set entirely by the amplitude, and oscillates back and forth between kinetic and potential form at angular frequency $2\omega_0$.

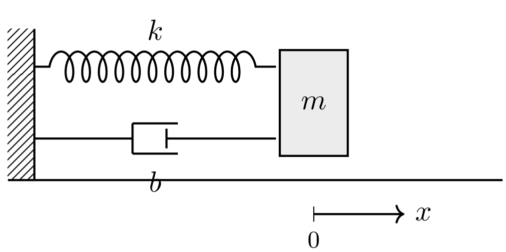
3 The Damped Harmonic Oscillator

3.1 Physical setup and equation of motion

Add to the previous system a linear drag force

$$F_{\mathrm{drag}} = -b\,\dot{x}, \qquad b > 0, \tag{35}$$

opposing the velocity. (Physically: viscous friction, valid as the leading-order model at low speeds; a dashpot in the mechanical idealization.)



Newton's second law now reads $m\ddot{x} = -kx - b\dot{x}$, i.e.

$$m\ddot{x} + b\dot{x} + kx = 0. \tag{36}$$

Convention 3.1 (Damping constant)**.** Define

$$\boxed{2\beta \equiv \frac{b}{m}} \qquad [\beta] = \frac{1}{[\text{time}]}. \tag{37}$$

The factor of 2 is chosen purely so that the roots of the characteristic equation below come out without factors of $\frac{1}{2}$.

Dividing by m, the standard form is

$$\boxed{\ddot{x} + 2\beta\dot{x} + \omega_0^2\, x = 0} \tag{38}$$

3.2 Characteristic equation and the discriminant

The same exponential ansatz $x = \mathrm{e}^{\lambda t}$ applies verbatim — this is the payoff of having set up the method in §2.4: nothing new is needed. Substituting,

$$\left(\lambda^2 + 2\beta\lambda + \omega_0^2\right)\mathrm{e}^{\lambda t} = 0 \quad \iff \quad \lambda^2 + 2\beta\lambda + \omega_0^2 = 0, \tag{39}$$

with roots (quadratic formula)

$$\boxed{\lambda_\pm = -\beta \pm \sqrt{\beta^2 - \omega_0^2}} \tag{40}$$

The qualitative behavior is decided by the sign of the discriminant $\beta^2 - \omega_0^2$, giving *three cases*:

Case	Condition	Roots
underdamped	$\beta < \omega_0$	complex conjugate pair, $\lambda_\pm = -\beta \pm \mathrm{i}\omega_d$
critically damped	$\beta = \omega_0$	real double root, $\lambda = -\beta$
overdamped	$\beta > \omega_0$	two distinct negative real roots

3.3 Case I: underdamped ($\beta < \omega_0$)

The discriminant is negative; write

$$\sqrt{\beta^2 - \omega_0^2} = \mathrm{i}\sqrt{\omega_0^2 - \beta^2} \qquad \text{and define} \qquad \boxed{\omega_d \equiv \sqrt{\omega_0^2 - \beta^2}} \tag{41}$$

(the *damped frequency*; note $0 < \omega_d < \omega_0$). Then $\lambda_\pm = -\beta \pm \mathrm{i}\omega_d$ and the general complex solution is

$$\tilde{z}(t) = \hat{c}_1\, \mathrm{e}^{(-\beta+\mathrm{i}\omega_d)t} + \hat{c}_2\, \mathrm{e}^{(-\beta-\mathrm{i}\omega_d)t} = \mathrm{e}^{-\beta t}\left(\hat{c}_1\, \mathrm{e}^{\mathrm{i}\omega_d t} + \hat{c}_2\, \mathrm{e}^{-\mathrm{i}\omega_d t}\right). \tag{42}$$

The bracket is exactly the SHO structure of (25) with $\omega_0 \to \omega_d$, so the reality condition and polar parametrization go through unchanged ($\hat{c}_2 = \hat{c}_1^*$, $\hat{c}_1 = \frac{A}{2}\mathrm{e}^{-\mathrm{i}\phi}$):

$$\boxed{x(t) = A\, \mathrm{e}^{-\beta t} \cos(\omega_d t - \phi)} \tag{43}$$

An oscillation at the *reduced* frequency ω_d, inside an exponentially decaying envelope $\pm A\mathrm{e}^{-\beta t}$. Two remarks:

- The zero crossings are still equally spaced (period $T_d = 2\pi/\omega_d$), but $T_d > 2\pi/\omega_0$: damping slows the oscillation.
- Over one period the amplitude shrinks by the fixed factor $e^{-\beta T_d}$; the *logarithmic decrement* $\Lambda \equiv \beta T_d = 2\pi\beta/\omega_d$ is a standard experimental measure of damping — for weak damping $\Lambda \approx \pi/Q$, tying this time-domain measure to the linewidth measure of §5.3.

Initial-value problem. With $x(0) = x_0$, $\dot{x}(0) = v_0$, it is easiest to use the form $x = e^{-\beta t}(B_1 \cos\omega_d t + B_2 \sin\omega_d t)$:

$$x(0) = B_1 = x_0, \tag{44}$$

$$\dot{x}(t) = -\beta e^{-\beta t}\big(B_1 \cos\omega_d t + B_2 \sin\omega_d t\big) + e^{-\beta t}\big(-B_1\omega_d \sin\omega_d t + B_2\omega_d \cos\omega_d t\big),$$

$$\dot{x}(0) = -\beta B_1 + \omega_d B_2 = v_0 \implies B_2 = \frac{v_0 + \beta x_0}{\omega_d}. \tag{45}$$

$$\boxed{x(t) = e^{-\beta t}\Big[x_0 \cos\omega_d t + \frac{v_0 + \beta x_0}{\omega_d}\,\sin\omega_d t\Big]} \tag{46}$$

3.4 Case II: overdamped ($\beta > \omega_0$)

Both roots are real; write $\gamma \equiv \sqrt{\beta^2 - \omega_0^2} > 0$, so

$$\lambda_\pm = -\beta \pm \gamma, \qquad \lambda_- < -\beta < \lambda_+ < 0. \tag{47}$$

(That $\lambda_+ < 0$: $\gamma = \sqrt{\beta^2 - \omega_0^2} < \beta$ because $\omega_0^2 > 0$.) The general solution is a sum of two pure decays, no oscillation:

$$\boxed{x(t) = C_+\, e^{\lambda_+ t} + C_-\, e^{\lambda_- t}} \tag{48}$$

Imposing $x(0) = x_0$, $\dot{x}(0) = v_0$ gives the linear system

$$C_+ + C_- = x_0, \qquad \lambda_+ C_+ + \lambda_- C_- = v_0. \tag{49}$$

Solving (multiply the first equation by λ_- and subtract, then by λ_+ and subtract):

$$C_+ = \frac{v_0 - \lambda_- x_0}{\lambda_+ - \lambda_-}, \qquad C_- = \frac{\lambda_+ x_0 - v_0}{\lambda_+ - \lambda_-}. \tag{50}$$

For example, released from rest ($v_0 = 0$),

$$x(t) = \frac{\lambda_+\, e^{\lambda_- t} - \lambda_-\, e^{\lambda_+ t}}{\lambda_+ - \lambda_-}\, x_0, \tag{51}$$

which one verifies directly: at $t = 0$ the numerator is $\lambda_+ - \lambda_-$, giving $x(0) = x_0$; the derivative's numerator at $t = 0$ is $\lambda_+\lambda_- - \lambda_-\lambda_+ = 0$, giving $\dot{x}(0) = 0$.

Remark 3.2 (Long-time behavior). Since $|\lambda_+| < |\lambda_-|$, the $e^{\lambda_+ t}$ term decays more slowly and dominates at late times: the approach to equilibrium is governed by the *smaller* decay rate $|\lambda_+| = \beta - \gamma$. Increasing the damping β beyond ω_0 actually makes the late-time return to equilibrium *slower*: as $\beta \to \infty$, $\beta - \sqrt{\beta^2 - \omega_0^2} = \frac{\omega_0^2}{\beta + \sqrt{\beta^2 - \omega_0^2}} \to \frac{\omega_0^2}{2\beta} \to 0$.

Remark 3.3 (At most one zero crossing). $x(t) = 0$ requires $C_+ e^{\lambda_+ t} = -C_- e^{\lambda_- t}$, i.e. $e^{(\lambda_+ - \lambda_-)t} = -C_-/C_+$. The left side is strictly monotone in t, so there is at most one solution: an overdamped oscillator can pass through equilibrium at most once.

3.5 Case III: critically damped ($\beta = \omega_0$)

The characteristic equation has the *double* root $\lambda = -\beta$, so the ansatz produces only one solution, $\mathrm{e}^{-\beta t}$. The solution space is two-dimensional, so a second, linearly independent solution must exist; we find it by *reduction of order*. Substitute

$$x(t) = u(t)\,\mathrm{e}^{-\beta t} \tag{52}$$

with u to be determined. Compute the derivatives:

$$\dot{x} = \dot{u}\,\mathrm{e}^{-\beta t} - \beta u\,\mathrm{e}^{-\beta t} = (\dot{u} - \beta u)\,\mathrm{e}^{-\beta t}, \tag{53}$$

$$\ddot{x} = (\ddot{u} - \beta\dot{u})\,\mathrm{e}^{-\beta t} - \beta(\dot{u} - \beta u)\,\mathrm{e}^{-\beta t} = (\ddot{u} - 2\beta\dot{u} + \beta^2 u)\,\mathrm{e}^{-\beta t}. \tag{54}$$

Substitute into (38) with $\omega_0^2 = \beta^2$:

$$\begin{aligned} 0 &= \ddot{x} + 2\beta\dot{x} + \beta^2 x \\ &= \Big[(\ddot{u} - 2\beta\dot{u} + \beta^2 u) + 2\beta(\dot{u} - \beta u) + \beta^2 u\Big]\mathrm{e}^{-\beta t} \\ &= \Big[\ddot{u} + (-2\beta + 2\beta)\dot{u} + (\beta^2 - 2\beta^2 + \beta^2)u\Big]\mathrm{e}^{-\beta t} = \ddot{u}\,\mathrm{e}^{-\beta t}. \end{aligned} \tag{55}$$

Every term in u and $\dot{u}$ cancels *exactly because $\lambda = -\beta$ is a double root*, leaving simply

$$\ddot{u} = 0 \quad\Longrightarrow\quad u(t) = C_1 + C_2\,t. \tag{56}$$

Hence the general solution in the critical case is

$$\boxed{x(t) = (C_1 + C_2\,t)\,\mathrm{e}^{-\beta t}} \tag{57}$$

The second solution $t\,\mathrm{e}^{-\beta t}$ is not an exponential — it is the degenerate limit in which the two exponentials $\mathrm{e}^{\lambda_\pm t}$ merge. (Indeed $\lim_{\gamma\to 0} \frac{\mathrm{e}^{(-\beta+\gamma)t} - \mathrm{e}^{(-\beta-\gamma)t}}{2\gamma} = t\,\mathrm{e}^{-\beta t}$, a difference quotient in the root itself.)

Initial conditions: $x(0) = C_1 = x_0$ and $\dot{x}(0) = C_2 - \beta C_1 = v_0 \Rightarrow C_2 = v_0 + \beta x_0$:

$$x(t) = \big[x_0 + (v_0 + \beta x_0)\,t\big]\,\mathrm{e}^{-\beta t}. \tag{58}$$

Remark 3.4 (Fastest return). Among the three regimes at fixed ω_0, compare the decay rate of the slowest-decaying term in each solution: β for the underdamped envelope, $\beta - \gamma$ for the overdamped case, and ω_0 (up to the polynomial factor t) for critical damping. Since $\beta < \omega_0$ in the first case and $\beta - \gamma < \omega_0$ in the second (both strict), the critical rate is the largest — and critical damping is the only regime that combines this fastest decay with no zero crossings. This is why measuring instruments and door closers are tuned near critical damping.

3.6 Energy decay and the quality factor

With drag, energy is no longer conserved. Differentiating $E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}kx^2$ along a solution of $m\ddot{x} = -kx - b\dot{x}$:

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \dot{x}\,(m\ddot{x} + kx) = \dot{x}\,(-b\dot{x}) = -\,b\,\dot{x}^2 \;\le\; 0. \tag{59}$$

Energy decreases monotonically (except at the isolated instants where $\dot{x} = 0$), at exactly the rate at which the drag force does negative work.

For weak damping ($\beta \ll \omega_0$), the envelope of (43) gives $x_{\max} \propto \mathrm{e}^{-\beta t}$, and since $E \propto A^2$,

$$E(t) \approx E(0)\,\mathrm{e}^{-2\beta t}. \tag{60}$$

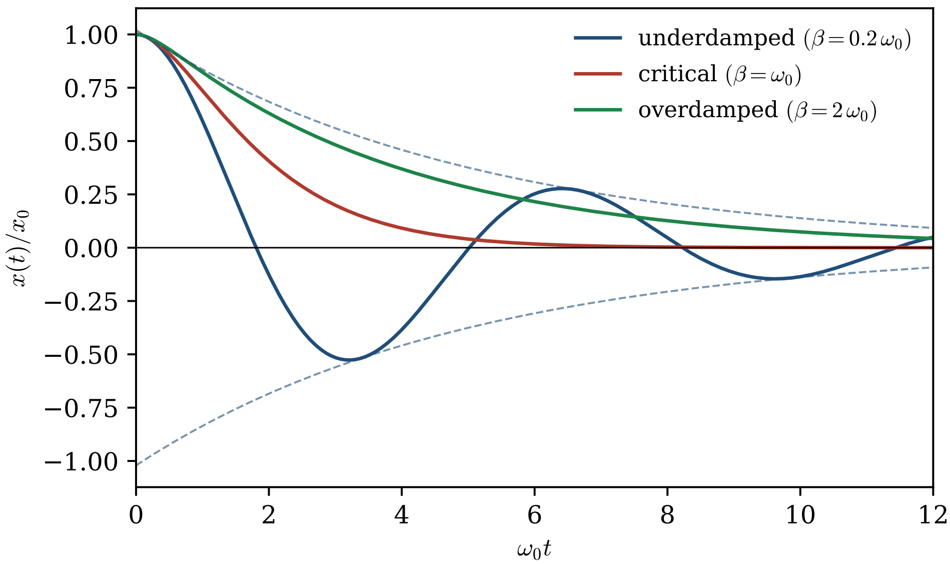


Figure 1: The three damping regimes for the same initial condition $x(0) = x_0$, $\dot{x}(0) = 0$. Underdamped motion (with its envelope $\pm\text{const}\,e^{-\beta t}$, dashed), critical damping, and overdamped motion.

Convention 3.5 (Quality factor)**.**

$$\boxed{Q \equiv \frac{\omega_0}{2\beta}} \tag{61}$$

Interpretation: the energy decays by e^{-1} in a time $1/(2\beta)$, during which the oscillator completes $\frac{\omega_0}{2\pi} \cdot \frac{1}{2\beta} = \frac{Q}{2\pi}$ cycles. Large Q = many oscillations before the energy is dissipated. The same Q will reappear in §5 as the sharpness of the resonance peak — a nontrivial consistency check.

4 The Forced Harmonic Oscillator

We now apply an external driving force $F(t) = F_0 \cos \omega t$ to the mass. Note carefully: ω is the *driving* frequency, an external knob we control; it is distinct from the system parameters ω_0 and β. Define the force per unit mass

$$\boxed{f_0 \equiv \frac{F_0}{m}} \qquad [f_0] = \frac{[\text{length}]}{[\text{time}]^2}. \tag{62}$$

We treat the two cases separately, in increasing order of realism:

§4.1 **Case A: undamped, driven** — cleanest formulas; exhibits beats and the resonance catastrophe.

§4.2 **Case B: damped, driven** — the physically complete case; exhibits transients, steady state, and finite resonance.

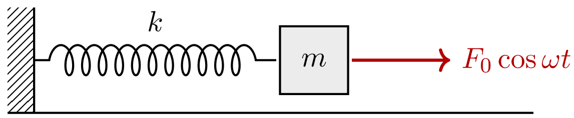
Structure of the solution space

Both cases are *inhomogeneous* linear equations, $L[x] = f_0 \cos \omega t$. If x_p is any one (*particular*) solution and x_h is the general solution of the homogeneous equation $L[x] = 0$, then $x_p + x_h$ solves the inhomogeneous equation, and conversely any two solutions differ by a homogeneous solution ($L[x_1 - x_2] = f - f = 0$). Hence

$$\boxed{\text{general solution} \;=\; \text{particular solution} \;+\; \text{general homogeneous solution}} \tag{63}$$

The homogeneous solutions are exactly those found in the previous two sections. The new task in this section is to construct one particular solution — and this is where the complex method of Convention 1.1 shows its value.

4.1 Case A: undamped and driven



The equation of motion is

$$\boxed{\ddot{x} + \omega_0^2\, x = f_0 \cos \omega t} \tag{64}$$

Particular solution by complexification. Following Convention 1.1, consider the complexified problem

$$\ddot{\tilde{z}} + \omega_0^2\, \tilde{z} = f_0\, \mathrm{e}^{\mathrm{i}\omega t}, \tag{65}$$

whose right-hand side has real part $f_0 \cos \omega t$; by Lemma 2.3(ii), $x_p = \operatorname{Re} \tilde{z}_p$ will solve (64). Since the right-hand side is an exponential and the left-hand side does not change the form of exponentials, try

$$\tilde{z}_p(t) = \hat{z}\, \mathrm{e}^{\mathrm{i}\omega t}, \qquad \hat{z} \in \mathbb{C} \text{ constant.} \tag{66}$$

Then $\ddot{\tilde{z}}_p = -\omega^2 \hat{z}\, \mathrm{e}^{\mathrm{i}\omega t}$, and substituting into (65):

$$\left(-\omega^2 + \omega_0^2\right) \hat{z}\, \mathrm{e}^{\mathrm{i}\omega t} = f_0\, \mathrm{e}^{\mathrm{i}\omega t} \quad \Longrightarrow \quad \hat{z} = \frac{f_0}{\omega_0^2 - \omega^2} \qquad (\omega \neq \omega_0). \tag{67}$$

The differential equation has been reduced to a one-line algebraic equation. Here $\hat{z}$ happens to be real, so

$$\boxed{x_p(t) = \frac{f_0}{\omega_0^2 - \omega^2} \cos \omega t \qquad (\omega \neq \omega_0)} \tag{68}$$

Observations before adding the homogeneous part:

- The particular solution oscillates at the *driving* frequency ω, not at ω_0.
- For $\omega < \omega_0$ the response is *in phase* with the drive (positive coefficient); for $\omega > \omega_0$ it is *exactly out of phase* (negative coefficient, i.e. phase lag π). The switch is abrupt — damping will smooth it out in Case B.
- The amplitude blows up as $\omega \to \omega_0$. The excluded case $\omega = \omega_0$ is treated below.

4.1.1 Sub-case A1: off resonance ($\omega \neq \omega_0$), beats

By (63) the general solution is

$$x(t) = \frac{f_0}{\omega_0^2 - \omega^2}\cos\omega t + B_1\cos\omega_0 t + B_2\sin\omega_0 t. \tag{69}$$

Take the oscillator to start *from rest at equilibrium*: $x(0) = 0$, $\dot{x}(0) = 0$. Then

$$x(0) = \frac{f_0}{\omega_0^2 - \omega^2} + B_1 = 0 \qquad \Longrightarrow \qquad B_1 = -\frac{f_0}{\omega_0^2 - \omega^2}, \tag{70}$$

$$\dot{x}(0) = 0 + B_2\,\omega_0 = 0 \qquad \Longrightarrow \qquad B_2 = 0, \tag{71}$$

so

$$x(t) = \frac{f_0}{\omega_0^2 - \omega^2}\big(\cos\omega t - \cos\omega_0 t\big). \tag{72}$$

To expose the structure, use the product identity. From the addition formulas,

$$\cos(P+Q) = \cos P\cos Q - \sin P\sin Q, \tag{73}$$

$$\cos(P-Q) = \cos P\cos Q + \sin P\sin Q, \tag{74}$$

subtracting gives $\cos(P-Q) - \cos(P+Q) = 2\sin P\sin Q$. Setting $P = \frac{(\omega_0+\omega)t}{2}$, $Q = \frac{(\omega_0-\omega)t}{2}$, so that $P - Q = \omega t$ and $P + Q = \omega_0 t$:

$$\cos\omega t - \cos\omega_0 t = 2\,\sin\!\Big(\tfrac{(\omega_0+\omega)t}{2}\Big)\,\sin\!\Big(\tfrac{(\omega_0-\omega)t}{2}\Big). \tag{75}$$

Therefore

$$\boxed{x(t) = \underbrace{\frac{2f_0}{\omega_0^2 - \omega^2}\,\sin\!\Big(\tfrac{(\omega_0-\omega)t}{2}\Big)}_{\text{slow envelope}} \cdot \underbrace{\sin\!\Big(\tfrac{(\omega_0+\omega)t}{2}\Big)}_{\text{fast oscillation}}} \tag{76}$$

When the drive is close to resonance, $\omega \approx \omega_0$, the first factor oscillates slowly (frequency $\frac{|\omega_0-\omega|}{2}$) and acts as a modulating envelope on the fast oscillation (frequency $\approx \omega_0$): the amplitude repeatedly builds up and dies back down. These are *beats* (Fig. 2). The maximum displacement, $\left|\frac{2f_0}{\omega_0^2-\omega^2}\right|$, grows as $\omega \to \omega_0$ — the beats become taller and slower, foreshadowing the resonant case.

4.1.2 Sub-case A2: on resonance ($\omega = \omega_0$)

At $\omega = \omega_0$ the ansatz $\hat{z}\,\mathrm{e}^{\mathrm{i}\omega_0 t}$ fails — $\mathrm{e}^{\mathrm{i}\omega_0 t}$ already solves the homogeneous equation, so the left-hand side annihilates it and no constant $\hat{z}$ can produce the required right-hand side. We give two independent derivations of the correct particular solution.

Derivation 1: limit of the off-resonance solution. Fix t and let $\omega \to \omega_0$ in (72). Both numerator and denominator vanish, so apply l'Hôpital's rule *in the variable* ω:

$$\lim_{\omega\to\omega_0}\frac{f_0\,(\cos\omega t - \cos\omega_0 t)}{\omega_0^2 - \omega^2} = \lim_{\omega\to\omega_0}\frac{f_0\cdot\big(-t\sin\omega t\big)}{-2\omega} = \frac{f_0\,t\sin\omega_0 t}{2\omega_0}. \tag{77}$$

So the from-rest resonant solution is

$$\boxed{x(t) = \frac{f_0}{2\omega_0}\,t\,\sin\omega_0 t} \tag{78}$$

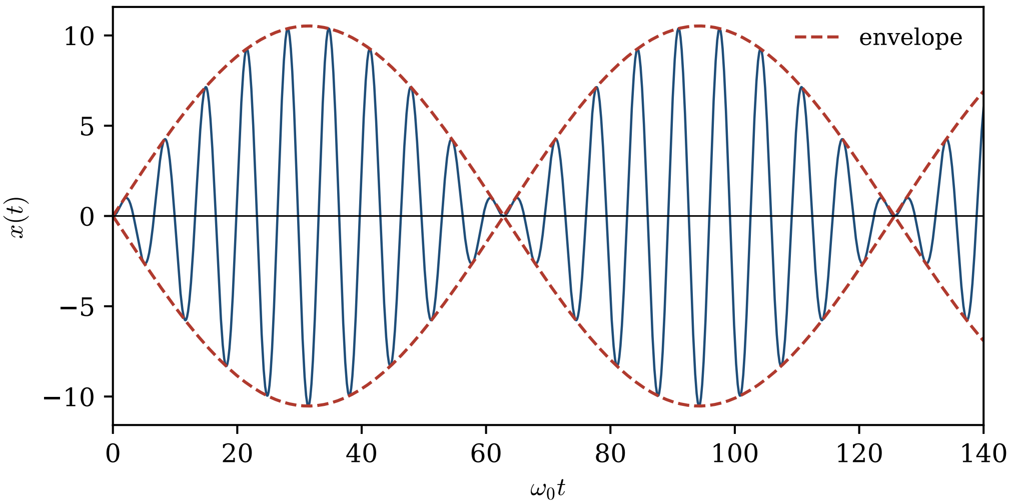


Figure 2: Undamped driven oscillator started from rest, $\omega = 0.9\,\omega_0$: beats. Dashed: the slow envelope from Eq. (76).

Derivation 2: direct complex ansatz. Guided by the critical-damping experience of §3.5 (when a root is "used up," multiply by t), try

$$\tilde{z}_p(t) = \hat{c}\, t\, \mathrm{e}^{\mathrm{i}\omega_0 t}, \qquad \hat{c} \in \mathbb{C}. \tag{79}$$

Compute the derivatives carefully:

$$\dot{\tilde{z}}_p = \hat{c}\,\mathrm{e}^{\mathrm{i}\omega_0 t} + \hat{c}\,t\,(\mathrm{i}\omega_0)\mathrm{e}^{\mathrm{i}\omega_0 t} = \hat{c}\,(1 + \mathrm{i}\omega_0 t)\,\mathrm{e}^{\mathrm{i}\omega_0 t}, \tag{80}$$

$$\ddot{\tilde{z}}_p = \hat{c}\,(\mathrm{i}\omega_0)\,\mathrm{e}^{\mathrm{i}\omega_0 t} + \hat{c}\,(1 + \mathrm{i}\omega_0 t)(\mathrm{i}\omega_0)\,\mathrm{e}^{\mathrm{i}\omega_0 t} = \hat{c}\,\left(2\mathrm{i}\omega_0 - \omega_0^2\,t\right)\mathrm{e}^{\mathrm{i}\omega_0 t}. \tag{81}$$

Substitute into $\ddot{\tilde{z}} + \omega_0^2 \tilde{z} = f_0 \mathrm{e}^{\mathrm{i}\omega_0 t}$:

$$\hat{c}\big(2\mathrm{i}\omega_0 - \omega_0^2 t\big)\mathrm{e}^{\mathrm{i}\omega_0 t} + \omega_0^2\,\hat{c}\,t\,\mathrm{e}^{\mathrm{i}\omega_0 t} = \hat{c}\cdot 2\mathrm{i}\omega_0\,\mathrm{e}^{\mathrm{i}\omega_0 t} \stackrel{!}{=} f_0\,\mathrm{e}^{\mathrm{i}\omega_0 t}. \tag{82}$$

The t-proportional terms cancel identically (same mechanism as in §3.5), leaving

$$\hat{c} = \frac{f_0}{2\mathrm{i}\omega_0} = -\frac{\mathrm{i}f_0}{2\omega_0}. \tag{83}$$

Extract the real part:

$$x_p = \mathrm{Re}\Big[-\tfrac{\mathrm{i}f_0}{2\omega_0}\,t\,\big(\cos\omega_0 t + \mathrm{i}\sin\omega_0 t\big)\Big] = \frac{f_0}{2\omega_0}\,t\,\sin\omega_0 t, \tag{84}$$

in agreement with Derivation 1. (This particular solution happens to satisfy $x_p(0) = \dot{x}_p(0) = 0$, so it *is* the from-rest solution; in general one would still add homogeneous terms.)

The displacement grows *linearly without bound* (Fig. 3): the drive, phase-locked a quarter cycle ahead of the displacement, does positive work on every cycle and there is no dissipation to absorb it. This is the *resonance catastrophe* — the purest form of resonance, visible here without any damping to obscure it. Any real system either has damping (Case B) or leaves the linear regime.

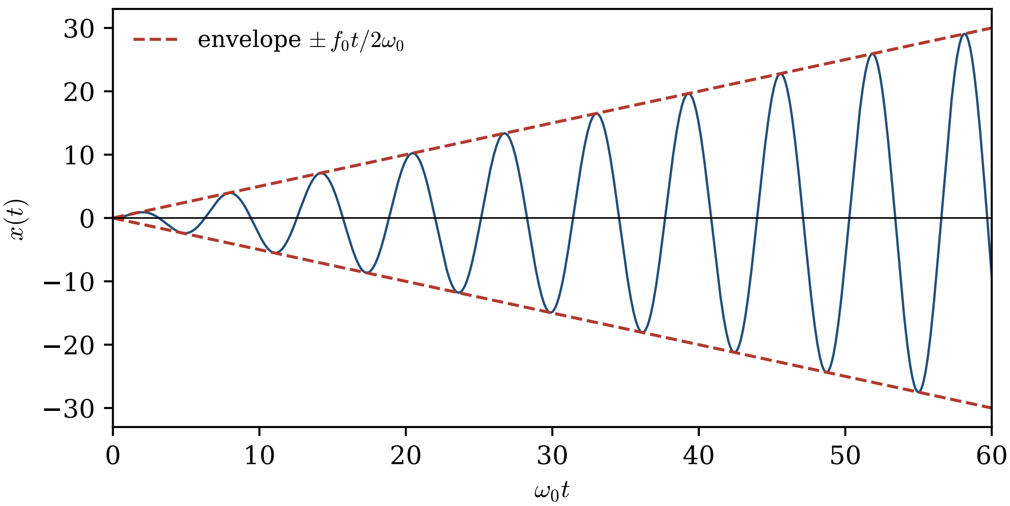


Figure 3: Resonant driving of the undamped oscillator, $\omega = \omega_0$: linear growth $x = \frac{f_0}{2\omega_0}\, t \sin \omega_0 t$.

4.2 Case B: damped and driven

The full equation of motion:

$$\boxed{\ddot{x} + 2\beta\dot{x} + \omega_0^2\, x = f_0 \cos \omega t} \tag{85}$$

Particular solution by complexification. Complexify as before:

$$\ddot{\tilde{z}} + 2\beta\dot{\tilde{z}} + \omega_0^2 \tilde{z} = f_0\, \mathrm{e}^{\mathrm{i}\omega t}, \qquad \tilde{z}_p = \hat{z}\, \mathrm{e}^{\mathrm{i}\omega t}. \tag{86}$$

Each derivative pulls down a factor of $\mathrm{i}\omega$:

$$\left(-\omega^2 + 2\mathrm{i}\beta\omega + \omega_0^2\right)\hat{z}\, \mathrm{e}^{\mathrm{i}\omega t} = f_0\, \mathrm{e}^{\mathrm{i}\omega t} \quad \Longrightarrow \quad \boxed{\hat{z} = \frac{f_0}{\omega_0^2 - \omega^2 + 2\mathrm{i}\beta\omega}} \tag{87}$$

One line. Note that the denominator now has a nonzero imaginary part for all $\omega > 0$, so it never vanishes: with damping there is no excluded frequency, and this single formula covers everything.

Extracting amplitude and phase. Write the denominator in polar form. With

$$\hat{D}(\omega) \equiv \left(\omega_0^2 - \omega^2\right) + \mathrm{i}\left(2\beta\omega\right) = |\hat{D}|\, \mathrm{e}^{\mathrm{i}\delta}, \tag{88}$$

the modulus and argument are

$$|\hat{D}(\omega)| = \sqrt{\left(\omega_0^2 - \omega^2\right)^2 + 4\beta^2\omega^2}, \qquad \cos\delta = \frac{\omega_0^2 - \omega^2}{|\hat{D}|}, \quad \sin\delta = \frac{2\beta\omega}{|\hat{D}|}. \tag{89}$$

Since $2\beta\omega > 0$, we have $\sin\delta > 0$, so $\delta \in (0, \pi)$: the argument is pinned to the upper half plane, and the often-quoted formula

$$\tan\delta = \frac{2\beta\omega}{\omega_0^2 - \omega^2} \tag{90}$$

must be read with this quadrant convention ($\delta \in (0, \frac{\pi}{2})$ for $\omega < \omega_0$; $\delta \in (\frac{\pi}{2}, \pi)$ for $\omega > \omega_0$; naively applying arctan would wrongly return a negative angle above resonance). Then

$$\hat{z} = \frac{f_0}{|\hat{D}|\, \mathrm{e}^{\mathrm{i}\delta}} = \frac{f_0}{|\hat{D}|}\, \mathrm{e}^{-\mathrm{i}\delta}, \qquad \tilde{z}_p(t) = \frac{f_0}{|\hat{D}|}\, \mathrm{e}^{\mathrm{i}(\omega t - \delta)}, \tag{91}$$

and taking the real part (Lemma 2.3(ii)):

$$\boxed{x_p(t) = A(\omega)\, \cos\big(\omega t - \delta(\omega)\big), \qquad A(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}} \tag{92}$$

The steady-state response is a sinusoid at the driving frequency, with amplitude $A(\omega)$ and *phase lag* $\delta(\omega)$ behind the drive. The entire frequency response — both numbers — was carried by the single complex quantity $\hat{z}$: modulus = amplitude, minus argument = phase. This is the punchline of the complex method.

Limiting behaviors.

Regime	Amplitude	Phase
$\omega \to 0$	$A \to f_0/\omega_0^2 = F_0/k$ (static/stiffness-controlled)	$\delta \to 0$ (in phase)
$\omega = \omega_0$	$A = \dfrac{f_0}{2\beta\omega_0}$	$\delta = \dfrac{\pi}{2}$ (quadrature)
$\omega \to \infty$	$A \approx f_0/\omega^2 \to 0$ (mass-controlled)	$\delta \to \pi$ (antiphase)

At low frequency the mass adiabatically tracks the force ($kx \approx F$); at high frequency the mass cannot keep up and $m\ddot{x} \approx F$ forces the response into antiphase; at ω_0 the response lags by exactly a quarter period regardless of the damping strength.

4.2.1 Contrast: the real undetermined-coefficients route

For comparison, solve the same problem without complex numbers. Since derivatives of $\cos\omega t$ generate $\sin\omega t$ and vice versa, the minimal real ansatz needs both:

$$x_p = a\cos\omega t + b\sin\omega t. \tag{93}$$

Then

$$\dot{x}_p = -a\omega\sin\omega t + b\omega\cos\omega t, \tag{94}$$
$$\ddot{x}_p = -a\omega^2\cos\omega t - b\omega^2\sin\omega t. \tag{95}$$

Substitute into (85) and collect the coefficients of $\cos\omega t$ and $\sin\omega t$ separately (they are linearly independent functions, so each must match):

$$\begin{aligned} \cos\omega t: &\qquad -a\omega^2 + 2\beta\omega b + \omega_0^2 a = f_0, \\ \sin\omega t: &\qquad -b\omega^2 - 2\beta\omega a + \omega_0^2 b = 0, \end{aligned} \tag{96}$$

i.e. the 2×2 linear system

$$\begin{pmatrix} \omega_0^2 - \omega^2 & 2\beta\omega \\ -2\beta\omega & \omega_0^2 - \omega^2 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} f_0 \\ 0 \end{pmatrix}. \tag{97}$$

The determinant is $\Delta = (\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2 = |\hat{D}|^2$, and Cramer's rule gives

$$a = \frac{f_0\,(\omega_0^2 - \omega^2)}{\Delta}, \qquad b = \frac{2\beta\omega f_0}{\Delta}. \tag{98}$$

Converting (a, b) to amplitude–phase form via §2.3 reproduces (92) exactly: $A = \sqrt{a^2 + b^2} = f_0\sqrt{\Delta}/\Delta = f_0/|\hat{D}|$ and $\tan\delta = b/a = 2\beta\omega/(\omega_0^2 - \omega^2)$.

Remark 4.1. Both routes are elementary, but note what the real route had to do: double the ansatz, expand derivatives into cross terms, sort two trigonometric families, solve a 2×2 system, and reassemble amplitude and phase at the end. The complex route did all of this implicitly, because $\mathrm{e}^{\mathrm{i}\omega t}$ is an eigenfunction of $\mathrm{d}/\mathrm{d}t$ while $\{\cos, \sin\}$ merely span an invariant 2-dimensional subspace that $\mathrm{d}/\mathrm{d}t$ mixes. In the coupled systems of §7 the same bookkeeping would multiply per mode; we will use the complex method without further comment.

4.2.2 Transient plus steady state

By (63), the *general* solution of (85) in the underdamped case is

$$x(t) = \underbrace{A\cos(\omega t - \delta)}_{\text{steady state}} + \underbrace{\mathrm{e}^{-\beta t}\big(B_1\cos\omega_d t + B_2\sin\omega_d t\big)}_{\text{transient}}, \tag{99}$$

with A, δ fixed by (92) and B_1, B_2 fixed by the initial conditions. Whatever B_1, B_2 are, the second term dies out on the timescale $1/\beta$:

$$\boxed{\text{after a few multiples of } 1/\beta, \quad x(t) \to A(\omega)\cos\big(\omega t - \delta(\omega)\big), \text{ independent of initial conditions.}} \tag{100}$$

The system forgets how it started; only the drive's imprint survives. This is why $A(\omega)$ and $\delta(\omega)$ are *the* experimentally meaningful response functions — and their reach is wider than a single sinusoid: the equation is linear, so a drive assembled as a sum of sinusoids produces the sum of the corresponding steady states. Knowing $A(\omega)$ and $\delta(\omega)$ for every ω is knowing the long-time response to any drive that can be so decomposed (§9.4 develops the decomposition idea itself, for initial data).

Worked example: starting from rest. Impose $x(0) = 0$, $\dot{x}(0) = 0$ on (99).

$$x(0) = A\cos\delta + B_1 = 0 \implies B_1 = -A\cos\delta. \tag{101}$$

$$\dot{x}(t) = -A\omega\sin(\omega t - \delta) + \mathrm{e}^{-\beta t}\Big[-\beta\big(B_1\cos\omega_d t + B_2\sin\omega_d t\big) + \omega_d\big(-B_1\sin\omega_d t + B_2\cos\omega_d t\big)\Big],$$

$$\dot{x}(0) = A\omega\sin\delta - \beta B_1 + \omega_d B_2 = 0 \implies B_2 = -\frac{A\big(\omega\sin\delta + \beta\cos\delta\big)}{\omega_d}, \tag{102}$$

where $-\beta B_1 = +\beta A\cos\delta$ was used. The response therefore rings at ω_d while settling into the drive; interference between the two frequencies produces transient beats when $\omega \approx \omega_d$, which then decay — the damped remnant of Fig. 2.

5 Resonance

This section analyzes the steady-state amplitude (92) as a function of the driving frequency:

$$A(\omega) = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}. \tag{103}$$

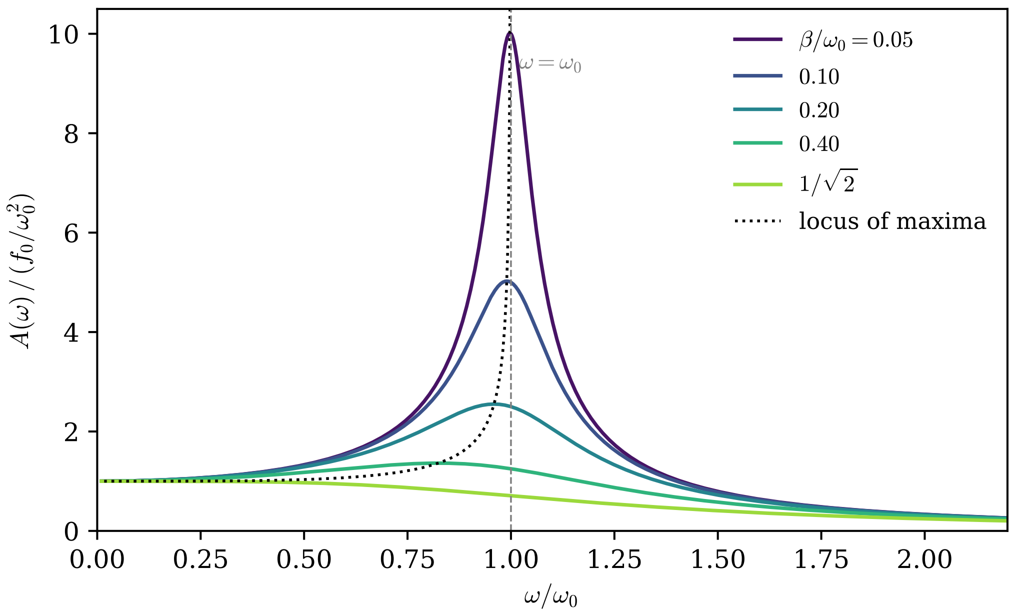


Figure 4: Amplitude response $A(\omega)$ for several damping strengths (amplitude in units of the static response f_0/ω_0^2). The dotted curve is the locus of maxima $\big(\omega_{\text{peak}}, A_{\max}\big)$ as β varies; the peak moves *left* of ω_0 and disappears at $\beta = \omega_0/\sqrt{2}$.

5.1 Peak frequency and peak amplitude

Maximizing A is equivalent to minimizing the radicand. Substitute $u \equiv \omega^2 \geq 0$ and define

$$g(u) = (\omega_0^2 - u)^2 + 4\beta^2 u. \tag{104}$$

Then

$$g'(u) = -2(\omega_0^2 - u) + 4\beta^2, \qquad g''(u) = 2 > 0, \tag{105}$$

so g is a convex parabola in u with unique minimum where $g'(u) = 0$:

$$u_* = \omega_0^2 - 2\beta^2. \tag{106}$$

Two cases:

- If $\beta < \omega_0/\sqrt{2}$, then $u_* > 0$ and the amplitude peaks at

$$\boxed{\omega_{\text{peak}} = \sqrt{\omega_0^2 - 2\beta^2}} \tag{107}$$

- If $\beta \geq \omega_0/\sqrt{2}$, then $u_* \leq 0$: on the physical domain $u \geq 0$ the function g is increasing (since $g'(0) = -2\omega_0^2 + 4\beta^2 \geq 0$), so $A(\omega)$ is monotonically decreasing and there is *no resonance peak at all* — the maximum response is the static one at $\omega = 0$.

Peak height, for $\beta < \omega_0/\sqrt{2}$: substitute u_* back,

$$\begin{aligned} g(u_*) &= \big(\omega_0^2 - (\omega_0^2 - 2\beta^2)\big)^2 + 4\beta^2\big(\omega_0^2 - 2\beta^2\big) \\ &= 4\beta^4 + 4\beta^2\omega_0^2 - 8\beta^4 = 4\beta^2\big(\omega_0^2 - \beta^2\big) = 4\beta^2\omega_d^2, \end{aligned} \tag{108}$$

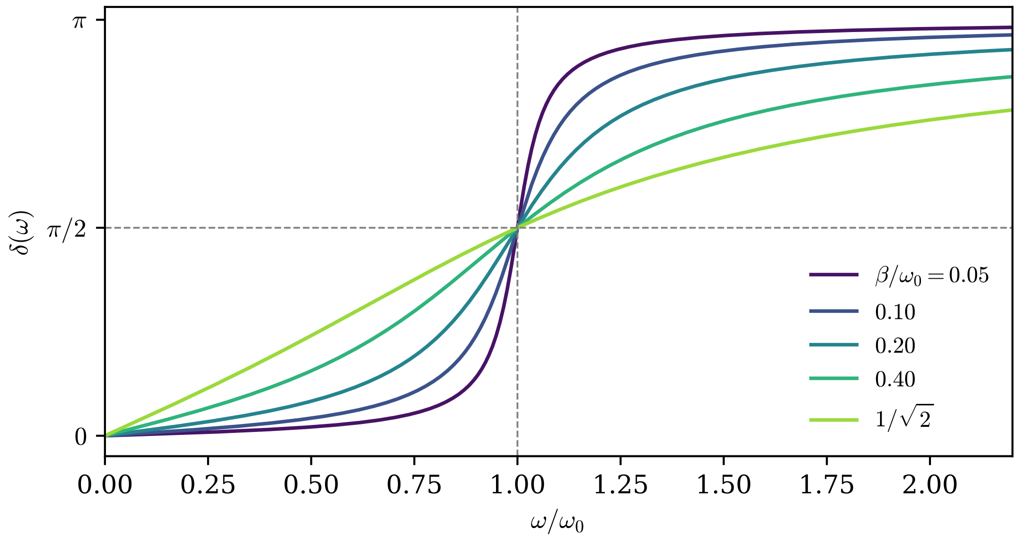


Figure 5: Phase lag $\delta(\omega)$ for the same damping strengths. All curves pass through $\delta = \pi/2$ exactly at $\omega = \omega_0$; the crossing sharpens as $\beta \to 0$, approaching the undamped step of §4.1.

so

$$\boxed{A_{\max} = \frac{f_0}{2\beta\,\omega_d} = \frac{f_0}{2\beta\sqrt{\omega_0^2 - \beta^2}}} \tag{109}$$

For weak damping, $A_{\max} \approx \dfrac{f_0}{2\beta\omega_0} = Q \cdot \dfrac{f_0}{\omega_0^2}$: *the resonant response exceeds the static response by the factor Q.*

5.2 Three distinct frequencies

It is worth displaying explicitly the hierarchy that has now emerged — three frequencies that coincide only in the undamped limit and are frequently conflated:

$$\boxed{\omega_{\text{peak}}^2 = \omega_0^2 - 2\beta^2 \;<\; \omega_d^2 = \omega_0^2 - \beta^2 \;<\; \omega_0^2} \tag{110}$$

Frequency	Where it appears	Role
ω_0	§2–§4	natural frequency; phase lag crosses $\pi/2$ here
ω_d	§3.3	frequency of *free* damped oscillation
ω_{peak}	this section	driving frequency of maximum *amplitude*

(A fourth, the frequency of maximum *velocity* response, sits exactly at ω_0; see §5.5.) The differences are $O(\beta^2)$ and thus invisible at weak damping, which is why the distinctions are easy to miss.

5.3 Width of the resonance and the quality factor

Assume weak damping, $\beta \ll \omega_0$, and examine $A(\omega)$ near the peak. For ω near ω_0, factor the resonant denominator:

$$\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega) \approx 2\omega_0\,(\omega_0 - \omega), \qquad 4\beta^2\omega^2 \approx 4\beta^2\omega_0^2, \tag{111}$$

where in each factor we replaced $\omega \to \omega_0$ *except* in the difference $\omega_0 - \omega$, which is the small quantity being tracked. Then

$$A^2(\omega) \approx \frac{f_0^2}{4\omega_0^2(\omega_0-\omega)^2 + 4\beta^2\omega_0^2} = \frac{f_0^2}{4\omega_0^2} \cdot \frac{1}{(\omega-\omega_0)^2+\beta^2}. \tag{112}$$

As a function of ω, the squared amplitude (equivalently the stored energy) is a *Lorentzian* centered at ω_0. Its maximum is $\frac{f_0^2}{4\omega_0^2\beta^2}$; it falls to half that value when

$$(\omega-\omega_0)^2 = \beta^2 \quad \Longleftrightarrow \quad \omega = \omega_0 \pm \beta, \tag{113}$$

so the *full width at half maximum* of the energy response is

$$\boxed{\Delta\omega_{\text{FWHM}} = 2\beta \qquad \Longrightarrow \qquad \frac{\omega_0}{\Delta\omega_{\text{FWHM}}} = \frac{\omega_0}{2\beta} = Q} \tag{114}$$

The same Q defined in §3 from the free decay rate reappears as the dimensionless sharpness of the driven resonance:

$$Q = 2\pi\,\frac{\text{energy stored}}{\text{energy lost per cycle}} = \frac{\text{center frequency}}{\text{linewidth}}. \tag{115}$$

(The first equality, for weak damping: $E \propto \mathrm{e}^{-2\beta t}$ loses $\Delta E \approx 2\beta T\,E$ per period $T \approx 2\pi/\omega_0$, so $2\pi E/\Delta E = \omega_0/(2\beta) = Q$.) Slow decay in time ⇔ narrow peak in frequency — a time–frequency tradeoff.

5.4 Phase behavior near resonance

From (89), at $\omega = \omega_0$ the real part of $\hat{D}$ vanishes, so $\delta = \pi/2$ *exactly*, for every β. For the slope, expand the cotangent near ω_0 ($\omega = \omega_0 + \varepsilon$):

$$\cot\delta = \frac{\omega_0^2-\omega^2}{2\beta\omega} \approx \frac{-2\omega_0\varepsilon}{2\beta\omega_0} = -\frac{\varepsilon}{\beta}. \tag{116}$$

Since $\cot\delta = -\tan(\delta - \frac{\pi}{2})$, and $\tan s \approx s$ for small s,

$$\delta(\omega) \approx \frac{\pi}{2} + \frac{\omega-\omega_0}{\beta}, \qquad \left.\frac{\mathrm{d}\delta}{\mathrm{d}\omega}\right|_{\omega_0} = \frac{1}{\beta}. \tag{117}$$

The phase sweeps through $\pi/2$ with slope $1/\beta$: the weaker the damping, the more abrupt the crossing (Fig. 5), degenerating into the step function of the undamped Case A.

5.5 Aside: velocity resonance and power absorption

Not the main thread, but worth recording because it explains why "resonance at ω_0" is also correct — for a different response variable. The steady-state *velocity* amplitude is $V(\omega) = \omega A(\omega)$. Dividing inside the square root by ω^2:

$$V(\omega) = \frac{f_0\,\omega}{\sqrt{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}} = \frac{f_0}{\sqrt{\left(\frac{\omega_0^2}{\omega} - \omega\right)^2 + 4\beta^2}}. \tag{118}$$

The only ω-dependence left is the term $\left(\frac{\omega_0^2}{\omega} - \omega\right)^2 \geq 0$, which vanishes iff $\omega = \omega_0$: velocity resonance sits *exactly* at ω_0 for any damping. The time-averaged power delivered by the drive,

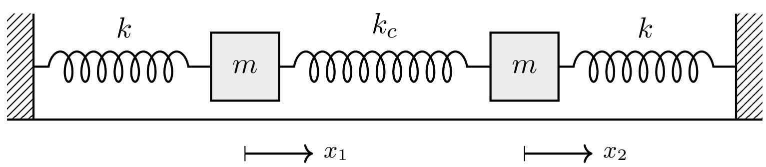
$$\langle P \rangle = \big\langle F(t)\, \dot{x}(t) \big\rangle = \big\langle F_0 \cos\omega t \cdot \big(-\omega A \sin(\omega t - \delta)\big) \big\rangle = \tfrac{1}{2} F_0\, \omega A \sin\delta, \tag{119}$$

(using $\sin(\omega t - \delta) = \sin\omega t \cos\delta - \cos\omega t \sin\delta$, $\langle \cos\omega t \sin\omega t \rangle = 0$, $\langle \cos^2\omega t \rangle = \frac{1}{2}$) is likewise maximized at ω_0, where $\sin\delta = 1$ and V peaks simultaneously. In steady state this input power exactly balances the dissipation $\langle b\dot{x}^2 \rangle$.

6 Coupled Harmonic Oscillators

6.1 Physical setup and equations of motion

Two identical blocks of mass m slide on a frictionless surface between two walls. Each block is attached to its wall by a spring of stiffness k, and the two blocks are connected to each other by a *coupling spring* of stiffness k_c. Let x_1, x_2 be the displacements of blocks 1, 2 from their respective equilibrium positions (both measured positive to the right).



Free-body analysis (the coupling spring is where sign errors happen, so we do it slowly). The *extension* of the coupling spring beyond its relaxed length is

$$\Delta_c = x_2 - x_1 \tag{120}$$

(block 2 moving right stretches it; block 1 moving right compresses it). A stretched spring pulls its two ends toward each other:

- on block 1 it pulls to the *right*: force $+k_c(x_2 - x_1)$;
- on block 2 it pulls to the *left*: force $-k_c(x_2 - x_1)$.

(Check with $\Delta_c < 0$, compressed: the spring pushes the blocks apart, and both formulas flip sign correctly. Also note the two forces are equal and opposite — Newton's third law is built in.) The wall springs contribute $-kx_1$ and $-kx_2$ as before. Newton's second law for each block:

$$\boxed{\begin{aligned} m\ddot{x}_1 &= -kx_1 + k_c\,(x_2 - x_1), \\ m\ddot{x}_2 &= -kx_2 - k_c\,(x_2 - x_1). \end{aligned}} \tag{121}$$

The equations are *coupled*: $\ddot{x}_1$ depends on x_2 and vice versa. Neither equation can be solved alone.

Convention 6.1 (Coupling frequency)**.**

$$\boxed{\omega_c^2 \equiv \frac{k_c}{m}} \qquad \text{(and as before } \omega_0^2 = k/m\text{).} \tag{122}$$

6.2 Matrix form

Divide (121) by m and collect:

$$\begin{aligned} \ddot{x}_1 &= -(\omega_0^2 + \omega_c^2)\, x_1 + \omega_c^2\, x_2, \\ \ddot{x}_2 &= \quad\ \omega_c^2\, x_1 - (\omega_0^2 + \omega_c^2)\, x_2. \end{aligned} \tag{123}$$

Introduce the column vector $\vec{x} = \binom{x_1}{x_2}$; then

$$\boxed{\ddot{\vec{x}} = -\,\mathsf{A}\,\vec{x}, \qquad \mathsf{A} = \begin{pmatrix} \omega_0^2 + \omega_c^2 & -\omega_c^2 \\ -\omega_c^2 & \omega_0^2 + \omega_c^2 \end{pmatrix}} \tag{124}$$

This is formally the SHO equation $\ddot{x} = -\omega_0^2 x$ with the scalar ω_0^2 promoted to the real symmetric matrix A. The strategy is clear: find the directions in which A acts like a scalar — its eigenvectors — and the system will fall apart into independent SHOs.

6.3 Normal modes as an eigenvalue problem

Try a solution in which *both* coordinates oscillate at one common frequency with a fixed amplitude ratio — the same exponential ansatz as always, now vector-valued:

$$\tilde{\vec{x}}(t) = \vec{v}\,\mathrm{e}^{\mathrm{i}\omega t}, \qquad \vec{v} \in \mathbb{R}^2 \text{ constant} \tag{125}$$

(complex stand-in as usual; the physical motion is its real part $\vec{v}\cos\omega t$). Substituting into (124):

$$-\omega^2\,\vec{v}\,\mathrm{e}^{\mathrm{i}\omega t} = -\mathsf{A}\,\vec{v}\,\mathrm{e}^{\mathrm{i}\omega t} \quad \Longleftrightarrow \quad \boxed{\mathsf{A}\,\vec{v} = \omega^2\,\vec{v}} \tag{126}$$

— an eigenvalue problem: the allowed frequencies squared are the eigenvalues of A, and the corresponding motion patterns are its eigenvectors. Nontrivial $\vec{v}$ exists iff

$$\det\big(\mathsf{A} - \omega^2\,\mathbb{1}\big) = 0. \tag{127}$$

Compute, abbreviating $s \equiv \omega_0^2 + \omega_c^2$:

$$\det\begin{pmatrix} s - \omega^2 & -\omega_c^2 \\ -\omega_c^2 & s - \omega^2 \end{pmatrix} = (s - \omega^2)^2 - \omega_c^4 = 0 \quad \Longleftrightarrow \quad s - \omega^2 = \pm\,\omega_c^2. \tag{128}$$

Hence the two *normal frequencies*:

$$\boxed{\omega_1^2 = s - \omega_c^2 = \omega_0^2, \qquad \omega_2^2 = s + \omega_c^2 = \omega_0^2 + 2\omega_c^2} \tag{129}$$

Eigenvectors: for $\omega^2 = \omega_1^2$, the equation $(\mathsf{A} - \omega_1^2\mathbb{1})\vec{v} = 0$ reads

$$\begin{pmatrix} \omega_c^2 & -\omega_c^2 \\ -\omega_c^2 & \omega_c^2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v_1 = v_2 \implies \vec{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}; \tag{130}$$

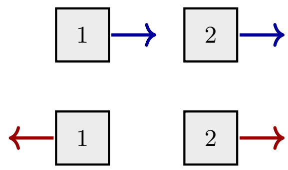
for $\omega^2 = \omega_2^2$,

$$\begin{pmatrix} -\omega_c^2 & -\omega_c^2 \\ -\omega_c^2 & -\omega_c^2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \implies v_1 = -v_2 \implies \vec{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{131}$$

Physical interpretation.

Mode 1 ($\omega_1 = \omega_0$):

Mode 2 ($\omega_2 = \sqrt{\omega_0^2 + 2\omega_c^2}$):



in-phase: coupling spring keeps its length, exerts no force $\Rightarrow$ each mass oscillates as if alone, at ω_0.

anti-phase: coupling spring is stretched/compressed maximally, adds restoring force $\Rightarrow$ stiffer, faster.

That mode 1 comes out at exactly ω_0, untouched by k_c, and that mode 2 sees the coupling spring *twice* ($2\omega_c^2$: each mass moves against a spring whose other end is moving the opposite way, doubling the relative displacement) are both visible in the picture before any algebra.

6.4 Decoupling by a change of coordinates

The eigenvector directions suggest new coordinates. Define the *normal coordinates*

$$\boxed{q_1 \equiv \frac{x_1 + x_2}{2}, \qquad q_2 \equiv \frac{x_1 - x_2}{2}} \qquad \Longleftrightarrow \qquad x_1 = q_1 + q_2, \quad x_2 = q_1 - q_2. \tag{132}$$

(The normalization $\frac{1}{2}$ is a convention; it makes the inverse map coefficient-free.) Now *add* and *subtract* the two equations (121) divided by m:

$$\begin{aligned} \text{sum:} \qquad & \ddot{x}_1 + \ddot{x}_2 = -\omega_0^2(x_1 + x_2) + \omega_c^2(x_2 - x_1) - \omega_c^2(x_2 - x_1) = -\omega_0^2\,(x_1 + x_2), \\ \text{difference:} \qquad & \ddot{x}_1 - \ddot{x}_2 = -\omega_0^2(x_1 - x_2) + \omega_c^2(x_2 - x_1) + \omega_c^2(x_2 - x_1) = -(\omega_0^2 + 2\omega_c^2)\,(x_1 - x_2). \end{aligned} \tag{133}$$

Dividing by 2, the equations for q_1, q_2 close on themselves:

$$\boxed{\ddot{q}_1 + \omega_1^2\, q_1 = 0, \qquad \ddot{q}_2 + \omega_2^2\, q_2 = 0} \qquad \big(\omega_1^2 = \omega_0^2,\ \ \omega_2^2 = \omega_0^2 + 2\omega_c^2\big). \tag{134}$$

The coupled pair has become two *independent* simple harmonic oscillators — everything from §2 applies verbatim to each. In matrix language: the change of basis to eigenvector coordinates has diagonalized A,

$$\mathsf{A} = S \begin{pmatrix} \omega_1^2 & 0 \\ 0 & \omega_2^2 \end{pmatrix} S^{-1}, \qquad S = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \tag{135}$$

and in the new coordinates the matrix equation $\ddot{\vec{x}} = -\mathsf{A}\vec{x}$ falls apart row by row.

6.5 General solution and initial conditions

Each normal coordinate carries the full SHO solution (16):

$$q_1(t) = A_1 \cos(\omega_1 t - \phi_1), \qquad q_2(t) = A_2 \cos(\omega_2 t - \phi_2), \tag{136}$$

so the general motion of the blocks is the superposition

$$\boxed{\begin{aligned} x_1(t) &= A_1 \cos(\omega_1 t - \phi_1) + A_2 \cos(\omega_2 t - \phi_2), \\ x_2(t) &= A_1 \cos(\omega_1 t - \phi_1) - A_2 \cos(\omega_2 t - \phi_2), \end{aligned}} \tag{137}$$

with four constants $(A_1, \phi_1, A_2, \phi_2)$ matching the four initial data $x_1(0), x_2(0), \dot{x}_1(0), \dot{x}_2(0)$. Generic motion is *not* periodic (unless ω_2/ω_1 is rational): only the pure modes oscillate at a single frequency.

6.6 Worked example: energy exchange (beats again)

Displace block 1 by a and release everything from rest:

$$x_1(0) = a, \quad x_2(0) = 0, \quad \dot{x}_1(0) = \dot{x}_2(0) = 0. \tag{138}$$

In normal coordinates: $q_1(0) = \frac{a}{2}$, $q_2(0) = \frac{a}{2}$, $\dot{q}_1(0) = \dot{q}_2(0) = 0$. Each mode starts at rest, so ((11) with $v_0 = 0$) $q_i(t) = q_i(0)\cos\omega_i t$:

$$q_1 = \frac{a}{2}\cos\omega_1 t, \qquad q_2 = \frac{a}{2}\cos\omega_2 t. \tag{139}$$

Transforming back:

$$x_1(t) = \frac{a}{2}\big(\cos\omega_1 t + \cos\omega_2 t\big), \qquad x_2(t) = \frac{a}{2}\big(\cos\omega_1 t - \cos\omega_2 t\big). \tag{140}$$

Apply the product identities (sum and difference forms; the difference form was derived in §4.1, the sum form follows the same way from adding the two addition formulas):

$$\boxed{x_1(t) = a\,\cos\!\Big(\tfrac{\omega_2-\omega_1}{2}t\Big)\cos\!\Big(\tfrac{\omega_1+\omega_2}{2}t\Big), \qquad x_2(t) = a\,\sin\!\Big(\tfrac{\omega_2-\omega_1}{2}t\Big)\sin\!\Big(\tfrac{\omega_1+\omega_2}{2}t\Big)} \tag{141}$$

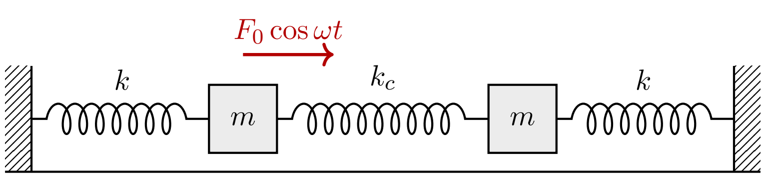
For weak coupling ($\omega_c \ll \omega_0$, so $\omega_2 - \omega_1 \approx \omega_c^2/\omega_0$ is small), both blocks oscillate fast at $\approx \omega_0$ inside *slow, complementary* envelopes: $|\cos|$ for block 1 and $|\sin|$ for block 2 (Fig. 6). When block 1's envelope is at a node, block 2's is at an antinode — the energy sloshes back and forth through the coupling spring with period $2\pi/(\omega_2 - \omega_1)$, complete transfer occurring precisely because the two modes were excited with equal amplitude.

Remark 6.2 (Why this always works). Nothing here depended on luck. A is real and symmetric (this traces back to Newton's third law acting through the coupling spring), and every real symmetric matrix is diagonalizable by an orthogonal change of basis with real eigenvalues [stated; the spectral theorem of linear algebra]; positive-definiteness of A (all spring constants positive) makes the eigenvalues positive, so each is legitimately an ω_i^2. The recipe — write $\ddot{\vec{x}} = -\mathsf{A}\vec{x}$, find eigenpairs, decouple — therefore extends to N masses, unequal parameters, and beyond. For unequal masses one writes $\mathsf{M}\ddot{\vec{x}} = -\mathsf{K}\vec{x}$ and diagonalizes $\mathsf{M}^{-1}\mathsf{K}$ (symmetric with respect to the mass-weighted inner product) [stated; same theorem, adapted inner product]; we will not need this generality here.

7 The Forced Coupled Oscillator

7.1 Setup

Drive *block 1 only* with the force $F_0 \cos\omega t$; block 2 is driven purely through the coupling. We treat the undamped system in the main text (formulas stay exact and transparent); the figures overlay a small damping to show what a real measurement would produce.



Adding $f_0 \cos\omega t$ (with $f_0 = F_0/m$) to the first equation of motion:

$$\boxed{\ddot{\vec{x}} + \mathsf{A}\,\vec{x} = f_0\cos\omega t\,\vec{e}_1, \qquad \vec{e}_1 = \begin{pmatrix}1\\0\end{pmatrix}} \tag{142}$$

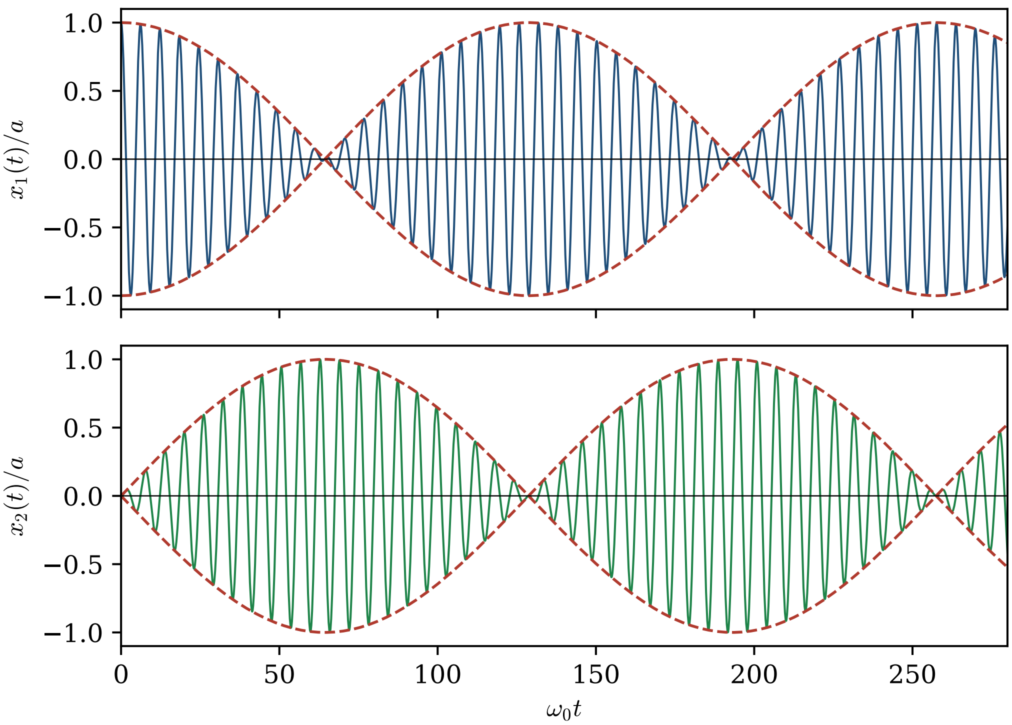


Figure 6: Weakly coupled oscillators ($\omega_c^2 = 0.05\,\omega_0^2$), block 1 displaced and released from rest. The envelopes (dashed) are complementary: energy migrates fully from block 1 to block 2 and back.

7.2 Reduction to two independent forced oscillators

Pass to the normal coordinates of §6.4. Add and subtract the two component equations of (142) (only the first has a drive) and divide by 2:

$$\begin{aligned} \tfrac{1}{2}(\mathrm{eq}_1 + \mathrm{eq}_2): \qquad & \ddot{q}_1 + \omega_1^2\, q_1 = \tfrac{f_0}{2}\cos\omega t, \\ \tfrac{1}{2}(\mathrm{eq}_1 - \mathrm{eq}_2): \qquad & \ddot{q}_2 + \omega_2^2\, q_2 = \tfrac{f_0}{2}\cos\omega t. \end{aligned} \tag{143}$$

Interpretation: a force applied to block 1 alone decomposes as $\vec{e}_1 = \frac{1}{2}\binom{1}{1} + \frac{1}{2}\binom{1}{-1}$ — it pushes on *both* modes, each with half strength. Each line of (143) is exactly Case A of §4.1, already solved: the steady-state (particular) responses are

$$q_{1,p}(t) = \frac{f_0/2}{\omega_1^2 - \omega^2}\,\cos\omega t, \qquad q_{2,p}(t) = \frac{f_0/2}{\omega_2^2 - \omega^2}\,\cos\omega t. \tag{144}$$

Every piece of machinery built in §4 is being reused; the only new step was the change of basis.

7.3 Response of the physical coordinates; two resonances

Transform back ($x_1 = q_1 + q_2$, $x_2 = q_1 - q_2$):

$$x_{1,p}(t) = X_1(\omega)\cos\omega t, \qquad x_{2,p}(t) = X_2(\omega)\cos\omega t, \tag{145}$$

with the response functions

$$\boxed{X_1(\omega) = \frac{f_0}{2}\left[\frac{1}{\omega_1^2-\omega^2} + \frac{1}{\omega_2^2-\omega^2}\right], \qquad X_2(\omega) = \frac{f_0}{2}\left[\frac{1}{\omega_1^2-\omega^2} - \frac{1}{\omega_2^2-\omega^2}\right]} \tag{146}$$

Both responses diverge at $\omega = \omega_1$ *and* at $\omega = \omega_2$: driving at either normal frequency resonantly excites the corresponding mode, and both blocks participate in each mode. *The resonances of a coupled system sit at its normal frequencies, not at the frequencies of its parts.* With small damping the divergences become the two finite peaks in Fig. 7.

7.4 Antiresonance

Between the two resonances, something less familiar happens to the *driven* block. Combine the two fractions in X_1:

$$X_1(\omega) = \frac{f_0}{2}\cdot\frac{(\omega_2^2-\omega^2)+(\omega_1^2-\omega^2)}{(\omega_1^2-\omega^2)(\omega_2^2-\omega^2)} = \frac{f_0}{2}\cdot\frac{\omega_1^2+\omega_2^2-2\omega^2}{(\omega_1^2-\omega^2)(\omega_2^2-\omega^2)}. \tag{147}$$

The numerator vanishes at

$$\omega_a^2 = \frac{\omega_1^2+\omega_2^2}{2} = \frac{\omega_0^2+(\omega_0^2+2\omega_c^2)}{2} \quad\Longrightarrow\quad \boxed{\omega_a = \sqrt{\omega_0^2+\omega_c^2}} \tag{148}$$

which lies strictly between ω_1 and ω_2. At this *antiresonance* frequency, the block being pushed *does not move at all* (in steady state), while the undriven block does. Indeed, evaluating X_2 at ω_a: since $\omega_1^2-\omega_a^2 = -\omega_c^2$ and $\omega_2^2-\omega_a^2 = +\omega_c^2$,

$$X_2(\omega_a) = \frac{f_0}{2}\left[\frac{1}{-\omega_c^2} - \frac{1}{\omega_c^2}\right] = -\frac{f_0}{\omega_c^2} \neq 0. \tag{149}$$

How can the driven block stand still? Check force balance on block 1 directly. If $x_1 \equiv 0$ and $x_2 = -\frac{f_0}{\omega_c^2}\cos\omega t$, the coupling spring exerts on block 1 the force

$$k_c\,(x_2-x_1) = k_c\,x_2 = m\omega_c^2\cdot\left(-\frac{f_0}{\omega_c^2}\right)\cos\omega t = -\,mf_0\cos\omega t = -\,F_0\cos\omega t, \tag{150}$$

which cancels the applied drive *exactly*, at every instant. Block 2 has spontaneously arranged itself in antiphase with just the right amplitude to shield block 1 from its own drive. (Consistency: with $x_1 \equiv 0$, block 2 is a free oscillator bound by springs k and k_c in parallel, natural frequency $\sqrt{(k+k_c)/m} = \sqrt{\omega_0^2+\omega_c^2} = \omega_a$ — the antiresonance of block 1 is the natural frequency of block 2 *with block 1 clamped*.)

This is the operating principle of the *dynamic vibration absorber* (tuned mass damper): to protect a structure from vibration at a known frequency, attach an auxiliary mass–spring system tuned to that frequency, and it absorbs the motion. The skyscraper pendulum is this calculation at scale.

Remark 7.1 (On the undamped idealization). In the undamped system the homogeneous solutions (free oscillations of the two modes) never decay, so strictly there is no unique long-time steady state: the particular solution above is *the* steady state only in the presence of some damping, however small, which kills the transients exactly as in §4.2.2 — mode by mode. This is the sense in which the undamped formulas (146) should be read: as the $\beta \to 0^+$ limit of the damped response, which is what the dashed curves of Fig. 7 converge to.

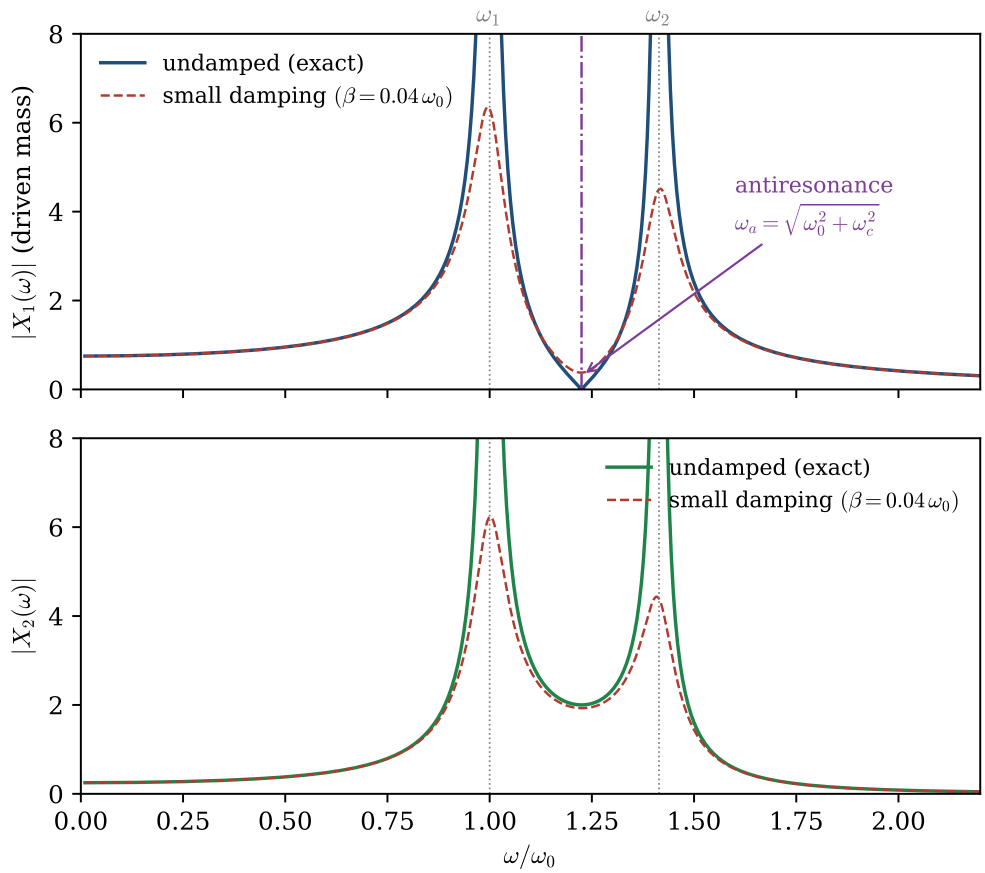


Figure 7: Steady-state amplitudes versus driving frequency for the forced coupled pair ($\omega_c^2 = 0.5\,\omega_0^2$, so $\omega_1 = \omega_0$, $\omega_2 = \sqrt{2}\,\omega_0$, $\omega_a = \sqrt{1.5}\,\omega_0$). Solid: exact undamped response (146) (clipped near the divergences). Dashed: the same system with small damping on both blocks, computed numerically — peaks become finite, the antiresonance zero of $|X_1|$ becomes a sharp dip.

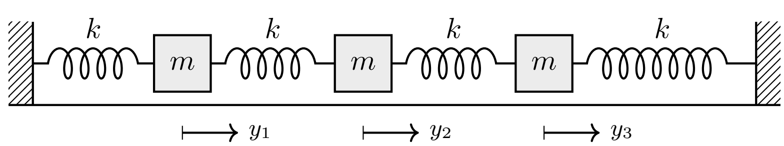
8 The N-Mass Chain

The two-mass system generalizes to a chain of N identical masses. From this section on we make two adjustments. First, *all* springs are taken identical with stiffness k (the special case $k_c = k$ of §6); $\omega_0^2 = k/m$ as always. Second, we rename the displacements $x_j \to y_j$, reserving the letter x for *position along the chain* — we will need it in the continuum limit of §9.

We build the general case by writing out $N = 3$, 4, 5 explicitly and letting the pattern emerge.

8.1 Three, four, five masses

$N = 3$. Three masses, four springs, walls at both ends:



Free-body analysis is the same computation as in §6, mass by mass. Each mass feels its two neighboring springs; the spring between sites j and $j+1$ has extension $y_{j+1} - y_j$ and pulls mass j with force $+k(y_{j+1} - y_j)$ and mass $j+1$ with force $-k(y_{j+1} - y_j)$. The end masses feel a wall spring, e.g. $-ky_1$ on mass 1. Dividing by m:

$$\begin{aligned} \ddot{y}_1 &= \omega_0^2 \big(\quad - 2y_1 + y_2 \big), \\ \ddot{y}_2 &= \omega_0^2 \big(y_1 - 2y_2 + y_3 \big), \\ \ddot{y}_3 &= \omega_0^2 \big(y_2 - 2y_3 \quad\quad \big), \end{aligned} \qquad \mathsf{A}_3 = \omega_0^2 \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}. \tag{151}$$

$N = 4$**.** Identical reasoning, one more interior mass:

$$\begin{aligned} \ddot{y}_1 &= \omega_0^2 \,(-2y_1 + y_2), \\ \ddot{y}_2 &= \omega_0^2 \,(y_1 - 2y_2 + y_3), \\ \ddot{y}_3 &= \omega_0^2 \,(y_2 - 2y_3 + y_4), \\ \ddot{y}_4 &= \omega_0^2 \,(y_3 - 2y_4), \end{aligned} \qquad \mathsf{A}_4 = \omega_0^2 \begin{pmatrix} 2 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{pmatrix}. \tag{152}$$

$N = 5$**.**

$$\begin{aligned} \ddot{y}_1 &= \omega_0^2 \,(-2y_1 + y_2), \\ \ddot{y}_2 &= \omega_0^2 \,(y_1 - 2y_2 + y_3), \\ \ddot{y}_3 &= \omega_0^2 \,(y_2 - 2y_3 + y_4), \\ \ddot{y}_4 &= \omega_0^2 \,(y_3 - 2y_4 + y_5), \\ \ddot{y}_5 &= \omega_0^2 \,(y_4 - 2y_5), \end{aligned} \qquad \mathsf{A}_5 = \omega_0^2 \begin{pmatrix} 2 & -1 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & -1 & 2 \end{pmatrix}. \tag{153}$$

The pattern. Three observations, each visible above:

1. Every *interior* mass obeys the same equation, $\ddot{y}_j = \omega_0^2 \,(y_{j-1} - 2y_j + y_{j+1})$: the chain is translation-invariant in its bulk.

2. The *end* equations look truncated — but they are the interior equation in disguise. Introduce *phantom sites* $j = 0$ and $j = N+1$ at the walls and declare

$$\boxed{y_0(t) \equiv 0, \qquad y_{N+1}(t) \equiv 0} \qquad \text{(fixed-wall boundary conditions).} \tag{154}$$

   Then $\ddot{y}_1 = \omega_0^2(y_0 - 2y_1 + y_2)$ reproduces the first row exactly, and likewise at the far end. One rule, all rows.

3. The matrix is *tridiagonal*: 2's on the diagonal, -1's on the two adjacent diagonals, zeros elsewhere — each mass talks only to its neighbors.

8.2 The general chain

The induction is now immediate: for any N, the equations of motion are

$$\boxed{\ddot{y}_j = \omega_0^2 \big(y_{j-1} - 2y_j + y_{j+1} \big), \qquad j = 1, \ldots, N, \qquad y_0 = y_{N+1} = 0} \tag{155}$$

or in matrix form $\ddot{\vec{y}} = -\mathsf{A}_N\,\vec{y}$ with

$$\mathsf{A}_N = \omega_0^2 \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix}_{N\times N} \qquad \text{(blank entries are 0).} \tag{156}$$

A_N is real symmetric, so the program of §6 applies: N normal modes with real frequencies, mutually decoupled. For $N = 2$ the determinant was a quadratic; for general N we need a better idea than expanding an $N \times N$ determinant.

8.3 Normal modes: an exponential ansatz in space

Substituting $\tilde{\vec{y}}(t) = \vec{v}\,\mathrm{e}^{\mathrm{i}\omega t}$ as in §6, the eigenvalue problem $\mathsf{A}_N\vec{v} = \omega^2\vec{v}$ reads, component by component,

$$\omega_0^2\left(2v_j - v_{j-1} - v_{j+1}\right) = \omega^2\, v_j, \qquad j = 1, \ldots, N, \qquad v_0 = v_{N+1} = 0. \tag{157}$$

Look at what this is: a *linear recursion in j with constant coefficients* — the same structure our differential equations had in t, with the site index playing the role of time. In §2.4 the winning move for such equations was the exponential ansatz; try its discrete counterpart, a geometric sequence in the site index:

$$v_j = \mathrm{e}^{\mathrm{i}j\theta}, \qquad \theta \text{ constant.} \tag{158}$$

Substitute into the interior equation (157) and divide by $\mathrm{e}^{\mathrm{i}j\theta} \neq 0$:

$$\omega_0^2\bigl(2 - \mathrm{e}^{-\mathrm{i}\theta} - \mathrm{e}^{+\mathrm{i}\theta}\bigr) = \omega^2 \quad \Longrightarrow \quad \omega^2 = 2\omega_0^2\left(1 - \cos\theta\right), \tag{159}$$

using $\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta} = 2\cos\theta$ (23). With the half-angle identity $1 - \cos\theta = 2\sin^2(\theta/2)$:

$$\boxed{\omega^2 = 4\omega_0^2 \sin^2\frac{\theta}{2}} \tag{160}$$

This is the *dispersion relation* of the chain: it converts a spatial oscillation rate θ (radians of phase per site) into a temporal frequency ω. Note what has and has not been fixed: *every* θ solves the interior recursion — the analog of "every root of the characteristic equation gives a solution." It is the *boundary conditions* that select the allowed values, exactly as initial conditions selected constants before.

Imposing the boundaries. Since ω^2 in (160) depends only on $\cos\theta$, the two sequences $\mathrm{e}^{\pm\mathrm{i}j\theta}$ carry the *same* frequency, and any combination of them still solves the interior recursion. The combination

$$v_j = \frac{\mathrm{e}^{\mathrm{i}j\theta} - \mathrm{e}^{-\mathrm{i}j\theta}}{2\mathrm{i}} = \sin(j\theta) \tag{161}$$

satisfies the left boundary automatically: $v_0 = \sin 0 = 0$. The right boundary then quantizes θ:

$$v_{N+1} = \sin\bigl((N+1)\theta\bigr) = 0 \quad \Longleftrightarrow \quad (N+1)\theta = n\pi \quad \Longleftrightarrow \quad \boxed{\theta_n = \frac{n\pi}{N+1}} \tag{162}$$

with n an integer. Which integers give genuinely distinct modes? $n = 0$ and $n = N + 1$ make $v_j = \sin(j\theta_n)$ vanish identically (all sites at a node); $n' = 2(N+1) - n$ gives $\sin(j\theta_{n'}) = -\sin(j\theta_n)$, the same mode up to overall sign; larger n repeats periodically. The distinct modes are exactly

$$n = 1, 2, \ldots, N, \tag{163}$$

— N of them, matching the dimension of the problem, so we have found *all* the normal modes without ever expanding a determinant:

$$\boxed{v_j^{(n)} = \sin\Big(\frac{jn\pi}{N+1}\Big), \qquad \omega_n = 2\omega_0 \sin\Big(\frac{n\pi}{2(N+1)}\Big), \qquad n = 1, \ldots, N} \tag{164}$$

Consistency check against $N = 2$. Formula (164) with $N = 2$ (so $\theta_n = n\pi/3$):

$$\omega_1^2 = 4\omega_0^2 \sin^2\frac{\pi}{6} = 4\omega_0^2 \cdot \tfrac{1}{4} = \omega_0^2, \qquad \vec{v}^{(1)} = \Big(\sin\tfrac{\pi}{3},\, \sin\tfrac{2\pi}{3}\Big) = \tfrac{\sqrt{3}}{2}\,(1,\, 1), \tag{165}$$

$$\omega_2^2 = 4\omega_0^2 \sin^2\frac{\pi}{3} = 4\omega_0^2 \cdot \tfrac{3}{4} = 3\,\omega_0^2, \qquad \vec{v}^{(2)} = \Big(\sin\tfrac{2\pi}{3},\, \sin\tfrac{4\pi}{3}\Big) = \tfrac{\sqrt{3}}{2}\,(1,\, -1). \tag{166}$$

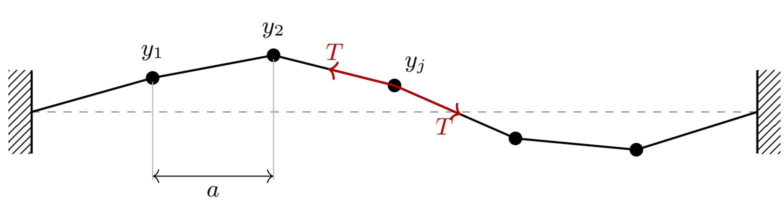
Compare §6 with $k_c = k$: there $\omega_1^2 = \omega_0^2$ and $\omega_2^2 = \omega_0^2 + 2\omega_c^2 = 3\omega_0^2$, with modes $(1, 1)$ and $(1, -1)$. Exact agreement.

Remark 8.1 (Reading the mode shapes). Low n: long, smooth waves; all the low-frequency physics. High n: rapidly alternating displacements, approaching the zigzag pattern at $\theta = \pi$ where each mass moves opposite to its neighbors and every spring is maximally worked — hence the maximum frequency $2\omega_0$. The chain has a *fastest* oscillation; there is no pattern it can support above $2\omega_0$. A continuous string will have no such ceiling, and comparing the two limits is instructive: the ceiling is the memory of the graininess.

9 The Continuum Limit: From Chain to Rope

9.1 The rope as a bead chain

The chain of §8 was longitudinal (masses displace along the line of springs). The classic continuum system — a stretched rope or string — is *transverse*, and it turns out to be the same mathematics. Consider N beads of mass m, spaced a apart on a massless string stretched to tension T between fixed walls; let y_j now denote the small *transverse* displacement of bead j.



Transverse force on bead j. The string segment from bead j to bead $j+1$ makes a small angle φ_{j+} with the horizontal, with $\tan\varphi_{j+} = (y_{j+1} - y_j)/a$. It pulls bead j along itself with magnitude T; the transverse component is

$$T\sin\varphi_{j+} \approx T\tan\varphi_{j+} = \frac{T}{a}\,(y_{j+1} - y_j) \tag{167}$$

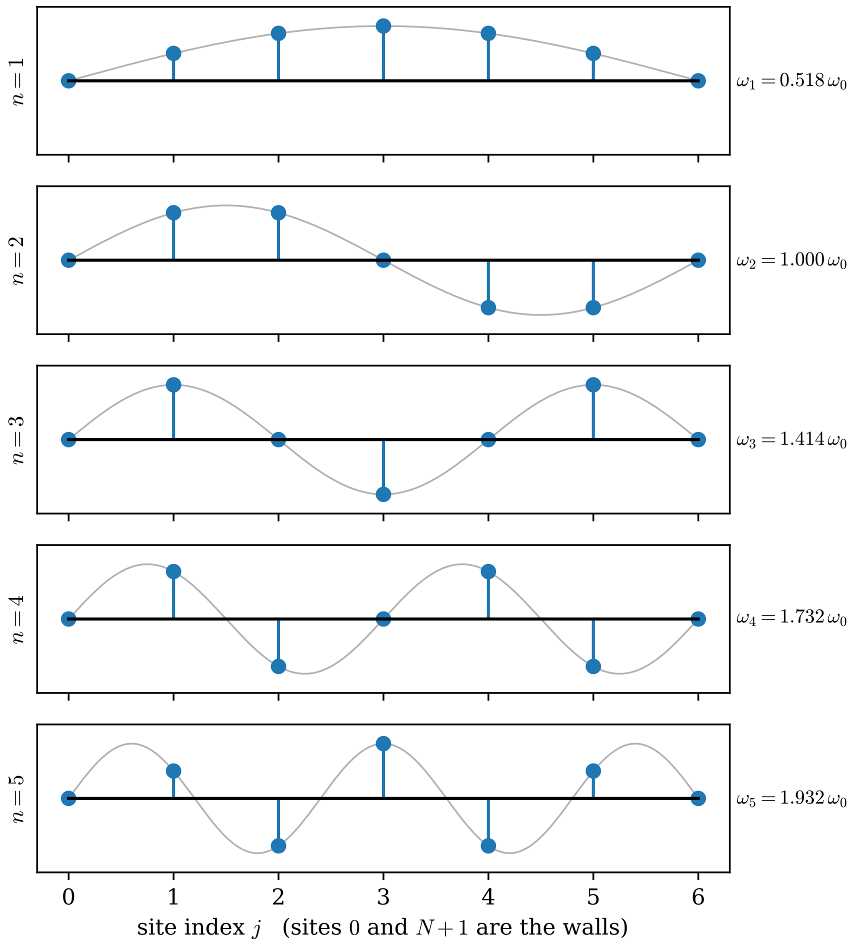


Figure 8: The five normal modes of the $N = 5$ chain. Dots: the mode vector components $v_j^{(n)} = \sin(jn\pi/6)$ at the sites (the phantom wall sites $j = 0, 6$ included). Gray curve: the interpolating function $\sin(n\pi x/6)$ — the masses sample a smooth standing wave, a fact that becomes the whole story in §9.

(small angles: $\sin\varphi \approx \tan\varphi \approx \varphi$; to this order the tension itself is unchanged). The left segment likewise contributes $\frac{T}{a}(y_{j-1} - y_j)$. Newton's second law transversely:

$$m\ddot{y}_j = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big). \tag{168}$$

This is exactly (155) with the substitution

$$\boxed{k \longrightarrow \frac{T}{a}, \qquad \omega_0^2 = \frac{T}{ma}} \tag{169}$$

Everything from §8 — modes, dispersion, the $2\omega_0$ ceiling — transfers verbatim. From here on we work with the beads.

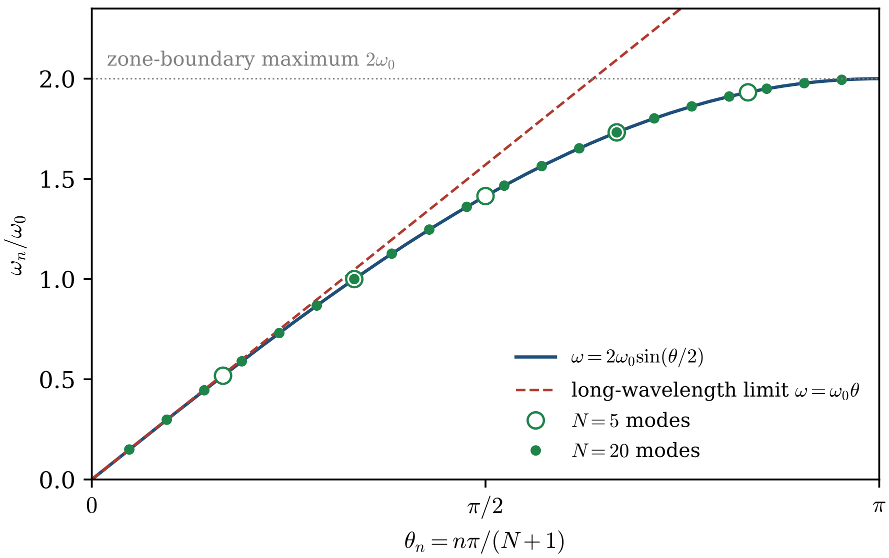


Figure 9: The dispersion relation (160). Allowed modes sit at $\theta_n = n\pi/(N+1)$: shown for $N = 5$ (open circles) and $N = 20$ (filled dots) — as N grows the modes fill in the curve. Dashed: the long-wavelength tangent $\omega = \omega_0\theta$, the seed of the continuum limit. The curve saturates at $2\omega_0$ at $\theta = \pi$, where adjacent masses move in exact opposition — the stiffest, fastest pattern the chain supports.

9.2 Taking the limit

Refine the chain: more beads, closer together, keeping the *physical* rope fixed. Precisely, let

$$a \to 0, \qquad N \to \infty, \qquad \text{holding fixed} \quad L = (N+1)\,a \;\; \text{(length)}, \quad \mu = \frac{m}{a} \;\; \text{(mass per length)}, \quad T. \tag{170}$$

Each bead's mass $m = \mu a \to 0$; the discrete displacements merge into a field: postulate a smooth function $y(x,t)$ interpolating them,

$$y_j(t) = y(x_j, t), \qquad x_j = j\,a. \tag{171}$$

The neighbors sit at $x_j \pm a$; Taylor-expand in the small spacing:

$$y_{j\pm1} = y(x_j \pm a, t) = y \;\pm\; a\,\frac{\partial y}{\partial x} \;+\; \frac{a^2}{2}\frac{\partial^2 y}{\partial x^2} \;\pm\; \frac{a^3}{6}\frac{\partial^3 y}{\partial x^3} \;+\; O(a^4), \tag{172}$$

all derivatives evaluated at (x_j, t). Add the two expansions: the odd powers of a cancel in pairs,

$$y_{j+1} + y_{j-1} - 2y_j = a^2\,\frac{\partial^2 y}{\partial x^2} + O(a^4). \tag{173}$$

(The combination $y_{j-1} - 2y_j + y_{j+1}$ that has followed us since the $N = 3$ matrix is a *second difference*: it was the discrete second derivative all along, and the tridiagonal matrix was the second-derivative operator in disguise.) Substitute into (168) and divide by $m = \mu a$:

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu a^2}\left(a^2\frac{\partial^2 y}{\partial x^2} + O(a^4)\right) = \frac{T}{\mu}\,\frac{\partial^2 y}{\partial x^2} + O(a^2). \tag{174}$$

As $a \to 0$ the correction vanishes and, defining the constant

$$\boxed{c^2 \equiv \frac{T}{\mu}} \qquad [c] = \sqrt{\frac{[\text{force}]}{[\text{mass}]/[\text{length}]}} = \frac{[\text{length}]}{[\text{time}]}, \tag{175}$$

we obtain the *wave equation*:

$$\boxed{\frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}} \qquad y(0,t) = y(L,t) = 0. \tag{176}$$

The boundary conditions are the phantom sites, carried into the limit: $y_0 = 0$ at $x_0 = 0$ and $y_{N+1} = 0$ at $x_{N+1} = L$. (For the longitudinal spring chain the identical computation gives $c^2 = ka^2/m$; only the expression for c in terms of microscopic parameters changes, not the equation.)

9.3 What happens to the modes

Track each object of §8.3 through the limit. Define the *wavenumber* carried by mode n,

$$k_n \equiv \frac{\theta_n}{a} = \frac{n\pi}{(N+1)a} = \frac{n\pi}{L} \qquad [\text{radians of phase per unit length}], \tag{177}$$

which stays *finite* in the limit — $\theta_n \to 0$ and $a \to 0$ at matched rates for fixed n.

Mode shapes.

$$v_j^{(n)} = \sin\!\Big(\frac{jn\pi}{N+1}\Big) = \sin\!\Big(\frac{n\pi\,(ja)}{L}\Big) = \sin\!\big(k_n\,x_j\big) \quad\longrightarrow\quad \sin\!\Big(\frac{n\pi x}{L}\Big) : \tag{178}$$

the sampled sinusoids of Fig. 8 become the standing waves of a string — n counts the antinodes.

Frequencies. Expand the dispersion relation (160) for small argument, $\sin\varepsilon = \varepsilon + O(\varepsilon^3)$:

$$\omega_n = 2\omega_0 \sin\!\Big(\frac{k_n a}{2}\Big) = 2\sqrt{\frac{T}{\mu a^2}}\,\Big(\frac{k_n a}{2} + O(a^3)\Big) = \sqrt{\frac{T}{\mu}}\;k_n + O(a^2) \quad\longrightarrow\quad c\,k_n, \tag{179}$$

where $\omega_0^2 = T/(ma) = T/(\mu a^2)$ was used. The dispersion relation *linearizes*:

$$\boxed{\omega_n = c\,k_n = \frac{n\pi c}{L}, \qquad n = 1, 2, 3, \ldots} \tag{180}$$

This is the harmonic series — overtones at integer multiples of the fundamental $\pi c/L$, the arithmetic that underlies music. On Fig. 9 this is the dashed tangent line at the origin: the continuum keeps only the long-wavelength foot of the chain's dispersion curve, where the graininess is invisible. The frequency ceiling $2\omega_0 = 2c/a$ escapes to infinity as $a \to 0$: the ideal string has arbitrarily fast modes, and any real string reveals its microscopic structure precisely where the ideal model puts none.

9.4 General solution and the Fourier expansion

Superposition of modes. Each standing wave, set oscillating at its own frequency, solves the wave equation: for any single n,

$$y_n(x,t) = \sin(k_n x)\big[A_n \cos\omega_n t + B_n \sin\omega_n t\big], \qquad \omega_n = c\,k_n, \tag{181}$$

satisfies (176) — verify in one line: ∂_t^2 produces a factor $-\omega_n^2$, $c^2\partial_x^2$ produces $-c^2k_n^2$, and these are equal by the dispersion relation; the boundary conditions hold term by term since $\sin(k_n \cdot 0) = \sin(k_n L) = 0$. The wave equation is linear and homogeneous, so superposition applies exactly as it has since §2.2, and the general solution is

$$\boxed{y(x,t) = \sum_{n=1}^{\infty} \sin\Big(\frac{n\pi x}{L}\Big)\big[A_n \cos\omega_n t + B_n \sin\omega_n t\big]} \tag{182}$$

— the continuum version of "general motion = sum over normal modes," now with infinitely many modes. In the chain this expansion was guaranteed by linear algebra: A_N is real symmetric, its eigenvectors span $\mathbb{R}^N$, so *any* initial configuration is a combination of modes. The new question in the continuum is whether that survives $N \to \infty$.

Initial data. Evaluate (182) and its time derivative at $t = 0$. If the string starts with shape $f(x)$ and velocity profile $g(x)$,

$$f(x) = \sum_{n=1}^{\infty} A_n \sin\Big(\frac{n\pi x}{L}\Big), \qquad g(x) = \sum_{n=1}^{\infty} B_n\,\omega_n \sin\Big(\frac{n\pi x}{L}\Big). \tag{183}$$

Fitting the motion to initial data therefore requires expanding *arbitrary* functions in the mode shapes. That this is possible — every sufficiently well-behaved f with $f(0) = f(L) = 0$ admits such an expansion — is *Fourier's theorem* [stated; the completeness proof belongs to analysis, not to this document], and the right-hand side of (183) is the *Fourier sine series* of f. What we *can* prove here is how to compute the coefficients, and that the answer is unique.

Orthogonality. The key computation. For integers $n, m \geq 1$, use the product-to-sum identity $\sin P \sin Q = \frac{1}{2}[\cos(P-Q) - \cos(P+Q)]$:

$$\int_0^L \sin\Big(\frac{n\pi x}{L}\Big)\sin\Big(\frac{m\pi x}{L}\Big)\mathrm{d}x = \frac{1}{2}\int_0^L \Big[\cos\Big(\frac{(n-m)\pi x}{L}\Big) - \cos\Big(\frac{(n+m)\pi x}{L}\Big)\Big]\mathrm{d}x. \tag{184}$$

For any nonzero integer p, $\int_0^L \cos(p\pi x/L)\,\mathrm{d}x = \big[\frac{L}{p\pi}\sin(p\pi x/L)\big]_0^L = 0$ since $\sin(p\pi) = 0$. If $n \neq m$ both cosine integrals vanish. If $n = m$ the first integrand is $\cos 0 = 1$, giving L, while the second still vanishes. Hence

$$\boxed{\int_0^L \sin\Big(\frac{n\pi x}{L}\Big)\sin\Big(\frac{m\pi x}{L}\Big)\mathrm{d}x = \frac{L}{2}\,\delta_{nm}} \tag{185}$$

The mode shapes are *orthogonal* — the continuum analog of the orthogonality of eigenvectors of the symmetric matrix A_N, with the integral playing the role of the dot product. (Indeed the chain modes obey the discrete identity $\sum_{j=1}^{N} \sin(j\theta_n)\sin(j\theta_m) = \frac{N+1}{2}\,\delta_{nm}$ [stated; it follows from the same product-to-sum identity applied to finite geometric sums, and nothing below depends on it]. Multiplying by the spacing a and letting $a \to 0$ turns the sum into the integral and $\frac{(N+1)a}{2}$ into $\frac{L}{2}$ — the two statements are one statement.)

Extracting the coefficients. Multiply the first equation of (183) by $\sin(m\pi x/L)$ and integrate over $[0, L]$; orthogonality kills every term of the sum except $n = m$:

$$\int_0^L f(x)\sin\Bigl(\frac{m\pi x}{L}\Bigr)\mathrm{d}x = \sum_{n=1}^{\infty} A_n \cdot \frac{L}{2}\,\delta_{nm} = \frac{L}{2}\,A_m, \tag{186}$$

so

$$\boxed{A_m = \frac{2}{L}\int_0^L f(x)\sin\Bigl(\frac{m\pi x}{L}\Bigr)\mathrm{d}x, \qquad B_m = \frac{2}{L\,\omega_m}\int_0^L g(x)\sin\Bigl(\frac{m\pi x}{L}\Bigr)\mathrm{d}x} \tag{187}$$

(the B_m formula obtained the same way from the second equation). Each mode's amplitude is read off by projecting the initial data onto that mode — exactly the operation "decompose the initial condition into normal coordinates" of §6.6, promoted to infinitely many coordinates. Uniqueness is automatic: the coefficients are *determined* by these integrals. Physically, the A_n of a plucked string decide how much of each overtone sounds — the Fourier coefficients of the pluck shape *are* the timbre.

9.5 Traveling waves: d'Alembert's form

The mode expansion answered the fixed-end problem. The wave equation has a second canonical face, best seen with the boundaries removed (the infinite line). Factor the operator,

$$\frac{\partial^2}{\partial t^2} - c^2\frac{\partial^2}{\partial x^2} = \Bigl(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x}\Bigr)\Bigl(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x}\Bigr), \tag{188}$$

and pass to the characteristic variables $\xi = x - ct$, $\eta = x + ct$. By the chain rule, $\partial_\xi = \frac{1}{2}(\partial_x - c^{-1}\partial_t)$ and $\partial_\eta = \frac{1}{2}(\partial_x + c^{-1}\partial_t)$, so

$$\frac{\partial^2 y}{\partial\xi\,\partial\eta} = -\,\frac{1}{4c^2}\left(\frac{\partial^2 y}{\partial t^2} - c^2\frac{\partial^2 y}{\partial x^2}\right), \tag{189}$$

and the wave equation is exactly $\partial^2 y/\partial\xi\,\partial\eta = 0$. Integrate twice: $\partial y/\partial\eta$ is independent of ξ, hence a function of η alone; integrating it in η and adding the "constant" (a function of ξ),

$$\boxed{y(x,t) = f(x - ct) + g(x + ct)} \tag{190}$$

with f, g arbitrary twice-differentiable profiles — and this is *every* solution, by the construction just run. A right-moving and a left-moving shape, each translating rigidly at speed c (verify forward: each term is annihilated by one factor of the operator, by the chain rule).

The two faces are two bases for one solution space. Each standing wave is a pair of counter-propagating traveling waves,

$$\sin(kx)\cos(\omega t) = \tfrac{1}{2}\Bigl[\sin\bigl(k(x-ct)\bigr) + \sin\bigl(k(x+ct)\bigr)\Bigr] \qquad (\omega = ck), \tag{191}$$

by the product-to-sum identity of §4.1. The mode basis fits fixed ends; the traveling basis fits pulses and the open line. (On the finite string the traveling picture survives as the method of reflections; not developed here [stated].)

9.6 The dictionary, and the arc completed

	Chain (N masses)	String (continuum)
configuration	vector $\vec{y} = (y_1, \dots, y_N)$	field $y(x,t)$
dynamics	$\ddot{\vec{y}} = -\mathsf{A}_N\,\vec{y}$	$\partial_t^2 y = c^2\,\partial_x^2 y$
coupling	tridiagonal matrix A_N	operator $-c^2\,\partial_x^2$
second difference	$y_{j-1} - 2y_j + y_{j+1}$	$a^2\,\partial_x^2 y$
boundary	phantom sites $y_0 = y_{N+1} = 0$	$y(0,t) = y(L,t) = 0$
mode label	$\theta_n = n\pi/(N+1)$, $n \le N$	$k_n = n\pi/L$, n unbounded
mode shape	$v_j^{(n)} = \sin(j\theta_n)$	$\sin(k_n x)$
initial data	expansion in eigenvectors of A_N	Fourier sine series
inner product	$\sum_j u_j v_j$	$\int_0^L u(x)v(x)\,\mathrm{d}x$
dispersion	$\omega = 2\omega_0\sin(\theta/2)$, ceiling $2\omega_0$	$\omega = ck$, no ceiling

System	What it added
SHO	the complex ansatz $\mathrm{e}^{\lambda t}$; amplitude–phase in one number
damped	complex roots classify three regimes; Q from decay
forced (undamped)	particular + homogeneous; beats; resonance catastrophe
forced (damped)	$\hat{z} = f_0/(\omega_0^2 - \omega^2 + 2\mathrm{i}\beta\omega)$; transients vs. steady state
resonance	peak at $\sqrt{\omega_0^2 - 2\beta^2}$; width 2β; Q from linewidth
coupled	eigenvalue problem; normal coordinates decouple
forced coupled	one forced HO per mode; two resonances; antiresonance
N-mass chain	exponential ansatz *in space*; dispersion relation
continuum limit	difference → derivative; the wave equation, $c^2 = T/\mu$
general solution	mode superposition; initial data by Fourier expansion

Every step reused the previous ones: a single ansatz — exponential in time, then again in space — one lemma about real parts, and one change of basis carry the subject from a mass on a spring to the wave equation.

10 Outlook: Waves in Two and Three Dimensions

Everything so far lived on a line. This closing section is an *introduction only*: we exhibit the higher-dimensional wave equation, indicate where it comes from, and point at what is new — the systematic treatment belongs to a later course.

Where the equation comes from. Repeat the bead construction on a plane: a square grid of masses with spacing a, tension T, each mass coupled to its *four* nearest neighbors, displacing by $u_{ij}(t)$ perpendicular to the grid. (We write u for the displacement field now that x and y are both needed as coordinates.) The transverse force analysis of §9.1 applies along each grid direction separately, so the two directions simply add:

$$m\,\ddot{u}_{ij} = \frac{T}{a}\underbrace{\left(u_{i+1,j} - 2u_{ij} + u_{i-1,j}\right)}_{\text{second difference in } x} + \frac{T}{a}\underbrace{\left(u_{i,j+1} - 2u_{ij} + u_{i,j-1}\right)}_{\text{second difference in } y}. \tag{192}$$

Each second difference becomes a^2 times a second partial derivative in the continuum limit, exactly as in (173), so

$$m\,\frac{\partial^2 u}{\partial t^2} = \frac{T}{a}\cdot a^2\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + O(a^4). \tag{193}$$

The quantities to hold fixed are the two-dimensional analogs of μ and T: the *areal density* $\sigma \equiv m/a^2$ (mass per area) and the *membrane tension* $\tau \equiv T/a$ (force per unit length of edge — in a sheet, tension pulls across lines, not at points). Dividing by $m = \sigma a^2$:

$$\frac{\partial^2 u}{\partial t^2} = \frac{Ta}{m}\,\nabla^2 u + O(a^2) = \frac{\tau}{\sigma}\,\nabla^2 u + O(a^2) \qquad \Longrightarrow \qquad \boxed{\frac{\partial^2 u}{\partial t^2} = c^2\,\nabla^2 u, \qquad c^2 = \frac{\tau}{\sigma}} \tag{194}$$

where $\nabla^2 \equiv \partial_x^2 + \partial_y^2$, the *Laplacian*, is the continuum's "sum of second differences over all directions." The formula $c^2 = \tau/\sigma$ is the exact structural echo of $c^2 = T/\mu$: restoring agent per unit of geometry, over inertia per unit of geometry. In three dimensions the same equation with $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$ governs sound in air and, in a deeper development, light.

What survives, and what is new. Every structural element of this document transfers. The equation is linear: superposition holds. Fixed-boundary modes exist, oscillate independently, and are orthogonal; initial data is expanded in them by projection — Fourier's idea, now in two variables. For a rectangular membrane ($0 \le x \le L_x$, $0 \le y \le L_y$, edges clamped), the mode shapes are products of the familiar one-dimensional ones, and one verifies by direct substitution that

$$u_{nm}(x,y,t) = \sin\!\Big(\frac{n\pi x}{L_x}\Big)\sin\!\Big(\frac{m\pi y}{L_y}\Big)\cos\!\big(\omega_{nm} t\big), \qquad \omega_{nm} = c\,\pi\sqrt{\frac{n^2}{L_x^2} + \frac{m^2}{L_y^2}} \tag{195}$$

solves the equation with the correct boundary values (each factor contributes its $-k^2$; the Laplacian adds them; the frequency is c times the length of the wavevector (k_x, k_y)).

Three genuinely new phenomena are already visible in these formulas (and in Fig. 10):

- **Inharmonic overtones.** The frequencies ω_{nm} are *not* integer multiples of the lowest one — no analog of the harmonic series survives. This is audible: a string sings a pitch, a drum thuds.
- **Nodal lines.** A one-dimensional mode has nodal *points*; a membrane mode is silent along nodal *curves* (here the grid lines where either sine vanishes), the patterns made visible by sand in Chladni's classic experiment.
- **Degeneracy.** On a square membrane ($L_x = L_y$), the modes (n,m) and (m,n) share a frequency: distinct motion patterns, identical pitch. This is no accident: reflecting the square across its diagonal ($x \leftrightarrow y$) maps solutions to solutions and interchanges the pair $(n,m) \leftrightarrow (m,n)$, so the two must share a frequency — degeneracy forced by the symmetry of the square, a first glimpse of the link between symmetry and degeneracy that becomes a central theme in quantum mechanics.

None of this requires new methods — an ansatz, a boundary condition, a projection — only richer geometry for them to act on. That is where the subject goes next.

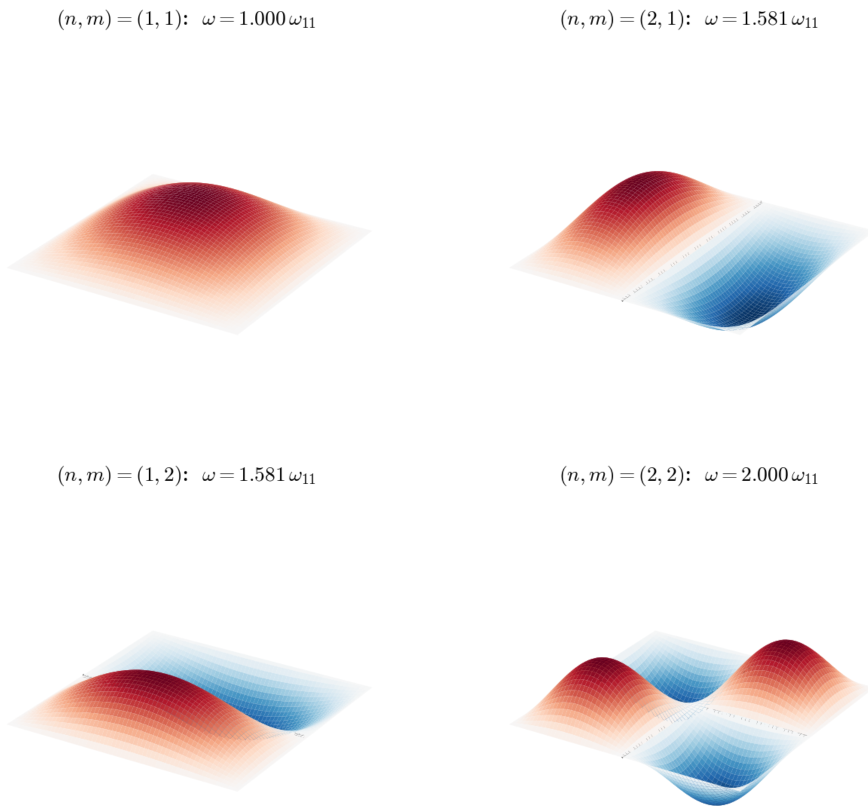


Figure 10: The four lowest modes of a square membrane ($L_x = L_y$), rendered as surfaces; dashed lines mark the interior nodal lines, where the membrane never moves. The frequency ratios $1 : \sqrt{5/2} : \sqrt{5/2} : 2$ display all three phenomena discussed below: they are not integer multiples of the fundamental, the nodes form curves rather than points, and the pair $(2,1)$, $(1,2)$ is degenerate.

Part II
Variational and Hamiltonian Mechanics

A Brief Course in Variational Methods and Analytical Mechanics

From the brachistochrone to Hamilton's equations

This document develops Lagrangian and Hamiltonian mechanics from the calculus of variations, with every step of the main logical chain displayed. Part I opens with two problems that ordinary calculus cannot pose — the brachistochrone and the catenary — builds the variational framework rigorously (functionals, variations, the Euler–Lagrange equation, first integrals, constrained problems), solves both opening problems in full, and then turns to mechanics: Hamilton's principle, generalized coordinates and constraints, conservation laws and Noether's theorem, the pendulum, and the double pendulum at small angles, whose normal modes require and receive the general theory of the generalized eigenvalue problem. Part II introduces Hamiltonian mechanics: the Legendre transform and Hamilton's equations (derived twice); then, as the stress test of the formalism, the pendulum and the double pendulum are solved a second time, completely, in Hamiltonian form — the exact pendulum reduced to its elliptic quadrature, the double pendulum's exact Hamiltonian constructed, its linearized flow decoupled by mode variables, and its general small-angle solution given in closed form — followed by Poisson brackets and phase-space structure including Liouville's theorem. Physical motivation is kept in view throughout: each formal step is preceded by the question it answers. Proofs whose details are purely technical are placed in appendices; every statement used in the main text is either proved there or explicitly pointed to its appendix. The main text keeps all key steps, all logic, and all physics.

1 Conventions and Notation

Dots denote time derivatives; primes denote derivatives with respect to the spatial variable of a curve, $y' = \mathrm{d}y/\mathrm{d}x$. Vectors carry arrows ($\vec{q}$, $\vec{p}$), matrices are sans-serif (M, K). Indices i, j run over generalized coordinates, $1, \ldots, n$; the summation convention is *not* used — sums are written out.

Convention 1.1 (Functionals)**.** A *functional* is a map that eats a whole function and returns a number; we write square brackets, $J[y]$, to distinguish it from a function $f(x)$ of finitely many variables. The domain will always be a space of sufficiently smooth functions with prescribed boundary behavior, stated when it matters:

$$\mathcal{C} = \left\{\, y \in C^2([a,b]) \;:\; y(a) = y_A,\; y(b) = y_B \,\right\}. \tag{1}$$

Convention 1.2 (Variations, and what δ means)**.** Physics texts write δy for "an infinitesimal variation of the function y." We give this a precise meaning once, and then use the physicists' notation freely. Fix $y \in \mathcal{C}$ and a *test function*

$$\eta \in C^2([a,b]) \quad \text{with} \quad \eta(a) = \eta(b) = 0 \tag{2}$$

(so that $y + \varepsilon\eta$ stays in $\mathcal{C}$ for every real ε: the endpoints are pinned). The *first variation* of J at y in the direction η is the ordinary derivative

$$\delta J[y;\eta] \;\equiv\; \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\right|_{\varepsilon=0} J[y + \varepsilon\eta] \tag{3}$$

— a *Gateaux derivative*: one real parameter, ordinary calculus. The symbol δy is shorthand for $\varepsilon\eta$ at first order, and an equation like $\delta J = 0$ means: $\delta J[y;\eta] = 0$ *for every admissible test function* η. All manipulations with δ below are theorems about this definition, not a new calculus.

Convention 1.3 (Stationary, not minimal)**.** A function y at which $\delta J[y;\eta] = 0$ for all η is a *stationary point* (or *extremal*) of J. Whether it is a minimum, maximum, or saddle is a separate question, governed by the second variation, and largely irrelevant to mechanics: the physical principle below demands stationarity only. The traditional name "principle of least action" overstates; we will say *stationary action.*

Citations across the series. References to the companion documents use the bracketed register [Osc. §n] (the oscillator reference) and [QM. §n] (the quantum mechanics notes), always against the cited document's current numbering.

Part I
Variational Calculus and Lagrangian Mechanics

2 The Calculus of Variations

2.1 Two problems that ordinary calculus cannot pose

Ordinary calculus optimizes over *numbers*: find x minimizing $f(x)$. The following two classical problems ask for an optimum over *curves* — infinitely many degrees of freedom — and launched the subject.

Problem 1: the brachistochrone (Johann Bernoulli, 1696). A bead slides without friction along a wire from point A to a lower point B, released from rest, under uniform gravity. *Along which shape of wire is the travel time least?* Set A at the origin, measure y *downward*, and describe the wire by a function $y(x)$ with $y(0) = 0$, $y(x_B) = y_B$. Two ingredients convert the question into a formula:

- **Energy conservation** (the wire is frictionless and does no work on the bead — the constraint force is perpendicular to the motion): starting from rest, $\frac{1}{2}mv^2 = mgy$, so the speed depends only on the depth,
$$v = \sqrt{2gy}\,. \tag{4}$$
- **Arc length**: a step $\mathrm{d}x$ along the curve covers distance $\mathrm{d}s = \sqrt{1 + y'^2}\,\mathrm{d}x$.

Time is distance over speed, summed along the curve:

$$\boxed{T[y] = \int_0^{x_B} \frac{\mathrm{d}s}{v} = \int_0^{x_B} \sqrt{\frac{1 + y'(x)^2}{2g\,y(x)}}\,\mathrm{d}x} \tag{5}$$

The unknown is the *function* y. Note the genuine tension in the problem: a steep start builds speed early but lengthens the path; a straight line is shortest but slow at the start. Neither extreme wins (Fig. 1), and no finite list of parameters captures "all curves."

Problem 2: the catenary (Galileo's wrong guess, resolved 1691). A chain of fixed length L and uniform mass per length μ hangs between two pegs. *What shape does it assume?* The physics: the chain settles into the shape of *least potential energy* compatible with its fixed length. With height $y(x)$ (now measured upward) and the same arc length element,

$$\boxed{U[y] = \mu g \int_a^b y\,\sqrt{1 + y'^2}\,\mathrm{d}x \quad \text{to be minimized subject to} \quad \Lambda[y] = \int_a^b \sqrt{1 + y'^2}\,\mathrm{d}x = L} \tag{6}$$

— an optimization over curves *with a constraint that is itself a functional.* (Galileo guessed a parabola; Fig. 2 shows how close and how wrong that is.)

A third problem, already solved by nature. Both problems have a famous older sibling: *Fermat's principle* (1662) — light traveling between two points takes the path of least *time*. In a medium with position-dependent refractive index n, light moves at speed c/n, so the principle reads: minimize $\int n\,\mathrm{d}s/c$ — again a functional of the path. Every law of geometric optics (straight rays, reflection, Snell's law of refraction) is a consequence, and this is the prototype of a lesson physics has since learned repeatedly: *a law of nature can be packaged not as a rule applied instant by instant, but as an optimization over entire histories.* Part of this document's business is to show that all of mechanics admits exactly such a packaging.

Meanwhile, the structure shared by all three problems is one and the same: extremize $J[y] = \int F(x, y, y')\,\mathrm{d}x$ over curves with pinned endpoints. We now build the tool that solves them all — and then discover that mechanics itself has been secretly posed in this form all along.

2.2 The Euler–Lagrange equation

Theorem 2.1 (Euler–Lagrange)**.** *Let $F(x, u, w)$ be C^2 in its three arguments and let*

$$J[y] = \int_a^b F\big(x,\, y(x),\, y'(x)\big)\,\mathrm{d}x \tag{7}$$

on the class $\mathcal{C}$ of Convention 1.1. If $y \in \mathcal{C}$ is stationary — $\delta J[y;\eta] = 0$ for all test functions η — then y satisfies

$$\boxed{\frac{\partial F}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial F}{\partial y'}\right) = 0 \qquad \textit{for all } x \in [a,b].} \tag{8}$$

Here $\partial F/\partial y$ and $\partial F/\partial y'$ denote partial derivatives of $F(x, u, w)$ with respect to its second and third slots, evaluated along the curve.

Proof. The proof is four moves; each is displayed.

Move 1: reduce to one variable. Fix a test function η and consider the one-parameter family of competitor curves $y + \varepsilon\eta$ (each in $\mathcal{C}$: the pinned endpoints are undisturbed since $\eta(a) = \eta(b) = 0$). Along this family the functional becomes an *ordinary function of one real variable*,

$$\Phi(\varepsilon) \equiv J[y + \varepsilon\eta] = \int_a^b F\big(x,\; y(x) + \varepsilon\eta(x),\; y'(x) + \varepsilon\eta'(x)\big)\,\mathrm{d}x, \tag{9}$$

and the hypothesis $\delta J[y;\eta] = 0$ is, by Convention 1.2, precisely the statement $\Phi'(0) = 0$. The infinite-dimensional problem has been probed along a line; all of ordinary calculus now applies to Φ.

Move 2: differentiate under the integral. Two things must be done: compute the ε-derivative of the integrand, and justify pulling $\mathrm{d}/\mathrm{d}\varepsilon$ inside the integral. For the first, fix x and apply the chain rule to $\varepsilon \mapsto F(x, u(\varepsilon), w(\varepsilon))$ with $u = y + \varepsilon\eta$, $w = y' + \varepsilon\eta'$ (so $\mathrm{d}u/\mathrm{d}\varepsilon = \eta(x)$, $\mathrm{d}w/\mathrm{d}\varepsilon = \eta'(x)$):

$$\frac{\partial}{\partial\varepsilon} F\big(x,\, y + \varepsilon\eta,\, y' + \varepsilon\eta'\big) = \frac{\partial F}{\partial u}\big(x,\, y + \varepsilon\eta,\, y' + \varepsilon\eta'\big)\,\eta(x) \;+\; \frac{\partial F}{\partial w}\big(x,\, y + \varepsilon\eta,\, y' + \varepsilon\eta'\big)\,\eta'(x). \tag{10}$$

For the second: F is C^2 and y, η are C^2, so the right side of (10) is continuous in (x, ε) on the compact rectangle $[a,b] \times [-1,1]$; this is exactly the hypothesis of the Leibniz rule (stated and proved as Appendix A), which licenses

$$\Phi'(\varepsilon) = \int_a^b \frac{\partial}{\partial\varepsilon} F\big(x, y + \varepsilon\eta, y' + \varepsilon\eta'\big)\,\mathrm{d}x. \tag{11}$$

Now set $\varepsilon = 0$: the partial derivatives in (10) are then evaluated at $(x, y(x), y'(x))$ — i.e. *along the original curve* — and we adopt the standing shorthand $\partial F/\partial y$, $\partial F/\partial y'$ for exactly these evaluated slot-derivatives (as announced in the theorem statement):

$$\Phi'(0) = \int_a^b \left[\frac{\partial F}{\partial y}\, \eta(x) + \frac{\partial F}{\partial y'}\, \eta'(x) \right] \mathrm{d}x. \tag{12}$$

This quantity — linear in the probe η — is the *first variation* of J; the notation $\delta J = \int (\dots)\, \delta y \, \mathrm{d}x$ of physics texts is (12) with $\delta y = \varepsilon\eta$.

Move 3: integrate by parts to strip the derivative off η. The two terms of (12) probe the curve differently — one through η, one through η' — and cannot be compared until they speak the same language. The conversion is the product rule read backwards: along the curve, the function $x \mapsto \frac{\partial F}{\partial y'}\, \eta$ is C^1 (here the hypotheses do their work: $y \in C^2$ and $F \in C^2$ make $x \mapsto \partial F/\partial y'$ a C^1 function — its derivative is the chain-rule expansion (16) below, every term of which is continuous), and

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial F}{\partial y'}\, \eta \right) = \left[\frac{\mathrm{d}}{\mathrm{d}x} \frac{\partial F}{\partial y'} \right] \eta + \frac{\partial F}{\partial y'}\, \eta'. \tag{13}$$

Integrate both sides over $[a, b]$; the left side is handled by the fundamental theorem of calculus, and rearranging,

$$\int_a^b \frac{\partial F}{\partial y'}\, \eta' \, \mathrm{d}x = \underbrace{\left[\frac{\partial F}{\partial y'}\, \eta \right]_a^b}_{=0 \text{ since } \eta(a)=\eta(b)=0} - \int_a^b \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial F}{\partial y'} \right) \eta \, \mathrm{d}x. \tag{14}$$

The boundary term dies *because the endpoints are pinned* — this is where the boundary conditions enter the theory, and the only place. Substituting,

$$0 = \Phi'(0) = \int_a^b \left[\frac{\partial F}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial F}{\partial y'} \right) \right] \eta(x) \, \mathrm{d}x \qquad \text{for } \textit{every} \text{ test function } \eta. \tag{15}$$

Move 4: localize. From “the integral against every test function vanishes” we want “the bracket vanishes pointwise.” That step is the

> **Fundamental Lemma of the Calculus of Variations.** *If g is continuous on $[a, b]$ and $\int_a^b g(x)\, \eta(x)\, \mathrm{d}x = 0$ for every $\eta \in C^2$ vanishing at the endpoints, then $g \equiv 0$.*

The logic of its proof is worth carrying in the main text even though the construction lives in Appendix B: if g were nonzero somewhere, say $g(x_0) > 0$, then by continuity $g > 0$ on a small interval around x_0; choosing η to be a smooth *bump* — positive inside that interval, zero outside — makes the integral strictly positive, a contradiction. Test functions are probes; enough probes pin g down everywhere. The bracket in (15) is continuous (exactly what Move 3 checked), so the lemma applies to it, and the conclusion of the theorem is reached: for every $x \in [a, b]$,

$$\boxed{\frac{\partial F}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial F}{\partial y'} \right) = 0.}$$

□

Remark 2.2 (What kind of object (8) is)*.* The Euler–Lagrange equation converts a problem over an infinite-dimensional space of curves into an ordinary differential equation — generally second order, since $\frac{\mathrm{d}}{\mathrm{d}x}\big(\partial F/\partial y'\big)$ produces a y'' through the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}x} \frac{\partial F}{\partial y'} = \frac{\partial^2 F}{\partial x\, \partial y'} + \frac{\partial^2 F}{\partial y\, \partial y'}\, y' + \frac{\partial^2 F}{\partial y'^2}\, y''. \tag{16}$$

Stationarity is a *local, differential* condition on the curve — this is the first deep lesson of the subject: a globally posed optimization is answered by a pointwise law. It is also a *necessary* condition only; Convention 1.3 applies.

Remark 2.3 (Several unknown functions). If the functional depends on n functions, $J[\vec{q}] = \int_a^b F(x, q_1, \dots, q_n, q_1', \dots,$ run the same proof with independent test functions η_i attached to each q_i: choosing all but one η_i zero isolates one equation at a time, giving the *system*

$$\boxed{\frac{\partial F}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial F}{\partial q_i'} = 0, \qquad i = 1, \dots, n.} \tag{17}$$

Nothing else changes. Mechanics will use exactly this form, with $x \to t$.

Example 2.4 (Calibration: the shortest path is a straight line)**.** Before the hard problems, the sanity check. The length of a graph $y(x)$ between two points is $J[y] = \int_a^b \sqrt{1 + y'^2}\,\mathrm{d}x$, so $F = \sqrt{1 + y'^2}$ with

$$\frac{\partial F}{\partial y} = 0, \qquad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{1 + y'^2}}. \tag{18}$$

The Euler–Lagrange equation (8) says $\frac{\mathrm{d}}{\mathrm{d}x}\big(y'/\sqrt{1 + y'^2}\big) = 0$, i.e. $y'/\sqrt{1 + y'^2} = c$ constant; solving, $y'^2 = c^2/(1 - c^2)$ is constant, so y' is constant: a straight line, through the two pinned endpoints. The machinery returns the right answer on the one problem where we already knew it — the same calibration discipline as checking a resonance formula at $\omega \to 0$.

With the equation in hand, the section returns to its two opening problems. Both opening functionals share one feature the general theory rewards: their integrands carry *no explicit x-dependence*. The next subsection converts that feature into a ready-made first integration, and the two subsections after it collect the brachistochrone and the catenary in full — the promised payoff of the machinery just built.

2.3 First integrals

Two structural shortcuts, both indispensable, both provable in a few lines.

Proposition 2.5 (Cyclic variable)**.** *If F does not depend explicitly on y (i.e. $\partial F/\partial y \equiv 0$), then along any extremal*

$$\frac{\partial F}{\partial y'} = \textit{const.} \tag{19}$$

Proof. Equation (8) collapses to $\frac{\mathrm{d}}{\mathrm{d}x}\big(\partial F/\partial y'\big) = 0$. □

Proposition 2.6 (Beltrami identity)**.** *If F does not depend explicitly on x (i.e. $\partial F/\partial x \equiv 0$), then along any extremal*

$$\boxed{F - y'\,\frac{\partial F}{\partial y'} = \textit{const.}} \tag{20}$$

Proof. Differentiate the left side along the curve and watch everything cancel. By the chain rule (no explicit x-dependence),

$$\frac{\mathrm{d}F}{\mathrm{d}x} = \frac{\partial F}{\partial y}\,y' + \frac{\partial F}{\partial y'}\,y'', \tag{21}$$

while

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y'\,\frac{\partial F}{\partial y'}\right) = y''\,\frac{\partial F}{\partial y'} + y'\,\frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial F}{\partial y'}. \tag{22}$$

Subtracting,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(F - y'\frac{\partial F}{\partial y'}\right) = y'\left[\frac{\partial F}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial F}{\partial y'}\right] = y'\cdot 0 = 0 \tag{23}$$

by the Euler–Lagrange equation. (This identity will return in §3.3 in mechanical form: it *is* energy conservation.) □

The practical payoff: both opening problems have integrands with no explicit x-dependence, so both are governed by the *first-order* equation (20) instead of the second-order (8) — one integration is handed to us for free.

2.4 Solution of the brachistochrone

The problem that opened the section can now be solved. Apply Beltrami to (5). Drop the constant $1/\sqrt{2g}$ and take

$$F(y,y') = \sqrt{\frac{1+y'^2}{y}}, \qquad \frac{\partial F}{\partial y'} = \frac{y'}{\sqrt{y}\,\sqrt{1+y'^2}}. \tag{24}$$

Then

$$F - y'\frac{\partial F}{\partial y'} = \frac{1+y'^2}{\sqrt{y}\sqrt{1+y'^2}} - \frac{y'^2}{\sqrt{y}\sqrt{1+y'^2}} = \frac{1}{\sqrt{y\,(1+y'^2)}} = \text{const}, \tag{25}$$

i.e.

$$y\left(1+y'^2\right) = 2a \qquad (2a \text{ a positive constant}). \tag{26}$$

Solve by a substitution tailored to the structure: since $y \le 2a$ along the curve, write

$$y = a\,(1-\cos\theta) = 2a\sin^2\tfrac{\theta}{2}, \qquad \theta\in[0,\pi], \tag{27}$$

with θ a parameter along the curve. From (26),

$$y'^2 = \frac{2a-y}{y} = \frac{2a\cos^2(\theta/2)}{2a\sin^2(\theta/2)} \quad\Longrightarrow\quad y' = \cot\tfrac{\theta}{2} \tag{28}$$

(positive branch: the curve initially descends with our downward-y convention). Then, using $\mathrm{d}y = a\sin\theta\,\mathrm{d}\theta$,

$$\mathrm{d}x = \frac{\mathrm{d}y}{y'} = a\sin\theta\,\tan\tfrac{\theta}{2}\,\mathrm{d}\theta = 2a\sin\tfrac{\theta}{2}\cos\tfrac{\theta}{2}\cdot\frac{\sin(\theta/2)}{\cos(\theta/2)}\,\mathrm{d}\theta = 2a\sin^2\tfrac{\theta}{2}\,\mathrm{d}\theta = a(1-\cos\theta)\,\mathrm{d}\theta, \tag{29}$$

which integrates immediately (constant fixed by A at $\theta = 0$):

$$\boxed{x(\theta) = a\,(\theta - \sin\theta), \qquad y(\theta) = a\,(1-\cos\theta)} \tag{30}$$

— a *cycloid*: the curve traced by a point on the rim of a wheel of radius a rolling under the line $y = 0$. The two constants (a, θ_B) are fixed by requiring the curve to pass through B.

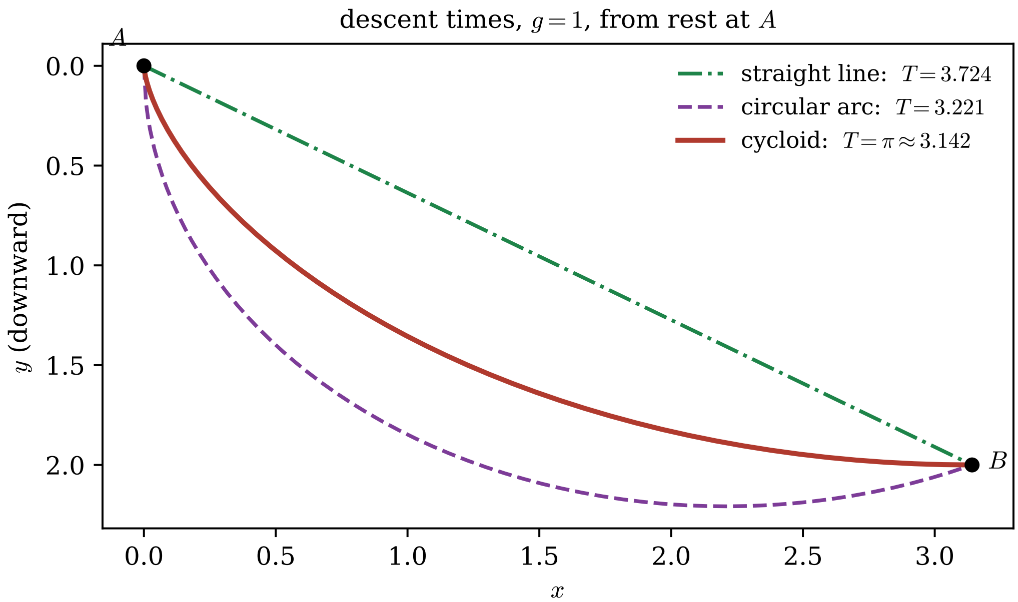


Figure 1: Three wires from A to $B = (\pi, 2)$ and their descent times (computed numerically from (5) with $g = 1$; the cycloid's time is exactly $\pi\sqrt{a/g}$). The straight line is shortest and slowest; the cycloid trades extra length for early speed and wins.

Remark 2.7 (Two bonuses of the parametrization). (i) Along the cycloid the travel-time element collapses beautifully: with $\mathrm{d}s = 2a\sin(\theta/2)\,\mathrm{d}\theta$ and $v = \sqrt{2gy} = 2\sqrt{ga}\,\sin(\theta/2)$,

$$\mathrm{d}T = \frac{\mathrm{d}s}{v} = \sqrt{\frac{a}{g}}\,\mathrm{d}\theta \quad\Longrightarrow\quad T = \sqrt{\frac{a}{g}}\,\theta_B : \tag{31}$$

the bead consumes *equal times for equal parameter intervals.* (ii) Releasing the bead from rest anywhere on the same cycloid, the time to reach the bottom is the same — the cycloid is also the *tautochrone*, the fact Huygens built into pendulum clocks. We state (ii) without proof; it is a short exercise from (i)'s machinery.

Remark 2.8 (Bernoulli's own solution: physics borrowed from optics). Johann Bernoulli solved the problem in 1697 without the Euler–Lagrange equation (which did not yet exist) by a physical translation: the bead's speed $v = \sqrt{2gy}$ depends only on depth, exactly like light in a stratified medium with index $n(y) \propto 1/v \propto 1/\sqrt{y}$. By Fermat's principle (§2.1), the least-time path of the bead must be the path such light would take — and Snell's law, $\sin\alpha/v = \text{const}$ layer by layer, is then a ready-made first integral. Writing $\sin\alpha = 1/\sqrt{1+y'^2}$ for a graph reproduces precisely our Beltrami result $y(1+y'^2) = \text{const}$. The calculus of variations is the common language in which the optical and mechanical statements are the same statement.

2.5 Constrained problems and the catenary

One opening problem remains, and it is the one the plain machinery cannot yet touch: the catenary (6) adds a new structural feature — the constraint is itself a functional (an *isoperimetric* constraint) — and closing it will add the last general-purpose tool of the section. The resolution is the infinite-dimensional version of Lagrange multipliers.

Theorem 2.9 (Isoperimetric multiplier rule)**.** *Let* $J[y] = \int_a^b F\,\mathrm{d}x$ *and* $\Lambda[y] = \int_a^b G\,\mathrm{d}x$ *with* F, G *of class* C^2*, and suppose* y_* *is stationary for* J *among curves in* $\mathcal{C}$ *satisfying* $\Lambda[y] = L$*. If* y_* *is* not *an unconstrained extremal of* Λ *itself, then there exists a constant* $\lambda \in \mathbb{R}$ *such that* y_* *satisfies the Euler–Lagrange equation of the* combined *integrand*

$$\widetilde{F} \;=\; F + \lambda\, G. \tag{32}$$

The logic (full proof: Appendix D): with a constraint, one cannot vary in every direction η — only in directions that keep Λ fixed to first order. A two-parameter variation $y + \varepsilon_1\eta_1 + \varepsilon_2\eta_2$, with η_2 chosen to "repair" the constraint violation caused by η_1, shows that δJ must be proportional to $\delta\Lambda$ direction by direction; the proportionality constant is λ. This is exactly the finite-dimensional picture — $\nabla f \parallel \nabla g$ on a constraint surface — promoted to function space. The hypothesis that y_* not extremize Λ is the nondegeneracy condition ($\nabla g \neq 0$) in disguise.

Solving the catenary. With $F = \mu g\, y\sqrt{1+y'^2}$ and $G = \sqrt{1+y'^2}$, absorb constants and set

$$\widetilde{F} = (y+\lambda)\,\sqrt{1+y'^2}, \tag{33}$$

(the multiplier shifts the height — already a hint of its physical meaning). No explicit x: Beltrami applies.

$$\frac{\partial\widetilde{F}}{\partial y'} = \frac{(y+\lambda)\,y'}{\sqrt{1+y'^2}}, \qquad \widetilde{F} - y'\frac{\partial\widetilde{F}}{\partial y'} = (y+\lambda)\left[\sqrt{1+y'^2} - \frac{y'^2}{\sqrt{1+y'^2}}\right] = \frac{y+\lambda}{\sqrt{1+y'^2}} = C. \tag{34}$$

Hence $(y+\lambda)^2 = C^2(1+y'^2)$, i.e.

$$y' = \sqrt{\frac{(y+\lambda)^2}{C^2} - 1}\,. \tag{35}$$

Substitute $y + \lambda = C\cosh u$: then $\mathrm{d}y = C\sinh u\,\mathrm{d}u$ and $y' = \sqrt{\cosh^2 u - 1} = \sinh u$, so

$$C\sinh u\,\frac{\mathrm{d}u}{\mathrm{d}x} = \sinh u \quad\Longrightarrow\quad \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{C} \quad\Longrightarrow\quad u = \frac{x-x_0}{C}, \tag{36}$$

$$\boxed{y(x) = C\cosh\!\left(\frac{x-x_0}{C}\right) - \lambda} \tag{37}$$

— the *catenary.* The three constants (C, x_0, λ) are fixed by the two endpoint conditions and the length constraint $\Lambda[y] = L$. Not a parabola: cosh and a parabola agree to quadratic order at the bottom and then part ways (Fig. 2), which is why Galileo's guess survived casual inspection for half a century.

Remark 2.10 (The physical meaning of the multiplier). λ entered as bookkeeping and exits with a job: comparing with the statics of a hanging chain, $C = H/(\mu g)$ where H is the horizontal component of the chain's tension (constant along the chain), and λ fixes the reference height at which the potential-plus-constraint total balances. Multipliers of constraints are *forces of constraint* in embryo — a theme §3.2 makes precise.

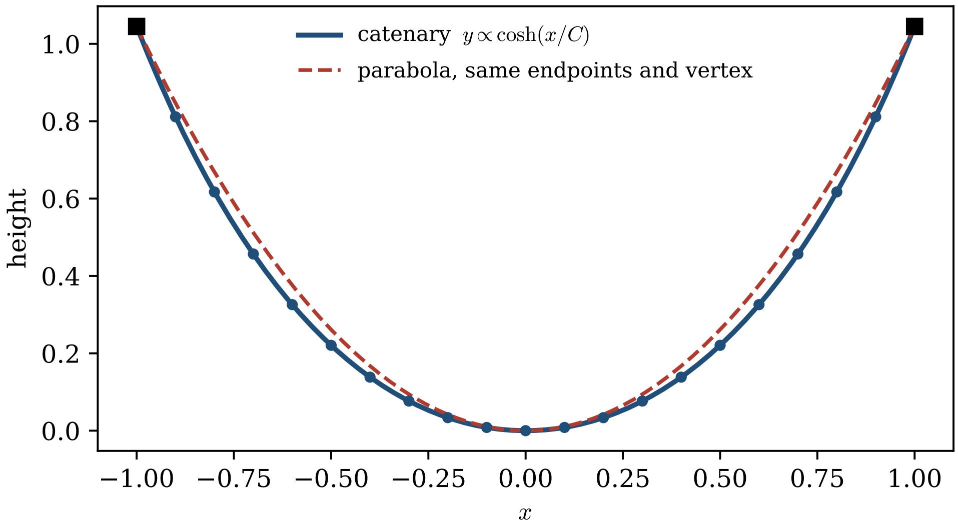


Figure 2: The catenary against the parabola with the same endpoints and vertex: close near the bottom (they share the quadratic term), visibly different toward the supports.

2.6 Covariance: why arbitrary coordinates are allowed

One more structural theorem before mechanics — the one that makes everything that follows practical.

Theorem 2.11 (Covariance of the Euler–Lagrange equations)**.** *Let $Q_i = Q_i(q_1, \dots, q_n, t)$, $i = 1, \dots, n$, be a C^2 change of coordinates with C^2 inverse, and let $\widetilde{L}$ be L rewritten in the new coordinates: $\widetilde{L}(Q, \dot{Q}, t) = L\big(q(Q,t), \dot{q}(Q,\dot{Q},t), t\big)$. Then a curve satisfies the Euler–Lagrange equations for L in the q's* if and only if *its image satisfies the Euler–Lagrange equations for $\widetilde{L}$ in the Q's.*

Proof in Appendix C (two chain-rule identities and a cancellation). The content is one sentence: *the Euler–Lagrange equations do not care which coordinates you use.* Newton's equations, by contrast, hold in Cartesian coordinates and mutate under coordinate change (centrifugal and Coriolis terms in polar coordinates, etc.). The variational formulation is coordinate-blind because the underlying object — "this curve makes $\int L\,dt$ stationary" — never mentions coordinates at all: stationarity is a property of the *curve*, and (8) is merely its expression in whichever chart is convenient. This theorem is the license behind the phrase *generalized coordinates*: angles, distances, differences of angles — anything that smoothly labels configurations is as good as x, y, z.

3 From Newton to Lagrange

Why rebuild mechanics, when $\vec{F} = m\vec{a}$ already works? Because Newton's formulation, for all its power, has three practical irritations and hides one deep structure. The irritations: it is wedded to *vectors in Cartesian coordinates* (change to polar coordinates and fictitious terms sprout); it forces you to carry *constraint forces* — rod tensions, normal forces — as unknowns you must solve for and rarely want; and its symmetries are *invisible* — nothing in the notation $m\ddot{\vec{r}} = \vec{F}$ announces that momentum conservation is the same fact as the homogeneity of space. The variational reformulation answers all three at once: Theorem 2.11 makes any coordinates legal, the treatment of constraints below makes constraint forces vanish before the calculation starts, and Noether's theorem (§3.3) turns symmetry into conservation by a two-line argument. The deep structure it exposes — one scalar function L generating all equations of motion, the dynamics as a stationarity condition on whole histories — is the form in which mechanics generalizes: field theory, relativity, and quantum mechanics are all written in this language, not in Newton's. The cost is one new postulate, which we now state and immediately check against Newton.

3.1 Hamilton's principle

Definition 3.1 (Lagrangian and action)**.** For a mechanical system described by coordinates $\vec{q} = (q_1, \ldots, q_n)$, the *Lagrangian* is

$$\boxed{L(\vec{q}, \dot{\vec{q}}, t) = T - V} \tag{38}$$

kinetic minus potential energy, and the *action* of a trajectory $\vec{q}(t)$ on $[t_0, t_1]$ is the functional

$$\boxed{S[\vec{q}] = \int_{t_0}^{t_1} L\big(\vec{q}(t), \dot{\vec{q}}(t), t\big)\,\mathrm{d}t.} \tag{39}$$

Postulate 3.2 (Hamilton's principle of stationary action)**.** The physical trajectory between fixed configurations $\vec{q}(t_0) = \vec{q}_A$ and $\vec{q}(t_1) = \vec{q}_B$ is a stationary point of the action: $\delta S = 0$.

By Theorem 2.1 in its several-function form (Remark 2.3, with $x \to t$), the postulate is equivalent to the *Euler–Lagrange equations of motion*

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = 0, \qquad i = 1, \ldots, n.} \tag{40}$$

First duty: recover Newton. A postulate must first reproduce what is known. Take one particle in Cartesian coordinates, $V = V(\vec{r})$:

$$L = \tfrac{1}{2} m\big(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\big) - V(x, y, z). \tag{41}$$

For the x-equation:

$$\frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{x}} = m\ddot{x}, \qquad \frac{\partial L}{\partial x} = -\frac{\partial V}{\partial x} = F_x, \tag{42}$$

so (40) reads $m\ddot{x} = F_x$, and likewise for y, z:

$$m\ddot{\vec{r}} = -\nabla V = \vec{F}. \tag{43}$$

Newton's second law, exactly. *For unconstrained conservative systems in Cartesian coordinates, Hamilton's principle and Newton's law are the same statement.* Everything the new formulation adds — and it adds a great deal — comes from the two theorems already in hand: covariance (Theorem 2.11: use any coordinates) and the treatment of constraints (next).

Remark 3.3 (Why $T - V$? The accounting, plus an intuition)*.* The *minus* sign is not derivable from anything prior in this document: $L = T - V$ is part of the postulate, and its justification is exactly the check just performed — it is the choice that reproduces Newton. (Try $T + V$: the sign of the force flips.) That said, the combination can be *felt*. Take a thrown ball, and grant for intuition's sake that its short arcs *minimize* the action $\int (T - V)\,\mathrm{d}t$ (true for sufficiently short arcs;

Convention 1.3 guards the general statement). The integrand is lowered by spending time where V is *high* — so the path bulges upward, lingering near the top of its flight — but height requires speed on the way up and down, and speed raises the kinetic term. Bulge too little and the $-V$ reduction is wasted; bulge too much and the kinetic cost dominates. The parabola is the balance point of this trade-off, and the Euler–Lagrange equation is the statement that the balance holds at every instant separately. This is intuition, not derivation — the derivation is the check above — but it is the right picture to carry.

3.2 Constraints and generalized coordinates

Real systems are full of constraints: the pendulum bob stays on its rod, the bead stays on its wire, the chain link stays attached to its neighbor. The Newtonian treatment must carry the *constraint forces* (rod tension, wire normal force) as unknowns and solve for them. The Lagrangian treatment makes them vanish from the outset. Here is the physics of why, kept in full in the main text because it *is* the key step.

Setup. Let a system of N particles $\vec{r}_a$ $(a = 1, \ldots, N)$ be subject to *holonomic* constraints — conditions on positions (possibly time), $f_\alpha(\vec{r}_1, \ldots, \vec{r}_N, t) = 0$ — which are enforced by constraint forces $\vec{C}_a$, so that Newton reads

$$m_a\ddot{\vec{r}}_a = \vec{F}_a + \vec{C}_a \qquad (\vec{F}_a = -\nabla_a V \text{ the applied forces}). \tag{44}$$

Suppose the constraint surface is smoothly parametrized by n *generalized coordinates*: $\vec{r}_a = \vec{r}_a(q_1, \ldots, q_n, t)$.

Postulate 3.4 (Ideal constraints / d'Alembert)**.** The constraint forces do no work under any *virtual displacement* — any infinitesimal displacement consistent with the constraints at frozen time:

$$\sum_{a=1}^{N} \vec{C}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = 0 \qquad \text{for each } i. \tag{45}$$

This is a physical assumption, and an excellent one: rods, frictionless wires, and rigid links push only *perpendicular* to the motions they permit — that is what "rigid and frictionless" means. (Sliding friction violates it; such forces must be kept explicitly.)

The computation. Take Newton (44), dot each equation with $\partial\vec{r}_a/\partial q_i$, and sum over particles. The constraint forces drop by Postulate 3.4:

$$\sum_a m_a\ddot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \sum_a \vec{F}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} \;\equiv\; \mathcal{Q}_i \qquad (\text{the } \textit{generalized force}). \tag{46}$$

The left side is reorganized by two kinetic identities. From the chain rule, $\dot{\vec{r}}_a = \sum_j \frac{\partial \vec{r}_a}{\partial q_j}\dot{q}_j + \frac{\partial \vec{r}_a}{\partial t}$, whence

$$\text{(K1)} \quad \frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_i} = \frac{\partial \vec{r}_a}{\partial q_i}, \qquad\qquad \text{(K2)} \quad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \vec{r}_a}{\partial q_i} = \frac{\partial \dot{\vec{r}}_a}{\partial q_i} \tag{47}$$

— (K1) because $\dot{q}_i$ enters $\dot{\vec{r}}_a$ linearly with coefficient $\partial\vec{r}_a/\partial q_i$; (K2) because both sides equal $\sum_j \frac{\partial^2 \vec{r}_a}{\partial q_j \partial q_i}\dot{q}_j + \frac{\partial^2 \vec{r}_a}{\partial t\, \partial q_i}$ (mixed partials commute for C^2 maps). Now write, by the product rule,

$$\ddot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\dot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i}\right) - \dot{\vec{r}}_a \cdot \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \vec{r}_a}{\partial q_i} \overset{\text{(K1),(K2)}}{=} \frac{\mathrm{d}}{\mathrm{d}t}\left(\dot{\vec{r}}_a \cdot \frac{\partial \dot{\vec{r}}_a}{\partial \dot{q}_i}\right) - \dot{\vec{r}}_a \cdot \frac{\partial \dot{\vec{r}}_a}{\partial q_i}. \tag{48}$$

Multiply by m_a, sum over a, and recognize the kinetic energy $T = \sum_a \frac{1}{2} m_a \dot{\vec{r}}_a^2$ on both terms:

$$\sum_a m_a \ddot{\vec{r}}_a \cdot \frac{\partial \vec{r}_a}{\partial q_i} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_i} - \frac{\partial T}{\partial q_i}. \tag{49}$$

If moreover the applied forces are conservative, $\vec{F}_a = -\nabla_a V$ with $V = V(\vec{r}_1, \dots, \vec{r}_N, t)$, then by the chain rule $\mathcal{Q}_i = -\partial V/\partial q_i$, and since V has no $\dot{q}$-dependence, (46) assembles into

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial (T-V)}{\partial \dot{q}_i} - \frac{\partial (T-V)}{\partial q_i} = 0. \tag{50}$$

Theorem 3.5 (Lagrange's equations for constrained systems)**.** *For a system with ideal holonomic constraints and conservative applied forces, the generalized coordinates obey the Euler–Lagrange equations* (40) *of $L = T - V$,* with the constraint forces nowhere appearing and the constraints automatically respected.

The working recipe of Lagrangian mechanics, now fully justified:

> **Recipe.** (1) Choose any n independent coordinates that label the allowed configurations. (2) Write T and V in them. (3) Form $L = T - V$ and turn the Euler–Lagrange crank. Constraint forces never enter; if you want them afterward (engineering loads!), reinstate the constrained coordinate with a multiplier — the multiplier *is* the force, the catenary's λ writ general.

3.3 Conservation laws

The variational structure converts symmetries into conserved quantities with almost no work. Three instances, in increasing generality.

Cyclic coordinates. Define the *generalized (canonical) momentum* conjugate to q_i:

$$\boxed{p_i \equiv \frac{\partial L}{\partial \dot{q}_i}} \tag{51}$$

If L does not contain q_i (q_i is *cyclic*), then (40) gives $\dot{p}_i = \partial L/\partial q_i = 0$:

$$\boxed{q_i \text{ cyclic} \implies p_i \text{ conserved.}} \tag{52}$$

(Proposition 2.5 in mechanical clothing. Example: a free particle, $L = \frac{1}{2} m \dot{x}^2$, has x cyclic and $p = m\dot{x}$ conserved.)

Energy. Define the *energy function*

$$\boxed{h(\vec{q}, \dot{\vec{q}}, t) \;\equiv\; \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} \;-\; L.} \tag{53}$$

Proposition 3.6. *Along any solution of* (40), $\dfrac{\mathrm{d}h}{\mathrm{d}t} = -\dfrac{\partial L}{\partial t}$. *In particular, if L has no explicit time dependence, h is conserved.*

Proof. Differentiate along the trajectory:

$$\begin{aligned}\frac{\mathrm{d}h}{\mathrm{d}t} &= \sum_i \ddot{q}_i \frac{\partial L}{\partial \dot{q}_i} + \sum_i \dot{q}_i \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \left[\sum_i \frac{\partial L}{\partial q_i}\dot{q}_i + \sum_i \frac{\partial L}{\partial \dot{q}_i}\ddot{q}_i + \frac{\partial L}{\partial t}\right] \\ &= \sum_i \dot{q}_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}\right] - \frac{\partial L}{\partial t} = -\frac{\partial L}{\partial t},\end{aligned} \tag{54}$$

the bracket vanishing by the equations of motion. (This is the Beltrami identity, Proposition 2.6, in n variables: "no explicit t" plays the role of "no explicit x.") Whether h equals $T+V$ is a separate question answered precisely in §6.3; and whether h "is" the Hamiltonian is a sharper question than it looks, answered precisely in Remark 6.2. □

Noether's theorem. The two facts above are shadows of one statement: *continuous symmetries of L produce conserved quantities.*

Theorem 3.7 (Noether, point-symmetry version)**.** *Let $\vec{q} \mapsto \vec{Q}(\vec{q}, s)$ be a smooth one-parameter family of transformations with $\vec{Q}(\vec{q}, 0) = \vec{q}$, and suppose it leaves the Lagrangian invariant:*

$$L\big(\vec{Q}(\vec{q}, s),\ \tfrac{\mathrm{d}}{\mathrm{d}t}\vec{Q}(\vec{q}, s),\ t\big) = L(\vec{q}, \dot{\vec{q}}, t) \qquad \textit{for all } s. \tag{55}$$

Then, writing $\eta_i(\vec{q}) = \left.\dfrac{\partial Q_i}{\partial s}\right|_{s=0}$ for the infinitesimal generator, the quantity

$$\boxed{I \;=\; \sum_i \frac{\partial L}{\partial \dot{q}_i}\, \eta_i(\vec{q})} \tag{56}$$

is conserved along every solution of the equations of motion.

Proof. Differentiate the invariance identity with respect to s at $s = 0$. Since along the transformed family $\frac{\partial}{\partial s}\dot{Q}_i\big|_{s=0} = \dot{\eta}_i$ (the s- and t-derivatives commute for smooth families),

$$0 = \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_0 L = \sum_i \frac{\partial L}{\partial q_i}\,\eta_i + \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\dot{\eta}_i. \tag{57}$$

So far this holds for any curve. Now evaluate *on a solution* and use the equations of motion to replace $\partial L/\partial q_i = \frac{\mathrm{d}}{\mathrm{d}t}(\partial L/\partial \dot{q}_i)$:

$$0 = \sum_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right]\eta_i + \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\dot{\eta}_i = \frac{\mathrm{d}}{\mathrm{d}t}\left[\sum_i \frac{\partial L}{\partial \dot{q}_i}\,\eta_i\right] = \frac{\mathrm{d}I}{\mathrm{d}t}. \tag{58}$$

□

Example 3.8 (The classic pairs)**.** (i) *Translation*: $Q = q + s$ for a coordinate absent from L; $\eta = 1$, $I = p$. Cyclic coordinates are Noether's theorem for translations in configuration space. (ii) *Rotation*: a particle in a central potential, $L = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2) - V(\sqrt{x^2 + y^2})$, is invariant under $(x, y) \mapsto (x\cos s - y\sin s,\ x\sin s + y\cos s)$; the generator is $\eta = (-y, x)$ and

$$I = m\dot{x}\,(-y) + m\dot{y}\,(x) = m(x\dot{y} - y\dot{x}) = L_z, \tag{59}$$

angular momentum. (iii) Time translation stands slightly apart — it shifts the curve's parameter rather than its values, so it does not fit the hypothesis of Theorem 3.7 as stated — and its conserved quantity is exactly the h of Proposition 3.6, whose proof is the time-translation analogue of the argument above. The moral is uniform: **momentum is the charge of spatial translation, angular momentum of rotation, energy of time translation.**

Example 3.9 (Central forces: the effective one-dimensional problem)**.** The cyclic-coordinate machinery at full power. A particle in a plane under $V = V(r)$ (planar without loss for a central force: angular momentum fixes the orbital plane [stated; it follows from conservation of the vector angular momentum $m\,\vec{r}\times\dot{\vec{r}}$]), in polar coordinates:

$$L = \tfrac{1}{2}m\big(\dot{r}^2 + r^2\dot{\varphi}^2\big) - V(r). \tag{60}$$

The angle is cyclic, so its momentum is conserved,

$$p_\varphi = \frac{\partial L}{\partial\dot{\varphi}} = mr^2\dot{\varphi} \equiv \ell = \text{const}, \tag{61}$$

and the conserved energy $h = T + V$ (constraints time-independent), with $\dot{\varphi}$ eliminated via $\dot{\varphi} = \ell/mr^2$, becomes

$$\boxed{E = \tfrac{1}{2}m\dot{r}^2 + \underbrace{\frac{\ell^2}{2mr^2} + V(r)}_{V_{\text{eff}}(r)}} \tag{62}$$

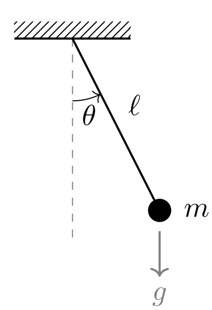
— a *one-dimensional* problem on the half-line $r > 0$, with the conserved angular momentum converted into a repulsive *centrifugal barrier* $\ell^2/2mr^2$. Circular orbits sit at $V'_{\text{eff}}(r_0) = 0$, stable where $V''_{\text{eff}} > 0$; the full radial motion is read off the one-dimensional energy diagram exactly as in §9. One warning, displayed because it is the step a reader gets wrong: eliminate $\dot{\varphi}$ in the *energy*, not in the Lagrangian — substituting $\dot{\varphi} = \ell/mr^2$ into L before varying flips the sign of the centrifugal term in the resulting equation of motion [the correct reduction at the Lagrangian level is the Routhian; stated]. (The quantum version of this reduction, with ℓ^2 replaced by $\ell(\ell+1)\hbar^2$, is the radial equation of the quantum companion.)

4 The Pendulum

The recipe of §3.2, run start to finish on the simplest constrained system — one rigid rod, one angle. The point of the exercise is partly the result (the harmonic oscillator's most venerable realization) but mostly the *silence*: watch for what never appears in the calculation.

4.1 Setup and equation of motion

A bob of mass m on a rigid massless rod of length ℓ, pivoting freely in a vertical plane; uniform gravity g.



The constraint (fixed distance ℓ from the pivot) is holonomic and ideal (the rod's force lies along itself, perpendicular to the circular motion). One generalized coordinate suffices: the angle θ from the downward vertical. Cartesian positions and velocities:

$$x = \ell \sin\theta, \quad y = -\ell\cos\theta \qquad \Longrightarrow \qquad \dot{x} = \ell\cos\theta\,\dot{\theta}, \quad \dot{y} = \ell\sin\theta\,\dot{\theta}, \tag{63}$$

$$T = \tfrac{1}{2}m(\dot{x}^2 + \dot{y}^2) = \tfrac{1}{2}m\ell^2\dot{\theta}^2\big(\cos^2\theta + \sin^2\theta\big) = \tfrac{1}{2}m\ell^2\dot{\theta}^2, \qquad V = mgy = -\,mg\ell\cos\theta. \tag{64}$$

$$\boxed{L(\theta,\dot{\theta}) = \tfrac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta} \tag{65}$$

Euler–Lagrange:

$$\frac{\partial L}{\partial\dot{\theta}} = m\ell^2\dot{\theta}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial\dot{\theta}} = m\ell^2\ddot{\theta}, \qquad \frac{\partial L}{\partial\theta} = -mg\ell\sin\theta, \tag{66}$$

$$\boxed{\ddot{\theta} = -\,\frac{g}{\ell}\,\sin\theta} \tag{67}$$

Note what did *not* happen: the rod tension was never introduced, never computed, never eliminated — the coordinate θ respects the constraint by construction, and Theorem 3.5 did the rest. (Compare the Newtonian route: decompose gravity along and across the rod, introduce the tension, project. Same result, more work, and the work multiplies with each additional constraint — as §5 is about to demonstrate by its absence.)

4.2 Small angles: the harmonic oscillator recovered

For $|\theta| \ll 1$, $\sin\theta = \theta + O(\theta^3)$, and (67) linearizes to

$$\ddot{\theta} = -\,\omega_0^2\,\theta, \qquad \omega_0 = \sqrt{\frac{g}{\ell}} \tag{68}$$

— the simple harmonic oscillator, whose complete theory (solution forms, energy, damping, driving, resonance) is the subject of the companion oscillator reference and is reused here wholesale. Period $T = 2\pi\sqrt{\ell/g}$, independent of amplitude and of mass.

Remark 4.1 (Beyond small angles). Equation (67) is exactly solvable, but not in elementary functions: conservation of h gives $\frac{1}{2}\ell\dot{\theta}^2 = g(\cos\theta - \cos\theta_{\max})$, and separating variables produces a period

$$T(\theta_{\max}) = 4\sqrt{\frac{\ell}{g}}\,K\!\Big(\sin\tfrac{\theta_{\max}}{2}\Big), \qquad K(k) = \int_0^{\pi/2}\frac{\mathrm{d}\varphi}{\sqrt{1-k^2\sin^2\varphi}}, \tag{69}$$

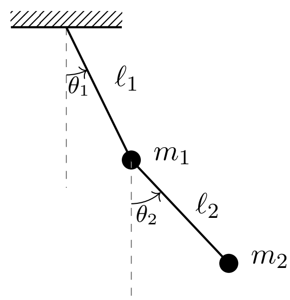
a complete elliptic integral. Expanding: $T = 2\pi\sqrt{\ell/g}\,\big(1 + \frac{1}{16}\theta_{\max}^2 + O(\theta_{\max}^4)\big)$ — the period grows with amplitude, by about 1.7% at $\theta_{\max} = 30°$. The isochronism of §4 is the θ^2-truncation speaking, not the pendulum. (The curve that *is* exactly isochronous is the cycloid of §2.4 — the tautochrone; the subject's opening problem and its first mechanical example are secretly the same curve.)

5 The Double Pendulum at Small Angles

The first system where the Lagrangian route is decisively easier. Attempt this system in Newtonian style and you will meet two unknown rod tensions, one of which acts on a bob that is itself accelerating in a curved path determined by the other — a genuinely unpleasant exercise in algebra. The Lagrangian route needs eleven lines of kinematics and two derivatives. The system is also the smallest one exhibiting two structural novelties: a kinetic energy that is *not* a sum of squares (the mass matrix acquires off-diagonal terms), which will force the correct generalization of the eigenvalue problem; and, beyond small angles, deterministic chaos (§11) — all from two rods and two bobs.

5.1 Setup and kinematics

Bob 1 (mass m_1) hangs from the pivot on a rod of length ℓ_1; bob 2 (mass m_2) hangs from bob 1 on a rod of length ℓ_2. Angles θ_1, θ_2 from the vertical.



Positions (y upward from the pivot):

$$x_1 = \ell_1 \sin\theta_1, \qquad y_1 = -\ell_1 \cos\theta_1, \tag{70}$$

$$x_2 = \ell_1 \sin\theta_1 + \ell_2 \sin\theta_2, \qquad y_2 = -\ell_1 \cos\theta_1 - \ell_2 \cos\theta_2. \tag{71}$$

Velocities by the chain rule:

$$\dot{x}_2 = \ell_1 \cos\theta_1\, \dot{\theta}_1 + \ell_2 \cos\theta_2\, \dot{\theta}_2, \qquad \dot{y}_2 = \ell_1 \sin\theta_1\, \dot{\theta}_1 + \ell_2 \sin\theta_2\, \dot{\theta}_2, \tag{72}$$

and the speed of bob 2, expanded carefully (the cross term is the whole story of this system):

$$\begin{aligned} \dot{x}_2^2 + \dot{y}_2^2 &= \ell_1^2\dot{\theta}_1^2\,(\cos^2\theta_1 + \sin^2\theta_1) + \ell_2^2\dot{\theta}_2^2\,(\cos^2\theta_2 + \sin^2\theta_2) \\ &\quad + 2\ell_1\ell_2\dot{\theta}_1\dot{\theta}_2\,(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2) \\ &= \ell_1^2\dot{\theta}_1^2 + \ell_2^2\dot{\theta}_2^2 + 2\ell_1\ell_2\ \cos(\theta_1 - \theta_2)\,\dot{\theta}_1\dot{\theta}_2. \end{aligned} \tag{73}$$

Hence

$$T = \tfrac{1}{2}m_1\ell_1^2\dot{\theta}_1^2 + \tfrac{1}{2}m_2\Big[\ell_1^2\dot{\theta}_1^2 + \ell_2^2\dot{\theta}_2^2 + 2\ell_1\ell_2\cos(\theta_1{-}\theta_2)\,\dot{\theta}_1\dot{\theta}_2\Big], \tag{74}$$

$$V = -\,(m_1{+}m_2)\,g\ell_1\cos\theta_1\ -\ m_2 g\ell_2\cos\theta_2, \tag{75}$$

and $L = T - V$. Pause on what was *not* needed: neither rod tension appears, no forces were decomposed, no free-body diagram of bob 2 in the accelerating frame of bob 1 was contemplated. The two constraints cost nothing.

5.2 Exact equations of motion

Write $\Delta \equiv \theta_1 - \theta_2$. For θ_1:

$$\frac{\partial L}{\partial \dot{\theta}_1} = (m_1{+}m_2)\ell_1^2\dot{\theta}_1 + m_2\ell_1\ell_2\cos\Delta\ \dot{\theta}_2, \tag{76}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{\theta}_1} = (m_1{+}m_2)\ell_1^2\ddot{\theta}_1 + m_2\ell_1\ell_2\cos\Delta\ \ddot{\theta}_2 - m_2\ell_1\ell_2\sin\Delta\ (\dot{\theta}_1 - \dot{\theta}_2)\,\dot{\theta}_2, \tag{77}$$

$$\frac{\partial L}{\partial \theta_1} = -\,m_2\ell_1\ell_2\sin\Delta\ \dot{\theta}_1\dot{\theta}_2\ -\ (m_1{+}m_2)\,g\ell_1\sin\theta_1, \tag{78}$$

and the Euler–Lagrange equation, after the two $\sin\Delta$ terms combine,

$$(m_1{+}m_2)\ell_1^2\,\ddot\theta_1 + m_2\ell_1\ell_2\cos\Delta\,\ddot\theta_2 + m_2\ell_1\ell_2\sin\Delta\,\dot\theta_2^{\,2} + (m_1{+}m_2)\,g\ell_1\sin\theta_1 = 0. \tag{79}$$

For θ_2, the same procedure ($\partial L/\partial\theta_2$ picks up $+m_2\ell_1\ell_2\sin\Delta\,\dot\theta_1\dot\theta_2$, note the sign) gives

$$m_2\ell_2^2\,\ddot\theta_2 + m_2\ell_1\ell_2\cos\Delta\,\ddot\theta_1 - m_2\ell_1\ell_2\sin\Delta\,\dot\theta_1^{\,2} + m_2\,g\ell_2\sin\theta_2 = 0. \tag{80}$$

These exact equations are nonlinear, coupled, and — as §11 notes — chaotic at large amplitude. Our target is their small-angle limit.

5.3 Linearization done in the right order

Small oscillations about the hanging equilibrium: θ_i, $\dot\theta_i$ all first-order small. There are two natural procedures — (a) linearize the exact equations (79)–(80), or (b) truncate L at quadratic order and *then* apply Euler–Lagrange — and it deserves one displayed sentence that they agree:

Lemma 5.1 (Expand-then-vary equals vary-then-expand)**.** *The Euler–Lagrange equations of the quadratic truncation L_2 of L about an equilibrium coincide with the linearization of the Euler–Lagrange equations of L about that equilibrium.*

Proof. Euler–Lagrange takes one derivative of L in each slot; hence terms of cubic and higher order in $(\vec\theta,\dot{\vec\theta})$ in L produce terms of quadratic and higher order in the equations, which the linearization discards. Conversely every quadratic term of L survives one differentiation as a linear term. The two truncations remove exactly the same terms. □

Route (b) is less error-prone (truncate once, differentiate a polynomial). Truncate: $\cos\Delta = 1{+}O(\Delta^2)$ multiplies $\dot\theta_1\dot\theta_2$, already second order, so the correction is fourth order — drop it; $\cos\theta_i = 1-\frac12\theta_i^2+O(\theta_i^4)$ in V; discard constants. The quadratic Lagrangian:

$$L_2 = \tfrac12(m_1{+}m_2)\ell_1^2\dot\theta_1^2 + \tfrac12 m_2\ell_2^2\dot\theta_2^2 + m_2\ell_1\ell_2\,\dot\theta_1\dot\theta_2 - \tfrac12(m_1{+}m_2)g\ell_1\,\theta_1^2 - \tfrac12 m_2 g\ell_2\,\theta_2^2. \tag{81}$$

Euler–Lagrange on this polynomial (each derivative now trivial):

$$\begin{aligned}(m_1{+}m_2)\ell_1^2\,\ddot\theta_1 + m_2\ell_1\ell_2\,\ddot\theta_2 &= -\,(m_1{+}m_2)g\ell_1\,\theta_1,\\ m_2\ell_1\ell_2\,\ddot\theta_1 + m_2\ell_2^2\,\ddot\theta_2 &= -\,m_2 g\ell_2\,\theta_2,\end{aligned} \tag{82}$$

i.e., in matrix form,

$$\boxed{\mathsf{M}\,\ddot{\vec\theta} = -\,\mathsf{K}\,\vec\theta,\qquad \mathsf{M} = \begin{pmatrix}(m_1{+}m_2)\ell_1^2 & m_2\ell_1\ell_2\\ m_2\ell_1\ell_2 & m_2\ell_2^2\end{pmatrix},\quad \mathsf{K} = \begin{pmatrix}(m_1{+}m_2)g\ell_1 & 0\\ 0 & m_2 g\ell_2\end{pmatrix}} \tag{83}$$

Both matrices are symmetric — M from the quadratic form T, K from the quadratic form V — and both are positive definite ($T>0$ for any nonzero velocity; V has a strict minimum at hanging equilibrium). But unlike every system in the oscillator companion, *the mass matrix is not diagonal*: the kinetic cross term $\dot\theta_1\dot\theta_2$ couples the coordinates through their *velocities*. The plain eigenvalue problem no longer applies; its correct generalization is next.

5.4 The generalized eigenvalue problem

Insert the oscillatory ansatz $\vec{\theta}(t) = \vec{v}\,\cos(\omega t - \phi)$ into (83):

$$-\,\omega^2\,\mathsf{M}\,\vec{v}\,\cos(\omega t-\phi) = -\,\mathsf{K}\,\vec{v}\,\cos(\omega t-\phi) \quad\iff\quad \boxed{\mathsf{K}\,\vec{v} = \omega^2\,\mathsf{M}\,\vec{v}} \tag{84}$$

— a *generalized* eigenvalue problem: M stands where the identity used to. Nontrivial $\vec{v}$ requires

$$\det\bigl(\mathsf{K} - \omega^2\mathsf{M}\bigr) = 0. \tag{85}$$

The general theory is exactly as strong as in the ordinary case:

Theorem 5.2 (Normal modes exist and diagonalize the system)**.** *Let M be symmetric positive definite and K symmetric ($n\times n$, real). Then (84) has n real eigenvalues $\omega_1^2 \le \cdots \le \omega_n^2$ (positive if K is positive definite) with eigenvectors $\vec{v}_1,\ldots,\vec{v}_n$ that form a basis and are orthogonal in the M-inner product, $\vec{v}_r^{\mathsf{T}}\mathsf{M}\,\vec{v}_s = 0$ for $r \neq s$. In the coordinates $\vec{\theta} = \sum_r \xi_r \vec{v}_r$, the system (83) decouples into n independent harmonic oscillators $\ddot{\xi}_r = -\omega_r^2\,\xi_r$.*

Proof in Appendix E; the idea is one line — since $\mathsf{M} > 0$ it has a symmetric square root, and the substitution $\vec{w} = \mathsf{M}^{1/2}\vec{v}$ turns (84) into an *ordinary* symmetric eigenvalue problem for $\mathsf{M}^{-1/2}\mathsf{K}\mathsf{M}^{-1/2}$, whose spectral theorem does all the work. The physics of the statement: *small oscillations of any constrained conservative system about a stable equilibrium are superpositions of independent normal modes* — the oscillator companion's central structure, now proved in the generality that mechanics actually delivers.

5.5 Equal masses and lengths: the explicit modes

Set $m_1 = m_2 = m$, $\ell_1 = \ell_2 = \ell$, and abbreviate $\lambda \equiv \omega^2\ell/g$ (dimensionless). Then

$$\mathsf{M} = m\ell^2\begin{pmatrix}2 & 1\\ 1 & 1\end{pmatrix}, \qquad \mathsf{K} = mg\ell\begin{pmatrix}2 & 0\\ 0 & 1\end{pmatrix}, \qquad \mathsf{K} - \omega^2\mathsf{M} = mg\ell\begin{pmatrix}2-2\lambda & -\lambda\\ -\lambda & 1-\lambda\end{pmatrix}. \tag{86}$$

Determinant:

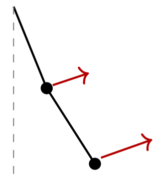
$$(2-2\lambda)(1-\lambda) - \lambda^2 = 2 - 4\lambda + 2\lambda^2 - \lambda^2 = \lambda^2 - 4\lambda + 2 = 0 \quad\Longrightarrow\quad \lambda_\pm = 2 \pm \sqrt{2}, \tag{87}$$

$$\boxed{\omega_-^2 = \bigl(2-\sqrt{2}\bigr)\frac{g}{\ell} \approx 0.586\,\frac{g}{\ell}, \qquad \omega_+^2 = \bigl(2+\sqrt{2}\bigr)\frac{g}{\ell} \approx 3.414\,\frac{g}{\ell}} \tag{88}$$

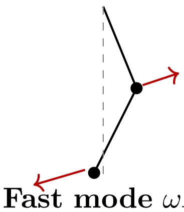
Eigenvectors from the first row, $(2-2\lambda)v_1 = \lambda v_2$:

$$\frac{v_2}{v_1} = \frac{2-2\lambda}{\lambda} = \begin{cases}+\sqrt{2} & (\lambda = 2-\sqrt{2}),\\ -\sqrt{2} & (\lambda = 2+\sqrt{2}),\end{cases} \tag{89}$$

(for $\lambda_- = 2-\sqrt{2}$: $\frac{2-2(2-\sqrt{2})}{2-\sqrt{2}} = \frac{2\sqrt{2}-2}{2-\sqrt{2}} = \frac{2(\sqrt{2}-1)(2+\sqrt{2})}{(2-\sqrt{2})(2+\sqrt{2})} = \frac{2(\sqrt{2}-1)(2+\sqrt{2})}{2} = (\sqrt{2}-1)(2+\sqrt{2}) = \sqrt{2}$; the other case is the mirror computation). Hence



Slow mode ω_-:
$\vec{v}_- = (1, \sqrt{2})$
in phase; the lower bob swings wider by $\sqrt{2}$



Fast mode ω_+:
$\vec{v}_+ = (1, -\sqrt{2})$
opposed; the rods scissor about a near-stationary point

Sanity checks, each one line: the frequencies bracket the single pendulum's, $\omega_-^2 < g/\ell < \omega_+^2$ (the in-phase mode is effectively a longer pendulum, the scissoring mode a stiffer one); M-orthogonality holds, $\vec{v}_-^{\mathsf{T}} \mathsf{M}\, \vec{v}_+ = m\ell^2\, (1, \sqrt{2}) \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right) \left(\begin{smallmatrix} 1 \\ -\sqrt{2} \end{smallmatrix}\right) = m\ell^2\, (1, \sqrt{2}) \cdot (2-\sqrt{2},\, 1-\sqrt{2})^{\mathsf{T}} = m\ell^2 \big(2-\sqrt{2}+\sqrt{2}-2\big) = 0$ ✓ (note: plain orthogonality fails, $(1, \sqrt{2}) \cdot (1, -\sqrt{2}) = -1 \neq 0$ — the M-inner product is not decorative); and by Theorem 5.2, the general small motion is

$$\vec{\theta}(t) = \xi_-(0\text{-data})\, \vec{v}_- \cos(\omega_- t - \phi_-) + \xi_+(0\text{-data})\, \vec{v}_+ \cos(\omega_+ t - \phi_+), \tag{90}$$

with beats, energy exchange, and every phenomenon of the oscillator companion's coupled chapter available on demand.

Part II
Hamiltonian Mechanics

6 The Legendre Transform and Hamilton's Equations

Lagrangian mechanics runs on $(\vec{q}, \dot{\vec{q}})$: positions and velocities. Hamiltonian mechanics changes one variable — velocity out, momentum in — and before the machinery, the case for bothering. First, the *shape* of the dynamics changes: n second-order equations become $2n$ *first-order* ones, so that a single point in the $2n$-dimensional space of $(\vec{q}, \vec{p})$ determines the entire future [existence and uniqueness for the smooth first-order system; stated] — the dynamics becomes a *flow*, and flows can be drawn, transported, and reasoned about globally (§9 will prove a theorem, Liouville's, that is invisible in the Lagrangian picture and is the opening axiom of statistical mechanics). Second, the choice of *momentum* as the fundamental variable is not arbitrary: p_i is what conservation laws conserve (§3.3), and in quantum mechanics it is $\hat{p}$, not $\hat{\dot{q}}$, that stands beside $\hat{q}$ in the commutation relations — classical mechanics in Hamiltonian form is the version that quantizes. Third, and deepest: putting $\vec{q}$ and $\vec{p}$ on *equal footing* unlocks a group of transformations (mixing coordinates with momenta!) vastly larger than the point transformations of Theorem 2.11, and with it the strongest solution techniques classical mechanics possesses (§11). Goldstein's summary is the standard one: the Hamiltonian formulation's advantage is not calculational convenience but structural insight — it is the formal skeleton on which the more modern theories of matter are built. The change of variable itself is not a cosmetic substitution but a *Legendre transform*, whose structure is worth stating exactly.

6.1 The transform

Definition 6.1 (Legendre transform of the Lagrangian)**.** Assume $L(\vec{q}, \dot{\vec{q}}, t)$ is C^2 and *convex in the velocities*: the Hessian $\left(\partial^2 L/\partial \dot{q}_i \partial \dot{q}_j\right)$ is positive definite. Define the *canonical momenta*

$$\boxed{p_i \;\equiv\; \frac{\partial L}{\partial \dot{q}_i}(\vec{q}, \dot{\vec{q}}, t)} \tag{91}$$

Convexity makes the map $\dot{\vec{q}} \mapsto \vec{p}$ invertible for each fixed $(\vec{q}, t)$ (its Jacobian is the positive-definite Hessian; the inverse function theorem applies), so velocities can be re-expressed as $\dot{\vec{q}} = \dot{\vec{q}}(\vec{q}, \vec{p}, t)$. The *Hamiltonian* is

$$\boxed{H(\vec{q}, \vec{p}, t) \;\equiv\; \sum_i p_i\, \dot{q}_i(\vec{q}, \vec{p}, t) \;-\; L\big(\vec{q}, \dot{\vec{q}}(\vec{q}, \vec{p}, t), t\big).} \tag{92}$$

For mechanical Lagrangians the convexity hypothesis is physical: the Hessian of T in the velocities is the mass matrix $\mathsf{M}(\vec{q})$ of §5.4, positive definite because kinetic energy is positive.

6.2 Hamilton's equations by two routes

Derivation 1: track the differential. Compute $\mathrm{d}H$ from (92), treating H as a function of $(\vec{q}, \vec{p}, t)$ but differentiating through the definition:

$$\begin{aligned}\mathrm{d}H &= \sum_i \big(p_i\,\mathrm{d}\dot{q}_i + \dot{q}_i\,\mathrm{d}p_i\big) - \sum_i \frac{\partial L}{\partial q_i}\,\mathrm{d}q_i - \sum_i \underbrace{\frac{\partial L}{\partial \dot{q}_i}}_{=p_i}\,\mathrm{d}\dot{q}_i - \frac{\partial L}{\partial t}\,\mathrm{d}t \\ &= \sum_i \dot{q}_i\,\mathrm{d}p_i \;-\; \sum_i \frac{\partial L}{\partial q_i}\,\mathrm{d}q_i \;-\; \frac{\partial L}{\partial t}\,\mathrm{d}t. \end{aligned} \tag{93}$$

The $\mathrm{d}\dot{q}_i$ terms cancel *identically* — this cancellation is the entire design of the Legendre transform: it removes the dependence on the old variable and leaves a function of the new one. Reading off the coefficients of the independent differentials $\mathrm{d}q_i, \mathrm{d}p_i, \mathrm{d}t$:

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i}, \qquad \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \tag{94}$$

So far, no dynamics — these are identities of the transform. Dynamics enters through Euler–Lagrange: *on a solution*, $\partial L/\partial q_i = \frac{\mathrm{d}}{\mathrm{d}t}(\partial L/\partial \dot{q}_i) = \dot{p}_i$, and the middle identity becomes an equation of motion:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}} \qquad (i = 1, \dots, n) \tag{95}$$

— *Hamilton's equations*: $2n$ *first-order* equations replacing n second-order ones, in a form so symmetric that q and p differ only by a sign.

Derivation 2: a variational principle in phase space. Hamilton's equations are themselves Euler–Lagrange equations — of the *phase-space action*

$$S[\vec{q}, \vec{p}] = \int_{t_0}^{t_1} \Big[\sum_i p_i\,\dot{q}_i - H(\vec{q}, \vec{p}, t)\Big]\,\mathrm{d}t, \tag{96}$$

in which $\vec{q}(t)$ and $\vec{p}(t)$ are varied *independently*, with the q's pinned at the endpoints and the p's entirely free. Vary $\vec{p}$ (no integration by parts needed, since $\dot{\vec{p}}$ never appears):

$$\delta_p S = \int \sum_i \Big[\dot{q}_i - \frac{\partial H}{\partial p_i}\Big]\,\delta p_i\,\mathrm{d}t = 0 \quad \Longrightarrow \quad \dot{q}_i = \frac{\partial H}{\partial p_i}, \tag{97}$$

by the fundamental lemma. Vary $\vec{q}$ (one integration by parts on $p_i\,\delta\dot{q}_i$; the boundary term dies on the pinned q's):

$$\delta_q S = \int \sum_i \Big[-\dot{p}_i - \frac{\partial H}{\partial q_i}\Big]\,\delta q_i\,\mathrm{d}t = 0 \quad \Longrightarrow \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}. \tag{98}$$

Both equations from one functional, on equal footing — the formulation that generalizes cleanly (to field theory, and to the path integral).

Energy bookkeeping. Along a solution of (95),

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \sum_i \Big[\frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial p_i}\dot{p}_i\Big] + \frac{\partial H}{\partial t} = \sum_i \Big[\frac{\partial H}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i}\frac{\partial H}{\partial q_i}\Big] + \frac{\partial H}{\partial t} \quad \Longrightarrow \quad \boxed{\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{\partial H}{\partial t}}: \tag{99}$$

if H has no explicit time dependence it is conserved.

Remark 6.2 (h versus H: the same number, not the same function). Comparing (92) with §3.3: along any state, H takes the *value* of the energy function h — $H(\vec{q}, \vec{p}, t) = h(\vec{q}, \dot{\vec{q}}, t)$ whenever $\vec{p}$ and $\dot{\vec{q}}$ correspond under (91). Are they then the same object? *No*, and the difference carries the whole formalism: they are different functions of *different variables* — h lives on $(\vec{q}, \dot{\vec{q}})$, H on $(\vec{q}, \vec{p})$ — and partial derivatives care about what is held fixed. Concretely, differentiate $h = \sum_j \dot{q}_j \, \partial L/\partial \dot{q}_j - L$ at fixed $\dot{\vec{q}}$ and compare with (94):

$$\left(\frac{\partial h}{\partial q_i}\right)_{\dot{\vec{q}}} = \sum_j \dot{q}_j \, \frac{\partial^2 L}{\partial q_i \, \partial \dot{q}_j} - \frac{\partial L}{\partial q_i}, \qquad \left(\frac{\partial H}{\partial q_i}\right)_{\vec{p}} = -\frac{\partial L}{\partial q_i} : \tag{100}$$

the two disagree by $\sum_j \dot{q}_j \, \partial^2 L/\partial q_i \partial \dot{q}_j$ — zero for a constant mass matrix, but *nonzero* whenever M depends on the coordinates, as in the double pendulum. This is why Hamilton's equations hold for H and not for h: the clean identity $\partial H/\partial q_i = -\partial L/\partial q_i$ is a property of differentiating *at fixed momentum*, and arranging for it is exactly what the Legendre transform's cancellation in (93) accomplished. The same re-reading applies to the momentum itself: in §3.3, $p_i = \partial L/\partial \dot{q}_i$ was a *derived bookkeeping quantity* — a combination evaluated along a trajectory, singled out because cyclic coordinates conserve it. Here the identical formula is re-read as a *change of variables*, and p_i is promoted to an independent coordinate of state space. Same formula, different ontological status; the Legendre transform is the promotion ceremony, and keeping the two names (h on the Lagrangian side, H on the Hamiltonian side, following Goldstein) keeps the two objects distinct.

Remark 6.3 (Involutivity). The transform undoes itself: from (94), $\dot{q}_i = \partial H/\partial p_i$, and then $\sum_i p_i \dot{q}_i - H = L$ by (92) — performing the Legendre transform of H in the p's returns L. Lagrangian and Hamiltonian mechanics are two charts on the same theory, with a lossless dictionary between them (given convexity).

6.3 When is $H = T + V$?

Proposition 6.4. *Suppose the constraints are time-independent (so that $\vec{r}_a = \vec{r}_a(\vec{q})$ and T is purely quadratic in the velocities, $T = \frac{1}{2}\sum_{ij} M_{ij}(\vec{q})\, \dot{q}_i \dot{q}_j$) and $V = V(\vec{q}, t)$ is velocity-independent. Then $H = T + V$.*

Proof. The one tool needed is *Euler's theorem on homogeneous functions* in its quadratic case, proved inline: with $T = \frac{1}{2}\sum_{ij} M_{ij} \dot{q}_i \dot{q}_j$ and $M_{ij} = M_{ji}$,

$$\frac{\partial T}{\partial \dot{q}_k} = \sum_j M_{kj}\, \dot{q}_j \quad \Longrightarrow \quad \sum_k \dot{q}_k \frac{\partial T}{\partial \dot{q}_k} = \sum_{kj} M_{kj}\, \dot{q}_k \dot{q}_j = 2\,T. \tag{101}$$

Since V carries no velocities, $p_k = \partial T/\partial \dot{q}_k$, and

$$H = \sum_k p_k \dot{q}_k - L = 2T - (T - V) = T + V. \tag{102}$$

□

Remark 6.5 (The rheonomic caveat). If the constraints move — $\vec{r}_a = \vec{r}_a(\vec{q}, t)$, e.g. a bead on a hoop *rotating at imposed rate* — then T acquires terms linear in $\dot{q}$ and a $\dot{q}$-free term, the computation above breaks, and $H \neq T + V$. H may still be conserved (if L ends up t-independent) while the mechanical energy $T + V$ is *not* — whatever drives the constraint does work. The two statements "H is conserved" and "H is the energy" have different hypotheses; conflating them is the standard trap.

6.4 Examples

Harmonic oscillator. $L = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2$; $p = m\dot{q}$, so $\dot{q} = p/m$ and

$$H = p \cdot \frac{p}{m} - \frac{p^2}{2m} + \frac{1}{2}kq^2 = \frac{p^2}{2m} + \frac{kq^2}{2}, \tag{103}$$

$$\dot{q} = \frac{\partial H}{\partial p} = \frac{p}{m}, \qquad \dot{p} = -\frac{\partial H}{\partial q} = -kq \quad \Longrightarrow \quad \ddot{q} = \frac{\dot{p}}{m} = -\frac{k}{m}q = -\omega_0^2 q. \checkmark \tag{104}$$

The pendulum and the double pendulum deserve more than a paragraph: the next section re-derives and *completely solves* both in the Hamiltonian formalism, as the proper stress test of the new machinery.

7 The Two Systems Revisited: Complete Hamiltonian Solutions

Part I solved the pendulum and the double pendulum by the Lagrangian recipe. We now solve both again, start to finish, in the Hamiltonian formalism — not as repetition but as calibration: the same physics must come out, and watching *where the same content sits in the two formalisms* is how a formalism is actually learned.

7.1 Complete solution of the pendulum

Setting up. From §4, $L = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta$. The canonical momentum and its inversion:

$$p = \frac{\partial L}{\partial \dot{\theta}} = m\ell^2\dot{\theta} \qquad \Longleftrightarrow \qquad \dot{\theta} = \frac{p}{m\ell^2} \tag{105}$$

— physically, p is the *angular momentum* about the pivot (moment of inertia $m\ell^2$ times angular velocity); the canonical momentum conjugate to an angle is always an angular momentum. The Hamiltonian, via (92) or directly via Proposition 6.4 ($T + V$; the constraint is time-independent):

$$\boxed{H(\theta, p) = \frac{p^2}{2m\ell^2} - mg\ell\cos\theta} \tag{106}$$

Hamilton's equations (95):

$$\dot{\theta} = \frac{\partial H}{\partial p} = \frac{p}{m\ell^2}, \qquad \dot{p} = -\frac{\partial H}{\partial \theta} = -\,mg\ell\sin\theta. \tag{107}$$

(Consistency: differentiate the first equation and substitute the second to recover $\ddot{\theta} = -(g/\ell)\sin\theta$, equation (67). The same physics redistributed: one second-order equation has become a pair of first-order ones, and the pair is what draws the phase portrait of §9.)

Small angles: the complete solution. Expand H to quadratic order ($\cos\theta = 1 - \frac{1}{2}\theta^2 + O(\theta^4)$; drop the constant):

$$H_2 = \frac{p^2}{2m\ell^2} + \frac{mg\ell}{2}\,\theta^2, \qquad \dot{\theta} = \frac{p}{m\ell^2}, \quad \dot{p} = -\,mg\ell\,\theta. \tag{108}$$

Differentiating the first equation and inserting the second gives $\ddot{\theta} = -(g/\ell)\theta = -\omega_0^2\theta$; then p follows from $p = m\ell^2\dot{\theta}$. With initial data (θ_0, p_0):

$$\boxed{\theta(t) = \theta_0 \cos\omega_0 t + \frac{p_0}{m\ell^2\omega_0}\,\sin\omega_0 t, \qquad p(t) = p_0\cos\omega_0 t - m\ell^2\omega_0\,\theta_0\sin\omega_0 t} \tag{109}$$

(Checks: at $t = 0$ the pair returns (θ_0, p_0) ✓; substituting both expressions into (107) verifies the equations at all times, term by term ✓.) *Every* small motion of the pendulum is (109) — completeness leans on uniqueness for the linear system, which the energy method proves in three lines: the difference of two solutions with the same initial data starts with $H_2 = 0$, keeps H_2 conserved, and H_2 is positive definite, so the difference vanishes identically. The problem is closed.

Geometry of the solution: eliminate t. Direct substitution (using $\cos^2 + \sin^2 = 1$) shows $\frac{1}{2m\ell^2}p(t)^2 + \frac{mg\ell}{2}\theta(t)^2 = H_2(\theta_0, p_0) \equiv E$ for all t, so the orbit is the ellipse

$$\frac{\theta^2}{2E/mg\ell} + \frac{p^2}{2m\ell^2 E} = 1 \tag{110}$$

with semi-axes $\sqrt{2E/mg\ell}$ and $\sqrt{2m\ell^2 E}$, traversed once per period — precisely the small ovals around the centers of Fig. 3. Its enclosed area is π times the product of the semi-axes,

$$\text{area} = \pi\sqrt{\frac{2E}{mg\ell}}\,\sqrt{2m\ell^2 E} = 2\pi E\sqrt{\frac{\ell}{g}} = \frac{2\pi E}{\omega_0} \tag{111}$$

— area $= 2\pi \times$ energy over frequency. The quantity E/ω_0 (enclosed area over 2π) is the *action variable* of Hamilton–Jacobi theory (§10, Remark 10.15), making its first, unforced appearance.

Large angles: complete reduction to quadrature. Beyond small angles the Hamiltonian route still delivers everything short of elementary functions, in three moves. *(i) Conservation solves for p*: on the energy surface $H = E$,

$$p(\theta) = \pm\sqrt{2m\ell^2\left(E + mg\ell\cos\theta\right)}. \tag{112}$$

(ii) The first Hamilton equation separates: from $\dot{\theta} = p/m\ell^2$,

$$t(\theta) = \int \frac{m\ell^2\,\mathrm{d}\theta}{\sqrt{2m\ell^2\left(E + mg\ell\cos\theta\right)}} \tag{113}$$

— the dynamics is *reduced to quadrature*: one integral stands between the initial data and the full trajectory. *(iii) The period, explicitly.* For libration with turning angle $\theta_{\max}$ (where $p = 0$): $E = -mg\ell\cos\theta_{\max}$, and a quarter period runs from $\theta = 0$ to $\theta_{\max}$:

$$\frac{T}{4} = \sqrt{\frac{\ell}{2g}}\int_0^{\theta_{\max}} \frac{\mathrm{d}\theta}{\sqrt{\cos\theta - \cos\theta_{\max}}}. \tag{114}$$

Convert with $\cos\theta = 1 - 2\sin^2(\theta/2)$, so that $\cos\theta - \cos\theta_{\max} = 2\big(\sin^2\frac{\theta_{\max}}{2} - \sin^2\frac{\theta}{2}\big)$; then substitute $\sin\frac{\theta}{2} = k\sin\varphi$ with $k \equiv \sin\frac{\theta_{\max}}{2}$ (so $\varphi : 0 \to \pi/2$ as $\theta : 0 \to \theta_{\max}$), whence $\frac{1}{2}\cos\frac{\theta}{2}\,\mathrm{d}\theta = k\cos\varphi\,\mathrm{d}\varphi$ and $\cos\frac{\theta}{2} = \sqrt{1 - k^2\sin^2\varphi}$:

$$\frac{T}{4} = \sqrt{\frac{\ell}{2g}}\int_0^{\pi/2} \frac{1}{\sqrt{2}\,\sqrt{1 - k^2\sin^2\varphi}} \cdot \frac{2k\cos\varphi\,\mathrm{d}\varphi}{k\cos\varphi} \quad\Longrightarrow\quad \boxed{T = 4\sqrt{\frac{\ell}{g}}\,K\Big(\sin\tfrac{\theta_{\max}}{2}\Big)} \tag{115}$$

— the elliptic-integral period stated in Remark 4.1, now derived. Every qualitative feature of the phase portrait becomes quantitative here: $K(k) \to \pi/2$ as $k \to 0$ recovers $T = 2\pi\sqrt{\ell/g}$, and $K(k) \to \infty$ as $k \to 1$ ($\theta_{\max} \to \pi$) is the diverging period at the separatrix.

7.2 Complete solution of the double pendulum

The Hamiltonian for any quadratic kinetic energy. One structural proposition organizes everything.

Proposition 7.1. *If* $L = \frac{1}{2}\,\dot{\vec{q}}^{\mathsf{T}}\mathsf{M}(\vec{q})\,\dot{\vec{q}} - V(\vec{q})$ *with* $\mathsf{M}(\vec{q})$ *symmetric positive definite, then*

$$\vec{p} = \mathsf{M}(\vec{q})\,\dot{\vec{q}}, \qquad \dot{\vec{q}} = \mathsf{M}(\vec{q})^{-1}\vec{p}, \qquad \boxed{H(\vec{q},\vec{p}) = \tfrac{1}{2}\,\vec{p}^{\mathsf{T}}\mathsf{M}(\vec{q})^{-1}\,\vec{p} + V(\vec{q}).} \tag{116}$$

Proof. $p_i = \partial L/\partial \dot{q}_i = \sum_j M_{ij}\dot{q}_j$ is $\vec{p} = \mathsf{M}\dot{\vec{q}}$; positive definiteness makes M invertible, giving the middle display. Then, using the symmetry of M^{-1},

$$H = \vec{p}^{\mathsf{T}}\dot{\vec{q}} - L = \vec{p}^{\mathsf{T}}\mathsf{M}^{-1}\vec{p} - \tfrac{1}{2}(\mathsf{M}^{-1}\vec{p})^{\mathsf{T}}\mathsf{M}\,(\mathsf{M}^{-1}\vec{p}) + V = \tfrac{1}{2}\,\vec{p}^{\mathsf{T}}\mathsf{M}^{-1}\vec{p} + V. \tag{117}$$

(This is Proposition 6.4 again — $H = T + V$ — with T now written in the momenta.) ☐

The exact double-pendulum Hamiltonian. From §5, with $\Delta = \theta_1 - \theta_2$,

$$\mathsf{M}(\vec{\theta}) = \begin{pmatrix} (m_1{+}m_2)\ell_1^2 & m_2\ell_1\ell_2\cos\Delta \\ m_2\ell_1\ell_2\cos\Delta & m_2\ell_2^2 \end{pmatrix}, \tag{118}$$

whose determinant deserves one display of its own:

$$\det\mathsf{M} = (m_1{+}m_2)\,m_2\,\ell_1^2\ell_2^2 - m_2^2\ell_1^2\ell_2^2\cos^2\Delta = m_2\,\ell_1^2\ell_2^2\left(m_1 + m_2\sin^2\Delta\right) \;>\; 0 \tag{119}$$

for *all* configurations — the mass matrix never degenerates (even with the rods aligned, $\sin\Delta = 0$, the m_1 term keeps it positive), so the Legendre transform is globally legitimate. Inverting the 2×2 matrix and applying Proposition 7.1:

$$\boxed{H = \frac{m_2\ell_2^2\,p_1^2 + (m_1{+}m_2)\ell_1^2\,p_2^2 - 2m_2\ell_1\ell_2\cos\Delta\;p_1p_2}{2\,m_2\,\ell_1^2\ell_2^2\left(m_1 + m_2\sin^2\Delta\right)} \;-\; (m_1{+}m_2)g\ell_1\cos\theta_1 - m_2g\ell_2\cos\theta_2} \tag{120}$$

with $p_1 = (m_1{+}m_2)\ell_1^2\dot{\theta}_1 + m_2\ell_1\ell_2\cos\Delta\,\dot{\theta}_2$ and $p_2 = m_2\ell_2^2\dot{\theta}_2 + m_2\ell_1\ell_2\cos\Delta\,\dot{\theta}_1$. The exact dynamics is the four first-order equations

$$\dot{\vec{\theta}} = \mathsf{M}(\vec{\theta})^{-1}\vec{p}, \qquad \dot{\vec{p}} = -\frac{\partial}{\partial\vec{\theta}}\Big[\tfrac{1}{2}\vec{p}^{\mathsf{T}}\mathsf{M}(\vec{\theta})^{-1}\vec{p}\Big] - \frac{\partial V}{\partial\vec{\theta}}, \tag{121}$$

where the configuration-dependence of the inverse is differentiated via $\partial_\theta\,\mathsf{M}^{-1} = -\,\mathsf{M}^{-1}(\partial_\theta\mathsf{M})\,\mathsf{M}^{-1}$ (differentiate $\mathsf{M}\mathsf{M}^{-1} = \mathbb{1}$ by the product rule and solve). These four equations are the complete dynamical content of the system — what a numerical integrator integrates, and the flow whose chaos §11 describes.

Small angles: decoupling and solving Hamilton's equations. Quadratic truncation: in the kinetic term $\vec{p}$ is already first-order small, so $\mathsf{M}(\vec{\theta})^{-1}$ may be evaluated at equilibrium ($\cos\Delta \to 1$; the correction enters at fourth order), giving the constant matrix M_0 of §5.3; the potential truncates as before. With K from (83),

$$H_2 = \tfrac{1}{2}\,\vec{p}^{\mathsf{T}}\mathsf{M}_0^{-1}\vec{p} + \tfrac{1}{2}\,\vec{\theta}^{\mathsf{T}}\mathsf{K}\,\vec{\theta}, \qquad \dot{\vec{\theta}} = \mathsf{M}_0^{-1}\vec{p}, \quad \dot{\vec{p}} = -\,\mathsf{K}\vec{\theta}. \tag{122}$$

Now the payoff of Theorem 5.2. Take the generalized eigenvectors $\mathsf{K}\vec{v}_r = \omega_r^2\mathsf{M}_0\vec{v}_r$, normalized to $\vec{v}_r^{\mathsf{T}}\mathsf{M}_0\vec{v}_s = \delta_{rs}$, and define *mode coordinates and mode momenta*

$$\xi_r \equiv \vec{v}_r^{\mathsf{T}}\,\mathsf{M}_0\,\vec{\theta}, \qquad \pi_r \equiv \vec{v}_r^{\mathsf{T}}\,\vec{p}. \tag{123}$$

Differentiate each along the flow (122):

$$\dot{\xi}_r = \vec{v}_r^{\mathsf{T}}\mathsf{M}_0\,\dot{\vec{\theta}} = \vec{v}_r^{\mathsf{T}}\mathsf{M}_0\,\mathsf{M}_0^{-1}\vec{p} = \vec{v}_r^{\mathsf{T}}\vec{p} = \pi_r, \tag{124}$$

$$\dot{\pi}_r = \vec{v}_r^{\mathsf{T}}\dot{\vec{p}} = -\,\vec{v}_r^{\mathsf{T}}\mathsf{K}\,\vec{\theta} = -\sum_s \xi_s\,\vec{v}_r^{\mathsf{T}}\mathsf{K}\vec{v}_s = -\,\omega_r^2\,\xi_r \tag{125}$$

(the last step expands $\vec{\theta} = \sum_s \xi_s\vec{v}_s$ — valid because the $\vec{v}_s$ are a basis with coefficients recovered by M_0-orthonormality — and uses $\vec{v}_r^{\mathsf{T}}\mathsf{K}\vec{v}_s = \omega_s^2\,\vec{v}_r^{\mathsf{T}}\mathsf{M}_0\vec{v}_s = \omega_s^2\delta_{rs}$). Each pair obeys

$$\dot{\xi}_r = \pi_r, \qquad \dot{\pi}_r = -\,\omega_r^2\,\xi_r \tag{126}$$

— Hamilton's equations of the unit-mass oscillator Hamiltonian $\frac{1}{2}\pi_r^2 + \frac{1}{2}\omega_r^2\xi_r^2$; and indeed substituting the expansions into H_2 collapses it (same two identities) to

$$\boxed{H_2 = \sum_r \Big[\tfrac{1}{2}\,\pi_r^2 + \tfrac{1}{2}\,\omega_r^2\,\xi_r^2\Big]} \tag{127}$$

— *two independent oscillators, additively.* Where the Lagrangian treatment decoupled the *equations* (Theorem 5.2), the Hamiltonian treatment decouples the *energy itself*: the modes carry separately conserved energies ($\dot{H}_r = 0$ each, by the one-oscillator computation). That the change of variables $(\vec{\theta}, \vec{p}) \to (\xi, \pi)$ preserves the Hamiltonian form of the equations makes it our first nontrivial *canonical transformation*; §8 will supply the certificate that verifies this in one line.

The complete general solution. Each mode is solved by (109) with $m\ell^2 \to 1$, $\omega_0 \to \omega_r$:

$$\xi_r(t) = \xi_r(0)\cos\omega_r t + \frac{\pi_r(0)}{\omega_r}\sin\omega_r t, \qquad \pi_r(t) = \pi_r(0)\cos\omega_r t - \omega_r\,\xi_r(0)\sin\omega_r t, \tag{128}$$

and translating back through the definitions (using $\vec{p}(0) = \mathsf{M}_0\dot{\vec{\theta}}(0)$, so $\pi_r(0) = \vec{v}_r^{\mathsf{T}}\mathsf{M}_0\dot{\vec{\theta}}(0)$),

$$\boxed{\vec{\theta}(t) = \sum_r \Big[\big(\vec{v}_r^{\mathsf{T}}\mathsf{M}_0\,\vec{\theta}(0)\big)\cos\omega_r t + \frac{\vec{v}_r^{\mathsf{T}}\mathsf{M}_0\,\dot{\vec{\theta}}(0)}{\omega_r}\sin\omega_r t\Big]\,\vec{v}_r} \tag{129}$$

Four numbers in, one motion out. For equal masses and lengths everything is explicit: $\omega_\mp^2 = (2 \mp \sqrt{2})\,g/\ell$, unnormalized modes $(1, \pm\sqrt{2})$, with M_0-norms

$$(1,\ \sqrt{2})\,\mathsf{M}_0\begin{pmatrix}1\\ \sqrt{2}\end{pmatrix} = m\ell^2\big(4 + 2\sqrt{2}\big), \qquad (1,\ -\sqrt{2})\,\mathsf{M}_0\begin{pmatrix}1\\ -\sqrt{2}\end{pmatrix} = m\ell^2\big(4 - 2\sqrt{2}\big), \tag{130}$$

so $\vec{v}_{\mp} = (1, \pm\sqrt{2})/\sqrt{m\ell^2(4 \pm 2\sqrt{2})}$ in (129). One concrete instance, fully evaluated: release from rest with the upper rod displaced, $\vec{\theta}(0) = (\alpha, 0)$, $\dot{\vec{\theta}}(0) = 0$. The mode amplitudes are $\vec{v}_{\mp}^{\mathsf{T}}\mathsf{M}_0\vec{\theta}(0)$ with $\mathsf{M}_0\vec{\theta}(0) = \alpha\, m\ell^2\,(2,1)^{\mathsf{T}}$, and assembling (129) (all $m\ell^2$ factors cancel between the amplitude and the normalized eigenvector):

$$\theta_1(t) = \frac{\alpha}{2}\Big[\cos\omega_- t + \cos\omega_+ t\Big], \qquad \theta_2(t) = \frac{\alpha}{\sqrt{2}}\Big[\cos\omega_- t - \cos\omega_+ t\Big] \tag{131}$$

(equivalently: expanding the initial condition in the eigenbasis, $(\alpha, 0) = \frac{\alpha}{2}(1, \sqrt{2}) + \frac{\alpha}{2}(1, -\sqrt{2})$, each mode released from rest evolves with its own cosine). Checks: $\theta_1(0) = \alpha$ ✓; $\theta_2(0) = 0$ ✓; both initial velocities vanish since every term is a cosine ✓; and the initial *accelerations* — the check sensitive to the mode weights — match Newton: differentiating twice at $t = 0$ gives $\ddot{\vec{\theta}}(0) = \big(-\frac{\alpha}{2}(\omega_-^2 + \omega_+^2),\, -\frac{\alpha}{\sqrt{2}}(\omega_-^2 - \omega_+^2)\big) = \big(-2\alpha g/\ell,\, +2\alpha g/\ell\big)$, in agreement with $\ddot{\vec{\theta}}(0) = -\mathsf{M}_0^{-1}\mathsf{K}\,\vec{\theta}(0)$ evaluated directly ✓. The lower bob (131) is a difference of cosines at frequency ratio $\omega_+/\omega_- = \sqrt{(2+\sqrt{2})/(2-\sqrt{2})} = 1 + \sqrt{2}$ — irrational: by the product identity of the oscillator companion the motion beats, energy migrating between the bobs, and because the ratio is irrational the motion never exactly repeats. Complete, explicit, checkable: the Hamiltonian route closes on the same physics the Lagrangian one opened.

8 Poisson Brackets

Hamilton's equations treat $\vec{q}$ and $\vec{p}$ symmetrically; the Poisson bracket is the algebraic structure that makes the symmetry total. The motivation is threefold: it packages the dynamics of *every* observable in one formula (not just coordinates and momenta); it converts questions about conservation into questions of pure algebra; and it supplies the practical certificate for *canonical* changes of variables — the one promised in §7.2. It is also, nearly symbol for symbol, the classical face of the quantum commutator.

8.1 Definition and algebra

Definition 8.1. For C^1 functions f, g on phase space (possibly time-dependent),

$$\boxed{\{f, g\} \;\equiv\; \sum_{i=1}^{n}\left(\frac{\partial f}{\partial q_i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q_i}\right).} \tag{132}$$

Three properties are immediate from the definition, one is not:

1. *Bilinearity*: $\{af + bg, h\} = a\{f, h\} + b\{g, h\}$ (a, b constants) — derivatives are linear.
2. *Antisymmetry*: $\{f, g\} = -\{g, f\}$; in particular $\{f, f\} = 0$ — swap the two terms.
3. *Leibniz rule*: $\{fg, h\} = f\{g, h\} + \{f, h\}\, g$ — apply the product rule inside each partial derivative of fg and regroup: $\partial_i(fg) = f\,\partial_i g + (\partial_i f)\, g$ in both slots. The bracket with h acts as a *derivation.*
4. *Jacobi identity*:

$$\boxed{\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0.} \tag{133}$$

Not obvious: each summand contains second derivatives, and the identity asserts a threefold cancellation. The full computation is Appendix F; the organizing idea (worth knowing) is to sort all terms by *which* function carries the second derivative and check that each of the three groups cancels internally, using only the symmetry of mixed partials.

The canonical variables themselves obey the *fundamental brackets* (direct evaluation: $\partial q_i/\partial q_j = \delta_{ij}$ etc.):

$$\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \boxed{\{q_i, p_j\} = \delta_{ij}} \tag{134}$$

Example 8.2 (The certificate promised in §7.2)**.** A change of variables is *canonical* — preserves the Hamiltonian form of the equations for every H — precisely when the new variables reproduce the fundamental brackets (we use the criterion here in advance; §10.1 states and proves it in the direction this example needs). For the mode variables of the double pendulum, $\xi_r = \vec{v}_r^{\mathsf{T}}\mathsf{M}_0\vec{\theta}$, $\pi_r = \vec{v}_r^{\mathsf{T}}\vec{p}$:

$$\{\xi_r, \pi_s\} = \sum_i \left[\frac{\partial \xi_r}{\partial \theta_i}\frac{\partial \pi_s}{\partial p_i} - \underbrace{\frac{\partial \xi_r}{\partial p_i}}_{=0}\frac{\partial \pi_s}{\partial \theta_i}\right] = \sum_i (\mathsf{M}_0\vec{v}_r)_i\,(\vec{v}_s)_i = \vec{v}_s^{\mathsf{T}}\mathsf{M}_0\,\vec{v}_r = \delta_{rs}, \tag{135}$$

and $\{\xi_r, \xi_s\} = \{\pi_r, \pi_s\} = 0$ similarly (each variable of a pair depends on only one half of the phase-space coordinates). The M_0-orthonormality of the modes — which entered §5.4 as linear algebra — is revealed as exactly the condition that mode variables are canonical.

8.2 Dynamics as bracket flow

Proposition 8.3. *Along any solution of Hamilton's equations, every C^1 phase-space function $f(\vec{q}, \vec{p}, t)$ evolves by*

$$\boxed{\frac{\mathrm{d}f}{\mathrm{d}t} = \{f, H\} + \frac{\partial f}{\partial t}.} \tag{136}$$

Proof. Chain rule along the trajectory, then Hamilton's equations (95):

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \sum_i \Big[\frac{\partial f}{\partial q_i}\dot{q}_i + \frac{\partial f}{\partial p_i}\dot{p}_i\Big] + \frac{\partial f}{\partial t} = \sum_i \Big[\frac{\partial f}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial H}{\partial q_i}\Big] + \frac{\partial f}{\partial t} = \{f, H\} + \frac{\partial f}{\partial t}. \tag{137}$$

□

Special cases assemble the whole formalism into one notation: taking $f = q_i$ and $f = p_i$ recovers Hamilton's equations themselves,

$$\dot{q}_i = \{q_i, H\}, \qquad \dot{p}_i = \{p_i, H\} \tag{138}$$

— coordinates and momenta on *identical* footing; taking $f = H$ recovers $\mathrm{d}H/\mathrm{d}t = \partial H/\partial t$ (antisymmetry kills $\{H, H\}$); and conservation acquires its final form:

$$f \text{ (time-independent) is conserved} \quad \Longleftrightarrow \quad \{f, H\} = 0. \tag{139}$$

Corollary 8.4 (Poisson's theorem)**.** *If f and g are conserved (time-independent, $\{f, H\} = \{g, H\} = 0$), so is $\{f, g\}$.*

Proof. By Jacobi with $h = H$: $\{\{f, g\}, H\} = -\{\{g, H\}, f\} - \{\{H, f\}, g\} = -\{0, f\} - \{0, g\} = 0$ (after shuffling with antisymmetry). New conserved quantities from old, by pure algebra — occasionally the bracket closes on the old ones (angular momenta: $\{L_x, L_y\} = L_z$), occasionally it produces genuinely new information. □

8.3 Noether revisited: conserved quantities generate their symmetries

Noether's theorem (§3.3) was a one-way street: symmetry in, conserved quantity out. The Hamiltonian formalism reveals the street runs both ways — and that the two directions are the *same equation*.

The key construction: any time-independent phase-space function $G(\vec{q}, \vec{p})$ can be installed in place of H and used to generate a *flow* — a one-parameter family of transformations obtained by solving Hamilton's equations with G as the generator,

$$\frac{\mathrm{d}q_i}{\mathrm{d}s} = \frac{\partial G}{\partial p_i}, \qquad \frac{\mathrm{d}p_i}{\mathrm{d}s} = -\frac{\partial G}{\partial q_i} \qquad \left(\text{so that } \frac{\mathrm{d}f}{\mathrm{d}s} = \{f, G\} \text{ for any } f\right). \tag{140}$$

Time evolution is then just the special case $G = H$, $s = t$. Now read one bracket in two directions:

$$\boxed{\underbrace{\{G, H\} = 0}_{G \text{ conserved under time evolution}} \qquad \Longleftrightarrow \qquad \underbrace{\{H, G\} = 0}_{H \text{ invariant under the flow of } G}} \tag{141}$$

— the equivalence is the antisymmetry of the bracket, nothing more. *A quantity is conserved precisely when the transformation it generates is a symmetry of the Hamiltonian.* Symmetry and conservation, which Noether's theorem related by a computation, are exposed as one relation viewed from its two ends.

The circle closes on §3.3 exactly: given a point symmetry with generator $\eta_i(\vec{q})$, Noether's conserved quantity was $I = \sum_i p_i \, \eta_i(\vec{q})$ — and the flow that *this* I generates is

$$\frac{\mathrm{d}q_i}{\mathrm{d}s} = \frac{\partial I}{\partial p_i} = \eta_i(\vec{q}), \tag{142}$$

the original symmetry transformation itself. Check two instances:

- $G = p$ (one degree of freedom): flow $\mathrm{d}q/\mathrm{d}s = 1$, $\mathrm{d}p/\mathrm{d}s = 0$, i.e. $q \mapsto q + s$ — *momentum generates translations.*
- $G = L_z = xp_y - yp_x$: flow $\mathrm{d}x/\mathrm{d}s = \partial L_z/\partial p_x = -y$, $\mathrm{d}y/\mathrm{d}s = \partial L_z/\partial p_y = x$ — the infinitesimal rotation, with generator $(-y, x)$ matching the η of the Noether example in §3.3 — *angular momentum generates rotations.*

This is why conservation laws are placed where they are in this document: the theorem and its proof live on the Lagrangian side, where symmetries of L are stated; but their deepest formulation — observables as generators, symmetry $\Leftrightarrow$ conservation as a single antisymmetric bracket — is Hamiltonian property, and it is the form that survives into quantum mechanics ($\hat{p}$ generates translations, $\hat{L}_z$ generates rotations, $\hat{H}$ generates time evolution).

Remark 8.5 (The structure that survives quantization). The triple (bilinear, antisymmetric, Jacobi) makes phase-space functions a *Lie algebra*, with the Leibniz rule tying the algebra to the geometry. Quantum mechanics keeps this entire structure and changes one thing: $\{f, g\} \to \frac{1}{\mathrm{i}\hbar}[\hat{f}, \hat{g}]$. In particular $\{q, p\} = 1$ becomes $[\hat{q}, \hat{p}] = \mathrm{i}\hbar$. The bracket is not a notational convenience; it is the load-bearing wall.

9 Phase Space

9.1 Portraits

The state of the system is the point $(\vec{q}, \vec{p})$; Hamilton's equations define a *flow* on this $2n$-dimensional space, and for one degree of freedom the entire global behavior can be drawn: since H is conserved, *trajectories are level curves of* H, and drawing dynamics reduces to drawing contour lines.

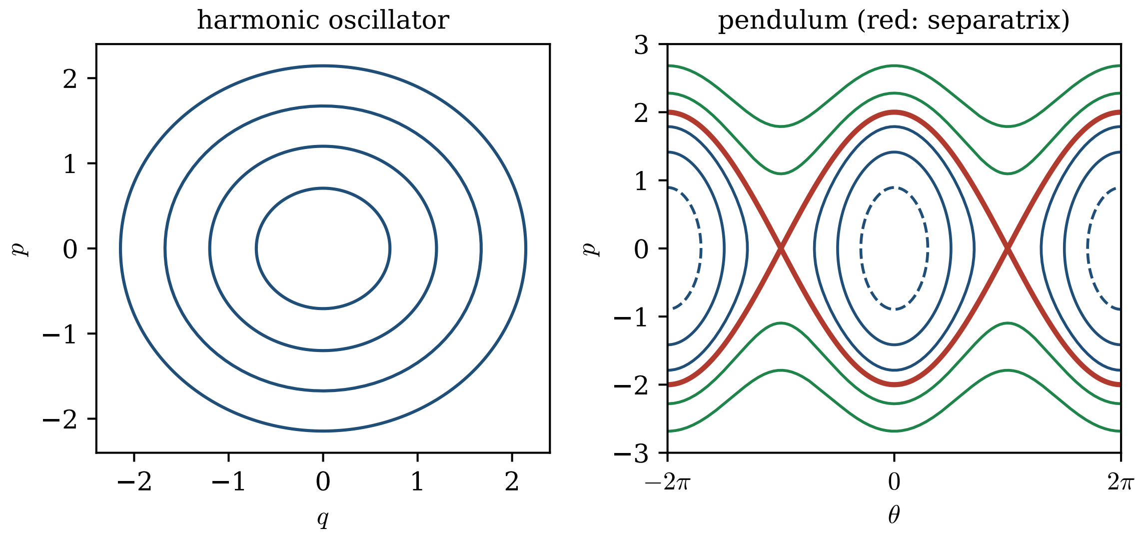


Figure 3: Left: the oscillator, $H = p^2/2m + kq^2/2$ — nested ellipses, one period each. Right: the pendulum (106) in normalized variables — closed curves (libration) inside the red *separatrix*, wavy open curves (rotation) outside it.

Read the pendulum portrait as a summary of everything the system can do:

- **Centers**: $(\theta, p) = (0, 0)$ (and its 2π translates) are stable equilibria; nearby level curves are small ellipses — the small-angle oscillator of §4, literally visible as the elliptical shape of low-energy contours.
- **Saddle**: $(\pm\pi, 0)$, the inverted pendulum, is an unstable equilibrium; level curves near it form an X.
- **Separatrix** (red): the level set $H = +mg\ell$ through the saddles — the watershed between *libration* (back-and-forth, closed curves) and *rotation* (over the top, open curves). A motion *on* the separatrix approaches the inverted position forever without arriving; the period of libration diverges as the amplitude approaches π (the elliptic K of Remark 4.1 blowing up as $k \to 1$).

One conserved function, one picture, the full qualitative theory — the economy that makes the Hamiltonian viewpoint the natural language for global and long-time questions.

9.2 Liouville's theorem

The flow does not merely move points; it moves them *incompressibly*.

Theorem 9.1 (Liouville)**.** *The Hamiltonian phase flow preserves phase-space volume: if a region* Ω_0 *of initial conditions is carried by the flow to* Ω_t*, then* $\mathrm{vol}(\Omega_t) = \mathrm{vol}(\Omega_0)$.

Proof. The flow is generated by the phase-space velocity field

$$X = \big(\dot q_1, \dots, \dot q_n,\ \dot p_1, \dots, \dot p_n\big) = \Big(\frac{\partial H}{\partial p_1}, \dots, \frac{\partial H}{\partial p_n},\ -\frac{\partial H}{\partial q_1}, \dots, -\frac{\partial H}{\partial q_n}\Big). \tag{143}$$

Its divergence vanishes *identically*:

$$\nabla \cdot X = \sum_{i=1}^{n} \left[\frac{\partial}{\partial q_i} \frac{\partial H}{\partial p_i} + \frac{\partial}{\partial p_i} \Big(-\frac{\partial H}{\partial q_i} \Big) \right] = \sum_{i=1}^{n} \left[\frac{\partial^2 H}{\partial q_i \, \partial p_i} - \frac{\partial^2 H}{\partial p_i \, \partial q_i} \right] = 0 \tag{144}$$

by equality of mixed partials ($H \in C^2$) — note the mechanism: the *minus sign* in Hamilton's equations pairs each expansion rate $\partial \dot{q}_i / \partial q_i$ with an equal and opposite contraction rate $\partial \dot{p}_i / \partial p_i$. For a region advected by a flow, the rate of change of volume is the flux of the velocity field through its boundary [the transport identity; stated — it is the change-of-variables formula for the flow map, differentiated at t], which by the divergence theorem is the integral of the divergence:

$$\frac{\mathrm{d}}{\mathrm{d}t} \operatorname{vol}(\Omega_t) = \oint_{\partial \Omega_t} X \cdot \vec{n} \, \mathrm{d}A = \int_{\Omega_t} \nabla \cdot X \, \mathrm{d}V = 0. \tag{145}$$

□

Remark 9.2 (What Liouville forbids and what breaks it)*.* Volume preservation forbids *attractors*: no open set of initial conditions in a Hamiltonian system can funnel into a point or a small cycle, because its volume would have to shrink. Contrast the damped oscillator of the companion document: adding $-b\dot{q}$ makes $\dot{p} = -kq - \frac{b}{m} p$, the divergence becomes $-b/m < 0$, and every region contracts exponentially onto the rest point — dissipation is, precisely, the failure of Liouville. The theorem is also the opening axiom of statistical mechanics: it is what allows a probability density on phase space to be transported like an incompressible fluid.

10 Canonical Transformations and Hamilton–Jacobi Theory

The story of this document has been a story about changes of variables. Theorem 2.11 established that the Lagrangian framework survives *any* smooth invertible change of the coordinates $\vec{q}$; normal coordinates (§5.5) showed how much a good choice delivers — the double pendulum fell apart into two clocks. But the Hamiltonian framework has twice as many variables, and §8 already met a change of variables that mixes them: the mode variables (ξ_r, π_r) of Example 8.2 are built from the θ's alone, but nothing in the bracket criterion required that. This section asks the question that example left open — *which* transformations of phase space preserve Hamiltonian mechanics — proves the criterion the example used, and then pushes the freedom to its logical end: a transformation so well adapted that the new Hamiltonian is *zero* and the new variables do not move at all. The condition for its generating function is a single partial differential equation, the *Hamilton–Jacobi equation*, and the function that satisfies it turns out to be an old acquaintance: the action itself.

10.1 Canonical transformations

Consider a smooth, invertible, possibly time-dependent change of phase-space variables

$$Q_i = Q_i(\vec{q}, \vec{p}, t), \qquad P_i = P_i(\vec{q}, \vec{p}, t), \qquad i = 1, \ldots, n. \tag{146}$$

Definition 10.1 (Canonical transformation)**.** The transformation (146) is *canonical* if for every Hamiltonian $H(\vec{q}, \vec{p}, t)$ there exists a function $K(\vec{Q}, \vec{P}, t)$ such that curves satisfying Hamilton's equations for $(\vec{q}, \vec{p}; H)$ are carried exactly to curves satisfying Hamilton's equations for $(\vec{Q}, \vec{P}; K)$:

$$\dot{Q}_i = \frac{\partial K}{\partial P_i}, \qquad \dot{P}_i = -\frac{\partial K}{\partial Q_i}. \tag{147}$$

The demand "for every H" is what gives the notion teeth: we are classifying changes of variables that respect the *form* of the theory, not tricks adapted to one system — the same standard Theorem 2.11 set on the Lagrangian side.

Two examples calibrate how large the class is. First, every point transformation $\vec{Q} = \vec{Q}(\vec{q}, t)$ extends to a canonical transformation (with P built from the covariant transformation of momenta); this is Theorem 2.11 read through the Legendre transform, and we will recover it from the generating-function machinery below. Second — and this is what point transformations can never do — the swap

$$Q = p, \qquad P = -q \tag{148}$$

is canonical. Check it in full for one degree of freedom: define $K(Q,P,t) \equiv H(q,p,t)$, i.e. $K(Q,P,t) = H(-P,Q,t)$. Then along a solution of the old equations,

$$\dot{Q} = \dot{p} = -\frac{\partial H}{\partial q} = -\frac{\partial H}{\partial q}(-P,Q,t) = \frac{\partial K}{\partial P}, \qquad \dot{P} = -\dot{q} = -\frac{\partial H}{\partial p} = -\frac{\partial K}{\partial Q}, \tag{149}$$

where the middle steps used the chain rule on $K(Q,P,t) = H(-P,Q,t)$: $\partial K/\partial P = \partial H/\partial q \cdot (-1)$ and $\partial K/\partial Q = \partial H/\partial p$. The new equations hold with the *same* Hamiltonian, relabeled. Coordinates and momenta are not two kinds of object in this formalism; they are two halves of one state, exchangeable at the cost of a sign — the symmetry that Hamilton's equations (95) wear on their face.

The bracket criterion, now proved. Example 8.2 used, without proof, the statement that a time-independent transformation is canonical precisely when it reproduces the fundamental brackets. Here is the proof, in the direction the example needed.

Proposition 10.2 (Fundamental brackets $\Rightarrow$ canonical; time-independent case)**.** *Let $Q_i(\vec{q},\vec{p})$, $P_i(\vec{q},\vec{p})$ be a smooth invertible time-independent transformation satisfying*

$$\{Q_i, Q_j\} = 0, \qquad \{P_i, P_j\} = 0, \qquad \{Q_i, P_j\} = \delta_{ij} \qquad (\textit{all brackets in } \vec{q}, \vec{p}). \tag{150}$$

Then the transformation is canonical with $K(\vec{Q}, \vec{P}, t)$ defined by substitution, $K(\vec{Q}(\vec{q},\vec{p}), \vec{P}(\vec{q},\vec{p}), t) = H(\vec{q}, \vec{p}, t)$.

Proof. The engine is Proposition 8.3: along any solution, $\dot{F} = \{F, H\}$ for every phase-space function F (with no $\partial F/\partial t$ term here, since Q_i, P_i are time-independent). Apply it to $F = Q_i$:

$$\dot{Q}_i = \{Q_i, H\}. \tag{151}$$

Now express H through the new variables, $H = K(\vec{Q}(\vec{q},\vec{p}), \vec{P}(\vec{q},\vec{p}), t)$, and expand the bracket by the chain rule. Writing it out from the definition — no shortcut, since this is the step that carries the proof:

$$\begin{aligned}
\{Q_i, H\} &= \sum_{a=1}^{n} \Big[\frac{\partial Q_i}{\partial q_a}\frac{\partial H}{\partial p_a} - \frac{\partial Q_i}{\partial p_a}\frac{\partial H}{\partial q_a}\Big] \\
&= \sum_{a=1}^{n} \left[\frac{\partial Q_i}{\partial q_a} \sum_{j=1}^{n} \Big(\frac{\partial K}{\partial Q_j}\frac{\partial Q_j}{\partial p_a} + \frac{\partial K}{\partial P_j}\frac{\partial P_j}{\partial p_a}\Big) - \frac{\partial Q_i}{\partial p_a} \sum_{j=1}^{n} \Big(\frac{\partial K}{\partial Q_j}\frac{\partial Q_j}{\partial q_a} + \frac{\partial K}{\partial P_j}\frac{\partial P_j}{\partial q_a}\Big)\right] \\
&= \sum_{j=1}^{n} \frac{\partial K}{\partial Q_j} \underbrace{\sum_{a=1}^{n}\Big[\frac{\partial Q_i}{\partial q_a}\frac{\partial Q_j}{\partial p_a} - \frac{\partial Q_i}{\partial p_a}\frac{\partial Q_j}{\partial q_a}\Big]}_{=\{Q_i, Q_j\}} + \sum_{j=1}^{n} \frac{\partial K}{\partial P_j} \underbrace{\sum_{a=1}^{n}\Big[\frac{\partial Q_i}{\partial q_a}\frac{\partial P_j}{\partial p_a} - \frac{\partial Q_i}{\partial p_a}\frac{\partial P_j}{\partial q_a}\Big]}_{=\{Q_i, P_j\}}.
\end{aligned} \tag{152}$$

The first line is the definition of the bracket; the second applies the ordinary chain rule to $\partial H/\partial p_a$ and $\partial H/\partial q_a$; the third merely regroups the four resulting double sums by which derivative of K they carry — and the coefficients that emerge are exactly the fundamental brackets. By hypothesis (150),

$$\dot{Q}_i = \sum_j \frac{\partial K}{\partial Q_j} \cdot 0 + \sum_j \frac{\partial K}{\partial P_j}\, \delta_{ij} = \frac{\partial K}{\partial P_i}. \tag{153}$$

The identical computation with $F = P_i$ gives

$$\dot{P}_i = \{P_i, H\} = \sum_j \frac{\partial K}{\partial Q_j}\, \{P_i, Q_j\} + \sum_j \frac{\partial K}{\partial P_j}\, \{P_i, P_j\} = \sum_j \frac{\partial K}{\partial Q_j}\, (-\delta_{ij}) = -\frac{\partial K}{\partial Q_i}, \tag{154}$$

using antisymmetry, $\{P_i, Q_j\} = -\{Q_j, P_i\} = -\delta_{ij}$. Both Hamilton equations hold in the new variables, for arbitrary H. $\square$

This closes the loop opened by Example 8.2: the mode variables of the double pendulum, certified there by their brackets, are now certified canonical by proof. Time-*dependent* transformations need one more idea — the new Hamiltonian acquires a correction term, $K \neq H$ in general — and that idea is the generating function.

10.2 Generating functions

The bracket criterion certifies a transformation after the fact. To *manufacture* canonical transformations — and eventually to search for the best one — we need a handle that produces them. The handle comes from the variational principle of §6.2, Derivation 2: Hamilton's equations for $(\vec{q}, \vec{p}; H)$ say exactly that the phase-space action

$$S[\vec{q}, \vec{p}] = \int_{t_0}^{t_1} \Big[\sum_i p_i\, \dot{q}_i - H(\vec{q}, \vec{p}, t)\Big] \mathrm{d}t \tag{155}$$

is stationary. Two integrands that differ by a total time derivative have the same interior variational content — so if the old and new integrands differ by one, the two sets of Hamilton equations must select the same curves. Making this precise is the next proposition; the care it takes is instructive.

Proposition 10.3 (Equivalence up to a total derivative)**.** *Let* (146) *be a smooth invertible transformation, and suppose there exist smooth functions $K(\vec{Q}, \vec{P}, t)$ and F (of the phase-space variables and t) such that along* every *smooth phase-space curve,*

$$\sum_i p_i\, \dot{q}_i - H \;=\; \sum_i P_i\, \dot{Q}_i - K \;+\; \frac{\mathrm{d}F}{\mathrm{d}t}. \tag{156}$$

Then a curve satisfies Hamilton's equations for $(\vec{q}, \vec{p}; H)$ if and only if its image satisfies Hamilton's equations for $(\vec{Q}, \vec{P}; K)$.

Proof. Integrate (156) along any curve from t_0 to t_1:

$$S[\vec{q}, \vec{p}] = S'[\vec{Q}, \vec{P}] + F\big|_{t_1} - F\big|_{t_0}, \tag{157}$$

where S' is the phase-space action built from $(\vec{Q}, \vec{P}; K)$. Now restrict attention to *compactly supported* variations: variations $(\delta\vec{q}, \delta\vec{p})$ that vanish identically on neighborhoods of t_0 and t_1 (the bump functions of Appendix B are of exactly this kind). Three observations, each doing one job:

(i) The boundary term drops out. A compactly supported variation leaves the curve unchanged near the endpoints, so $F|_{t_0}$ and $F|_{t_1}$ — functions of the curve's endpoint data — are unvaried. Varying (157),

$$\delta S = \delta S' \qquad \text{(compactly supported variations).} \tag{158}$$

(ii) Compact support is preserved by the transformation. Since (146) is smooth and invertible, a variation of $(\vec{q}, \vec{p})$ vanishing near the endpoints induces a variation of $(\vec{Q}, \vec{P})$ vanishing near the endpoints, and conversely. So "compactly supported" means the same thing on both sides of (158).

(iii) Stationarity under compactly supported variations is exactly Hamilton's equations. Rerun Derivation 2 of §6.2 with compactly supported variations: the $\vec{p}$-variation needs no boundary condition at all, and the $\vec{q}$-variation's single integration by parts produces a boundary term $[\,p_i\,\delta q_i\,]_{t_0}^{t_1}$ that vanishes because δq_i is identically zero near the ends. The fundamental lemma (Appendix B) requires only compactly supported test functions, so it applies verbatim, and delivers $\dot{q}_i = \partial H/\partial p_i$, $\dot{p}_i = -\partial H/\partial q_i$. The same argument runs for $(\vec{Q}, \vec{P}; K)$.

Combining: the curve solves the old equations $\iff$ $\delta S = 0$ for all compactly supported variations $\iff$ $\delta S' = 0$ for all compactly supported variations $\iff$ the image solves the new equations. □

Remark 10.4 (Why the usual endpoint phrasing is delicate). The textbook version of this argument pins $\vec{q}$ at the endpoints, pins $\vec{Q}$ at the endpoints, and declares both boundary terms dead. But those are two different conditions: when $\vec{Q}$ mixes coordinates and momenta, $\delta\vec{q}(t_0) = 0$ does *not* force $\delta\vec{Q}(t_0) = 0$. The variation of the boundary term is $\sum_i (p_i\,\delta q_i - P_i\,\delta Q_i)\big|_{t_0}^{t_1}$, and killing only its first half leaves the argument with a gap. Compactly supported variations close the gap wholesale: they kill every possible boundary term at once, and they are all that the fundamental lemma ever needed. The equations of motion are a statement about the *interior* of the time interval; it is fitting that the clean proof never touches the ends.

Type-2 generating functions. Proposition 10.3 converts the search for canonical transformations into the search for identities of the form (156) — and those can be manufactured from a single scalar function. Among the standard choices of independent variables we develop first the one Jacobi found most useful — a function of the *old coordinates and new momenta* — and then the remaining three, each in full.

Proposition 10.5 (The type-2 construction). *Let $F_2(\vec{q}, \vec{P}, t)$ be smooth on some domain, with*

$$\det\Big(\frac{\partial^2 F_2}{\partial q_i\,\partial P_j}\Big) \neq 0. \tag{159}$$

Define the transformation implicitly by

$$p_i = \frac{\partial F_2}{\partial q_i}(\vec{q}, \vec{P}, t), \qquad Q_i = \frac{\partial F_2}{\partial P_i}(\vec{q}, \vec{P}, t), \qquad K = H + \frac{\partial F_2}{\partial t}. \tag{160}$$

Then: (i) the first relation locally determines $\vec{P}(\vec{q}, \vec{p}, t)$, so (160) defines a genuine transformation $(\vec{q}, \vec{p}) \mapsto (\vec{Q}, \vec{P})$; and (ii) this transformation is canonical, with new Hamiltonian K.

Proof. (i) Fix t. The map $\vec{P} \mapsto \partial F_2/\partial\vec{q}$ has Jacobian matrix $\partial^2 F_2/\partial P_j\,\partial q_i$, whose determinant is nonzero by (159). The implicit function theorem — invoked exactly as in Appendix D — solves $p_i = \partial F_2/\partial q_i(\vec{q}, \vec{P}, t)$ locally for $\vec{P}(\vec{q}, \vec{p}, t)$, smoothly. Substituting into the second relation yields $\vec{Q}(\vec{q}, \vec{p}, t)$.

(ii) We verify the identity (156) with

$$F \;=\; F_2(\vec{q},\vec{P},t) - \sum_i P_i\,Q_i, \tag{161}$$

by direct differentiation along an arbitrary curve. Slowly, term by term:

$$\begin{aligned}\frac{\mathrm{d}F}{\mathrm{d}t} &= \sum_i \frac{\partial F_2}{\partial q_i}\,\dot{q}_i + \sum_i \frac{\partial F_2}{\partial P_i}\,\dot{P}_i + \frac{\partial F_2}{\partial t} - \sum_i \dot{P}_i\,Q_i - \sum_i P_i\,\dot{Q}_i \\ &= \sum_i p_i\,\dot{q}_i + \sum_i Q_i\,\dot{P}_i + (K-H) - \sum_i Q_i\,\dot{P}_i - \sum_i P_i\,\dot{Q}_i \\ &= \Big[\sum_i p_i\,\dot{q}_i - H\Big] - \Big[\sum_i P_i\,\dot{Q}_i - K\Big],\end{aligned} \tag{162}$$

where the second line substituted all three defining relations (160) (this is the only place they are used, and all three are used). Rearranged, (162) *is* (156), so Proposition 10.3 applies. □

Note what the third relation in (160) did: the mismatch between old and new Hamiltonian is exactly the explicit time dependence of the generating function. For time-independent F_2 the Hamiltonian carries over unchanged, $K = H$, recovering the setting of Proposition 10.2; for time-dependent F_2 we have gained the ability to *reshape* H — the crack of daylight that Hamilton–Jacobi theory will widen into a method.

Two calibrations. The choice $F_2 = \sum_i q_i P_i$ gives $p_i = P_i$, $Q_i = q_i$, $K = H$: the identity, with nondegeneracy matrix δ_{ij}. And $F_2 = \sum_i f_i(\vec{q},t)\,P_i$ gives $Q_i = f_i(\vec{q},t)$ with $p_i = \sum_j(\partial f_j/\partial q_i)P_j$ — every point transformation, with the momenta transforming by exactly the covariant rule, and $K = H + \sum_i P_i\,\partial f_i/\partial t$. Theorem 2.11's content reappears as the two easiest entries of a much larger catalogue.

Remark 10.6 (Near-identity transformations and the bracket, reunited). Perturb the identity: $F_2 = \sum_i q_i P_i + \epsilon\, G(\vec{q},\vec{P},t)$ with ϵ small. The relations (160) give $p_i = P_i + \epsilon\,\partial G/\partial q_i$ and $Q_i = q_i + \epsilon\,\partial G/\partial P_i$, so to first order in ϵ (where $\vec{P}$ may be replaced by $\vec{p}$ inside G),

$$Q_i = q_i + \epsilon\,\frac{\partial G}{\partial p_i} + O(\epsilon^2), \qquad P_i = p_i - \epsilon\,\frac{\partial G}{\partial q_i} + O(\epsilon^2) \quad\Longrightarrow\quad \delta q_i = \epsilon\,\{q_i, G\}, \quad \delta p_i = \epsilon\,\{p_i, G\}. \tag{163}$$

Infinitesimal canonical transformations are bracket flows: every function G *generates* a one-parameter family of canonical maps, moving phase space along $\{\,\cdot\,, G\}$. Read against Proposition 8.3, the case $G = H$, $\epsilon = \mathrm{d}t$ says: *time evolution itself is a canonical transformation, generated by the Hamiltonian.* §10.6 will exhibit its generating function in closed form.

The other three types, in full. The type-2 function is one entry in a catalogue of four, indexed by which mixed pair of variables — half old, half new — the scalar function depends on. The catalogue is not a convenience menu; it is forced, and the two easiest transformations in this section already show why. The *identity* has $\vec{Q} = \vec{q}$: old and new coordinates are functionally dependent, so no function of $(\vec{q},\vec{Q})$ can carry independent information about the transformation, and indeed no type-1 function generates it. The *swap* (148) has $\vec{P} = -\vec{q}$: old coordinates and new momenta are functionally dependent, and no type-2 function generates it — the nondegeneracy condition (159) cannot even be posed, because $(\vec{q},\vec{P})$ fails to be a coordinate system on the transformation's graph. Each type covers transformations the others cannot; a complete toolkit needs all four. We now construct the remaining three by the same two-part argument as Proposition 10.5 — an implicit function theorem step and a direct verification of (156) — writing each verification out in full, because the sign pattern is exactly where this subject sheds users.

Proposition 10.7 (The type-1 construction). *Let $F_1(\vec{q},\vec{Q},t)$ be smooth with $\det\bigl(\partial^2 F_1/\partial q_i\,\partial Q_j\bigr) \neq 0$. Define*

$$p_i = \frac{\partial F_1}{\partial q_i}(\vec{q},\vec{Q},t), \qquad P_i = -\frac{\partial F_1}{\partial Q_i}(\vec{q},\vec{Q},t), \qquad K = H + \frac{\partial F_1}{\partial t}. \tag{164}$$

Then the first relation locally determines $\vec{Q}(\vec{q},\vec{p},t)$, the second then evaluates $\vec{P}$, and the resulting transformation is canonical with new Hamiltonian K.

Proof. Solvability. At fixed t, the map $\vec{Q} \mapsto \partial F_1/\partial \vec{q}$ has Jacobian $\bigl(\partial^2 F_1/\partial Q_j\,\partial q_i\bigr)$, invertible by hypothesis; the implicit function theorem solves $p_i = \partial F_1/\partial q_i(\vec{q},\vec{Q},t)$ locally for $\vec{Q}(\vec{q},\vec{p},t)$, smoothly.

Verification. Here the generating function enters the identity (156) *bare*: take $F = F_1$. Along an arbitrary curve,

$$\begin{aligned} \frac{\mathrm{d}F_1}{\mathrm{d}t} &= \sum_i \frac{\partial F_1}{\partial q_i}\,\dot{q}_i + \sum_i \frac{\partial F_1}{\partial Q_i}\,\dot{Q}_i + \frac{\partial F_1}{\partial t} \\ &= \sum_i p_i\,\dot{q}_i + \sum_i (-P_i)\,\dot{Q}_i + (K - H) \;=\; \Bigl[\sum_i p_i\,\dot{q}_i - H\Bigr] - \Bigl[\sum_i P_i\,\dot{Q}_i - K\Bigr], \end{aligned} \tag{165}$$

substituting all three relations (164) in the second step. This is (156) rearranged; Proposition 10.3 finishes. Note where the minus sign in (164) comes from: F_1 needs no $\sum P_i Q_i$ correction (compare Proposition 10.5), so the $\dot{Q}_i$ coefficient of $\mathrm{d}F_1/\mathrm{d}t$ must *itself* be $-P_i$ for the identity to close. In the type-2 case that job was done by the subtracted $\sum P_i Q_i$; here it is done by a sign in the definition. Same identity, two ways to produce the same term. $\square$

Proposition 10.8 (The type-3 and type-4 constructions). *Let $F_3(\vec{p},\vec{Q},t)$ be smooth with $\det\bigl(\partial^2 F_3/\partial p_i\,\partial Q_j\bigr) \neq 0$, and define*

$$q_i = -\frac{\partial F_3}{\partial p_i}, \qquad P_i = -\frac{\partial F_3}{\partial Q_i}, \qquad K = H + \frac{\partial F_3}{\partial t}. \tag{166}$$

Let $F_4(\vec{p},\vec{P},t)$ be smooth with $\det\bigl(\partial^2 F_4/\partial p_i\,\partial P_j\bigr) \neq 0$, and define

$$q_i = -\frac{\partial F_4}{\partial p_i}, \qquad Q_i = \frac{\partial F_4}{\partial P_i}, \qquad K = H + \frac{\partial F_4}{\partial t}. \tag{167}$$

Each defines, locally, a canonical transformation with new Hamiltonian K.

Proof. Solvability, both cases, one sentence each. For F_3: the map $\vec{Q} \mapsto -\partial F_3/\partial \vec{p}$ has Jacobian $-\bigl(\partial^2 F_3/\partial Q_j\,\partial p_i\bigr)$, invertible, so $q_i = -\partial F_3/\partial p_i(\vec{p},\vec{Q},t)$ solves for $\vec{Q}(\vec{q},\vec{p},t)$. For F_4: the map $\vec{P} \mapsto -\partial F_4/\partial \vec{p}$ has Jacobian $-\bigl(\partial^2 F_4/\partial P_j\,\partial p_i\bigr)$, invertible, so the first relation solves for $\vec{P}(\vec{q},\vec{p},t)$.

Verification for F_3. The correction now runs on the *old* pair: take

$$F = F_3(\vec{p},\vec{Q},t) + \sum_i p_i\,q_i. \tag{168}$$

Differentiate along an arbitrary curve, term by term, substituting (166) where marked:

$$\begin{aligned} \frac{\mathrm{d}F}{\mathrm{d}t} &= \sum_i \frac{\partial F_3}{\partial p_i}\,\dot{p}_i + \sum_i \frac{\partial F_3}{\partial Q_i}\,\dot{Q}_i + \frac{\partial F_3}{\partial t} + \sum_i \dot{p}_i\,q_i + \sum_i p_i\,\dot{q}_i \\ &= \sum_i (-q_i)\,\dot{p}_i + \sum_i (-P_i)\,\dot{Q}_i + (K - H) + \sum_i q_i\,\dot{p}_i + \sum_i p_i\,\dot{q}_i \\ &= \Bigl[\sum_i p_i\,\dot{q}_i - H\Bigr] - \Bigl[\sum_i P_i\,\dot{Q}_i - K\Bigr] : \end{aligned} \tag{169}$$

the $q_i\,\dot p_i$ terms cancel in pairs — the added $\sum p_i q_i$ existed to produce exactly that cancellation — and the survivors assemble into (156).

Verification for F_4. Both pairs need correcting: take

$$F = F_4(\vec p, \vec P, t) + \sum_i p_i\,q_i - \sum_i P_i\,Q_i. \tag{170}$$

Again in full:

$$\begin{aligned}
\frac{\mathrm{d}F}{\mathrm{d}t} &= \sum_i \frac{\partial F_4}{\partial p_i}\,\dot p_i + \sum_i \frac{\partial F_4}{\partial P_i}\,\dot P_i + \frac{\partial F_4}{\partial t} + \sum_i \big(\dot p_i\,q_i + p_i\,\dot q_i\big) - \sum_i \big(\dot P_i\,Q_i + P_i\,\dot Q_i\big) \\
&= \sum_i (-q_i)\,\dot p_i + \sum_i Q_i\,\dot P_i + (K - H) + \sum_i q_i\,\dot p_i + \sum_i p_i\,\dot q_i - \sum_i Q_i\,\dot P_i - \sum_i P_i\,\dot Q_i \\
&= \Big[\sum_i p_i\,\dot q_i - H\Big] - \Big[\sum_i P_i\,\dot Q_i - K\Big],
\end{aligned} \tag{171}$$

with two cancellations this time ($q_i\,\dot p_i$ against its partner, $Q_i\,\dot P_i$ against its partner) — each correction term neutralizing the derivative that lands on the "wrong" member of its pair. Proposition 10.3 applies in both cases. ☐

Remark 10.9 (The catalogue, assembled). Side by side, with each type's nondegeneracy condition:

type	arguments	defining relations	nondegeneracy
F_1	$(\vec q, \vec Q, t)$	$p_i = \partial F_1/\partial q_i, \quad P_i = -\,\partial F_1/\partial Q_i$	$\det\big(\partial^2 F_1/\partial q\,\partial Q\big) \neq 0$
F_2	$(\vec q, \vec P, t)$	$p_i = \partial F_2/\partial q_i, \quad Q_i = \partial F_2/\partial P_i$	$\det\big(\partial^2 F_2/\partial q\,\partial P\big) \neq 0$
F_3	$(\vec p, \vec Q, t)$	$q_i = -\,\partial F_3/\partial p_i, \quad P_i = -\,\partial F_3/\partial Q_i$	$\det\big(\partial^2 F_3/\partial p\,\partial Q\big) \neq 0$
F_4	$(\vec p, \vec P, t)$	$q_i = -\,\partial F_4/\partial p_i, \quad Q_i = \partial F_4/\partial P_i$	$\det\big(\partial^2 F_4/\partial p\,\partial P\big) \neq 0$

In every row, $K = H + \partial F/\partial t$. The sign pattern is not arbitrary but algorithmic: a derivative with respect to a *coordinate*-type argument returns the conjugate momentum with a $+$ if the argument is old (p from q) and, wherever the $P\,\mathrm{d}Q$ side of the identity must be produced directly rather than through a correction term, with a $-$; a derivative with respect to a *momentum*-type argument returns the conjugate coordinate, with the sign again fixed by which correction terms were subtracted. The four types are related pairwise by Legendre transforms in the variables being swapped — e.g. $F_1 = F_2 - \sum_i P_i Q_i$ when both exist, with the equivalence verified by comparing (164) and (160) — the same construction that carried L to H in §6.1, here carrying one generating description to another. And a caution, to keep the hypotheses complete: the four *pure* types do not exhaust the possibilities. Nothing prevents a transformation from admitting, say, (q_1, P_2) but neither $(\vec q, \vec Q)$ nor $(\vec q, \vec P)$ as independent variables; mixed per-index choices are then required, and they work by the same argument [stated; the general statement is that locally *some* choice of n old and n new variables is independent, but which one is a property of the transformation, not a convention].

The catalogue in action. Two worked examples, chosen because each defeats one type and showcases another.

The swap, by F_1. Take $F_1 = \sum_i q_i\,Q_i$. The relations (164) give, walking through the signs,

$$p_i = \frac{\partial F_1}{\partial q_i} = Q_i, \qquad P_i = -\,\frac{\partial F_1}{\partial Q_i} = -\,q_i, \qquad K = H\,: \tag{172}$$

exactly the swap (148) that opened this section, with its awkward sign revealed as the type-1 minus sign, and nondegeneracy matrix $\partial^2 F_1/\partial q_i\,\partial Q_j = \delta_{ij}$. The identity and the swap now sit symmetrically: the simplest F_2 generates the one, the simplest F_1 generates the other, and neither transformation is reachable from the other's type.

The oscillator, by one lucky guess. Here is what a well-chosen canonical transformation can do *without* any partial differential equation — the trick that Hamilton–Jacobi theory will shortly systematize. For $H = p^2/2m + \frac{1}{2}m\omega^2 q^2$, try

$$F_1(q,Q) = \frac{m\omega\, q^2}{2}\,\cot Q. \tag{173}$$

Grind the relations (164), slowly:

$$p = \frac{\partial F_1}{\partial q} = m\omega\, q\,\cot Q, \qquad P = -\frac{\partial F_1}{\partial Q} = -\frac{m\omega q^2}{2}\cdot\Big(-\frac{1}{\sin^2 Q}\Big) = \frac{m\omega\, q^2}{2\sin^2 Q}. \tag{174}$$

Invert for the old variables: the second relation gives $q^2 = (2P/m\omega)\sin^2 Q$, so

$$q = \sqrt{\frac{2P}{m\omega}}\,\sin Q, \qquad p = m\omega\,\cot Q\,\sqrt{\frac{2P}{m\omega}}\,\sin Q = \sqrt{2\,m\omega P}\,\cos Q, \tag{175}$$

taking the positive root (a branch choice of the same kind as (197), and resolved the same way: the final formulas continue smoothly through it). Now substitute into H and watch:

$$H = \frac{(\sqrt{2m\omega P}\,\cos Q)^2}{2m} + \frac{m\omega^2}{2}\,\frac{2P}{m\omega}\,\sin^2 Q = \omega P\cos^2 Q + \omega P\sin^2 Q = \omega P. \tag{176}$$

The coordinate Q has vanished from the Hamiltonian entirely: in the new variables the oscillator is a *free* system with a cyclic coordinate (Proposition 2.5, meeting us again on the far side of the formalism). Since F_1 is time-independent, $K = H = \omega P$, and the new equations of motion solve themselves by inspection:

$$\dot P = -\frac{\partial K}{\partial Q} = 0, \qquad \dot Q = \frac{\partial K}{\partial P} = \omega \quad\Longrightarrow\quad P = \text{const} = \frac{E}{\omega}, \qquad Q = \omega t + Q_0. \tag{177}$$

Substituting back through (175),

$$q(t) = \sqrt{\frac{2E}{m\omega^2}}\,\sin(\omega t + Q_0), \tag{178}$$

in exact agreement with the solution that §10.5 will reach by systematic search (Eq. (202), under $Q_0 = \omega\beta$) — and with the constant new momentum $P = E/\omega$ recognizable on sight: it is the *action variable*, the enclosed-area-over-2π of §7.1, making its second unforced appearance (Remark 10.15 names the general construction). The nondegeneracy hypothesis, checked rather than assumed: $\partial^2 F_1/\partial q\,\partial Q = -m\omega q/\sin^2 Q$, nonzero away from $q = 0$ — the transformation is honest on each half-line and degenerates at the turning-phase instants — a degeneracy by now familiar from trigonometric charts. One closing observation, which is the hinge of this section: the obvious question — *where did* (173) *come from?* — has no answer inside this example. It is a guess, verified after the fact, and the Hamilton–Jacobi equation of §10.3 is precisely the machine that replaces the guess with a search. (§10.6 will add a postscript: the dynamics itself manufactures a function of exactly this shape.)

10.3 The Hamilton–Jacobi equation

Now push the freedom to its limit. The best coordinates the double pendulum admitted made H a sum of two independent pieces; the best coordinates *imaginable* would make H vanish identically, so that

$$\dot{Q}_i = \frac{\partial K}{\partial P_i} = 0, \qquad \dot{P}_i = -\frac{\partial K}{\partial Q_i} = 0 : \tag{179}$$

every new variable a constant of the motion, the entire dynamics transformed away. Nothing in §10.2 forbids the attempt. Take a type-2 generating function — rename it $S(\vec{q}, \vec{P}, t)$, in anticipation — and impose $K = 0$ in (160):

$$0 = K = H\Big(\vec{q}, \frac{\partial S}{\partial \vec{q}}, t\Big) + \frac{\partial S}{\partial t}, \tag{180}$$

where $\partial S/\partial \vec{q}$ replaced $\vec{p}$ by the first defining relation. The demand on S is therefore

$$\boxed{H\Big(q_1, \ldots, q_n,\ \frac{\partial S}{\partial q_1}, \ldots, \frac{\partial S}{\partial q_n},\ t\Big) + \frac{\partial S}{\partial t} = 0} \tag{181}$$

— the *Hamilton–Jacobi equation*: a single first-order partial differential equation for a single scalar function of $n+1$ variables, generically nonlinear (the momenta enter H quadratically already for a free particle).

Pause on the count, because it looks too good. Hamilton's equations are $2n$ coupled ordinary differential equations; (181) is *one* equation. The compression is real but not free, and the fine print is the notion the whole method rests on: we do not need the general solution of (181) — we need one solution *per family*, with enough parameters that the second relation of (160) can be inverted.

Definition 10.10 (Complete integral)**.** A *complete integral* of (181) is a solution $S(\vec{q}; \boldsymbol{\alpha}, t)$ depending smoothly on n parameters $\alpha_1, \ldots, \alpha_n$ and satisfying, on its domain,

$$\det\Big(\frac{\partial^2 S}{\partial q_i\, \partial \alpha_j}\Big) \neq 0. \tag{182}$$

The parameters are the new momenta of the construction: $P_i = \alpha_i$, constant because $K = 0$. Condition (182) is (159) verbatim, and it quietly enforces everything the informal phrase "n *independent, non-additive* constants" is meant to convey. In particular, since only derivatives of S enter (181), $S + c$ is a solution whenever S is; an additive constant is always available and never counts. Suppose one of the parameters, say α_n, entered only additively, $S = \tilde{S}(\vec{q}; \alpha_1, \ldots, \alpha_{n-1}, t) + \alpha_n$. Then $\partial^2 S/\partial q_i\, \partial \alpha_n = 0$ for every i: an entire column of the matrix in (182) vanishes, and the determinant with it. The nondegeneracy condition is not a technical afterthought bolted onto the definition; it *is* the definition, with the folklore exclusions as corollaries.

10.4 Jacobi's theorem, by two routes

Here is the payoff, stated with its hypotheses in full view.

Theorem 10.11 (Jacobi)**.** *Let $H(\vec{q}, \vec{p}, t)$ be C^1 and let $S(\vec{q}; \boldsymbol{\alpha}, t)$ be a C^2 complete integral of the Hamilton–Jacobi equation (181) on a domain where (182) holds. For each fixed $(\boldsymbol{\alpha}, \boldsymbol{\beta})$, the relations*

$$\beta_i = \frac{\partial S}{\partial \alpha_i}(\vec{q}; \boldsymbol{\alpha}, t), \qquad p_i = \frac{\partial S}{\partial q_i}(\vec{q}; \boldsymbol{\alpha}, t) \tag{183}$$

determine, locally in t, a curve $(\vec{q}(t), \vec{p}(t))$ — the first relation implicitly, the second by evaluation — and this curve satisfies Hamilton's equations. Conversely, every solution of Hamilton's equations passing through the domain arises this way, for suitable constants $(\boldsymbol{\alpha}, \boldsymbol{\beta})$.

We give two proofs of the forward direction. The first travels through the machinery of §§10.2–10.3 and shows *why* the theorem was inevitable; the second uses nothing but differentiation and shows the hypotheses doing their work line by line. A result this central deserves both.

Proof (Route 1: through the canonical machinery). S satisfies the nondegeneracy condition (159), so by Proposition 10.5 it generates a canonical transformation $(\vec{q},\vec{p}) \mapsto (\vec{Q},\vec{P}) = (\boldsymbol{\beta},\boldsymbol{\alpha})$ with new Hamiltonian $K = H + \partial S/\partial t$ — which vanishes, because S solves (181). In the new variables Hamilton's equations read $\dot{\beta}_i = \partial K/\partial\alpha_i = 0$ and $\dot{\alpha}_i = -\partial K/\partial\beta_i = 0$: the motion, viewed through S, is standing still. A trajectory is therefore specified by the $2n$ constants $(\boldsymbol{\alpha},\boldsymbol{\beta})$, and transforming back — i.e. solving (183) for $(\vec{q}(t),\vec{p}(t))$, which part (i) of Proposition 10.5 licenses — produces a solution of the original equations, by Proposition 10.3 read right to left. □

Proof (Route 2: direct verification). First, the curve exists: for fixed $(\boldsymbol{\alpha},\boldsymbol{\beta})$, the system $\beta_i = \partial S/\partial\alpha_i(\vec{q};\boldsymbol{\alpha},t)$ has Jacobian in $\vec{q}$ equal to $\big(\partial^2 S/\partial\alpha_i\,\partial q_j\big)$, invertible by (182); the implicit function theorem gives a unique smooth local solution $\vec{q}(t)$, and we set $p_i(t) = \partial S/\partial q_i(\vec{q}(t);\boldsymbol{\alpha},t)$.

First Hamilton equation. Differentiate the identity $\beta_i = \partial S/\partial\alpha_i(\vec{q}(t);\boldsymbol{\alpha},t)$ with respect to t (the left side is constant):

$$0 = \frac{\partial^2 S}{\partial t\,\partial\alpha_i} + \sum_{j=1}^{n} \frac{\partial^2 S}{\partial q_j\,\partial\alpha_i}\,\dot{q}_j. \tag{184}$$

Separately, S satisfies (181) *identically* in $(\vec{q},\boldsymbol{\alpha},t)$ — the parameters are honest variables of the solution — so we may differentiate (181) with respect to α_i at fixed $(\vec{q},t)$. The Hamiltonian depends on α_i only through its momentum slots $p_j = \partial S/\partial q_j$:

$$0 = \frac{\partial^2 S}{\partial\alpha_i\,\partial t} + \sum_{j=1}^{n} \frac{\partial H}{\partial p_j}\Big(\vec{q},\frac{\partial S}{\partial\vec{q}},t\Big)\,\frac{\partial^2 S}{\partial\alpha_i\,\partial q_j}. \tag{185}$$

Subtract (185) from (184); the mixed time derivatives cancel ($S \in C^2$, so the order of differentiation is immaterial — this is where that hypothesis works), leaving

$$\sum_{j=1}^{n} \frac{\partial^2 S}{\partial\alpha_i\,\partial q_j}\Big[\dot{q}_j - \frac{\partial H}{\partial p_j}\Big] = 0 \qquad (i = 1,\dots,n). \tag{186}$$

This is a square linear system with invertible coefficient matrix (182) acting on the bracketed vector; the only solution is the zero vector:

$$\dot{q}_j = \frac{\partial H}{\partial p_j} \qquad (j = 1,\dots,n). \tag{187}$$

Second Hamilton equation. Differentiate $p_i(t) = \partial S/\partial q_i(\vec{q}(t);\boldsymbol{\alpha},t)$ in t:

$$\dot{p}_i = \frac{\partial^2 S}{\partial t\,\partial q_i} + \sum_{j=1}^{n} \frac{\partial^2 S}{\partial q_j\,\partial q_i}\,\dot{q}_j. \tag{188}$$

Differentiate (181) with respect to q_i — now H depends on q_i both directly and through every momentum slot:

$$0 = \frac{\partial^2 S}{\partial q_i\,\partial t} + \frac{\partial H}{\partial q_i} + \sum_{j=1}^{n} \frac{\partial H}{\partial p_j}\,\frac{\partial^2 S}{\partial q_i\,\partial q_j}. \tag{189}$$

Solve (189) for $\partial^2 S/\partial q_i\,\partial t$, insert into (188), and use (187) to identify $\dot q_j = \partial H/\partial p_j$:

$$\dot p_i = -\frac{\partial H}{\partial q_i} - \sum_j \frac{\partial H}{\partial p_j}\frac{\partial^2 S}{\partial q_i\,\partial q_j} + \sum_j \frac{\partial^2 S}{\partial q_j\,\partial q_i}\frac{\partial H}{\partial p_j} = -\frac{\partial H}{\partial q_i}, \tag{190}$$

the two sums cancelling by equality of mixed partials once more.

Converse (completeness of the recipe). Let $(\bar{\vec q}(t), \bar{\vec p}(t))$ be any solution with data $(\vec q^{\,0}, \vec p^{\,0})$ at time t_0 inside the domain. Consider the system

$$p_i^0 = \frac{\partial S}{\partial q_i}(\vec q^{\,0}; \boldsymbol{\alpha}, t_0) \qquad (i = 1, \dots, n) \tag{191}$$

as equations for $\boldsymbol{\alpha}$: its Jacobian is $\big(\partial^2 S/\partial\alpha_j\,\partial q_i\big)$ — the transpose of the matrix in (182), hence also invertible — so the implicit function theorem produces (locally) a parameter vector $\boldsymbol{\alpha}$. Set $\beta_i = \partial S/\partial\alpha_i(\vec q^{\,0}; \boldsymbol{\alpha}, t_0)$. The trajectory that the forward direction builds from this $(\boldsymbol{\alpha}, \boldsymbol{\beta})$ passes through $(\vec q^{\,0}, \vec p^{\,0})$ at t_0 and solves Hamilton's equations; so does $(\bar{\vec q}, \bar{\vec p})$; and solutions of the smooth first-order system through a given point at a given time are unique [existence and uniqueness for the smooth first-order system; stated, as in §6]. The two coincide. $\square$

Remark 10.12 (What each route certifies). Route 1 is short because §10.2 did its work in advance; it explains the theorem — constants appear because the transformed dynamics is trivial. Route 2 is longer and self-contained; it locates every hypothesis — C^2 licenses the two exchanges of mixed partials, and nondegeneracy works three separate times (solvability for $\vec q(t)$, the linear-algebra step to (187), and the converse's solvability for $\boldsymbol{\alpha}$). The two proofs agreeing is not redundancy; it is the certificate that the machinery of generating functions and the bare calculus are describing the same mathematics.

Remark 10.13 (A particular family suffices). Nothing above asked for the general solution of the partial differential equation (181) — a hopeless demand — only for *some* n-parameter family satisfying a determinant condition. This is the method's entire pragmatics: one lucky guess with enough knobs, and Theorem 10.11 converts it into every trajectory. The next subsection makes the guess for the oscillator and watches the conversion happen.

10.5 The oscillator, one more time

Warm-up: the free particle. $H = p^2/2m$. The Hamilton–Jacobi equation reads $\frac{1}{2m}\big(\partial S/\partial q\big)^2 + \partial S/\partial t = 0$. Try $S = \alpha q - f(t)$: then $\alpha^2/2m = f'(t)$, so

$$S(q; \alpha, t) = \alpha q - \frac{\alpha^2}{2m}\,t. \tag{192}$$

Check completeness: $\partial^2 S/\partial q\,\partial\alpha = 1 \neq 0$. Turn Jacobi's crank:

$$\beta = \frac{\partial S}{\partial \alpha} = q - \frac{\alpha}{m}\,t \quad\Longrightarrow\quad q(t) = \beta + \frac{\alpha}{m}\,t, \qquad p = \frac{\partial S}{\partial q} = \alpha. \tag{193}$$

Uniform motion, with α the (conserved) momentum and β the initial position — the constants of Theorem 10.11 arriving pre-labeled with physical meaning.

Separating the time. For any time-independent Hamiltonian the same first move works. Insert the *ansatz*

$$S(\vec{q};\boldsymbol{\alpha},t) = W(\vec{q};\boldsymbol{\alpha}) - Et \tag{194}$$

into (181): the time derivative contributes the constant $-E$, every other term is time-independent, and the equation collapses to

$$H\Big(\vec{q}, \frac{\partial W}{\partial \vec{q}}\Big) = E \tag{195}$$

— the *reduced* (time-independent) Hamilton–Jacobi equation, with the separation constant E available as one of the parameters α_i. To be clear about the logic: (194) is not claimed to exhaust the solutions of (181); it is a guess, and by the previous remark a guess is all the theorem asks. If solving (195) yields a family with the right count and a nonvanishing determinant, Theorem 10.11 does not care how the family was found. Equation (195) also wears its meaning openly: the momentum field $\vec{p} = \partial W/\partial \vec{q}$ is constrained to the energy surface $H = E$ — the same surfaces that organized the phase portraits of §7.1.

The oscillator. $H = p^2/2m + \frac{1}{2}m\omega^2 q^2$, one degree of freedom, so the single parameter can be taken to be E itself ($\alpha \equiv E$). The reduced equation

$$\frac{1}{2m}\Big(\frac{\mathrm{d}W}{\mathrm{d}q}\Big)^2 + \frac{1}{2}m\omega^2 q^2 = E \tag{196}$$

is algebraic in W':

$$\frac{\mathrm{d}W}{\mathrm{d}q} = \pm\sqrt{2mE - m^2\omega^2 q^2}. \tag{197}$$

Here the derivation slows down, because a sign and a domain have just appeared. The radicand is nonnegative only for $|q| \le A$, where

$$A \equiv \sqrt{\frac{2E}{m\omega^2}} \tag{198}$$

— the classical turning points, at which $W' = 0$. Choose the $+$ branch on the open interval $-A < q < A$; the $-$ branch is the return sweep, and we will see in a moment that the final answer accounts for it without being asked. On the chosen branch,

$$W(q;E) = \int^q \sqrt{2mE - m^2\omega^2 s^2}\,\mathrm{d}s, \tag{199}$$

with the lower limit left unwritten: changing it shifts W by a constant, which the completeness discussion has already told us is invisible.

Now the labor-saving step. Jacobi's recipe wants $\beta = \partial S/\partial E$, and differentiating (199) *under the integral sign* (licensed by Appendix A; the integrand is smooth in (s,E) on the open branch) gives

$$\beta = \frac{\partial S}{\partial E} = \frac{\partial W}{\partial E} - t = \int^q \frac{m\,\mathrm{d}s}{\sqrt{2mE - m^2\omega^2 s^2}} - t, \tag{200}$$

— and the integral that remains is *easier* than (199): the recipe never needed W itself, only its E-derivative. Factor the radicand, $\sqrt{2mE - m^2\omega^2 s^2} = m\omega\sqrt{A^2 - s^2}$:

$$\beta + t = \frac{1}{\omega}\int^q \frac{\mathrm{d}s}{\sqrt{A^2 - s^2}} = \frac{1}{\omega}\arcsin\frac{q}{A} + \text{const}, \tag{201}$$

absorbing the constant into β (it shifts the origin of time, which is what β is for). The arcsin is the principal branch, valued in $[-\pi/2, \pi/2]$ — a single sweep from $-A$ to A, faithful to the single branch of (197) we integrated. Inverting,

$$\boxed{q(t) = A\,\sin\bigl(\omega(t+\beta)\bigr), \qquad A = \sqrt{\frac{2E}{m\omega^2}}} \tag{202}$$

— and the inversion is where the branch bookkeeping resolves itself: arcsin was single-valued, but sin is entire, and (202) happily continues through the turning points and back, executing the $-$ branch of (197) on the return sweep without further instruction. One can check the claim directly: $p = \partial S/\partial q = W'(q) = m\omega\sqrt{A^2 - q^2}$ on the $+$ branch, while (202) gives $m\dot{q} = m\omega A\cos(\omega(t+\beta))$, and the two agree in magnitude with the sign of the cosine selecting the branch — positive on the outbound sweep, negative on the return, switching exactly at $q = \pm A$.

Three closures, in order of increasing reach. *(i) The hypothesis, concretely:* on the open branch,

$$\frac{\partial^2 S}{\partial q\,\partial E} = \frac{\partial}{\partial E}\sqrt{2mE - m^2\omega^2 q^2} = \frac{m}{\sqrt{2mE - m^2\omega^2 q^2}} \neq 0, \tag{203}$$

finite and nonzero for $|q| < A$ — Definition 10.10 verified rather than assumed, with its failure at the turning points visible in the formula (the very points we excised from the branch). *(ii) Back-substitution:* (202) inserted into both Hamilton equations: $\dot{q} = \omega A\cos(\omega(t+\beta)) = p/m$ ✓, and $\dot{p} = \frac{\mathrm{d}}{\mathrm{d}t}\bigl[m\omega A\cos(\omega(t+\beta))\bigr] = -m\omega^2 A\sin(\omega(t+\beta)) = -m\omega^2 q$ ✓. *(iii) Against the series:* (202) is the amplitude–phase form of [Osc. §2.3] with amplitude $A = \sqrt{2E/m\omega^2}$, i.e. $E = \frac{1}{2}m\omega^2 A^2$ — which is precisely the constant total energy computed in [Osc. §2.5]. Three formalisms and two documents meet at one formula.

10.6 The action as the generating function

The Hamilton–Jacobi equation was manufactured abstractly: demand $K = 0$, obtain a PDE, hunt for complete integrals. This subsection shows that the dynamics has been solving that PDE all along — the solution it builds is the action, evaluated on true trajectories and read as a function of where they end.

Setup, with its hypothesis in the open. Fix an initial event $(\vec{q}_0, t_0)$. Suppose that for every endpoint $(\vec{q}, t)$ in some neighborhood there is a *unique* solution of the equations of motion running from $(\vec{q}_0, t_0)$ to $(\vec{q}, t)$, depending smoothly on the endpoint.[1] Define *Hamilton's principal function*

$$S(\vec{q}, t) \;\equiv\; \int_{t_0}^{t} L\bigl(\bar{\vec{q}}(\tau), \dot{\bar{\vec{q}}}(\tau)\bigr)\,\mathrm{d}\tau \qquad \text{along the true trajectory } \bar{\vec{q}} \text{ from } (\vec{q}_0, t_0) \text{ to } (\vec{q}, t). \tag{204}$$

The same integral that Hamilton's principle (§3.1) compared across *virtual* paths is here evaluated only on *real* ones and studied as a function of the endpoint. Everything below follows from the two derivatives of S.

[1] This is a genuine hypothesis, not a pleasantry. Existence and smooth dependence hold near a given trajectory for t close to t_0 [smooth dependence of ODE solutions on data and parameters; stated — the same import family as the uniqueness theorem used in §6]. Uniqueness of the *boundary-value* problem, unlike the initial-value problem, can genuinely fail at finite times: the oscillator below fails it at $\omega(t - t_0) = \pi$, visibly, in Fig. 4.

Spatial derivative. Vary the endpoint position at fixed time: $\vec q \to \vec q + \delta\vec q$. The trajectory family shifts, $\bar{\vec q} \to \bar{\vec q} + \delta\bar{\vec q}$ with $\delta\bar{\vec q}(t_0) = 0$ (the start is nailed down) and $\delta\bar{\vec q}(t) = \delta\vec q$. Run the variation exactly as in the proof of Theorem 2.1 — expand, integrate by parts — but keep the boundary term this time:

$$\delta S = \int_{t_0}^{t} \sum_i \underbrace{\left[\frac{\partial L}{\partial q_i} - \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial L}{\partial \dot q_i}\right]}_{=\,0 \text{ on a true trajectory}} \delta\bar q_i \,\mathrm{d}\tau \;+\; \Big[\sum_i \frac{\partial L}{\partial \dot q_i}\,\delta\bar q_i\Big]_{t_0}^{t}. \tag{205}$$

The bulk dies because $\bar{\vec q}$ solves Euler–Lagrange — in Hamilton's principle the *variation* was constrained and the bulk survived to become the equations of motion; here the *path* is constrained to be physical and the roles reverse: the boundary term, which §2.2 killed with pinned endpoints, is now the entire result. At t_0 it vanishes ($\delta\bar{\vec q}(t_0) = 0$); at t it reads $\sum_i p_i\,\delta q_i$ with $p_i = \partial L/\partial \dot q_i$ the terminal momentum. Hence

$$\frac{\partial S}{\partial q_i} = p_i. \tag{206}$$

Time derivative. Extend the endpoint *along the trajectory itself*: comparing S at $(\bar{\vec q}(t+\mathrm{d}t),\, t+\mathrm{d}t)$ with S at $(\bar{\vec q}(t), t)$ appends one sliver of integrand to (204),

$$\frac{\mathrm{d}}{\mathrm{d}t}\, S\big(\bar{\vec q}(t), t\big) = L. \tag{207}$$

But the left side, by the chain rule, is $\partial S/\partial t + \sum_i (\partial S/\partial q_i)\,\dot q_i \;=\; \partial S/\partial t + \sum_i p_i\,\dot q_i$, using (206). Therefore

$$\frac{\partial S}{\partial t} = L - \sum_i p_i\,\dot q_i = -\,H\Big(\vec q, \frac{\partial S}{\partial \vec q}, t\Big), \tag{208}$$

recognizing the Legendre transform (92) and eliminating $\vec p$ by (206). Assembling (206)–(208) into one statement:

$$\boxed{\mathrm{d}S = \sum_i p_i\,\mathrm{d}q_i - H\,\mathrm{d}t, \qquad \text{hence} \qquad H\Big(\vec q, \frac{\partial S}{\partial \vec q}, t\Big) + \frac{\partial S}{\partial t} = 0} \tag{209}$$

— *Hamilton's principal function solves the Hamilton–Jacobi equation.* The PDE we conjured by demanding $K = 0$ is satisfied by an object the theory already owned.

The motion is its own canonical transformation. Vary instead the *initial* point $\vec q_0$ at fixed $(\vec q, t, t_0)$: the same computation runs, but now the boundary term survives at t_0 with a minus sign (it is the lower limit) and dies at t:

$$\frac{\partial S}{\partial q_{0,i}} = -\,p_{0,i}, \tag{210}$$

the initial momentum. Read (206) and (210) together, treating $S(\vec q, \vec q_0, t)$ as a function of old coordinates $\vec q$ and "new coordinates" $\vec q_0$: they are exactly the defining relations (164) of a type-1 generating function (Proposition 10.7) — and the transformation it generates, $(\vec q, \vec p) \mapsto (\vec q_0, \vec p_0)$, is the map from the current state back to the initial data: *time evolution, run in reverse.* Remark 10.6 promised that time evolution is a canonical transformation and named its infinitesimal generator (H, through the bracket); here is the finite version, with the generating function exhibited in closed form: it is the classical action. The solvability condition for this reading is $\det(\partial^2 S/\partial q_i\,\partial q_{0,j}) \neq 0$ — watch it in the examples.

Two principal functions in closed form. *Free particle.* The true path from (q_0, t_0) to (q, t) is uniform, $\dot{\bar{q}} = (q - q_0)/(t - t_0)$, so

$$S = \int_{t_0}^{t} \frac{m\dot{\bar{q}}^2}{2}\,\mathrm{d}\tau = \frac{m\,(q-q_0)^2}{2\,(t-t_0)}. \tag{211}$$

Verify the Hamilton–Jacobi equation by substitution: $\partial S/\partial q = m(q-q_0)/(t-t_0)$ (which is indeed $m\dot{\bar{q}} = p$ ✓), and

$$\frac{\partial S}{\partial t} = -\frac{m(q-q_0)^2}{2(t-t_0)^2} = -\frac{1}{2m}\Big(\frac{\partial S}{\partial q}\Big)^2 \quad ✓ \tag{212}$$

The cross-derivative is $\partial^2 S/\partial q\,\partial q_0 = -m/(t-t_0)$: nonzero for all $t > t_0$. The free particle's endpoint problem never degenerates — one straight line per endpoint pair, always.

Oscillator. The boundary-value solution from (q_0, t_0) to (q, t), with $T \equiv t - t_0$ and $\sin\omega T \neq 0$, is

$$\bar{q}(\tau) = \frac{q_0\,\sin\omega(t-\tau) + q\,\sin\omega(\tau - t_0)}{\sin\omega T}, \tag{213}$$

checked in one look: it solves $\ddot{\bar{q}} = -\omega^2\bar{q}$ (each term is a sinusoid of frequency ω in τ), equals q_0 at $\tau = t_0$ and q at $\tau = t$. For the action, integrate by parts once and let the equation of motion clear the bulk — the same maneuver that proved (206), now used as a computational device:

$$S = \int_{t_0}^{t}\Big[\frac{m\dot{\bar{q}}^2}{2} - \frac{m\omega^2\bar{q}^2}{2}\Big]\mathrm{d}\tau = \frac{m}{2}\Big[\bar{q}\,\dot{\bar{q}}\Big]_{t_0}^{t} - \frac{m}{2}\int_{t_0}^{t}\bar{q}\,\underbrace{(\ddot{\bar{q}} + \omega^2\bar{q})}_{=0}\,\mathrm{d}\tau = \frac{m}{2}\big(q\,\dot{\bar{q}}(t) - q_0\,\dot{\bar{q}}(t_0)\big). \tag{214}$$

Differentiate (213) and evaluate at the two ends:

$$\dot{\bar{q}}(t) = \frac{\omega\,(q\cos\omega T - q_0)}{\sin\omega T}, \qquad \dot{\bar{q}}(t_0) = \frac{\omega\,(q - q_0\cos\omega T)}{\sin\omega T}, \tag{215}$$

so that

$$S = \frac{m\omega}{2\sin\omega T}\Big[\big(q^2 + q_0^2\big)\cos\omega T - 2\,q\,q_0\Big]. \tag{216}$$

Consistency checks, each one line. $\partial S/\partial q = \frac{m\omega}{\sin\omega T}(q\cos\omega T - q_0) = m\dot{\bar{q}}(t)$ ✓, matching (206). The limit $\omega \to 0$ at fixed T: $\cos\omega T \to 1$, $\sin\omega T \to \omega T$, and (216) $\to m\big[(q^2 + q_0^2) - 2qq_0\big]/2T = m(q-q_0)^2/2T$, recovering the free particle (211) — the general formula passing through its special case. And the promised postscript to §10.2: at $q_0 = 0$, (216) reduces to $(m\omega q^2/2)\cot\omega t$ — the *same function* as the lucky guess (173), evaluated at $Q = \omega t$. No coincidence: in the variables of that example the angle advances as $Q = \omega t + Q_0$, the orbits launched from $q = 0$ have $Q_0 = 0$, and both functions store the same information in their q-derivative — the momentum field $p = m\omega q\cot(\omega t)$ of that one family of orbits. The guess of §10.2 was the dynamics' own principal function, met before it had a name. That (216) satisfies the oscillator's Hamilton–Jacobi equation is verified symbolically in the figure script (`make_figures_lagrangian.py`) [verified here, by machine].

The cross-derivative of (216) is

$$\frac{\partial^2 S}{\partial q\,\partial q_0} = -\frac{m\omega}{\sin\omega T}: \tag{217}$$

nonzero whenever defined, but blowing up as $\omega T \to \pi$ — and at $\omega T = \pi$ the formula (213) fails outright. The geometry behind the failure is in Fig. 4: every trajectory launched from $q_0 = 0$, regardless of amplitude, returns to $q = 0$ at $\omega t = \pi$ (half a period — isochrony again, the same fact that made the small-angle clock honest in §4). At that instant the endpoint $(0, \pi/\omega)$ is reached

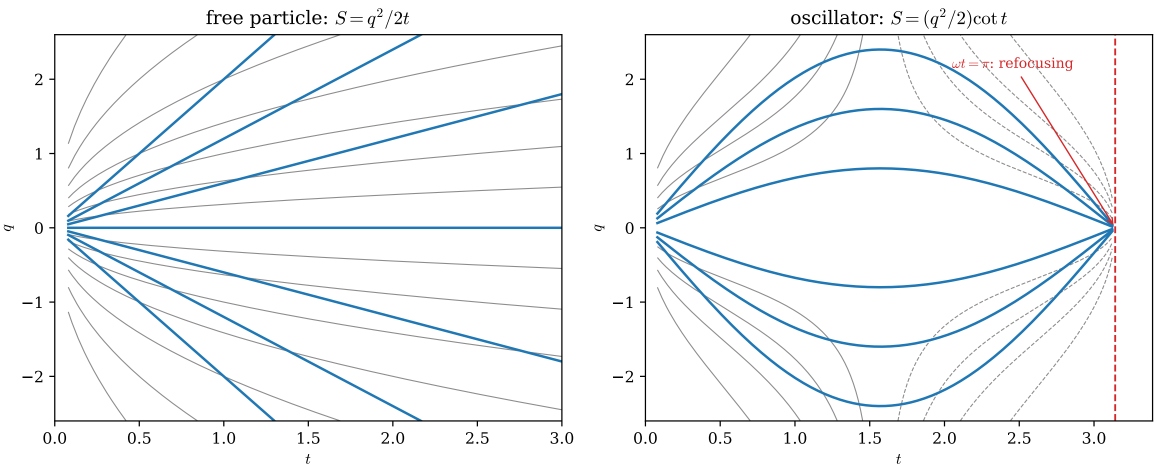


Figure 4: Hamilton's principal function as a field over endpoints (t, q), launched from $(q_0, t_0) = (0, 0)$; level sets of S (thin) with true trajectories (thick) crossing them. *Left:* free particle, $S = mq^2/2t$ (Eq. (211) with $q_0 = 0$), trajectories $q = vt$; every endpoint with $t > 0$ is reached by exactly one trajectory. *Right:* oscillator, $S = (m\omega q^2/2)\cot\omega t$ (Eq. (216) with $q_0 = 0$), trajectories $q = A\sin\omega t$; all trajectories refocus at $\omega t = \pi$, where the endpoint map degenerates ($\partial^2 S/\partial q\,\partial q_0 = -m\omega/\sin\omega t \to \infty$) and the principal function ceases to exist — the failure of the uniqueness hypothesis of §10.6, made visible. Parameters $m = \omega = 1$; both panels drawn from the exact formulas by the figure script, which also verifies both formulas against the Hamilton–Jacobi equation symbolically.

by *every* trajectory and the endpoints $(q \neq 0, \pi/\omega)$ by none: the unique-connection hypothesis collapses, and with it the principal function. Such refocusing points are called *conjugate points* (in optics, *caustics*), and they are the honest boundary of this subsection's construction [stated; the systematic theory belongs to the calculus of variations' second-variation chapter].

Remark 10.14 (The quantum pointer, made quantitative). Write a complex field from the principal function, $\psi(q, t) = e^{iS(q,t)/\hbar}$, with $\hbar$ a constant carrying the dimensions of action, and test it against the free Schrödinger operator. Differentiate carefully:

$$i\hbar\,\frac{\partial\psi}{\partial t} = -\frac{\partial S}{\partial t}\,\psi, \qquad -\frac{\hbar^2}{2m}\,\frac{\partial^2\psi}{\partial q^2} = \Big[\frac{1}{2m}\Big(\frac{\partial S}{\partial q}\Big)^2 - \frac{i\hbar}{2m}\,\frac{\partial^2 S}{\partial q^2}\Big]\psi, \tag{218}$$

(the first term of the second bracket from the square of $\partial_q S$, the second from the derivative acting twice — one power of $\hbar$ survives). Subtract, carrying a potential term $V\psi$ along on both sides:

$$i\hbar\,\frac{\partial\psi}{\partial t} - \Big[-\frac{\hbar^2}{2m}\partial_q^2 + V\Big]\psi = -\underbrace{\Big[\frac{\partial S}{\partial t} + \frac{1}{2m}\Big(\frac{\partial S}{\partial q}\Big)^2 + V\Big]}_{=\,0\text{ by Hamilton–Jacobi}}\psi \;+\; \frac{i\hbar}{2m}\,\frac{\partial^2 S}{\partial q^2}\,\psi = \frac{i\hbar}{2m}\,\frac{\partial^2 S}{\partial q^2}\,\psi. \tag{219}$$

The classical equation annihilates everything of order $\hbar^0$: the Hamilton–Jacobi equation is what the Schrödinger equation becomes when $\hbar$ is negligible against the actions in play, with the residue — one power of $\hbar$, proportional to $\partial_q^2 S$ — repaired by promoting the amplitude of ψ to a second unknown [stated; the systematic expansion is quantum mechanics' WKB approximation]. This is the precise sense in which classical mechanics is the stationary skeleton of a wave theory, and it dovetails with the path-integral route of the quantum companion: there, each path is weighted by exactly the phase $e^{iS/\hbar}$, and the classical trajectory emerges by stationary phase [QM. §10.3]. The function this section extracted from the endpoint variation is the phase of the quantum propagator.

Remark 10.15 (Beyond: action–angle variables, and the fence). Two directions continue from here; both are pointed at, not entered. *First*, for bounded periodic motion the natural constant is not E but the phase-space area enclosed by the orbit — the *action variable* $I = \oint p\,\mathrm{d}q/2\pi$ whose first, unforced appearance was the pendulum's E/ω_0 in §7.1, and whose second was the conserved new momentum $P = E/\omega$ of the oscillator example in §10.2. A complete integral built with I as the new momentum makes the conjugate *angle* advance uniformly, and the pair (I, θ) is the natural language for slow perturbations (adiabatic invariants) and for perturbation theory generally [stated; developed under the name action–angle variables]. *Second*, a fence: Jacobi's theorem is local, and its global reach is a genuine dynamical question. Complete integrals with global, single-valued W exist exactly when the system possesses n independent conserved quantities in involution — the *integrable* case [stated; the Liouville–Arnold theorem] — and the double pendulum at large amplitude (§11) is the standing reminder that this is the exception, not the rule: no choice of coordinates untangles a chaotic flow, and the Hamilton–Jacobi program, pushed against such a system, fails not by technique but by theorem. What survives universally is the local statement proved here, and the variational identity $\mathrm{d}S = \sum_i p_i\,\mathrm{d}q_i - H\,\mathrm{d}t$ that no amount of chaos revokes.

11 Outlook

Three continuations are visible from here; this section only points at them.

Beyond Hamilton–Jacobi. §10 built canonical transformations and pushed them to the $K = 0$ limit. The road continues: action–angle variables (the version of the construction adapted to bounded periodic motion, Remark 10.15), adiabatic invariants, and canonical perturbation theory — the machinery for systems *close* to integrable, and the road on which the failures of integrability (KAM theory, chaos) were first mapped [all stated at pointer level].

The double pendulum at large amplitude. Our §5 lives in the quadratic basin. The exact system (79)–(80) at higher energy is the standard laboratory example of *chaos*: deterministic, energy-conserving (Liouville and H-conservation hold exactly), yet with neighboring initial conditions separating exponentially fast, so that long-term prediction fails in practice while the phase-space structure theorems continue to hold. That these two facts coexist — perfect determinism, practical unpredictability — is a central lesson of twentieth-century mechanics, and the double pendulum is its cheapest demonstration: two rods, two bobs.

Not covered. Rigid-body motion and velocity-dependent potentials (the charged particle's $L = \frac{1}{2}m\dot{\vec{r}}^2 - q\phi + q\,\dot{\vec{r}}\cdot\vec{A}$, the classical seed of gauge coupling) fit the framework built here without modification; non-holonomic constraints (rolling) genuinely require more [all three stated at pointer level].

The bracket becomes a commutator. The remark closing §8 leads directly into quantum mechanics: the Hamiltonian formalism is the classical half of a correspondence in which Poisson brackets become commutators and the phase-space flow becomes unitary evolution. Nothing in Parts I–II has to be unlearned; it has to be *deformed*, with $\hbar$ as the deformation parameter.

A Differentiation Under the Integral Sign

Theorem A.1 (Leibniz rule, the case used in §2.2)**.** *Let $f(x,\varepsilon)$ and $\partial f/\partial\varepsilon$ be continuous on $[a,b]\times[-\delta,\delta]$. Then $\Phi(\varepsilon)=\int_a^b f(x,\varepsilon)\,\mathrm{d}x$ is differentiable and*

$$\Phi'(\varepsilon)=\int_a^b \frac{\partial f}{\partial\varepsilon}(x,\varepsilon)\,\mathrm{d}x. \tag{220}$$

Proof. Fix ε; for $h\neq 0$ small,

$$\frac{\Phi(\varepsilon+h)-\Phi(\varepsilon)}{h}-\int_a^b \frac{\partial f}{\partial\varepsilon}(x,\varepsilon)\,\mathrm{d}x=\int_a^b\left[\frac{f(x,\varepsilon+h)-f(x,\varepsilon)}{h}-\frac{\partial f}{\partial\varepsilon}(x,\varepsilon)\right]\mathrm{d}x. \tag{221}$$

By the mean value theorem in ε, for each x there is ξ_x between ε and $\varepsilon+h$ with

$$\frac{f(x,\varepsilon+h)-f(x,\varepsilon)}{h}=\frac{\partial f}{\partial\varepsilon}(x,\xi_x), \tag{222}$$

so the bracket equals $\partial_\varepsilon f(x,\xi_x)-\partial_\varepsilon f(x,\varepsilon)$ with $|\xi_x-\varepsilon|\le|h|$. Now the one analytic fact: $\partial_\varepsilon f$ is continuous on the *compact* set $[a,b]\times[-\delta,\delta]$, hence *uniformly* continuous there. Given $\epsilon'>0$ choose h small enough that $|\partial_\varepsilon f(x,\xi)-\partial_\varepsilon f(x,\varepsilon)|<\epsilon'$ whenever $|\xi-\varepsilon|\le|h|$, *uniformly in x*. The integral is then bounded by $(b-a)\,\epsilon'$, which proves the limit. □

In §2.2 this is applied with $f(x,\varepsilon)=F(x,\,y+\varepsilon\eta,\,y'+\varepsilon\eta')$: $F\in C^2$ and $y,\eta\in C^2$ make f and $\partial_\varepsilon f$ continuous on the required compact rectangle.

B The Fundamental Lemma: Full Proof

Lemma B.1. *Let g be continuous on $[a,b]$ and suppose $\int_a^b g\,\eta\,\mathrm{d}x=0$ for every $\eta\in C^2([a,b])$ with $\eta(a)=\eta(b)=0$. Then $g\equiv 0$ on $[a,b]$.*

Proof. Suppose not: $g(x_0)\neq 0$ for some x_0; without loss of generality $g(x_0)=2c>0$ (else replace g by $-g$, permissible since $-\eta$ is also a test function). By continuity there is a closed interval $[\alpha,\beta]\subset(a,b)$ containing x_0 on which $g\ge c>0$. Construct the explicit bump

$$\eta(x)=\begin{cases}(x-\alpha)^3\,(\beta-x)^3, & x\in[\alpha,\beta],\\ 0, & \text{otherwise.}\end{cases} \tag{223}$$

Admissibility: η vanishes at a and b (its support is inside (a,b)); it is C^2 across the seams $x=\alpha,\beta$ because each factor has a *triple* zero there, so η, η', η'' all vanish from inside, matching the zero function outside (a cubic-order zero leaves two derivatives intact). *Positivity*: $\eta>0$ on the open interval (α,β). Then

$$\int_a^b g\,\eta\,\mathrm{d}x=\int_\alpha^\beta g\,\eta\,\mathrm{d}x\ \ge\ c\int_\alpha^\beta \eta\,\mathrm{d}x\ >\ 0, \tag{224}$$

contradicting the hypothesis. Hence $g\equiv 0$. □

(If test functions are required merely continuous or C^1, the same bump works *a fortiori*; if C^∞ test functions are demanded, replace the cubic factors by the standard $\exp\bigl(-1/(x-\alpha)\bigr)\exp\bigl(-1/(\beta-x)\bigr)$ bump.)

C Covariance of the Euler–Lagrange Equations: Proof

Setting (Theorem 2.11): $q_i = q_i(\vec{Q}, t)$, a C^2 change of coordinates with invertible Jacobian $J_{ij} = \partial q_i/\partial Q_j$, and $\widetilde{L}(\vec{Q}, \dot{\vec{Q}}, t) = L\big(q(\vec{Q}, t),\, \dot{q}(\vec{Q}, \dot{\vec{Q}}, t),\, t\big)$, where velocities transform by the chain rule

$$\dot{q}_i = \sum_j \frac{\partial q_i}{\partial Q_j}\,\dot{Q}_j + \frac{\partial q_i}{\partial t}. \tag{225}$$

Proof. Two identities, precisely the (K1)–(K2) of §3.2 with $\vec{r} \to q$ and $q \to Q$, both read off (225):

$$\text{(K1)}\;\; \frac{\partial \dot{q}_i}{\partial \dot{Q}_j} = \frac{\partial q_i}{\partial Q_j}, \qquad \text{(K2)}\;\; \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial q_i}{\partial Q_j} = \frac{\partial \dot{q}_i}{\partial Q_j} \tag{226}$$

— (K1) because $\dot{Q}_j$ enters linearly; (K2) because both sides equal $\sum_k \frac{\partial^2 q_i}{\partial Q_k \partial Q_j}\dot{Q}_k + \frac{\partial^2 q_i}{\partial t\,\partial Q_j}$ (mixed partials of the C^2 map commute). Now compute the Q_j Euler–Lagrange expression of $\widetilde{L}$ by the chain rule. Momentum slot:

$$\frac{\partial \widetilde{L}}{\partial \dot{Q}_j} = \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\frac{\partial \dot{q}_i}{\partial \dot{Q}_j} \overset{\text{(K1)}}{=} \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\frac{\partial q_i}{\partial Q_j}, \tag{227}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \widetilde{L}}{\partial \dot{Q}_j} = \sum_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right]\frac{\partial q_i}{\partial Q_j} + \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial q_i}{\partial Q_j} \overset{\text{(K2)}}{=} \sum_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i}\right]\frac{\partial q_i}{\partial Q_j} + \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\frac{\partial \dot{q}_i}{\partial Q_j}. \tag{228}$$

Coordinate slot:

$$\frac{\partial \widetilde{L}}{\partial Q_j} = \sum_i \frac{\partial L}{\partial q_i}\,\frac{\partial q_i}{\partial Q_j} + \sum_i \frac{\partial L}{\partial \dot{q}_i}\,\frac{\partial \dot{q}_i}{\partial Q_j}. \tag{229}$$

Subtract: the $\partial L/\partial \dot{q}_i \cdot \partial \dot{q}_i/\partial Q_j$ sums cancel, leaving

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \widetilde{L}}{\partial \dot{Q}_j} - \frac{\partial \widetilde{L}}{\partial Q_j} = \sum_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i}\right]\frac{\partial q_i}{\partial Q_j}. \tag{230}$$

The new Euler–Lagrange expressions are the old ones contracted with the Jacobian. Since the Jacobian is invertible, the left side vanishes for all j *iff* the bracket vanishes for all i. ☐

Equation (230) says more than the theorem: the Euler–Lagrange expression transforms as a *covector* under point transformations — the differential-geometric identity behind the slogan "the equations are tensorial."

D The Isoperimetric Multiplier Rule: Proof

Setting (Theorem 2.9): y_* stationary for J subject to $\Lambda[y] = L$; y_* not an unconstrained extremal of Λ. Write, for test functions η,

$$A[\eta] \equiv \delta J[y_*; \eta], \qquad B[\eta] \equiv \delta\Lambda[y_*; \eta] \tag{231}$$

(both are integrals of the Euler–Lagrange–type bracket against η, by Moves 1–3 of Theorem 2.1's proof).

Proof. Since y_* is not an extremal of Λ, the fundamental lemma read backwards supplies a test function η_2 with $B[\eta_2] \neq 0$; normalize $B[\eta_2] = 1$. Fix an arbitrary test function η_1 and consider the two-parameter family

$$y_{\varepsilon_1,\varepsilon_2} = y_* + \varepsilon_1\eta_1 + \varepsilon_2\eta_2, \qquad \psi(\varepsilon_1,\varepsilon_2) \equiv \Lambda[y_{\varepsilon_1,\varepsilon_2}] - L. \tag{232}$$

Then $\psi(0,0) = 0$ and $\partial\psi/\partial\varepsilon_2(0,0) = B[\eta_2] = 1 \neq 0$, so the implicit function theorem yields a C^1 function $\varepsilon_2 = e(\varepsilon_1)$ near 0 with $e(0) = 0$, $\psi(\varepsilon_1, e(\varepsilon_1)) \equiv 0$, and

$$e'(0) = -\left.\frac{\partial\psi/\partial\varepsilon_1}{\partial\psi/\partial\varepsilon_2}\right|_{(0,0)} = -\,B[\eta_1]. \tag{233}$$

The curve $\varepsilon_1 \mapsto y_{\varepsilon_1,e(\varepsilon_1)}$ is a one-parameter family of *admissible* (constraint-respecting) competitors through y_*; stationarity of J among admissible curves gives

$$0 = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon_1}\right|_0 J\big[y_{\varepsilon_1,e(\varepsilon_1)}\big] = A[\eta_1] + A[\eta_2]\, e'(0) = A[\eta_1] - A[\eta_2]\, B[\eta_1]. \tag{234}$$

Set $\lambda \equiv -A[\eta_2]$ — a single constant, fixed once by η_2 and independent of η_1. The display reads

$$A[\eta_1] + \lambda\, B[\eta_1] = 0 \qquad \text{for } \textit{every} \text{ test function } \eta_1, \tag{235}$$

i.e. $\delta\big(J + \lambda\Lambda\big)[y_*;\eta_1] = 0$ for all η_1: y_* is an unconstrained stationary point of $\int(F + \lambda G)\,\mathrm{d}x$, and the Euler–Lagrange equation of $F + \lambda G$ follows by Theorem 2.1. ☐

E Simultaneous Diagonalization: Proof

Setting (Theorem 5.2): M symmetric positive definite, K symmetric, both real $n \times n$; the problem $\mathsf{K}\vec{v} = \omega^2\mathsf{M}\vec{v}$.

Proof. Step 1: the square root. By the spectral theorem [stated; the spectral theorem of linear algebra], $\mathsf{M} = O^{\mathsf{T}} D\, O$ with O orthogonal and $D = \mathrm{diag}(d_1,\ldots,d_n)$, $d_i > 0$ (positive definiteness). Define $\mathsf{M}^{1/2} \equiv O^{\mathsf{T}} D^{1/2} O$: symmetric, positive definite, invertible, and $(\mathsf{M}^{1/2})^2 = \mathsf{M}$.

Step 2: reduce to an ordinary symmetric problem. Substitute $\vec{w} = \mathsf{M}^{1/2}\vec{v}$ in the generalized problem:

$$\mathsf{K}\,\mathsf{M}^{-1/2}\vec{w} = \omega^2\,\mathsf{M}^{1/2}\vec{w} \quad\iff\quad \underbrace{\mathsf{M}^{-1/2}\,\mathsf{K}\,\mathsf{M}^{-1/2}}_{\equiv\mathsf{A}}\,\vec{w} = \omega^2\,\vec{w}, \tag{236}$$

and A is symmetric ($\mathsf{A}^{\mathsf{T}} = \mathsf{M}^{-1/2}\mathsf{K}^{\mathsf{T}}\mathsf{M}^{-1/2} = \mathsf{A}$ since $\mathsf{M}^{-1/2}$, K are). The spectral theorem, invoked once more, gives an orthonormal basis $\vec{w}_1,\ldots,\vec{w}_n$ of eigenvectors with real eigenvalues ω_r^2.

Step 3: translate back. Set $\vec{v}_r = \mathsf{M}^{-1/2}\vec{w}_r$: a basis (image of a basis under an invertible map), solving the generalized problem, with the claimed orthogonality

$$\vec{v}_r^{\mathsf{T}}\,\mathsf{M}\,\vec{v}_s = \vec{w}_r^{\mathsf{T}}\mathsf{M}^{-1/2}\,\mathsf{M}\,\mathsf{M}^{-1/2}\vec{w}_s = \vec{w}_r^{\mathsf{T}}\vec{w}_s = \delta_{rs}. \tag{237}$$

If K is positive definite, the eigenvalues are positive: $\omega_r^2 = \vec{w}_r^{\mathsf{T}}\mathsf{A}\,\vec{w}_r = (\mathsf{M}^{-1/2}\vec{w}_r)^{\mathsf{T}}\mathsf{K}\,(\mathsf{M}^{-1/2}\vec{w}_r) = \vec{v}_r^{\mathsf{T}}\mathsf{K}\,\vec{v}_r > 0$.

Step 4: decoupling. Expand $\vec{\theta}(t) = \sum_r \xi_r(t)\,\vec{v}_r$ and insert into $\mathsf{M}\ddot{\vec{\theta}} = -\mathsf{K}\vec{\theta}$; multiply on the left by $\vec{v}_s^{\mathsf{T}}$. Using $\mathsf{K}\vec{v}_r = \omega_r^2\mathsf{M}\vec{v}_r$ and then M-orthonormality,

$$\sum_r \ddot{\xi}_r\,\vec{v}_s^{\mathsf{T}}\mathsf{M}\vec{v}_r = -\sum_r \xi_r\,\omega_r^2\,\vec{v}_s^{\mathsf{T}}\mathsf{M}\vec{v}_r \quad\Longrightarrow\quad \ddot{\xi}_s = -\,\omega_s^2\,\xi_s \qquad (s = 1,\ldots,n). \tag{238}$$

☐

F The Jacobi Identity: Computation

We verify $\mathcal{J} \equiv \{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$ for one degree of freedom; the n-degree case is the identical computation repeated for each index pair, since the bracket is a sum over i of identical one-pair blocks. Write partials as subscripts ($f_q = \partial f/\partial q$, $f_{qp} = \partial^2 f/\partial q\,\partial p$, etc.), and recall $f_{qp} = f_{pq}$ throughout (C^2).

The organizing observation. Each double bracket expands into terms containing exactly *one* second derivative (of the innermost functions) times two first derivatives. Sort all terms of $\mathcal{J}$ into three groups by which function carries the second derivative; we show each group cancels on its own. By the cyclic symmetry of $\mathcal{J}$, it suffices to check the group of second derivatives of f — the other two follow by relabeling.

The expansion. From $\{g,h\} = g_q h_p - g_p h_q$:

$$\begin{aligned}\{f,\{g,h\}\} &= f_q\,\partial_p\big(g_q h_p - g_p h_q\big) - f_p\,\partial_q\big(g_q h_p - g_p h_q\big)\\ &= f_q\big(g_{qp}h_p + g_q h_{pp} - g_{pp}h_q - g_p h_{qp}\big) - f_p\big(g_{qq}h_p + g_q h_{pq} - g_{pq}h_q - g_p h_{qq}\big).\end{aligned} \tag{239}$$

No second derivative of f appears here — f sits outside. Second derivatives of f arise only from the two brackets in which f is inside:

$$\{g,\{h,f\}\}\Big|_{f''\text{-terms}} = g_q\big(h_q f_{pp} - h_p f_{qp}\big) - g_p\big(h_q f_{pq} - h_p f_{qq}\big), \tag{240}$$

$$\{h,\{f,g\}\}\Big|_{f''\text{-terms}} = h_q\big(f_{qp}\,g_p - f_{pp}\,g_q\big) - h_p\big(f_{qq}\,g_p - f_{pq}\,g_q\big). \tag{241}$$

Add and collect by second-derivative type:

$$\begin{aligned} f_{pp}: &\quad g_q h_q - h_q g_q = 0,\\ f_{qq}: &\quad g_p h_p - h_p g_p = 0,\\ f_{qp} = f_{pq}: &\quad \big(-g_q h_p\big) + \big(-g_p h_q\big) + \big(h_q g_p\big) + \big(h_p g_q\big) = 0.\end{aligned} \tag{242}$$

Every coefficient vanishes — crucially using $f_{qp} = f_{pq}$ to merge the mixed terms; the Jacobi identity is, at bottom, *the symmetry of second derivatives*, three times over. The g''- and h''-groups vanish by the cyclic relabelings $(f,g,h) \to (g,h,f) \to (h,f,g)$, and with all second-derivative groups zero and no other terms present, $\mathcal{J} = 0$.

Part III

Special Relativity and Classical Field Theory

From Spacetime to Fields:
Special Relativity and Classical Field Theory

From two postulates to the Maxwell Lagrangian

This document extends the series from mechanics to fields, with every step of the main logical chain displayed. Part I builds special relativity from its two postulates: the invariance of the spacetime interval is *derived*, the Lorentz transformation is obtained from it, and the kinematic consequences — relativity of simultaneity, time dilation, length contraction, velocity composition — follow with their operational definitions kept explicit. Minkowski geometry, four-vectors, and the structure of the Lorentz and Poincaré groups are then developed, and the relativistic particle is treated variationally: the action $S = -mc\int d\tau$ is motivated, its Euler–Lagrange equations solved, and the four-momentum obtained — not defined — as the Noether charge of spacetime translations, the same theorem of the companion mechanics notes run on a new symmetry group. Part II passes from particles to fields: the Lagrangian of the N-mass chain is taken to its continuum limit *at the level of the Lagrangian*, the variational framework of the mechanics notes is reused verbatim for fields, and the Klein–Gordon field appears as the simplest Lagrangian consistent with Lorentz invariance. Noether's theorem for fields localizes conservation laws into continuity equations and produces the energy-momentum tensor; the Hamiltonian formulation introduces the field momentum and the canonical bracket. Part III applies the entire apparatus to electromagnetism: the field tensor, the gauge-invariant Maxwell Lagrangian, all four Maxwell equations from variation and from the Bianchi identity, gauge freedom, and electromagnetic waves. The outlook closes the series: expanded in modes, every field is a family of harmonic oscillators — the opening object of the first document is the atom of field theory. Proofs whose details are purely technical are placed in appendices; every statement used in the main text is either proved there or explicitly pointed to its appendix.

1 Conventions and Notation

Dots denote time derivatives. Three-dimensional spatial vectors carry arrows ($\vec{x}$, $\vec{v}$); their components carry Latin indices i, j, k running over $1, 2, 3$. Spacetime indices are Greek, $\mu, \nu, \rho, \ldots$, running over $0, 1, 2, 3$, with $x^0 = ct$.

Convention 1.1 (Metric signature)**.** The Minkowski metric is

$$\eta_{\mu\nu} = \mathrm{diag}(+1,\ -1,\ -1,\ -1), \tag{1}$$

the "mostly-minus" convention of Schwartz and of Peskin–Schroeder. Squared timelike intervals are positive.

Convention 1.2 (Summation convention)**.** *This document breaks with the companion mechanics notes, where all sums were written out.* From here on, a repeated index — one upper, one lower — is summed over its range:

$$a^\mu b_\mu \ \equiv\ \sum_{\mu=0}^{3} a^\mu b_\mu = a^0 b_0 + a^1 b_1 + a^2 b_2 + a^3 b_3. \tag{2}$$

An index appearing twice is a *dummy* index and may be renamed freely; an index appearing once must appear, with the same placement, on both sides of every equation. Field-theory formulas become impractically long without this convention; the cost is vigilance about index placement, and the reader should expand a few contractions by hand early on until the shorthand is trusted.

Convention 1.3 (Units)**.** Sections 2–3 keep the speed of light c explicit: while the four-dimensional formalism is being assembled, c is the conversion factor between meters and seconds, and tracking it is a dimension check on every line. At the end of §3 we set $c = 1$ once and for all; the lost factors are recoverable uniquely by dimensional analysis.

The recurring symbols are collected here for reference; each is re-introduced where it first appears.

Symbol	Definition	First appears	Meaning
β	$\beta = v/c$	§2	dimensionless velocity
γ	$\gamma = (1-\beta^2)^{-1/2}$	§2	Lorentz factor
ϕ	$\tanh\phi = \beta$	§2	rapidity
s^2	$s^2 = c^2t^2 - \lvert\vec{x}\rvert^2$	§2	spacetime interval
$\eta_{\mu\nu}$	$\mathrm{diag}(+,-,-,-)$	§3	Minkowski metric
$\Lambda^\mu{}_\nu$	$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$	§3	Lorentz transformation
τ	$c^2\mathrm{d}\tau^2 = \mathrm{d}x^\mu \mathrm{d}x_\mu$	§3	proper time
∂_μ	$\partial/\partial x^\mu$	§3	spacetime gradient
$\Box$	$\Box = \partial_\mu\partial^\mu$	§3	d'Alembertian
$\phi(x)$	—	§6	scalar field (context separates it from rapidity)
$\pi(x)$	$\pi = \partial\mathcal{L}/\partial\dot{\phi}$	§10	field momentum density
$F_{\mu\nu}$	$\partial_\mu A_\nu - \partial_\nu A_\mu$	§11	electromagnetic field tensor

Cross-references to the companion documents are written as [Mech. §n] for the variational mechanics course and [Osc. §n] for the oscillator reference.

Part I
Spacetime

2 The Two Postulates and the Lorentz Transformation

2.1 A wave equation with a problem

The oscillator reference ended with the wave equation [Osc. §9]: the continuum limit of a chain of masses is a displacement field $u(x,t)$ obeying

$$\frac{\partial^2 u}{\partial t^2} = c_s^2 \frac{\partial^2 u}{\partial x^2}, \qquad c_s^2 = \frac{T}{\mu}, \tag{3}$$

with T the tension and μ the mass per length. Newtonian mechanics comes with a democracy of inertial frames: the laws take the same form for every observer in uniform motion, frames being related by the *Galilean transformation*

$$\bar{x} = x - vt, \qquad \bar{t} = t. \tag{4}$$

Is the wave equation a law in this sense? Chain rule: for $u(x,t) = \bar{u}(\bar{x},\bar{t})$,

$$\frac{\partial}{\partial x} = \frac{\partial}{\partial \bar{x}}, \qquad \frac{\partial}{\partial t} = \frac{\partial}{\partial \bar{t}} - v\,\frac{\partial}{\partial \bar{x}}, \tag{5}$$

so that

$$\frac{\partial^2 u}{\partial t^2} - c_s^2 \frac{\partial^2 u}{\partial x^2} \;=\; \frac{\partial^2 \bar{u}}{\partial \bar{t}^2} \;-\; 2v\,\frac{\partial^2 \bar{u}}{\partial \bar{x}\,\partial \bar{t}} \;+\; \left(v^2 - c_s^2\right)\frac{\partial^2 \bar{u}}{\partial \bar{x}^2}. \tag{6}$$

The right-hand side is *not* of the form $\bar{u}_{\bar{t}\bar{t}} - c_s^2\,\bar{u}_{\bar{x}\bar{x}}$: the wave equation changes form under a Galilean boost. For sound this is not a problem — the equation was derived in the rest frame of the *medium*, and the medium's rest frame is genuinely special. A moving observer measures different sound speeds upstream and downstream, and Eq. (6) is precisely the equation that says so.

The problem arrives with light. Maxwell's equations predict electromagnetic waves of speed $c = (\mu_0\varepsilon_0)^{-1/2} \approx 3.00 \times 10^8\,\mathrm{m/s}$ — a wave equation of the form (3) with $c_s \to c$, as Part III will derive. By the logic above, that equation should hold only in the rest frame of light's medium, and measuring the anisotropy of the speed of light should reveal the Earth's motion through it. The Michelson–Morley experiment (1887) and its successors looked and found nothing: the speed of light is the same in every direction, in every season, to (by now) parts in 10^{17}. [experimental input; we take it as given.]

Einstein's move (1905) was to stop treating the null result as a conspiracy to be explained and promote it to a law of nature. Two postulates:

Postulate 2.1 (Relativity)**.** The laws of physics take the same form in every inertial frame. No experiment performed inside a laboratory can detect the laboratory's uniform motion.

Postulate 2.2 (Invariance of the speed of light)**.** There is a finite speed c that is the same in every inertial frame, independent of the motion of the source. Light in vacuum travels at c.

Postulate 2.1 simply retains Galileo's principle. What is abandoned is the Galilean transformation (4) — because, as Eq. (6) shows, it is *incompatible* with Postulate 2.2: under (4) a speed-c signal in one frame has speed $c \pm v$ in another. The task of this section is to find the transformation law that the two postulates force, and the strategy is to first find what the postulates leave *invariant.*

2.2 Setup: events, frames, and the standard configuration

An *event* is a point of spacetime: something with a where and a when, idealized to a point — a firecracker, the coincidence of two particles, one tick of one clock. An inertial frame S assigns coordinates (t, x, y, z) to every event, operationally by a lattice of rulers and synchronized clocks at rest in S; we write the time coordinate as $x^0 = ct$ so that all four coordinates carry the same dimension.

Two frames S and S' are in *standard configuration* if their axes are parallel, S' moves with velocity v along the common x-axis, and the origins coincide: the event $(0, 0, 0, 0)$ has these coordinates in both frames. For the rest of this section we suppress the transverse coordinates y, z and work in the (ct, x) plane; the transverse directions are dealt with in Appendix A (they turn out to be spectators: $y' = y$, $z' = z$).

Two facts about the transformation $(ct, x) \mapsto (ct', x')$ come before the postulates proper:

Lemma 2.3 (Linearity)**.** *The transformation between two inertial frames in standard configuration is linear:*

$$\begin{pmatrix} ct' \\ x' \end{pmatrix} = \Lambda \begin{pmatrix} ct \\ x \end{pmatrix} \tag{7}$$

for a constant matrix Λ depending only on v.

The physics behind the lemma is the homogeneity of space and time — no event is privileged, so coordinate *differences* must transform the same way wherever they sit — plus continuity. The full argument is technical and lives in Appendix A. [stated; proved in Appendix A.]

2.3 The invariant: derivation of the interval

Define, for any pair of events with separations Δt, Δx in the frame S, the *interval*

$$\Delta s^2 \ = \ c^2 \, \Delta t^2 - \Delta x^2. \tag{8}$$

(No claim yet that this is a good idea; the definition is justified by Theorem 2.4 immediately below.) Since the transformation is linear, separations transform like coordinates, and it suffices to study $s^2 = c^2 t^2 - x^2$ for events relative to the common origin.

Theorem 2.4 (Invariance of the interval)**.** *For any two inertial frames in standard configuration and any pair of events,*

$$\boxed{c^2 \, \Delta t'^2 - \Delta x'^2 \ = \ c^2 \, \Delta t^2 - \Delta x^2.} \tag{9}$$

Proof. The proof runs in three steps: the postulates first pin down the *zero set* of s^2, then its scale, and isotropy kills the scale.

Step 1: null separations are preserved. Introduce the null coordinates

$$u = ct - x, \qquad w = ct + x, \qquad \text{so that} \qquad s^2 = u \, w. \tag{10}$$

A pair of events with $u = 0$ (i.e. $x = ct$) can be connected by a light signal moving in the $+x$ direction; a pair with $w = 0$, by one moving in the $-x$ direction. Postulate 2.2 says light moves at c in S' as well: therefore $u = 0 \Rightarrow u' = 0$ and $w = 0 \Rightarrow w' = 0$. (The boost does not swap the two light rays: at $v = 0$ the transformation is the identity, and by continuity in v the rightward ray cannot jump onto the leftward one.)

Step 2: the interval can change at most by an overall factor. In the null coordinates the linear transformation reads $u' = \alpha u + \beta_1 w$, $w' = \beta_2 u + \delta w$. Step 1 forces $\beta_1 = \beta_2 = 0$ (set $u = 0$, vary w, demand $u' = 0$; and conversely). Hence

$$u' = \alpha(v)\, u, \qquad w' = \delta(v)\, w, \qquad \Longrightarrow \qquad s'^2 = u'w' = \alpha(v)\,\delta(v)\, s^2 \;\equiv\; \lambda(v)\, s^2, \tag{11}$$

with $\alpha, \delta > 0$ for a transformation continuously connected to the identity that does not reverse the direction of time. The whole quadratic form is rescaled by a single velocity-dependent number $\lambda(v) > 0$.

Step 3: isotropy forces $\lambda = 1$. Two properties of λ:

- **Reciprocity.** The inverse transformation, from S' back to S, is the boost with velocity $-v$ (Postulate 2.1: S moves at $-v$ as seen from S', and neither frame is privileged). Applying (11) twice, $s^2 = \lambda(-v)\, s'^2 = \lambda(-v)\lambda(v)\, s^2$, so

$$\lambda(v)\,\lambda(-v) = 1. \tag{12}$$

- **Isotropy.** Space has no preferred direction: the physics of a boost to the right at speed $|v|$ is the mirror image of a boost to the left. Formally, let $P : (ct, x) \mapsto (ct, -x)$; then the boost of velocity $-v$ is $P\,\Lambda_v\,P$. Since P leaves s^2 unchanged, the scale factors of Λ_{-v} and Λ_v are equal:

$$\lambda(-v) = \lambda(v). \tag{13}$$

Together, $\lambda(v)^2 = 1$; positivity gives $\lambda(v) = 1$. □

Remark 2.5 (What just happened). Postulate 2.2 by itself only protects the light cone $s^2 = 0$; it is Postulate 2.1 — entering through reciprocity and isotropy — that rigidifies the whole quadratic form. The interval (8) is the replacement for the two quantities Newtonian physics held absolute separately: the time between two events and the distance between them are now frame-dependent, and only the combination $c^2\Delta t^2 - \Delta x^2$ is common property. Figure 1 shows the geometry: the events at fixed interval from the origin form hyperbolae, and every inertial frame's unit ticks lie on the *same* hyperbolae.

2.4 Two routes from the invariant to the transformation

Theorem 2.4 converts the search for the transformation law into a purely algebraic problem: *find all linear maps of the (ct, x) plane that preserve $c^2t^2 - x^2$*, then pick out the one that describes a boost of velocity v. In the spirit of the companion notes' double derivation of Hamilton's equations [Mech. §6.2], we solve it twice: once in coordinates adapted to the problem's structure, where the answer explains itself, and once by brute force in the physical coordinates, where every coefficient is worked out by hand.

Route 1: null coordinates. The work is already done. Any linear map that preserves $s^2 = uw$ in particular preserves its zero set, the two null lines, so Steps 1–2 of the proof of Theorem 2.4 apply verbatim: in the null coordinates (10) the map must have the diagonal form $u' = \alpha u$, $w' = \delta w$ with $\alpha, \delta > 0$ — and now, since the interval is preserved exactly,

$$\alpha\,\delta = \lambda = 1, \qquad \text{i.e.} \qquad \delta = \frac{1}{\alpha}. \tag{14}$$

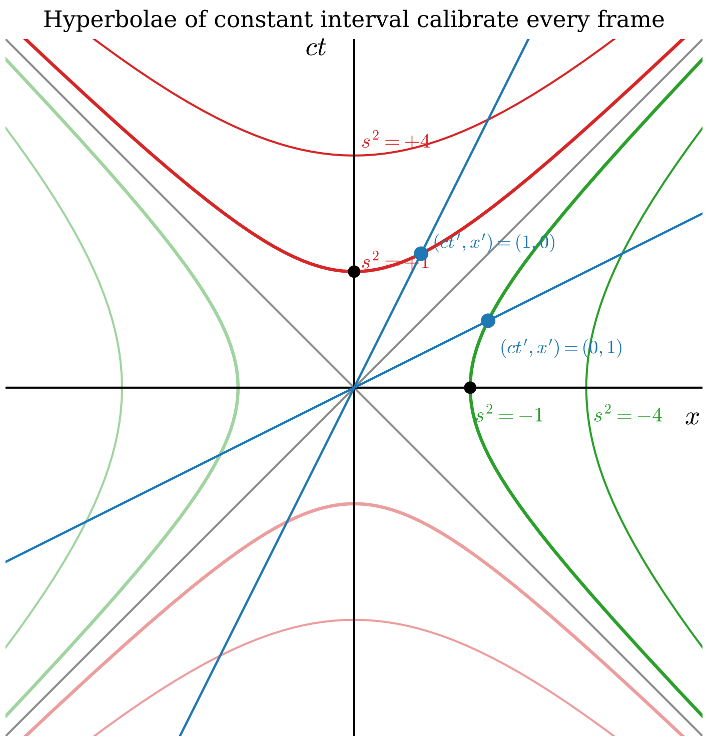


Figure 1: Curves of constant interval $s^2 = c^2t^2 - x^2$ (timelike in red, spacelike in green) and the axes of a boosted frame ($\beta = 0.5$, blue). The boosted frame's unit events — one tick of its clock at its origin, one meter along its ruler at its zero time — lie on the same unit hyperbolae as the unprimed frame's. The hyperbolae are drawn from the exact equations $ct = \pm\sqrt{s^2 + x^2}$ and $x = \pm\sqrt{s^2 + (ct)^2}$.

One free positive number α remains. Since the exponential is a bijection from $\mathbb{R}$ onto the positive reals, we may write $\alpha = \mathrm{e}^{\phi}$ for a unique $\phi \in \mathbb{R}$ — pure repackaging, $\phi = \ln \alpha$, but repackaging with a payoff: composing two such maps *multiplies* the α's, hence *adds* the ϕ's, so the parameter ϕ is the one in which the group law will look simplest (used in §2.5). The one-parameter family is

$$u' = \mathrm{e}^{\phi}\, u, \qquad w' = \mathrm{e}^{-\phi}\, w, \qquad \phi \in \mathbb{R} : \tag{15}$$

each null coordinate is stretched by exactly the factor that shrinks the other, and the product uw survives.

Now translate back. Inverting (10), $ct = \frac{1}{2}(w+u)$ and $x = \frac{1}{2}(w-u)$, and the same relations hold in the primed frame. Substitute (15), then $u = ct - x$, $w = ct + x$:

$$\begin{aligned} ct' = \tfrac{1}{2}\big(w' + u'\big) &= \tfrac{1}{2}\big(\mathrm{e}^{-\phi}(ct+x) + \mathrm{e}^{\phi}(ct-x)\big) \\ &= ct\,\frac{\mathrm{e}^{\phi} + \mathrm{e}^{-\phi}}{2} - x\,\frac{\mathrm{e}^{\phi} - \mathrm{e}^{-\phi}}{2} \;=\; ct\,\cosh\phi - x\,\sinh\phi, \\ x' = \tfrac{1}{2}\big(w' - u'\big) &= \tfrac{1}{2}\big(\mathrm{e}^{-\phi}(ct+x) - \mathrm{e}^{\phi}(ct-x)\big) \\ &= -\,ct\,\frac{\mathrm{e}^{\phi} - \mathrm{e}^{-\phi}}{2} + x\,\frac{\mathrm{e}^{\phi} + \mathrm{e}^{-\phi}}{2} \;=\; -\,ct\,\sinh\phi + x\,\cosh\phi, \end{aligned} \tag{16}$$

where the last step in each line is nothing but the definitions $\cosh\phi = \frac{1}{2}(\mathrm{e}^{\phi} + \mathrm{e}^{-\phi})$, $\sinh\phi = \frac{1}{2}(\mathrm{e}^{\phi} - \mathrm{e}^{-\phi})$: the hyperbolic functions are not imported; they arise directly as the symmetric and antisymmetric combinations of the two exponential stretch factors.

Remark 2.6 (Why hyperbolic functions were inevitable). Compare with the Euclidean plane, feature by feature:

	rotation	boost
preserved form	$x^2 + y^2$ (circles)	$c^2t^2 - x^2$ (hyperbolae)
matrix	$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$	$\begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix}$
parameter	angle θ, additive	rapidity ϕ, additive
level-set parametrization	$(\cos\theta, \sin\theta)$	$(\cosh\phi, \sinh\phi)$
function family	trigonometric	hyperbolic
eigenvalues	$\mathrm{e}^{\pm \mathrm{i}\theta}$ (complex)	$\mathrm{e}^{\pm\phi}$ (real; Remark 2.11)

A rotation slides points along the circles that its invariant form draws; Eq. (16) slides events along the hyperbolae of Fig. 1, which its invariant form draws — a *hyperbolic rotation*, with $\cosh^2\phi - \sinh^2\phi = 1$ playing the role of $\cos^2\theta + \sin^2\theta = 1$. Every row of the table follows from the single sign difference in the preserved form, and the last row — real eigenvalues where rotations have complex ones — is the difference with the most physical content, as Remark 2.11 will show.

It remains to identify the parameter ϕ, called the *rapidity*, in terms of the velocity. The origin of S' is the worldline $x' = 0$; by (16) that is the line $x = ct\tanh\phi$. But the origin of S' moves at velocity v by definition of the standard configuration, i.e. along $x = vt$. Hence

$$\boxed{\tanh\phi = \frac{v}{c} \equiv \beta, \qquad \cosh\phi = \frac{1}{\sqrt{1-\beta^2}} \equiv \gamma, \qquad \sinh\phi = \gamma\beta.} \tag{17}$$

(The last two follow from the first and $\cosh^2 - \sinh^2 = 1$.) Substituting into (16) gives the boxed result of Theorem 2.8 below.

Route 2: undetermined coefficients. Now the same problem with no change of variables: write the general linear map in the physical coordinates and let the invariant grind out the coefficients.

(1) Ansatz. By Lemma 2.3,

$$ct' = A\,ct + B\,x, \qquad x' = D\,ct + E\,x, \tag{18}$$

with four constants depending on β only.

(2) Motion of the moving origin. The worldline $x' = 0$ is, by definition of the standard configuration, the line $x = \beta\,ct$. Substituting: $0 = D\,ct + E\beta\,ct$ for all t, so $D = -\beta E$ and

$$x' = E\,(x - \beta\,ct). \tag{19}$$

One coefficient down, three to go.

(3) Balance the invariant. Demand $(ct')^2 - x'^2 = (ct)^2 - x^2$ *identically* in (ct, x). Expand the left side,

$$(A\,ct + Bx)^2 - E^2(x - \beta\,ct)^2, \tag{20}$$

and match the coefficients of $(ct)^2$, x^2, and the cross term $(ct)\,x$ separately — an identity between quadratic forms is three scalar equations:

$$\begin{aligned} A^2 - \beta^2 E^2 &= 1 && [(ct)^2 \text{ term}] \\ B^2 - E^2 &= -1 && [x^2 \text{ term}] \\ AB + \beta E^2 &= 0 && [(ct)\,x \text{ term}] \end{aligned} \tag{21}$$

(4) Solve. Square the third equation, $A^2B^2 = \beta^2E^4$, and substitute the first two in the forms $A^2 = 1 + \beta^2E^2$ and $B^2 = E^2 - 1$:

$$(1 + \beta^2E^2)(E^2 - 1) = \beta^2E^4 \quad\Longrightarrow\quad E^2 - 1 - \beta^2E^2 = 0 \quad\Longrightarrow\quad E^2 = \frac{1}{1 - \beta^2}. \tag{22}$$

So $E = \gamma$, and back-substituting,

$$A^2 = 1 + \beta^2\gamma^2 = (1 - \beta^2)\gamma^2 + \beta^2\gamma^2 = \gamma^2 \;\Rightarrow\; A = \gamma, \qquad B = -\frac{\beta E^2}{A} = -\gamma\beta. \tag{23}$$

The signs: the equations (21) are quadratic and admit sign flips of (A, B) and of E; the positive roots are selected by continuity to the identity at $\beta = 0$, the other choices being spatial reflection and/or time reversal, excluded by the standard configuration. The result is $ct' = \gamma(ct - \beta x)$, $x' = \gamma(x - \beta ct)$ — the two routes meet.

Remark 2.7 (What the change of coordinates achieved)*.* Route 2 is three coupled quadratic equations and an elimination; Route 1 is two decoupled one-line equations, (15). The difference is entirely in the coordinates: the null variables u, w are adapted to the preserved form ($s^2 = uw$ has no diagonal terms to balance), and in them the transformation is diagonal. This is the same simplification that normal coordinates achieve for coupled oscillators [Osc. §6]: pick the variables the structure prefers, and the coupled system falls apart. Remark 2.11 will name the sense in which the null coordinates are exactly the "normal coordinates" of a boost.

Collecting the result of both routes:

Theorem 2.8 (Lorentz transformation)**.** *Two inertial frames in standard configuration with relative velocity $v = \beta c$, $|\beta| < 1$, are related by*

$$\boxed{ct' = \gamma\,(ct - \beta x), \qquad x' = \gamma\,(x - \beta\,ct), \qquad \gamma = \frac{1}{\sqrt{1-\beta^2}},} \tag{24}$$

equivalently by the rapidity form (16) *with* $\tanh\phi = \beta$. *The inverse transformation is the same formula with* $\beta \to -\beta$.

Remark 2.9 (Sanity checks). (i) *Nonrelativistic limit.* For $\beta \ll 1$, $\gamma = 1 + \frac{1}{2}\beta^2 + O(\beta^4)$, and (24) becomes $x' = x - vt + O(\beta^2)$, $t' = t - vx/c^2 + O(\beta^2)$: the Galilean transformation is recovered in the regime where it was ever tested, with the first correction hitting the *time* coordinate — the term vx/c^2 is the relativity of simultaneity, previewed. (ii) *Light stays light.* If $x = ct$ then $x' = \gamma(ct - \beta ct) = ct'$, as built in. (iii) *The speed limit.* γ is real only for $|\beta| < 1$: no inertial frame moves at or above c relative to another. Rapidity makes the limit natural — ϕ ranges over all of $\mathbb{R}$ while $\beta = \tanh\phi$ is confined to $(-1, 1)$.

2.5 Composition: the group structure and velocity addition

What happens when boosts are stacked — a passenger walking in a moving train? Compose two transformations of the form (16) with rapidities ϕ_1, ϕ_2. The matrices are

$$\Lambda(\phi) = \begin{pmatrix} \cosh\phi & -\sinh\phi \\ -\sinh\phi & \cosh\phi \end{pmatrix}, \tag{25}$$

and the hyperbolic addition formulas $\cosh(\phi_1+\phi_2) = \cosh\phi_1\cosh\phi_2 + \sinh\phi_1\sinh\phi_2$, $\sinh(\phi_1+\phi_2) = \sinh\phi_1\cosh\phi_2 + \cosh\phi_1\sinh\phi_2$ give, by direct multiplication,

$$\boxed{\Lambda(\phi_2)\,\Lambda(\phi_1) = \Lambda(\phi_1+\phi_2)} \tag{26}$$

— boosts along a common axis form a one-parameter group, and *rapidities simply add*, exactly as rotation angles add in the plane. Velocities do not:

Corollary 2.10 (Velocity composition)**.** *If S' moves at β_1 relative to S, and a body moves at β_2 relative to S' (both along x), its velocity relative to S is*

$$\boxed{\beta_{12} = \tanh(\phi_1+\phi_2) = \frac{\beta_1+\beta_2}{1+\beta_1\beta_2}.} \tag{27}$$

Proof. Immediate from (26) and the addition formula for tanh. □

The denominator does the work: for $\beta_1, \beta_2 \in (0,1)$ it exceeds 1 just enough to keep $\beta_{12} < 1$, and if either input is 1 the output is 1 — light composed with anything is light. Figure 2 composes eight successive boosts of $\beta = 0.5$: the Galilean sum crosses the light speed on the second step; the true velocity approaches it and never arrives, while the rapidity grows linearly.

Remark 2.11 (The light cone is an eigendirection). Equation (15) was secretly an eigendecomposition. The characteristic polynomial of $\Lambda(\phi)$ is

$$\det\big(\Lambda(\phi) - \lambda\,1\big) = (\cosh\phi - \lambda)^2 - \sinh^2\phi = \lambda^2 - 2\lambda\cosh\phi + 1, \tag{28}$$

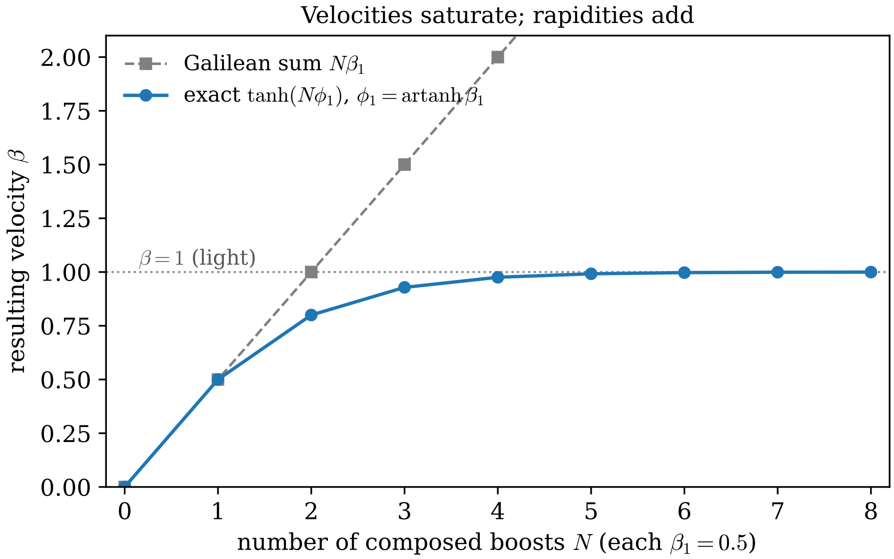


Figure 2: Composing N successive boosts of $\beta_1 = 0.5$. Gray squares: the Galilean prediction $N\beta_1$. Blue circles: the exact result $\tanh(N\phi_1)$ with $\phi_1 = \operatorname{artanh}(0.5)$, from Eq. (26).

with roots

$$\lambda_\pm = \cosh\phi \pm \sinh\phi = \mathrm{e}^{\pm\phi} : \tag{29}$$

real, positive, and reciprocal. The eigenvectors are the null directions: $(ct, x) \propto (1, -1)$, the line $w = 0$, carries $\lambda = \mathrm{e}^{\phi}$, and $(1, 1)$, the line $u = 0$, carries $\lambda = \mathrm{e}^{-\phi}$ — *the eigenvectors of a boost are the light cone.* The null coordinates of Route 1 are the eigencoordinates, Eq. (15) is the diagonalized boost, and the geometric action is now plain: stretch one light-ray direction by e^{ϕ}, compress the other by $\mathrm{e}^{-\phi}$, with $\det = \mathrm{e}^{\phi}\mathrm{e}^{-\phi} = 1$ — an area-preserving *squeeze* of the plane, sliding every event along its invariant hyperbola (Fig. 1) while the asymptotes stay put. Three payoffs:

- **Postulate 2.2, restated in linear algebra.** The light cone is the common eigendirection of every boost — the directions of spacetime on which all inertial observers agree. "The speed of light is invariant" and "light rays are eigenvectors" are the same sentence in two languages.

- **Composition, in one line.** Composing boosts multiplies eigenvalues on shared eigenvectors: $\mathrm{e}^{\phi_1}\mathrm{e}^{\phi_2} = \mathrm{e}^{\phi_1+\phi_2}$, which is Eq. (26) again — rapidity addition is the multiplicativity of eigenvalues, read on the light cone.

- **The eigenvalue is physical.** The stretch factor

$$\mathrm{e}^{\phi} = \cosh\phi + \sinh\phi = \gamma(1+\beta) = \sqrt{\frac{1+\beta}{1-\beta}} \tag{30}$$

  will return: it is the relativistic Doppler factor, the ratio by which a receding observer sees periods stretched, derived as such in §3. [stated; derived in §3.]

2.6 Consequences with operational care

Each of the three classic kinematic effects is a one-line consequence of Theorem 2.8 — *once the measurement being described is stated precisely.* Most of the traditional confusion comes from skipping that step.

Relativity of simultaneity. Two events are simultaneous in S' when $\Delta t' = 0$. By (24), $c\Delta t' = \gamma(c\Delta t - \beta\,\Delta x)$, so

$$\boxed{\Delta t' = 0 \quad \Longleftrightarrow \quad c\,\Delta t = \beta\,\Delta x.} \tag{31}$$

Events simultaneous in S' are *not* simultaneous in S unless they also happen at the same place. In the spacetime diagram (Fig. 3) the surfaces of simultaneity of S' are the tilted lines $ct = \beta x + \text{const}$: simultaneity is a frame-dependent slicing of spacetime, not a fact about spacetime itself. This effect is first order in β — it is the vx/c^2 term of Remark 2.9 — and it is the engine behind the other two effects, which are second order.

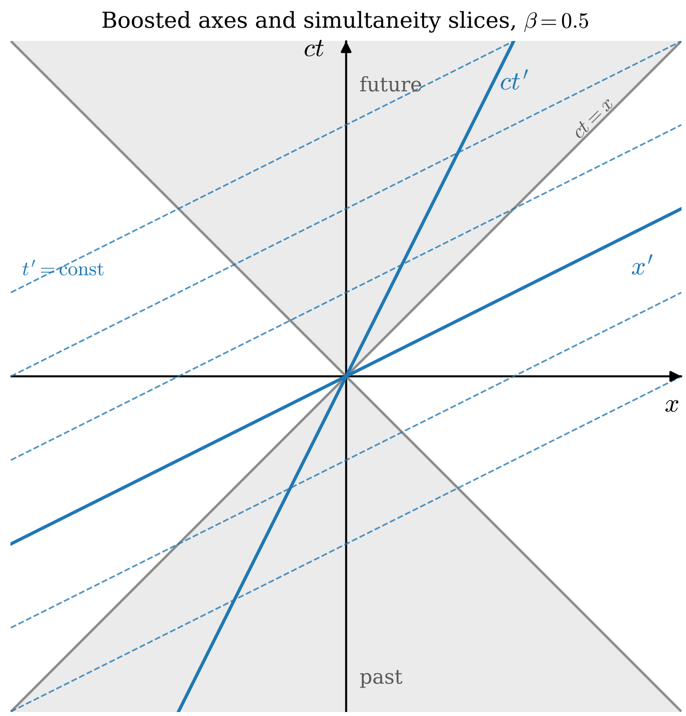


Figure 3: The (x, ct) plane with the light cone of the origin (gray) and the axes of a frame boosted to $\beta = 0.5$ (blue). The ct' axis is the worldline of the moving origin, $x = \beta\,ct$; the x' axis and its dashed translates, $ct = \beta x + \text{const}$, are the moving observer's surfaces of simultaneity, Eq. (31). Both axes close symmetrically onto the light cone, which is shared by all frames.

Time dilation. Take one clock at rest in S', sitting at $x' = 0$, and let it tick twice, the ticks separated by $\Delta t'$ — an interval of *proper time*, time measured by a clock present at both events. What does S record between the same two ticks? Use the inverse of (24) (i.e. $\beta \to -\beta$) with $\Delta x' = 0$:

$$\boxed{\Delta t = \gamma\,\Delta t' \;\geq\; \Delta t'.} \tag{32}$$

A moving clock ticks slowly by the factor γ. The statement is reciprocal — each frame sees the other's clocks run slow — and the reciprocity is consistent precisely because the two frames disagree about which distant events are simultaneous, Eq. (31): the two measurements compare one clock against *different pairs* of synchronized clocks.

Length contraction. Take a rod at rest in S', occupying $x' \in [0, L_0]$; L_0 is its *proper length.* To measure its length in S one must mark the positions of both ends *at the same S-time* — length measurement of a moving object is a simultaneity-laden operation, which is the warning that a frame-dependent answer is coming. Set $\Delta t = 0$ in the second equation of (24): $\Delta x' = \gamma\,\Delta x$, with $\Delta x' = L_0$ and $\Delta x = L$ the measured length. Hence

$$\boxed{L = \frac{L_0}{\gamma} \;\le\; L_0.} \tag{33}$$

A moving rod is short by the factor γ, along its direction of motion only (the transverse coordinates are untouched, Appendix A).

Example 2.12 (Cosmic-ray muons: the worked number)**.** Muons are produced by cosmic-ray collisions at altitude $h \approx 15\,\text{km}$, with proper lifetime $\tau = 2.197\,\mu\text{s}$, and arrive at the ground copiously — which, pre-relativity, is impossible. Take $\beta = 0.995$, so $\gamma = (1 - 0.995^2)^{-1/2} = 10.0$.

Without time dilation, the characteristic decay length is

$$\beta c\,\tau = 0.995 \times (2.998 \times 10^8\,\text{m/s}) \times (2.197 \times 10^{-6}\,\text{s}) = 655\,\text{m}, \tag{34}$$

and the surviving fraction after 15 km is $\exp(-15\,000/655) = \mathrm{e}^{-22.9} \approx 1.1 \times 10^{-10}$: effectively none.

With time dilation, the lab-frame lifetime is $\gamma\tau$ (Eq. (32): the muon's decay clock is a moving clock), the decay length is $\gamma\beta c\,\tau = 6.56\,\text{km}$, and the surviving fraction is $\exp(-15\,000/6560) = \mathrm{e}^{-2.29} \approx 0.10$: one in ten arrives, consistent with observation. The two exponential laws are plotted in Fig. 4; at ground level they disagree by *nine orders of magnitude.*

The same arithmetic in the muon's own frame is length contraction: the muon's clock runs normally, but the atmosphere is a moving object of proper thickness 15 km, contracted to $15\,\text{km}/\gamma = 1.5\,\text{km}$ (Eq. (33)) — about 2.3 decay lengths, and $\mathrm{e}^{-2.3} \approx 0.10$ again. One phenomenon, two accounts, one answer: the consistency is Theorem 2.4 at work.

2.7 Where this leaves the wave equation

Return to the opening problem. The wave equation (3) is not Galilean invariant — and now we know it was never supposed to be. The transformation that matters is (24), and under *it* the operator $\frac{1}{c^2}\partial_t^2 - \partial_x^2$ is form-invariant, as §3 will show in one line once the four-dimensional gradient is available. Two waves, two conclusions:

- **Light** obeys a wave equation with the invariant speed c. Lorentz invariance is exact, there is no medium, and the equation holds in the same form for every inertial observer.
- **Sound** obeys a wave equation with $c_s = \sqrt{T/\mu}$ — formally the same equation, and formally invariant under "Lorentz" transformations built with c_s in place of c. But this invariance is an accident of the long-wavelength description and is *physically broken*: the medium has a rest frame, detectable by any observer (measure the sound speed in two directions), and the microscopic chain [Osc. §8] is not invariant at all — its dispersion relation deviates from $\omega = c_s k$ at wavelengths approaching the lattice spacing.

This distinction — exact versus emergent, broken-by-the-medium symmetry — returns in §7 when the string is re-derived from a Lagrangian, and it sharpens the question Part II answers: *what does a field theory look like when Lorentz invariance is exact and fundamental?* The simplest answer is the Klein–Gordon field of §8.

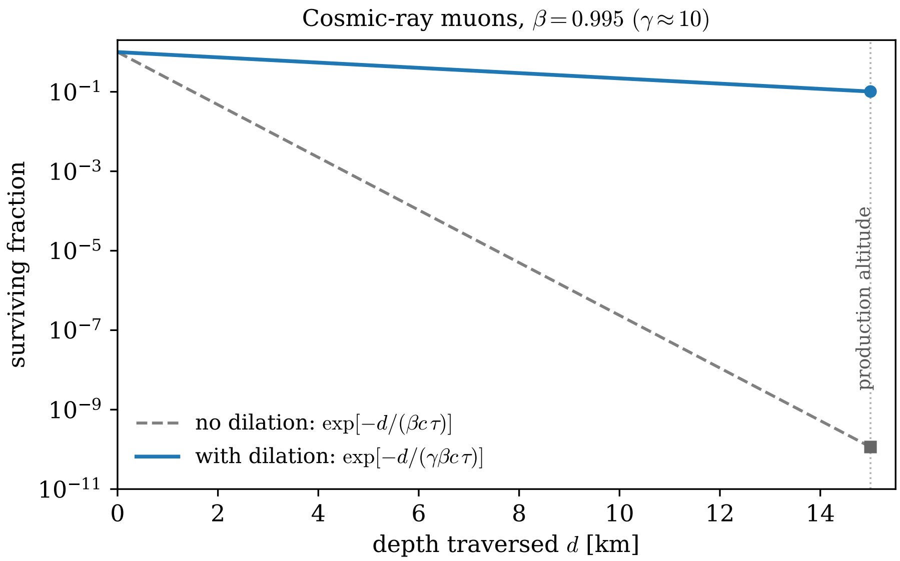


Figure 4: Surviving fraction of cosmic-ray muons ($\beta = 0.995$) versus traversed depth, from the exact exponential laws of Example 2.12. Note the logarithmic scale: at 15 km the prediction without time dilation (gray, dashed) is wrong by nine orders of magnitude.

3 Minkowski Geometry and Four-Vectors

Section 2 did everything in the (ct, x) plane with bare hands. This section builds the apparatus that makes the same physics effortless in four dimensions — the metric, index gymnastics, four-vectors, tensors — and uses it to deliver two results promised earlier: the invariance of the wave operator (§2.7), and the Doppler factor (Remark 2.11). It closes with the structure of the Lorentz group and the switch to natural units.

3.1 The metric and index gymnastics

An event is now a *four-vector* of coordinates,

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct,\ x,\ y,\ z), \tag{35}$$

the index written *upstairs*; such components are called *contravariant*. The interval of Theorem 2.4, extended to all four dimensions (the transverse directions enter with minus signs, being ordinary Euclidean lengths — Appendix A showed they are untouched by the boost), is encoded by the *Minkowski metric* of Convention 1.1:

$$\boxed{s^2 \ = \ \eta_{\mu\nu}\, x^\mu x^\nu \ = \ (x^0)^2 - (x^1)^2 - (x^2)^2 - (x^3)^2, \qquad \eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1).} \tag{36}$$

Following Convention 1.2, both μ and ν are summed; the reader meeting the convention for the first time should expand (36) once by hand — sixteen terms, twelve of which vanish because η is diagonal — and never again.

The metric converts upstairs indices to downstairs ones and back. Define the *covariant* components

$$x_\mu = \eta_{\mu\nu}\, x^\nu = (ct,\ -x,\ -y,\ -z), \tag{37}$$

— lowering an index costs a sign on the spatial components — and let $\eta^{\mu\nu}$ denote the matrix inverse of $\eta_{\mu\nu}$,

$$\eta^{\mu\rho}\,\eta_{\rho\nu} = \delta^{\mu}{}_{\nu}, \qquad \eta^{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1) \tag{38}$$

(numerically the same matrix in these coordinates — a coincidence of the diagonal form, not a general fact). *Inverse* is not a choice: demanding that raising undo lowering — $\eta^{\mu\nu}(\eta_{\nu\rho}x^{\rho}) = x^{\mu}$ for every x — is verbatim the statement $\eta^{\mu\nu}\eta_{\nu\rho} = \delta^{\mu}{}_{\rho}$, and the inverse exists, uniquely, because the metric is *nondegenerate*, $\det\eta = -1 \neq 0$ — a genuine requirement, since a degenerate metric would lower indices irreversibly. Then $\eta^{\mu\nu}$ raises indices, $x^{\mu} = \eta^{\mu\nu}x_{\nu}$, and the two operations are inverse to each other.

Remark 3.1 (What the names mean). "Contravariant" and "covariant" are not decorations; they name two opposite transformation behaviors, and the reference point of the "co-" and "contra-" is the *basis*. Expand a vector on basis vectors, $x = x^{\mu}e_{\mu}$; the vector itself is a geometric object that does not care about frames, so if the components change by $x'^{\mu} = \Lambda^{\mu}{}_{\nu}x^{\nu}$, the basis vectors must change by exactly the inverse, $e'_{\mu} = (\Lambda^{-1})^{\nu}{}_{\mu}\,e_{\nu}$, for the expansion to keep describing the same object. Quantities transforming like the basis — with Λ^{-1} — *co*-vary with it and are called covariant; quantities transforming with Λ vary *contra* to it. (§3.5 shows the gradient is the natural citizen of the covariant class, and §3.6 takes the two behaviors as the prototypes for the whole tensor hierarchy.) In special relativity the distinction is bookkeeping rather than content — the metric converts either version into the other at the cost of spatial signs, so x^{μ} and x_{μ} carry the same information — but the bookkeeping is load-bearing: Convention 1.2 sums only upper-lower pairs, and §3.6 shows that pairing is precisely the one whose transformation factors cancel.

The general *inner product* of four-vectors and its expansion in $3+1$ language:

$$\boxed{a\cdot b \;\equiv\; \eta_{\mu\nu}\,a^{\mu}b^{\nu} \;=\; a^{\mu}b_{\mu} \;=\; a_{\mu}b^{\mu} \;=\; a^0b^0 - \vec{a}\cdot\vec{b}.} \tag{39}$$

Which linear maps preserve this structure? Write

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu}\,x^{\nu}. \tag{40}$$

Demanding $s'^2 = s^2$ for *every* event x forces, by the polarization identity $a\cdot b = \frac{1}{4}\big[(a+b)^2 - (a-b)^2\big]$, the invariance of every inner product, and hence the condition on the matrix alone:

$$\boxed{\eta_{\mu\nu}\,\Lambda^{\mu}{}_{\rho}\,\Lambda^{\nu}{}_{\sigma} = \eta_{\rho\sigma} \qquad \text{(defining condition of a Lorentz transformation).}} \tag{41}$$

Three immediate consequences of (41) are collected here, before anything is built on top, because the entire calculus of Part I leans on them.

(i) Matrix form. With $\mathsf{\Lambda}$ the matrix $[\Lambda^{\mu}{}_{\nu}]$ (row μ, column ν) and $\boldsymbol{\eta}$ the matrix of the metric,

$$\boxed{\mathsf{\Lambda}^{\mathsf{T}}\,\boldsymbol{\eta}\,\mathsf{\Lambda} = \boldsymbol{\eta}} \tag{42}$$

— the spacetime analogue of $R^{\mathsf{T}}R = 1$ for rotations, with η replacing the identity as the preserved bilinear form. Note carefully: the *transpose*, not the inverse, sits on the left. The superficially similar $\mathsf{\Lambda}^{-1}\boldsymbol{\eta}\,\mathsf{\Lambda} = \boldsymbol{\eta}$ would assert that $\boldsymbol{\eta}$ *commutes* with every Lorentz transformation, which is already false for a single boost (check the 2×2 blocks: $\boldsymbol{\eta}\mathsf{\Lambda}$ flips the sign of $\mathsf{\Lambda}$'s spatial rows, $\mathsf{\Lambda}\boldsymbol{\eta}$ of its spatial columns, and a boost matrix is not invariant under that exchange).

Remark 3.2 (Rotations and Lorentz transformations: one template). Equation (42) is one instance of a template worth seeing whole. For any fixed invertible symmetric matrix G, the solutions of

$$\mathsf{M}^{\mathsf{T}}\,\mathsf{G}\,\mathsf{M} = \mathsf{G} \tag{43}$$

are exactly the linear maps preserving the bilinear form $\langle x, y\rangle = x^{\mathsf{T}}\mathsf{G}\,y$, and they always form a group. The choice $\mathsf{G} = \mathbb{1}$ in n dimensions gives $R^{\mathsf{T}}R = \mathbb{1}$ — the orthogonal group $O(n)$, rotations and reflections, preservers of Euclidean length. The choice $\mathsf{G} = \boldsymbol{\eta}$ gives the Lorentz group, in this classification named $O(1,3)$ — one plus sign, three minus signs — and the general signature $\mathsf{G} = \mathrm{diag}(+\cdots+,-\cdots-)$ gives the family $O(p,q)$. Two things are *shared* across the family, and everything else divides along the signs of G.

Shared: the constraint counting — the template is a symmetric matrix equation, $n(n{+}1)/2$ conditions on n^2 entries, leaving $n(n{-}1)/2$ parameters, one per coordinate plane. For $n=4$ that is 6 for the rotation group $O(4)$ and 6 for the Lorentz group alike: *same number of planes, same dimension* (the count is signature-blind; §3.8 spends the Lorentz six as three rotations plus three boosts). Shared too: $\det\mathsf{M} = \pm 1$, by taking determinants of the template.

Divided by the sign, with each division already witnessed in this document:

- **Bounded versus unbounded.** A rotation angle lives on a circle; the rapidity lives on $\mathbb{R}$ (Remark 2.9(iii)), $\cosh\phi$ is unbounded where $\cos\theta$ is trapped in $[-1,1]$, and correspondingly $O(4)$ is a *compact* group (closed and bounded as a set of matrices) while $O(1,3)$ is not — there is no largest boost, and the group is unbounded as a set of matrices. Most of what makes Lorentz representation theory harder than rotation representation theory descends from this single property. [stated; QFT-course territory.]

- **Eigenvalues.** Rotations have unit-modulus eigenvalues $\mathrm{e}^{\pm\mathrm{i}\theta}$; boosts have real reciprocal ones $\mathrm{e}^{\pm\phi}$, with null eigenvectors — Remark 2.11, now recognizable as the G-sign showing up in a characteristic polynomial.

- **How many pieces.** $O(n)$ has two connected components ($\det = \pm 1$); $O(1,3)$ has *four* (§3.8), and the doubling has a clean geometric cause: the Euclidean "unit sphere" $\{x : \langle x, x\rangle = 1\}$ is connected, but its Minkowski counterpart is the two-sheeted hyperboloid $t^2 - |\vec{x}|^2 = 1$ of Fig. 1 — and since a continuous family of transformations can never carry a unit timelike vector from one sheet to the other, "which sheet" is a second discrete invariant, alongside det. That is the invariant §3.8 will meet as $\mathrm{sgn}\,\Lambda^0{}_0$.

One more choice of G completes the family: take G *antisymmetric* and the template names the symplectic group — whose preserved pairing is exactly the (q,p) structure underlying the Poisson bracket, so that the canonical transformations of [Mech. §8] are this same template's third appearance in the series. [stated; not developed further here.]

(ii) The inverse, without inverting. Apply the raising and lowering machinery to Λ itself: define $\Lambda_\mu{}^\nu \equiv \eta_{\mu\rho}\,\Lambda^\rho{}_\sigma\,\eta^{\sigma\nu}$, and contract the defining condition (41) with $\eta^{\sigma\nu}$:

$$\eta_{\alpha\beta}\,\Lambda^\alpha{}_\rho\,\Lambda^\beta{}_\sigma\,\eta^{\sigma\nu} = \eta_{\rho\sigma}\,\eta^{\sigma\nu} = \delta_\rho{}^\nu \qquad \Longrightarrow \qquad \Lambda_\alpha{}^\nu\,\Lambda^\alpha{}_\rho = \delta^\nu{}_\rho, \tag{44}$$

i.e.

$$\boxed{\left(\Lambda^{-1}\right)^\nu{}_\mu \;=\; \Lambda_\mu{}^\nu\, :} \tag{45}$$

the inverse transformation is the index-lowered transpose, and no matrix inversion is ever needed. This document adopts (45) as standing notation: from here on the inverse is always written $\Lambda_\mu{}^\nu$, the index positions doing the inverting silently. The horizontal order of Λ's slots is therefore load-bearing — $\Lambda^\mu{}_\nu$ and $\Lambda_\nu{}^\mu$ are inverse-transposes of one another, not the same array, and stacking the indices vertically would destroy the distinction.

(iii) Lowering commutes with transforming. The covariant components $x_\mu = \eta_{\mu\nu} x^\nu$ were defined by the metric; do they inherit a coherent transformation law? Transform the lowered object, and rewrite using (45):

$$a'_\mu = \eta_{\mu\nu}\, a'^\nu = \eta_{\mu\nu}\, \Lambda^\nu{}_\rho\, \eta^{\rho\sigma} a_\sigma = \Lambda_\mu{}^\sigma\, a_\sigma : \tag{46}$$

lowered components transform with the *inverse* — with the basis, in the language of Remark 3.1, whose "covariant" label is hereby proved rather than assigned. Lowering first and transforming second agrees with transforming first and lowering second, and the agreement is (41) once more: one condition, read three ways — inner products preserved, the metric unmoved, index gymnastics frame-proof.

Section 3.8 mines (42) for the structure of the group. Meanwhile, a *four-vector* is, by definition, any quadruple a^μ that transforms like the coordinates, $a'^\mu = \Lambda^\mu{}_\nu a^\nu$ (the definition is systematized, and extended to tensors of arbitrary rank, in §3.6); Eq. (41) then guarantees that $a \cdot b$ is the same number in every frame — a *Lorentz invariant* (or *scalar*). The strategy for the rest of Part I is now set: *build the objects of mechanics out of four-vectors, and equations between four-vectors will hold in every frame at once.*

3.2 Causal structure and proper time

The sign of the interval sorts pairs of events into three invariant classes — invariant because Δs^2 is, Theorem 2.4:

class	condition	invariant meaning
timelike	$\Delta s^2 > 0$	a slower-than-light traveler can attend both events
lightlike (null)	$\Delta s^2 = 0$	exactly a light signal connects them
spacelike	$\Delta s^2 < 0$	no signal connects them; causally independent

Two useful normal forms, each a one-line boost exercise: for a timelike pair there is a frame in which both events happen *at the same place*, separated by pure time $\Delta t = \sqrt{\Delta s^2}/c$; for a spacelike pair, a frame in which they are *simultaneous*, separated by pure distance $\sqrt{-\Delta s^2}$. (For the timelike case, boost with $\beta = \Delta x/(c\Delta t)$, which is < 1 precisely because the pair is timelike; the spacelike case uses $\beta = c\Delta t/\Delta x$.) In the spacelike case the *time order* of the two events is frame-dependent — there are frames in which either precedes the other — which is why any signal between spacelike-separated events would be a causality violation for some observer, and why the light cone of Fig. 3 is the absolute causal boundary: the future and past cones are frame-independent regions, and "elsewhere" is negotiable only in time-ordering, never in reachability.

Now promote the interval from pairs of events to *worldlines.* A material particle traces a curve $x^\mu(t)$ whose every segment is timelike ($|\vec{v}| < c$); the interval along an infinitesimal segment,

$$c^2\, \mathrm{d}\tau^2 \;\equiv\; \mathrm{d}s^2 = c^2 \mathrm{d}t^2 - |\mathrm{d}\vec{x}|^2 = c^2 \mathrm{d}t^2 \left(1 - \beta(t)^2\right), \tag{47}$$

defines the *proper time* $\mathrm{d}\tau$: the time elapsed on an ideal clock riding the segment, since in the clock's momentary rest frame $\mathrm{d}\vec{x} = 0$ and $\mathrm{d}s^2 = c^2 \mathrm{d}t^2_{\text{rest}}$. Integrating,

$$\boxed{\tau \;=\; \int \mathrm{d}\tau \;=\; \int \frac{\mathrm{d}t}{\gamma(t)}, \qquad \gamma(t) = \frac{1}{\sqrt{1-\beta(t)^2}}.} \tag{48}$$

Proper time is a Lorentz scalar — all observers agree on what a given clock reads — and it is the natural parameter along any timelike worldline, a role it takes up in §3.3 and keeps for the rest of the document.

Remark 3.3 (The clock hypothesis). Equation (48) asserts that a clock's rate depends on its instantaneous *speed* and not separately on its acceleration. This is an additional physical assumption — special relativity as derived in §2 relates inertial frames only — and an experimentally supported one: muons circulating in storage rings at $\gamma \approx 29$, under centripetal accelerations of order $10^{18}\,g$, show lifetimes dilated by exactly γ with no acceleration correction. [experimental input.]

Remark 3.4 (The twin "paradox"). Between two fixed timelike-separated events A and B, different worldlines accumulate *different* proper times, and the straight (inertial) worldline accumulates the *most*. Proof: work in the frame where the straight worldline is at rest, so its proper time is the full coordinate time, $\tau_{\text{straight}} = \Delta t$; any other worldline between the same events has $\mathrm{d}\tau = \mathrm{d}t/\gamma \le \mathrm{d}t$ everywhere, with strict inequality wherever it moves. Hence the traveling twin — outbound, turn, return — ages less than the stay-at-home twin (Fig. 5): with $\beta = 0.6$ on both legs, $\tau_{\text{travel}} = \tau_{\text{home}}/\gamma = 0.8\,\tau_{\text{home}}$. This is the *reversed triangle inequality* of Minkowski geometry — the straight path is longest, the single sign in the metric again — and there is no paradox to resolve: the two worldlines are not symmetric, one is straight and one is kinked, and straightness (inertia) is frame-independent. The traveler's turnaround is the geometric fact every observer agrees on.

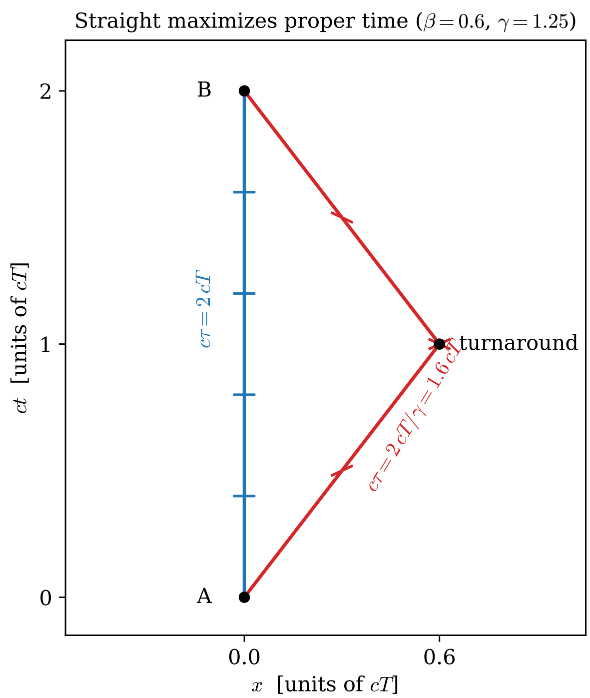


Figure 5: Two worldlines between the same events A and B. Tick marks are placed at equal intervals of *proper* time, $c\,\Delta\tau = 0.4\,cT$, computed from Eq. (48): five intervals fit on the straight worldline, four on the kinked one ($\beta = 0.6$, $\gamma = 1.25$).

Example 3.5 (GPS clocks: the worked number)**.** A GPS satellite orbits at $r = 26\,571\,\text{km}$ with speed $v = \sqrt{GM_{\oplus}/r} = 3.87\,\text{km/s}$, i.e. $\beta = 1.29 \times 10^{-5}$. By Eq. (48) its clock runs slow relative to ground clocks by the fraction

$$1 - \frac{1}{\gamma} \approx \frac{\beta^2}{2} = 8.3 \times 10^{-11}, \tag{49}$$

which over one day (86 400 s) accumulates to 7.2 μs — and a timing error of 7.2 μs is a ranging error of $c \times 7.2\,\mu\text{s} \approx 2\,\text{km}$. GPS would be useless within hours if this were not corrected. (Gravitational time dilation — general relativity — pushes the other way and harder, +45.9 μs per day, for a net satellite clock *gain* of 38.7 μs per day; the system corrects for both. [stated; the gravitational part is outside this document's scope.])

3.3 Four-velocity and four-acceleration

The ordinary velocity $\mathrm{d}x^\mu/\mathrm{d}t$ is *not* a four-vector: t is the zeroth *component* of x^μ, not a scalar, so dividing by $\mathrm{d}t$ ruins the transformation law. The fix is to differentiate with respect to the scalar the worldline carries with it — proper time:

$$\boxed{u^\mu \;\equiv\; \frac{\mathrm{d}x^\mu}{\mathrm{d}\tau} = \gamma\,\frac{\mathrm{d}x^\mu}{\mathrm{d}t} = \gamma\,(c,\,\vec{v}),} \tag{50}$$

using $\mathrm{d}t = \gamma\,\mathrm{d}\tau$ from (48). Its norm is fixed once and for all:

$$u \cdot u = \gamma^2\big(c^2 - |\vec{v}|^2\big) = \gamma^2 c^2\big(1-\beta^2\big) = c^2. \tag{51}$$

Every four-velocity is a unit timelike vector (in units of c): the "speed through spacetime" is not a degree of freedom, and the constraint has a derivative worth keeping. Defining the four-acceleration $a^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau$ and differentiating (51) along the worldline:

$$0 = \frac{\mathrm{d}}{\mathrm{d}\tau}(u \cdot u) = 2\,a \cdot u \qquad \Longrightarrow \qquad \boxed{a \cdot u = 0\,:} \tag{52}$$

four-acceleration is always Minkowski-orthogonal to four-velocity — acceleration can rotate the four-velocity on its hyperboloid $u \cdot u = c^2$ but never change its length. Both facts return in §4, where mu^μ becomes the four-momentum and (51) becomes the mass shell.

3.4 The Doppler effect from the null eigenvalue

Remark 2.11 stated without proof that the boost eigenvalue e^ϕ is the relativistic Doppler factor. The derivation is two lines precisely *because* it runs along the eigendirection.

A source at rest at the origin of S emits light pulses toward $+x$ at proper period T: emission events $(ct, x) = (cnT, 0)$, $n = 0, 1, 2, \ldots$ Each pulse travels the ray $x = x_0 + c(t - t_0)$, along which the null coordinate $u = ct - x$ is *constant*; the n-th pulse is the line $u = cnT$, so successive pulses are separated by

$$\Delta u = cT. \tag{53}$$

An observer receding at velocity β is at rest in the boosted frame S'; by Eq. (15), the pulse spacing in her null coordinate is the eigenvalue-stretched

$$\Delta u' = \mathrm{e}^\phi\,\Delta u = \mathrm{e}^\phi\,cT. \tag{54}$$

She receives the pulses at a fixed position, $\Delta x' = 0$, so the period she measures is $c\,\Delta t'_{\text{rec}} = \Delta(ct' - x') = \Delta u'$. Hence

$$\boxed{\frac{T_{\text{obs}}}{T_{\text{src}}} = \mathrm{e}^\phi = \sqrt{\frac{1+\beta}{1-\beta}}, \qquad \frac{f_{\text{obs}}}{f_{\text{src}}} = \mathrm{e}^{-\phi} = \sqrt{\frac{1-\beta}{1+\beta}}} \tag{55}$$

— redshift for recession ($\phi > 0$), blueshift for approach ($\beta \to -\beta$ flips the exponent), and successive recessions compound by *multiplying* the factors, i.e. adding rapidities: Doppler shifts of collinear relative motions compose exactly as Remark 2.11 promised eigenvalues would.

Remark 3.6 (Against the nonrelativistic Doppler formulas). The classical Doppler effect for sound has *two* formulas — moving source, $f_{\mathrm{obs}} = f_{\mathrm{src}}/(1+\beta_s)$, versus moving observer, $f_{\mathrm{obs}} = f_{\mathrm{src}}(1-\beta_o)$ — distinguishable because the medium's rest frame settles who is "really" moving. Equation (55) is a single formula depending only on the *relative* velocity, and expanding it, $\mathrm{e}^{-\phi} = 1-\beta+\frac{1}{2}\beta^2-\ldots$, it interpolates between the two classical answers, splitting the difference at second order. With no medium, no such distinction survives: this is Postulate 2.1 showing up in a lineshape. There is also a purely relativistic *transverse* Doppler shift, a factor $1/\gamma$ at closest approach with no classical analogue at all — it is time dilation (32) observed directly.

3.5 The spacetime gradient and the d'Alembertian

One more object completes the kit: differentiation. Define

$$\partial_\mu \;\equiv\; \frac{\partial}{\partial x^\mu} = \left(\frac{1}{c}\frac{\partial}{\partial t},\; \nabla\right). \tag{56}$$

Why the *downstairs* index? Chain rule: under $x'^\mu = \Lambda^\mu{}_\nu x^\nu$,

$$\partial'_\mu = \frac{\partial}{\partial x'^\mu} = \frac{\partial x^\nu}{\partial x'^\mu}\frac{\partial}{\partial x^\nu} = \Lambda_\mu{}^\nu\,\partial_\nu : \tag{57}$$

the gradient transforms with the *inverse* matrix, in the notation of (45) — precisely the behavior (46) established for lowered components, so the index placement is derived, not decorative. (Check the logic of (57): the old coordinates as functions of the new are $x^\nu = \Lambda_\mu{}^\nu\, x'^\mu$, whence the Jacobian.) Raising the index, $\partial^\mu = \eta^{\mu\nu}\partial_\nu = (\frac{1}{c}\partial_t, -\nabla)$, and contracting gradient with gradient produces a scalar operator, the *d'Alembertian*:

$$\boxed{\Box \;\equiv\; \partial_\mu\partial^\mu = \frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \nabla^2.} \tag{58}$$

Because it is a full contraction of Lorentz-covariant objects, $\Box$ is the same operator in every inertial frame — the promise of §2.7 is now a one-line theorem. The wave equation for light, $\Box\,\varphi = 0$, is manifestly form-invariant under Lorentz transformations, in exactly the way Eq. (6) showed it fails to be under Galilean ones; and when field theory needs a relativistic equation of motion in §8, $\Box$ will be the only two-derivative scalar operator available.

3.6 Scalars, vectors, tensors, and the covariance principle

The words "scalar" and "four-vector" have so far been used case-by-case. Before dynamics begins, it is worth assembling the full hierarchy once, systematically — because the central objects ahead carry *two* indices (the field tensor $F_{\mu\nu}$ unifying $\vec{E}$ and $\vec{B}$, the energy-momentum tensor $T^{\mu\nu}$), and the reasoning that will certify them deserves to be set up properly rather than improvised on arrival.

The two prototype behaviors. Everything descends from two objects already in hand. The coordinate displacement transforms with Λ,

$$\mathrm{d}x'^\mu = \Lambda^\mu{}_\nu\,\mathrm{d}x^\nu \qquad \text{(contravariant prototype)}, \tag{59}$$

and the gradient transforms with the inverse, Eq. (57),

$$\partial'_\mu = \Lambda_\mu{}^\nu\,\partial_\nu \qquad \text{(covariant prototype)}. \tag{60}$$

Every transformation behavior in the theory is built from copies of these two — one copy per index, upstairs indices behaving like $\mathrm{d}x^\mu$, downstairs like ∂_μ.

Definition 3.7 (Lorentz scalars, vectors, tensors)**.** Under $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$:

- a **scalar** (rank 0) is a single number assigned the same value in every frame: $s' = s$. Already met: the interval s^2, proper time τ, the norm $u \cdot u$.

- a **contravariant vector** (type $(1,0)$) is a quadruple transforming like the displacement: $a'^{\mu} = \Lambda^{\mu}{}_{\nu}\, a^{\nu}$. Already met: x^{μ}, u^{μ}.

- a **covariant vector** (type $(0,1)$) transforms like the gradient: $b'_{\mu} = \Lambda_{\mu}{}^{\nu}\, b_{\nu}$ (the inverse, per (45)). Already met: ∂_{μ}, and $x_{\mu} = \eta_{\mu\nu} x^{\nu}$.

- a **tensor of type** (p,q) carries p upstairs and q downstairs indices, each transforming with its prototype's factor:

$$T'^{\mu_1 \cdots \mu_p}{}_{\nu_1 \cdots \nu_q} = \Lambda^{\mu_1}{}_{\rho_1} \cdots \Lambda^{\mu_p}{}_{\rho_p}\, \Lambda_{\nu_1}{}^{\sigma_1} \cdots \Lambda_{\nu_q}{}^{\sigma_q}\, T^{\rho_1 \cdots \rho_p}{}_{\sigma_1 \cdots \sigma_q}. \tag{61}$$

  Scalars and vectors are the ranks $(0,0)$, $(1,0)$, $(0,1)$ of the one definition.

Why this law and no other. Equation (61) is not an arbitrary decoration; it is the *unique* linear transformation behavior under which index contraction produces frame-independent physics. The engine is one cancellation:

Proposition 3.8 (Contraction)**.** *Setting one upstairs index of a tensor equal to one downstairs index and summing yields a tensor of type* $(p-1, q-1)$*. In particular, a full contraction — no free indices — is a scalar.*

Proof. It suffices to watch one contracted pair; spectator indices carry their factors along unchanged. For a $(1,1)$ tensor, contract and transform:

$$T'^{\mu}{}_{\mu} = \Lambda^{\mu}{}_{\rho}\, \Lambda_{\mu}{}^{\sigma}\, T^{\rho}{}_{\sigma} = \delta_{\rho}{}^{\sigma}\, T^{\rho}{}_{\sigma} = T^{\rho}{}_{\rho}: \tag{62}$$

the Λ of the upstairs slot annihilates the $\Lambda_{\mu}{}^{\sigma}$ of the downstairs slot, by (45). This is why every invariant met so far — $a \cdot b = a^{\mu} b_{\mu}$, s^2, $u \cdot u$, $\Box$ acting on a scalar — was an upper-lower contraction: the pairing of the two prototype behaviors is precisely the pairing that cancels. ☐

Read backwards, this *derives* the definition: demand that upstairs and downstairs indices exist, that contracting them yields invariants, and that the behavior be linear — then each upstairs index must carry Λ exactly where each downstairs index carries the inverse $\Lambda_{\nu}{}^{\sigma}$, which is Eq. (61).

Where higher ranks come from. Two mechanisms manufacture tensors of rank ≥ 2, and both will be used. *Outer products*: if a^{μ} and b^{ν} are vectors, the sixteen products $a^{\mu} b^{\nu}$ transform as a $(2,0)$ tensor — immediate from the definition. *Linear physical laws*: whenever physics linearly relates one vector to another, the array of coefficients is forced to be a tensor. That is the content of:

Proposition 3.9 (Quotient rule)**.** *Let $T_{\mu\nu}$ be an array defined in every frame, and suppose that for* every *four-vector a^{ν} the contraction $c_{\mu} \equiv T_{\mu\nu} a^{\nu}$ transforms as a covariant vector. Then $T_{\mu\nu}$ is a $(0,2)$ tensor. (Analogous statements hold for every index structure.)*

Proof. Write what is assumed, in the primed frame and with the primed ingredients: $T'_{\mu\nu}a'^{\nu} = c'_{\mu}$. Express the right side by the known laws for c and then a:

$$T'_{\mu\nu}\,\Lambda^{\nu}{}_{\sigma}\,a^{\sigma} = \Lambda_{\mu}{}^{\rho}\,c_{\rho} = \Lambda_{\mu}{}^{\rho}\,T_{\rho\sigma}\,a^{\sigma}. \tag{63}$$

This holds for *arbitrary* a^{σ}, so the coefficients of a^{σ} agree: $T'_{\mu\nu}\Lambda^{\nu}{}_{\sigma} = \Lambda_{\mu}{}^{\rho}\,T_{\rho\sigma}$. Multiply by $\Lambda_{\kappa}{}^{\sigma}$ to free the last index (using (45) in the form $\Lambda^{\nu}{}_{\sigma}\Lambda_{\kappa}{}^{\sigma} = \delta^{\nu}{}_{\kappa}$): $T'_{\mu\kappa} = \Lambda_{\mu}{}^{\rho}\,\Lambda_{\kappa}{}^{\sigma}\,T_{\rho\sigma}$ — the $(0,2)$ transformation law. □

The quotient rule is how tensors are *detected* in practice: one never checks sixteen transformation equations directly. Part III runs it verbatim — the Lorentz force on a charge is linear in the charge's four-velocity, so the coefficient array $F_{\mu\nu}$ is a tensor by Proposition 3.9, and the transformation rules of $\vec{E}$ and $\vec{B}$ fall out of Eq. (61) for free. [Forward pointer: §11.]

Sums of same-type tensors and products with scalars stay in type; with outer products and contraction, this closes the algebra of operations, and raising or lowering an index — contraction with $\eta^{\mu\nu}$ or $\eta_{\mu\nu}$ — is a special case rather than a new rule.

The invariant tensors. Two tensors have the same components in every frame. The metric: $\eta'_{\rho\sigma} = \Lambda_{\rho}{}^{\mu}\,\Lambda_{\sigma}{}^{\nu}\,\eta_{\mu\nu} = \eta_{\rho\sigma}$, which is the defining condition (41) rearranged — *a Lorentz transformation is precisely one that leaves the components of the metric unchanged.* And the Kronecker delta $\delta^{\mu}{}_{\nu}$, whose invariance is the statement $\Lambda\Lambda^{-1} = 1$. These are the index-shuffling machines available in every frame for free; a third, the totally antisymmetric $\epsilon^{\mu\nu\rho\sigma}$ (invariant under the proper orthochronous component only), joins them in Part III when first needed. [stated; introduced properly in §11.]

Index gymnastics is frame-proof. One structural fact, already proved in pieces, deserves its general statement. Because $\eta_{\mu\nu}$ and $\eta^{\mu\nu}$ are invariant tensors, *raising or lowering any index of any tensor commutes with the Lorentz transformation law*: gymnastics before the boost and gymnastics after the boost give the same components — for vectors this was Eq. (46), and the general case is the same computation with spectator indices riding along. The index notation's entire frame-independence rests on this commutativity, and the commutativity rests on (41): one condition behind the whole calculus.

Fields: a tensor at every event. Part II needs one more notion, planted here. A *scalar field* assigns a scalar to each event: one function ϕ with

$$\phi'(x') = \phi(x) \qquad \text{whenever } x' = \Lambda x \tag{64}$$

— the moved observer, evaluating her function at her label for the *same event*, reads the same number. A vector or tensor field attaches the corresponding object to each event, components transforming by Eq. (61) on top of the relabeling (64). One consequence is used repeatedly later: if ϕ is a scalar field, then $\partial_{\mu}\phi$ is a covariant vector field and $\Box\phi$ is again a scalar field — differentiation respects the hierarchy, which is what makes it possible to write field equations covariantly at all.

Remark 3.10 (The covariance principle, and a terminological hazard). The payoff of the whole subsection is one sentence. *If an equation sets equal two tensors of the same type, and it holds in one inertial frame, it holds in every inertial frame* — for the difference of its two sides is a tensor whose components vanish in one frame, and Eq. (61), being linear, sends zero to zero. This is the operational content of Postulate 2.1 and the standing strategy of the rest of the document: laws of physics will be written as equations between same-type tensors ("in covariant form"), whereupon their frame independence is automatic rather than checked. Terminology, regrettably, overloads one word: a *covariant index* is a downstairs index, while a *covariant equation* is one written in tensor form and hence form-invariant. Both usages are standard; context separates them.

3.7 The grammar of indices

The hierarchy of Definition 3.7 is complete, and everything ahead computes inside it constantly. The computations obey a small, closed set of rules — scattered so far across conventions and propositions; collected here once, as grammar, with the standing traps named. Everything in this subsection is bookkeeping for structures already established; nothing is new physics.

Rule 1: free and dummy indices. In any single term, an index appears either *once* (a *free* index, labeling which component equation is meant) or *exactly twice, once up and once down* (a *dummy* index, summed by Convention 1.2). Grammar of equations: every term of an equation must carry the same free indices, same letters, *same heights.* Dummies are private to their term and may be renamed at will — to any letter not already in use there.

$$\begin{array}{cc} \underbrace{a^\mu = b^\mu + \Lambda^\mu{}_\nu c^\nu}_{\text{legal}} & \underbrace{a^\mu = b_\mu}_{\text{illegal: heights}} \\ \underbrace{a^\mu = b^\nu}_{\text{illegal: free mismatch}} & \underbrace{a^\mu b_\mu c_\mu}_{\text{illegal: }\mu\text{ thrice}} \end{array} \tag{65}$$

An index appearing three times, or twice at the same height, is not "summed anyway" — it is a syntax error, and in practice almost always the fossil of a forgotten rename.

Rule 2: raising and lowering. $\eta_{\mu\nu}$ lowers, $\eta^{\mu\nu}$ raises, one slot at a time, spectator indices untouched; in this signature the operation costs a sign on each spatial component and nothing on the time component. Lower-then-raise is the identity (§3.1: the inverse property is exactly this demand). Contraction with the metric *is* raising/lowering — one mechanism, not two rules. *Example:* for the four-momentum $p^\mu = (E, \vec{p})$, lowering gives $p_\mu = (E, -\vec{p})$, and the contraction is immediately useful: $p^\mu p_\mu = E^2 - |\vec{p}|^2$, the mass shell (83) in one line.

Rule 3: slot order matters. The horizontal position of a slot is part of a tensor's identity: $T^\mu{}_\nu$ means $\eta^{\mu\rho}T_{\rho\nu}$, while $T_\nu{}^\mu$ means $\eta^{\mu\rho}T_{\nu\rho}$, and the two agree only when $T_{\mu\nu}$ is symmetric. The document's standing examples sit at the two extremes:

$$\begin{gathered} \eta^\mu{}_\nu = \eta_\nu{}^\mu = \delta^\mu{}_\nu \quad \text{(symmetric: stagger harmless)}, \\ F^\mu{}_\nu = -F_\nu{}^\mu \quad \text{(antisymmetric: stagger load-bearing; } F \text{ arrives in §5.3)}, \end{gathered} \tag{66}$$

and for the non-symmetric Λ the distinction is an inverse (Eq. (45)). Stacking indices vertically is therefore banned for any tensor not known to be symmetric. *Example:* let $T_{\mu\nu}$ have a single nonzero component, $T_{01} = 1$. Then $T^0{}_1 = \eta^{00}\,T_{01} = +1$, while $T_1{}^0 = T_{1\rho}\,\eta^{\rho 0} = T_{10}\,\eta^{00} = 0$: same letters, same heights, different slots, different numbers.

Rule 4: contraction pairs one up with one down. Only upper–lower pairs may be contracted — that pairing is the one whose transformation factors cancel (Proposition 3.8 proved it; here it is grammar). The would-be "contraction" $a^\mu b^\mu$ is not a scalar — it transforms with two uncancelled Λ's — and in this signature it is not even the Euclidean dot product of anything; the invariant pairing of two upper-index vectors is $\eta_{\mu\nu}a^\mu b^\nu$, the metric supplying the missing lower slot. Corollary worth memorizing: the Kronecker delta is an *index substitution machine*, $\delta^\mu{}_\nu\, a^\nu = a^\mu$, and its self-contraction counts dimensions,

$$\delta^\mu{}_\mu = 4 \qquad \text{(not 1)}, \tag{67}$$

— a very common numerical slip in the calculus. *Example:* take the event $(t,x) = (1,0)$ and boost with $\beta = 0.6$ ($\gamma = 1.25$): the new coordinates are $(t', x') = (1.25, -0.75)$. The illegal same-height sum changes, $t'^2 + x'^2 = 2.1875 \neq 1 = t^2 + x^2$, while the lawful contraction holds, $t'^2 - x'^2 = 1.5625 - 0.5625 = 1 = t^2 - x^2$ — the grammar rule and Theorem 2.4 are the same fact.

Rule 5: symmetric contracted with antisymmetric vanishes.

Proposition 3.11. *If $S^{\mu\nu} = S^{\nu\mu}$ and $A_{\mu\nu} = -A_{\nu\mu}$, then $S^{\mu\nu}A_{\mu\nu} = 0$.*

Proof. Rename the dummies, then use both symmetries: $S^{\mu\nu}A_{\mu\nu} = S^{\nu\mu}A_{\nu\mu} = (+S^{\mu\nu})(-A_{\mu\nu}) = -\,S^{\mu\nu}A_{\mu\nu}$; a number equal to its own negative vanishes. □

Example: for any vector v^μ, the outer product $v^\mu v^\nu$ is symmetric, so $A_{\mu\nu}\, v^\mu v^\nu = 0$ for every antisymmetric A — in particular $F_{\mu\nu}\, u^\mu u^\nu = 0$, which is why the Lorentz force (100) automatically preserves the normalization of the four-velocity: $u_\mu\, \mathrm{d}u^\mu/\mathrm{d}\tau = (q/m)\, F_{\mu\nu}u^\mu u^\nu = 0$, so $u \cdot u = 1$ is maintained along the motion. This three-line fact recurs at essential points later: second partials $\partial_\mu\partial_\nu$ are symmetric, so their contraction with anything antisymmetric vanishes identically — the mechanism behind gauge invariance of $F_{\mu\nu}$ (§5.3) and behind the automatic charge conservation of §12, both of which are this proposition wearing field-theory clothes.

Rule 6: differentiation. The coordinate derivative carries a genuine lower index, ∂_μ (derived in §3.5), and the basic derivative is

$$\partial_\mu x^\nu = \delta_\mu{}^\nu. \tag{68}$$

The one trap: *before differentiating with respect to an indexed variable, rename every dummy that collides with the free index of the derivative.* Canonical example — differentiate a quadratic form:

$$\frac{\partial}{\partial a_\nu}\big(a_\mu a^\mu\big) = \frac{\partial}{\partial a_\nu}\big(\eta^{\rho\sigma} a_\rho a_\sigma\big) = \eta^{\rho\sigma}\big(\delta_\rho{}^\nu\, a_\sigma + a_\rho\, \delta_\sigma{}^\nu\big) = 2\, a^\nu, \tag{69}$$

both slots contributing — the computation quoted at Eq. (138) for $\partial\mathcal{L}/\partial(\partial_\mu\phi)$ is exactly this, with $a \to \partial\phi$. Attempting it without the rename ($\partial(a_\mu a^\mu)/\partial a_\mu$ — three μ's) is Rule 1's syntax error in its most tempting form.

Fluency test, recommended once with pen and paper: expand $T^\mu{}_\nu\, a^\nu b_\mu$ fully for diagonal η (sixteen terms, four survivors), then verify each rule above was used and no step needed anything beyond them. The grammar is small; the entire covariant calculus of this document is sentences in it.

3.8 The Lorentz and Poincaré groups

Return to the matrix form (42) of the defining condition, $\Lambda^{\mathsf{T}}\boldsymbol{\eta}\,\Lambda = \boldsymbol{\eta}$. Its solutions form a group — the *Lorentz group*, $O(1,3)$: closure is one line (transpose a product), and the inverse is not only guaranteed but already computed, Eq. (45) — and it satisfies the defining condition itself (sandwich (42) between $(\Lambda^{\mathsf{T}})^{-1}$ and Λ^{-1}). Two discrete invariants follow immediately:

- Taking det of (42): $(\det\Lambda)^2 = 1$, so $\det\Lambda = \pm 1$.
- Taking the $\rho = \sigma = 0$ component of (41): $(\Lambda^0{}_0)^2 = 1 + \sum_i (\Lambda^i{}_0)^2 \geq 1$, so $\Lambda^0{}_0 \geq 1$ or $\Lambda^0{}_0 \leq -1$ — a transformation either preserves the direction of time (*orthochronous*) or reverses it, with no middle ground.

Neither sign can change along a continuous path, so the group falls into **four connected components**, labeled by the pair $(\det\Lambda,\ \operatorname{sgn}\Lambda^0{}_0)$ and represented by four familiar transformations:

component	$\det\Lambda$	$\operatorname{sgn}\Lambda^0{}_0$	representative
proper orthochronous	$+1$	$+$	$\mathbb{1}$ (boosts, rotations)
improper orthochronous	-1	$+$	P: $\vec{x} \to -\vec{x}$
improper non-orthochronous	-1	$-$	T: $t \to -t$
proper non-orthochronous	$+1$	$-$	PT: $x^\mu \to -x^\mu$

Only the first component, $SO^+(1,3)$, contains the identity, and everything derived by continuity arguments in §2 lives there; the other three are the first component times P, T, or PT. Whether P and T are symmetries of nature is a question about *dynamics*, not geometry — the weak interaction famously violates both — and the geometric statement here is only that they are disconnected from the identity: no accumulation of small boosts and rotations ever mirrors space or reverses time.

Dimension: the six planes of spacetime. How many continuous parameters does $SO^+(1,3)$ have? Count constraints: Eq. (42) is a symmetric 4×4 matrix equation — 10 independent conditions on the 16 entries of Λ — leaving $\boxed{6}$ parameters. The six have names: a transformation preserving the metric can rotate any of the six coordinate *planes* of spacetime,

$$\underbrace{(xy),\ (yz),\ (zx)}_{\text{3 rotations, angles }\theta} \qquad \underbrace{(tx),\ (ty),\ (tz)}_{\text{3 boosts, rapidities }\phi} \tag{70}$$

— a boost is a rotation that mixes time with one spatial axis, hyperbolically because of the metric's sign there, which is the whole content of Remark 2.6 said group-theoretically. Each plane generates a one-parameter subgroup with additive parameter (angles add; rapidities add, Eq. (26)).

Remark 3.12 (Boosts alone do not close). The three rotations form a subgroup, $SO(3)$. The three boosts do *not*: composing boosts along different axes yields a boost *times a rotation* (the Wigner rotation, physically observable as Thomas precession of accelerated spins). The clean algebraic statement — rotations and boosts as generators, their commutation relations, and why [boost, boost] = rotation — is Lie algebra material and is deferred. [stated; the Lorentz algebra is standard QFT-course territory, cf. Schwartz Ch. 10.]

The Poincaré group. The full symmetry of Minkowski spacetime adds what homogeneity always permitted: translations,

$$x'^\mu = \Lambda^\mu{}_\nu\, x^\nu + a^\mu, \qquad \boxed{6 + 4 = 10 \text{ parameters.}} \tag{71}$$

The number ten is worth remembering now and using later: by Noether's theorem [Mech. §3.3], every continuous symmetry parameter yields a conservation law, and the ten parameters of the Poincaré group will produce exactly the ten conserved quantities of relativistic physics — energy (1), momentum (3), angular momentum (3), and the three boost charges — for the particle in §4 and again, localized, for fields in §9.

3.9 Natural units from here on

The apparatus is assembled, every factor of c has served its purpose as a dimension check, and Convention 1.3 now takes effect.

Convention 3.13 (Natural units)**.** From this point on, $c = 1$: time and length are measured in the same unit, velocities are dimensionless (β and v are now the same symbol), and mass, energy, and momentum share a dimension. Any equation can be restored to SI form uniquely: insert the power of c that makes the dimensions balance. Example: $E = m$ restores to $E = mc^2$ — energy over mass has dimensions of speed squared, and c is the only speed in the formalism. The four-velocity becomes $u^\mu = \gamma(1, \vec{v})$ with $u \cdot u = 1$; the interval, $s^2 = t^2 - |\vec{x}|^2$; the d'Alembertian, $\Box = \partial_t^2 - \nabla^2$.

With geometry and units in hand, dynamics can start: what does a free particle *do* in Minkowski spacetime, and what action principle says so?

4 The Relativistic Particle

This section is where the two companion threads meet: the variational machinery of the mechanics notes runs, unmodified, on the geometry of §§2–3. The plan: find the action a free particle is forced to have, extract its equations of motion (twice — once in lab coordinates for contact with [Mech. §3], once covariantly), and then let Noether's theorem *derive* energy and momentum, which combine into a four-vector. Natural units throughout (Convention 3.13).

4.1 The action, and why it is forced

What action can a free particle in Minkowski spacetime possibly have? Constraints, in decreasing order of negotiability:

(i) S must be a **Lorentz scalar**. By the covariance principle (Remark 3.10), a scalar action yields equations of motion valid in every frame at once — Postulate 2.1 implemented rather than checked.

(ii) S must be built from the **worldline alone**. The particle is free: no fields, no external structure, no preferred events are available as ingredients.

(iii) S must reproduce **Newtonian mechanics** at $v \ll 1$ — fixing normalization, not form.

Conditions (i) and (ii) are extremely restrictive: from the worldline and its *first* derivative, the only reparametrization-invariant scalar functional is the Minkowski arc length — the proper time of Eq. (48). (Scalars built from higher derivatives do exist — the proper acceleration $\sqrt{-a \cdot a}$ is one — but actions containing them yield equations of motion of higher than second order, outside the mechanical framework of [Mech. §2]; "first derivatives only" is the standard assumption of the whole series, made explicit here.) So, up to the constant that (iii) will fix,

$$\boxed{S[x] \;=\; -m\int \mathrm{d}\tau \;=\; -m\int \sqrt{1-v^2}\,\mathrm{d}t, \qquad L(\vec{v}) = -m\sqrt{1-v^2}.} \tag{72}$$

The second form parametrizes the worldline by lab time and reads off the Lagrangian; it is the form on which the machinery of [Mech. §2–3] operates directly.

Fixing the constant. Expand for small speeds:

$$L = -m\sqrt{1-v^2} = -\,m \;+\; \tfrac{1}{2}mv^2 \;+\; \tfrac{1}{8}mv^4 + \ldots \tag{73}$$

The additive constant $-m$ never enters the Euler–Lagrange equations; the next term is the Newtonian kinetic energy, provided the prefactor of $\int \mathrm{d}\tau$ is $-m$ with m the particle's mass — minus sign included, since $\sqrt{1-v^2}$ *decreases* with speed. (Restoring units in (73): $-mc^2 + \frac{1}{2}mv^2 + \ldots$ — the rest energy appears as an additive constant before dynamics has even started.)

Remark 4.1 (Stationary action is stationary proper time). Since $S = -m\tau$, extremizing the action extremizes proper time, and by Remark 3.4 the straight worldline is the proper-time *maximum* among all worldlines between fixed events. So the variational principle (72) says: *a free particle moves so that its own clock runs the longest* — inertial motion is the stay-at-home twin's path. The equations of motion below merely spell out what Fig. 5 already shows.

4.2 Two routes to the equations of motion

Route 1: lab coordinates. Apply the Euler–Lagrange equations [Mech. §3.1] to $L = -m\sqrt{1-v^2}$, a function of $\vec{v}$ only. The canonical momentum:

$$\vec{p} \;=\; \frac{\partial L}{\partial \vec{v}} = \frac{m\,\vec{v}}{\sqrt{1-v^2}} = \gamma m\,\vec{v}, \tag{74}$$

— the Newtonian $m\vec{v}$ dressed by γ — and since $\partial L/\partial \vec{x} = 0$,

$$\frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = 0: \tag{75}$$

$\gamma m\vec{v}$ is constant, hence v and $\vec{v}$ are, and the worldline is straight. Free particles move uniformly — reassuring, and required: Lemma 2.3 leaned on it.

Route 2: covariant variation. The same, without ever choosing a frame. Parametrize the worldline arbitrarily, $x^\mu(\lambda)$, and write the action as the arc length it is:

$$S = -m\int \sqrt{\eta_{\mu\nu}\,\dot{x}^\mu \dot{x}^\nu}\;\mathrm{d}\lambda, \qquad \dot{x}^\mu \equiv \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}. \tag{76}$$

Vary the worldline, $x^\mu \to x^\mu + \delta x^\mu$ with endpoints pinned, exactly as in the Gateaux framework of [Mech. §2.2]:

$$\delta S = -m\int \frac{\eta_{\mu\nu}\,\dot{x}^\mu\,\delta\dot{x}^\nu}{\sqrt{\dot{x}\cdot\dot{x}}}\;\mathrm{d}\lambda = +\,m\int \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\dot{x}_\nu}{\sqrt{\dot{x}\cdot\dot{x}}}\right)\delta x^\nu\;\mathrm{d}\lambda, \tag{77}$$

integrating by parts and dropping the vanishing boundary term. The variation δx^ν is arbitrary, so by the fundamental lemma [Mech. App. B] the bracket has zero derivative. But $\sqrt{\dot{x}\cdot\dot{x}}\,\mathrm{d}\lambda = \mathrm{d}\tau$, so $\dot{x}^\mu/\sqrt{\dot{x}\cdot\dot{x}} = \mathrm{d}x^\mu/\mathrm{d}\tau = u^\mu$, and the equation of motion is

$$\boxed{\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} = 0 \qquad \Longleftrightarrow \qquad x^\mu(\tau) = x^\mu(0) + u^\mu\,\tau:} \tag{78}$$

straight worldlines, now as a manifestly covariant statement — true in all frames because written in four-vectors, per Remark 3.10.

Remark 4.2 (An equation with a built-in redundancy). Equation (78) looks like four equations but carries only three: contracting with u_μ gives $u \cdot \mathrm{d}u/\mathrm{d}\tau = \frac{1}{2}\,\mathrm{d}(u \cdot u)/\mathrm{d}\tau = 0$ identically, by the normalization (51) — one component is an empty identity, not a law. The missing content is exactly the freedom the description carried from the start: the parameter λ was arbitrary (the action, being arc length, is *reparametrization invariant*), and a redundant coordinate in the description yields an empty equation of motion. File the pattern away: in Part III the potential A^μ carries a redundancy of precisely this flavor — gauge freedom — with precisely this consequence (§13).

4.3 Energy and momentum from Noether's theorem

Here is the section's centerpiece, and the reason this document derives four-momentum rather than defining it. The Lagrangian (72) depends on neither t nor $\vec{x}$: the action is invariant under the four spacetime translations, and Noether's theorem [Mech. §3.3] converts each invariance into a conserved quantity.

Time translation → energy. Invariance under $t \to t+\epsilon$ conserves the Jacobi function $h = \vec{p}\cdot\vec{v} - L$ [Mech. §3.3, §6.3]:

$$E \;\equiv\; \vec{p}\cdot\vec{v} - L = \frac{mv^2}{\sqrt{1-v^2}} + m\sqrt{1-v^2} = m\,\frac{v^2 + (1-v^2)}{\sqrt{1-v^2}} = \frac{m}{\sqrt{1-v^2}} = \gamma m. \tag{79}$$

Space translations → momentum. Invariance under $\vec{x} \to \vec{x} + \vec{\epsilon}$ conserves the canonical momentum (74): $\vec{p} = \gamma m \vec{v}$.

Assembly. Stack the four conserved numbers:

$$\big(E,\ \vec{p}\big) = \big(\gamma m,\ \gamma m \vec{v}\big) = m\,\gamma\,(1, \vec{v}) = m\,u^\mu. \tag{80}$$

The right-hand side is a scalar times a four-vector (Eq. (50)) — so the four Noether charges of the four translations are not merely four conserved numbers; they are the components of a conserved *four-vector*,

$$\boxed{p^\mu \;=\; m\,u^\mu \;=\; \big(E,\ \vec{p}\big) = \big(\gamma m,\ \gamma m\vec{v}\big), \qquad \frac{\mathrm{d}p^\mu}{\mathrm{d}\tau} = 0.} \tag{81}$$

Energy is the *time component* of a four-vector: under a boost, E and $\vec{p}$ mix by Eq. (24) exactly as t and $\vec{x}$ do. That one sentence encodes much of relativistic kinematics — and it was derived, not postulated: it follows from translation symmetry through Noether's theorem, the same derivation as in [Mech. §3.3], run on a new Lagrangian.

Expanding the energy for small speeds,

$$E = \gamma m = m + \tfrac{1}{2}mv^2 + \tfrac{3}{8}mv^4 + \ldots \tag{82}$$

— Newtonian kinetic energy riding on a constant that Newtonian physics could never see, because Newtonian physics only ever measured energy *differences*. Restoring units, the constant is $E_{\text{rest}} = mc^2$: mass is the energy a particle cannot shed by slowing down.

The mass shell. The norm of (81) is fixed by (51): $p \cdot p = m^2\, u \cdot u = m^2$, i.e.

$$\boxed{E^2 - |\vec{p}|^2 = m^2 \qquad \Longleftrightarrow \qquad E = \sqrt{|\vec{p}|^2 + m^2}.} \tag{83}$$

In momentum space $(E, \vec{p})$ this is a hyperboloid — the *mass shell* (Fig. 6) — the same invariant-hyperbola geometry as Fig. 1, now with the particle's identity m as the invariant. Useful limits: at rest, $E = m$; and as $m \to 0$ the shell collapses onto the light cone of momentum space, $E = |\vec{p}|$, with $v = \partial E/\partial|\vec{p}| \to 1$ — massless particles move at c and have momentum with no mass to show for it. (The action (72) itself degenerates at $m = 0$, where $\mathrm{d}\tau = 0$ along the worldline; the mass-shell relation survives the limit and is taken as the massless particle's definition. [stated; the proper massless action is a QFT-course topic.])

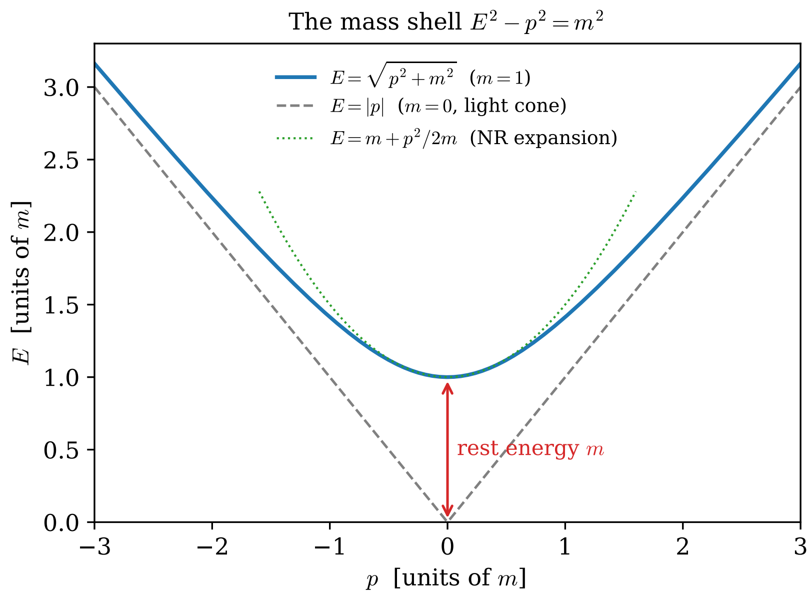


Figure 6: The mass shell (83) in 1+1 momentum space: the exact relativistic dispersion (blue), its massless light-cone asymptote (gray, dashed), and the nonrelativistic parabola $E = m + p^2/2m$ of Eq. (82) (green, dotted), which hugs the shell near $p = 0$ and departs from it at large p. This figure returns, with the axes relabeled (ω, k), as the Klein–Gordon dispersion relation in §8.

Example 4.3 (LHC protons: the worked number)**.** An LHC beam proton has $E = 6.8\,\mathrm{TeV}$ and $m = 0.938\,\mathrm{GeV}$, so

$$\gamma = \frac{E}{m} = 7.25 \times 10^3, \qquad 1 - \beta \approx \frac{1}{2\gamma^2} = 9.5 \times 10^{-9}\,: \tag{84}$$

the proton trails a light pulse by about $c\,(1-\beta) \approx 2.9\,\mathrm{m/s}$ — walking pace — while carrying 7250 times its rest energy. On Fig. 6 it sits far up the asymptote, where the shell and the light cone are indistinguishable to one part in 10^8.

Remark 4.4 (The other six charges). Section 3.8 promised ten conservation laws for the ten Poincaré parameters; translations have now supplied four. The three rotations conserve angular momentum $\vec{L} = \vec{x} \times \vec{p}$, exactly as in [Mech. §3.3] — nothing relativistic changes in that derivation. The three boosts conserve the less famous $\vec{K} = E\,\vec{x} - t\,\vec{p}$: indeed $\mathrm{d}\vec{K}/\mathrm{d}t = E\vec{v} - \vec{p} - t\,\dot{\vec{p}} = \gamma m\vec{v} - \gamma m\vec{v} - 0 = 0$. Solving $\vec{K} = \text{const}$ for $\vec{x}$ shows its meaning: the point $\vec{x} = (\vec{K} + t\vec{p})/E$ — the center of energy — moves uniformly. Boost symmetry conserves the *uniformity of the center-of-energy motion*, the relativistic upgrade of the center-of-mass theorem. All six reappear, localized into densities, in §9.

4.4 The relativistic Hamiltonian

The Legendre transform [Mech. §6] completes the particle's description. Invert (74) for the velocity: squaring, $|\vec{p}|^2(1 - v^2) = m^2 v^2$ gives

$$\vec{v} = \frac{\vec{p}}{\sqrt{|\vec{p}|^2 + m^2}}, \tag{85}$$

and the Hamiltonian is

$$H(\vec{p}) = \vec{p}\cdot\vec{v} - L = \frac{|\vec{p}|^2}{\sqrt{|\vec{p}|^2 + m^2}} + \frac{m^2}{\sqrt{|\vec{p}|^2 + m^2}} = \boxed{\sqrt{|\vec{p}|^2 + m^2}} \tag{86}$$

— the mass-shell function itself, as it must be: L has no explicit time dependence, so H equals the conserved energy [Mech. §6.3], and the conserved energy (79) lives on the shell (83). Hamilton's equations close the loop:

$$\dot{\vec{x}} = \frac{\partial H}{\partial \vec{p}} = \frac{\vec{p}}{\sqrt{|\vec{p}|^2 + m^2}} = \vec{v}, \qquad \dot{\vec{p}} = -\frac{\partial H}{\partial \vec{x}} = 0. \tag{87}$$

Remark 4.5 (Is H covariant? No — and instructively so). Route 2 delivered the manifestly covariant Eq. (78); the Hamiltonian route cannot deliver its like, and the loss happens at the first step: the Legendre transform differentiates with respect to *somebody's* time, so writing a Hamiltonian means choosing a slicing of spacetime into moments. Nor is H itself a scalar — it is the *time component* of p^μ, so different inertial observers use different Hamiltonians, related by boosts. Covariance is not broken, but it is demoted from a property visible by inspection to a theorem requiring proof: the ten charges of Remark 4.4 must close, under Poisson brackets, into the algebra of the Poincaré group. In that algebra H and $\vec{K}$ are the generators that *move the time slice*, and the fact that they do not commute is already on display: $\vec{K} = E\vec{x} - t\vec{p}$ carries explicit time dependence, and its conservation is a cancellation,

$$\frac{\mathrm{d}K_i}{\mathrm{d}t} = \frac{\partial K_i}{\partial t} + \{K_i, H\} = -\,p_i + E\,\frac{\partial H}{\partial p_i} = -\,p_i + p_i = 0, \tag{88}$$

with $\{K_i, H\} = p_i \neq 0$ — the classical fingerprint of the commutator $[\hat{K}_i, \hat{H}] \neq 0$ that canonical quantum field theory must live with. The same trade returns at field level in §10: there $H = \int T^{00}\,\mathrm{d}^3x$ is again a 00-component integrated over a chosen slice, canonical quantization inherits the non-manifest covariance, and "the quantum theory is Lorentz invariant" becomes a statement about the algebra of operator charges, verified rather than seen. (A manifestly covariant Hamiltonian formulation of the particle does exist: take the worldline parameter itself as time, whereupon H vanishes identically and the dynamics hides inside the constraint $p\cdot p = m^2$ — the Hamiltonian face of Remark 4.2's reparametrization redundancy. Constrained-dynamics territory; deferred. [stated.])

Remark 4.6 (The square-root problem). $H = \sqrt{|\vec{p}|^2 + m^2}$ is an unpleasant object to promote to a quantum operator: the square root of a differential operator is nonlocal, and awkward to work with. The historical escape was to *square* the relation — demand $E^2 = |\vec{p}|^2 + m^2$ of a wave $\mathrm{e}^{-\mathrm{i}(Et - \vec{p}\cdot\vec{x})}$ — which yields not a Schrödinger-type equation but the *Klein–Gordon equation*, a field equation. That road, from the awkward square root to relativistic fields, is exactly the road this document travels: §8 arrives at Klein–Gordon from the Lagrangian side, and Fig. 6 will be waiting there with its axes relabeled.

5 The Charged Particle

The free particle exhausted what a bare worldline can do. To make anything else happen, the particle needs an environment — and the simplest relativistic environment is a four-vector field $A_\mu(x)$ defined over spacetime, coupled in the simplest way the covariance principle permits. This short section builds that coupling, finds that its redundancy and its equation of motion are both familiar — gauge freedom and the Lorentz force — and establishes two structures that Part III will need: the antisymmetric derivative combination $F_{\mu\nu}$, and the split between canonical and mechanical momentum.

5.1 The coupling

What scalar can be built that is *linear* in a vector field A_μ and local to the worldline? The field must be evaluated on the worldline and contracted with something the worldline provides; the worldline provides $\mathrm{d}x^\mu$. Hence, appended to the free action (72):

$$\boxed{S \;=\; -m\int \mathrm{d}\tau \;-\; q\int A_\mu(x)\,\mathrm{d}x^\mu,} \tag{89}$$

with a coupling constant q — the *charge* — measuring how strongly this particular particle couples to this particular field. The interaction term is a Lorentz scalar by construction (one upper index contracted against one lower), and it is reparametrization invariant without help: $\mathrm{d}x^\mu$ needs no $\mathrm{d}\tau$, no metric, no clock. To recognize it, parametrize by lab time. With the traditional names for the components, $A^\mu = (\varphi, \vec{A})$ — so $A_\mu = (\varphi, -\vec{A})$ — the integrand is $A_\mu \mathrm{d}x^\mu = \varphi\,\mathrm{d}t - \vec{A}\cdot\mathrm{d}\vec{x}$, and

$$L = -\,m\sqrt{1-v^2} \;-\; q\,\varphi(t,\vec{x}) \;+\; q\,\vec{A}(t,\vec{x})\cdot\vec{v}. \tag{90}$$

The scalar potential enters like a potential energy; the vector potential enters through a velocity-dependent term — the first Lagrangian of the series in which force will depend on velocity.

5.2 Gauge invariance of the coupling

The field A_μ enters (89) with a redundancy. Shift it by the four-gradient of any scalar function:

$$\boxed{A_\mu \;\longrightarrow\; A_\mu + \partial_\mu\chi.} \tag{91}$$

The interaction changes by

$$-\,q\int \partial_\mu\chi\,\mathrm{d}x^\mu = -\,q\int \mathrm{d}\chi = -\,q\left[\chi(x_B) - \chi(x_A)\right] \tag{92}$$

— a pure endpoint term, which variations with pinned endpoints never see [Mech. §2.2]. Every choice of χ yields the *same* equations of motion: the four functions A_μ overdescribe the physics, and the overdescription is exactly one scalar function's worth. This is the pattern of Remark 4.2 in mirror image — there, redundancy in the *worldline's* description produced an empty equation; here, the redundancy sits in the *background's* description. What the equations of motion can depend on is only the gauge-invariant content of A_μ, and the next subsection watches the variational principle extract precisely that content on its own.

5.3 Equations of motion: the Lorentz force

Route 1: lab coordinates. Run Euler–Lagrange on (90). The canonical momentum picks up a field contribution:

$$\boxed{\vec{P} \;\equiv\; \frac{\partial L}{\partial \vec{v}} \;=\; \gamma m\vec{v} + q\vec{A} \;=\; \vec{p} + q\vec{A},} \tag{93}$$

where $\vec{p} = \gamma m\vec{v}$ is the *mechanical* momentum of (74) — the distinction matters and returns in §5.4. The coordinate derivative of L:

$$\frac{\partial L}{\partial \vec{x}} = -\,q\nabla\varphi + q\,\nabla(\vec{A}\cdot\vec{v}), \tag{94}$$

the gradient acting on the field arguments only ($\vec{v}$ is an independent slot of the Lagrangian, not a field on space). The Euler–Lagrange equation $\mathrm{d}\vec{P}/\mathrm{d}t = \partial L/\partial\vec{x}$ needs the total time derivative of $\vec{A}$ *along the moving particle*, which is the convective derivative:

$$\frac{\mathrm{d}\vec{A}}{\mathrm{d}t} = \frac{\partial\vec{A}}{\partial t} + (\vec{v}\cdot\nabla)\vec{A} \tag{95}$$

— the particle feels the field change both because the field changes and because the particle moves through it. Assembling:

$$\frac{\mathrm{d}\vec{p}}{\mathrm{d}t} = -\,q\left(\nabla\varphi + \frac{\partial\vec{A}}{\partial t}\right) + q\left[\nabla(\vec{A}\cdot\vec{v}) - (\vec{v}\cdot\nabla)\vec{A}\right]. \tag{96}$$

The last bracket is the curl identity $\vec{v}\times(\nabla\times\vec{A}) = \nabla(\vec{A}\cdot\vec{v}) - (\vec{v}\cdot\nabla)\vec{A}$ (valid because $\vec{v}$ carries no position dependence). The equation of motion has organized its right side into two derivative combinations of the potentials, and we *name* them:

$$\vec{E} \;\equiv\; -\nabla\varphi - \frac{\partial\vec{A}}{\partial t}, \qquad \vec{B} \;\equiv\; \nabla\times\vec{A}, \tag{97}$$

whereupon

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\big(\gamma m\vec{v}\big) \;=\; q\,\big(\vec{E} + \vec{v}\times\vec{B}\big)\,:} \tag{98}$$

the Lorentz force law, derived rather than postulated. Two things deserve to be said about the derivation rather than the result. First, $\vec{E}$ and $\vec{B}$ were not inputs: they are the combinations of potential derivatives that the variational principle assembled on its own, and both are inert under the gauge shift (91) — $\nabla\times\nabla\chi = 0$ kills it in $\vec{B}$, and $-\nabla\dot{\chi} - \partial_t\nabla\chi = 0$ in $\vec{E}$ — exactly as §5.2 predicted the equations of motion must arrange. Second, the left side is the *mechanical* momentum: what the force pushes is $\gamma m\vec{v}$, not the canonical $\vec{P}$.

Route 2: covariant form. The same variation without choosing a frame produces the central object of Part III. Vary the interaction in worldline form, $S_{\text{int}} = -q\int A_\mu(x)\,\dot{x}^\mu\,\mathrm{d}\lambda$:

$$\begin{aligned}
\delta S_{\text{int}} &= -\,q\int\Big[\big(\partial_\nu A_\mu\big)\,\delta x^\nu\,\dot{x}^\mu + A_\nu\,\delta\dot{x}^\nu\Big]\mathrm{d}\lambda \\
&= -\,q\int\Big[\partial_\nu A_\mu\,\dot{x}^\mu - \frac{\mathrm{d}}{\mathrm{d}\lambda}A_\nu\Big]\,\delta x^\nu\,\mathrm{d}\lambda = -\,q\int\Big[\partial_\nu A_\mu - \partial_\mu A_\nu\Big]\,\dot{x}^\mu\,\delta x^\nu\,\mathrm{d}\lambda,
\end{aligned} \tag{99}$$

integrating by parts in the second step and using $\mathrm{d}A_\nu/\mathrm{d}\lambda = \dot{x}^\mu \partial_\mu A_\nu$ (the covariant convective derivative) in the third. Adding the free variation from §4.2 and demanding stationarity:

$$\boxed{m\,\frac{\mathrm{d}u^\mu}{\mathrm{d}\tau} \;=\; q\,F^\mu{}_\nu\,u^\nu, \qquad F_{\mu\nu} \;\equiv\; \partial_\mu A_\nu - \partial_\nu A_\mu.} \tag{100}$$

The variation *manufactured* the antisymmetric combination $F_{\mu\nu}$ — the only part of $\partial_\mu A_\nu$ that survives, and manifestly gauge invariant, since $\partial_\mu\partial_\nu\chi - \partial_\nu\partial_\mu\chi = 0$. And its tensor character comes with no extra work: the force (100) is linear in the four-vector u^ν with a four-vector result, so by the quotient rule (Proposition 3.9) $F^\mu{}_\nu$ *is* a $(1,1)$ tensor — as promised in §3.6. Its six independent components (antisymmetry kills the diagonal and pairs the rest) are exactly the six numbers (97): $\vec{E}$ and $\vec{B}$ are one tensor seen in $3+1$ pieces, which Part III displays in full (§11).

Example 5.1 (The LHC proton, bent: the worked number)**.** For circular motion at constant speed, γ is constant, and (98) in a pure magnetic field reads $\gamma m\,v^2/r = qvB$, so

$$r = \frac{\gamma m v}{qB} = \frac{p}{qB}\;: \tag{101}$$

the bending radius is set by the *mechanical momentum*, γ included. The proton of Example 4.3 ($p \approx E = 6.8\,\mathrm{TeV}$) rides a dipole field $B = 8.1\,\mathrm{T}$ (the magnets' $8.33\,\mathrm{T}$ design value scaled to the running energy), so

$$r = \frac{6800\,\mathrm{GeV}}{0.300\times 8.1\,\mathrm{T}}\,\frac{\mathrm{m}}{\mathrm{GeV/T}} \approx 2.80\,\mathrm{km}, \tag{102}$$

the machine's actual bending radius. The cyclotron frequency $\omega = qB/(\gamma m)$ falls with energy — which is why a fixed-frequency cyclotron fails relativistically and the LHC is a *synchro*tron, its fields ramped in step with γ.

Example 5.2 (Constant field: hyperbolic motion)**.** The simplest exact solution of (100): a uniform, static $\vec{E} = E\,\hat{x}$, no $\vec{B}$, charge starting from rest. The potentials $\varphi = -Ex$, $\vec{A} = 0$ give (check against (97): $-\nabla\varphi = E\hat{x}$) the single nonzero pair $F_{01} = \partial_0 A_1 - \partial_1 A_0 = E$, whence $F^0{}_1 = \eta^{00}F_{01} = E$ and $F^1{}_0 = \eta^{11}F_{10} = E$. The covariant force law reduces to two coupled components, with $\alpha \equiv qE/m$:

$$\frac{\mathrm{d}u^0}{\mathrm{d}\tau} = \alpha\,u^1, \qquad \frac{\mathrm{d}u^1}{\mathrm{d}\tau} = \alpha\,u^0 \qquad \Longrightarrow \qquad \frac{\mathrm{d}}{\mathrm{d}\tau}\big(u^0 \pm u^1\big) = \pm\,\alpha\,\big(u^0 \pm u^1\big)\;: \tag{103}$$

the *null combinations* of §2.4 diagonalize the dynamics exactly as they diagonalized the boost. Starting from rest, $u^\mu(0) = (1,0)$, so $u^0 \pm u^1 = \mathrm{e}^{\pm\alpha\tau}$ and

$$u^0 = \cosh\alpha\tau, \qquad u^1 = \sinh\alpha\tau\;: \tag{104}$$

the *rapidity grows linearly in proper time*, $\phi = \alpha\tau$ — each proper second adds the same rapidity increment, and $\beta = \tanh\phi$ approaches but never reaches 1, which is Remark 2.9(iii) in dynamical form. Integrating once more,

$$t = \alpha^{-1}\sinh\alpha\tau, \qquad x = \alpha^{-1}\big(\cosh\alpha\tau - 1\big) \qquad \Longrightarrow \qquad \big(x + \alpha^{-1}\big)^2 - t^2 = \alpha^{-2}\;: \tag{105}$$

the worldline is one of the calibration hyperbolae of Fig. 1 — hence *hyperbolic motion*. The four-acceleration $a^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau = \alpha\,(\sinh\alpha\tau,\,\cosh\alpha\tau)$ satisfies $a\cdot u = 0$ by inspection and has constant invariant norm $\sqrt{-a\cdot a} = \alpha = qE/m$: constant *proper* acceleration — the higher-derivative scalar mentioned in §4.1, realized here by the simplest field. One number, units restored: at proper acceleration g, one year of proper time gives $\phi = g\tau/c \approx 1.03$, hence $\beta \approx 0.77$, with about 1.19 years of coordinate time elapsed and 0.56 light-years covered.

5.4 Canonical versus mechanical momentum

Equation (93) split the momentum in two, and the two halves divide the labor:

	mechanical $\vec{p} = \gamma m\vec{v}$	canonical $\vec{P} = \vec{p} + q\vec{A}$
gauge behavior	invariant	shifts by $q\nabla\chi$
what force acts on	Eq. (98)	—
Noether charge of $\vec{x} \to \vec{x} + \vec{\epsilon}$	—	conserved (when A_μ is $\vec{x}$-independent)
enters Hamilton's equations	—	$\vec{P}$ is the phase-space variable

Neither column can do the other's job: the measurable, gauge-invariant $\vec{p}$ is not the conserved quantity; the conserved $\vec{P}$ is not gauge invariant, and by itself not measurable. The Hamiltonian makes the division explicit. Legendre-transforming (90) with respect to $\vec{P}$ (the inversion runs exactly as in §4.4 with $\vec{p} = \vec{P} - q\vec{A}$):

$$\boxed{H(\vec{x}, \vec{P}) = \sqrt{\big(\vec{P} - q\vec{A}\big)^2 + m^2} \ + \ q\,\varphi.} \tag{106}$$

Comparing with the free $H = \sqrt{|\vec{P}|^2 + m^2}$: the entire interaction is the substitution

$$\vec{P} \to \vec{P} - q\vec{A}, \qquad H \to H - q\varphi \qquad \text{i.e.} \qquad p^\mu \ \to \ p^\mu - qA^\mu \tag{107}$$

— *minimal coupling*, the rule quantum mechanics will inherit verbatim ($\hat{\vec{p}} \to \hat{\vec{p}} - q\vec{A}$ in the Pauli and Dirac equations) and quantum field theory will re-derive from gauge symmetry. That the whole of electromagnetism's action on matter compresses into one substitution is not an accident of this Lagrangian; it is the classical shadow of the gauge principle, and the outlook (§14) says one more word about it.

Part I is complete: kinematics from two postulates, dynamics from one scalar action, energy and momentum from Noether, and a first field — entering as a fixed background to which the particle couples. The obvious next question is the reverse one: what dynamics does A_μ itself obey, and what action prescribes it? Answering requires learning to vary *fields* rather than worldlines, and that is Part II's business — built, like everything in this series, from a chain of masses.

Part II
Classical Field Theory

6 From Chain to Field

Part I closed by asking for the dynamics of a field. This section builds the concept from the series' oldest object: the bead chain of [Osc. §8–9]. There, the continuum limit was taken in the *equations of motion* — a limit in each equation. Here it is taken once, in the *Lagrangian*, and the single limit does more than reproduce the wave equation: it exposes the structural pivot on which all of field theory turns — the demotion of position from dynamical variable to label.

6.1 The chain acquires a Lagrangian

Recall the setting of [Osc. §9.1]: N beads of mass m, spaced a apart on a massless string under tension T between fixed walls, with small transverse displacements $y_j(t)$, $j = 1, \ldots, N$ (and $y_0 = y_{N+1} = 0$ at the walls). There, Newton delivered

$$m\ddot{y}_j = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big). \tag{108}$$

The Lagrangian route needs the one ingredient the free-body diagram never wrote down: the potential energy. It is the work done against tension by the *extra length* of the displaced string. The segment from bead j to bead $j+1$ has length

$$\sqrt{a^2 + (y_{j+1} - y_j)^2} = a\sqrt{1 + \left(\tfrac{y_{j+1}-y_j}{a}\right)^2} = a + \frac{(y_{j+1} - y_j)^2}{2a} + O\big((\Delta y)^4\big), \tag{109}$$

so, to the same small-slope order as (108),

$$U = T \times (\text{excess length}) = \sum_{j=0}^{N} \frac{T}{2a}\,\big(y_{j+1} - y_j\big)^2, \qquad L = \sum_{j=1}^{N} \tfrac{1}{2}\, m\, \dot{y}_j^2 \;-\; \sum_{j=0}^{N} \frac{T}{2a}\,\big(y_{j+1} - y_j\big)^2. \tag{110}$$

Check against (108): the coordinate y_j appears in *two* terms of the potential sum — the spring on its left and the spring on its right —

$$\frac{\partial L}{\partial y_j} = -\frac{T}{a}\Big[\big(y_j - y_{j-1}\big) - \big(y_{j+1} - y_j\big)\Big] = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big), \tag{111}$$

and Euler–Lagrange, $\mathrm{d}(\partial L/\partial \dot{y}_j)/\mathrm{d}t = \partial L/\partial y_j$, reproduces (108) exactly. Note for later the mechanism of (111): differencing with *both* neighbors is what turned one difference into a second difference. It is the discrete shadow of an integration by parts, and §7 meets its continuum original.

6.2 Taking the limit once

Refine as in [Osc. §9.2]: $a \to 0$, $N \to \infty$, holding fixed the physical rope — its length $L_{\text{rope}} = (N+1)a$, its mass per length $\mu = m/a$, and its tension T. The displacements merge into a smooth profile, and here is the moment the notation should mark: write

$$y_j(t) \;=\; \phi(x_j, t), \qquad x_j = ja, \tag{112}$$

renaming $y \to \phi$ as the displacement sheds its origin (a transverse direction) and becomes an abstract *field value* — the letter ϕ serves for the rest of the document, and no confusion with the rapidity survives §3's $c = 1$ switch. Then $\dot{y}_j = \dot{\phi}(x_j, t)$ and $(y_{j+1} - y_j)/a \to \partial\phi/\partial x \equiv \phi'$, and the Lagrangian (110), rewritten to expose one factor of a per term,

$$L = \sum_j a \left[\frac{1}{2} \frac{m}{a} \dot{y}_j^2 - \frac{1}{2} T \Big(\frac{y_{j+1} - y_j}{a} \Big)^2 \right], \tag{113}$$

converges as a Riemann sum, $\sum_j a \to \int \mathrm{d}x$, to

$$\boxed{L = \int_0^{L_{\text{rope}}} \mathrm{d}x\, \mathcal{L}\big(\dot{\phi}, \phi'\big), \qquad \mathcal{L} = \tfrac{1}{2}\,\mu\,\dot{\phi}^2 - \tfrac{1}{2}\,T\,\phi'^2.} \tag{114}$$

$\mathcal{L}$ is the *Lagrangian density* — Lagrangian per unit length — and its two terms are exactly the kinetic and potential energy densities of the rope. One limit, taken in one object; the equations of motion will be extracted from it once, for all x simultaneously, in §7.

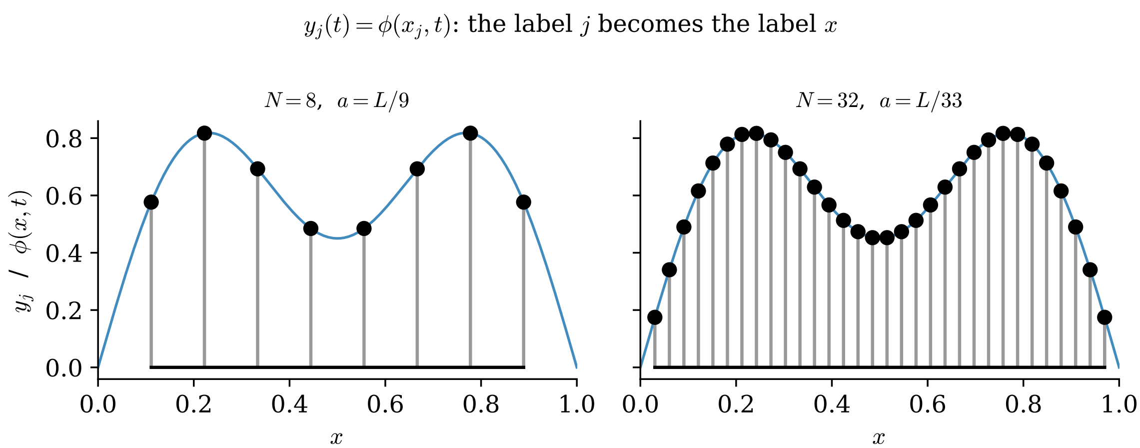


Figure 7: The same smooth profile sampled by two chains ($N = 8$ and $N = 32$ beads). The dynamical variables are the heights y_j; the site index j — equivalently the position $x_j = ja$ — merely *labels* them. Refining the chain makes the label continuous and the collection of heights a function $\phi(x, t)$; nothing about the label's role changes.

6.3 The pivot: x is demoted from variable to label

Now look at what happened to the letter x — the point of the whole construction, easy to use forever without once being said aloud.

In mechanics, position is what the theory is *about*: the $q_i(t)$ are the dynamical variables, and t is the label parametrizing their evolution. In (114) the dynamical variable is ϕ — the field's *value* — and x stands beside t as a second label, saying *which* degree of freedom is meant, exactly as the index j did on the chain (Fig. 7). The chain makes the demotion undramatic: j was never a dynamical variable, and $x_j = ja$ is just j wearing units. A field theory is a mechanical system with a continuous infinity of degrees of freedom, one per point of space, and the whole translation is a dictionary:

	mechanics / chain	field theory
labels degrees of freedom	index i	position x
dynamical variable	$q_i(t)$	$\phi(x,t)$
aggregation	$\sum_i$	$\int \mathrm{d}x$
Lagrangian	$L(q,\dot{q})$	$L = \int \mathrm{d}x\, \mathcal{L}(\phi,\dot{\phi},\phi')$
sensitivity to one d.o.f.	$\dfrac{\partial L}{\partial q_i}$	$\dfrac{\delta L}{\delta \phi(x)}$ (§7)
coupling structure	matrix K_{ij} (near-diagonal)	differential operator $-T\,\partial_x^2$ (local)
normal modes	eigenvectors of K [Osc. §8]	Fourier modes

Two of the rows remain to be justified. The functional derivative $\delta L/\delta\phi(x)$ — "the partial derivative of L with respect to the single degree of freedom sitting at x" — is defined properly in §7. And the mode row is already verified: [Osc. §9.3] showed the chain's normal modes turning into the string's Fourier modes, with the dispersion $\omega = 2\sqrt{T/(ma)}\,\sin(ka/2)$ flattening to $\omega = c_s k$, $c_s^2 = T/\mu$, as $ka \to 0$ — the dictionary's last row, verified before the dictionary existed.

Remark 6.1 (Why relativity insists on fields). Part I gives the pivot a second, independent motivation. Under a boost, t and x mix (Theorem 2.8); but mechanics hardwires an asymmetry between them — t a label, position a dynamical variable — so a mechanics of particles can be made relativistic only against the grain (the covariant formalism of §4 had to demote t itself to a function $x^0(\lambda)$ of an unphysical label to restore the symmetry, and Remark 4.2 was the consequence). Field theory dissolves the asymmetry at the root: t and x sit on the same side of the dictionary, joint labels x^μ on one dynamical variable ϕ, and Lorentz transformations act on labels exactly as §3.6's tensor fields require. The string itself is not relativistic — its $c_s = \sqrt{T/\mu}$ is no invariant speed, and §2.7's conclusion stands — but the *shape* of its description is precisely the shape relativistic dynamics needs, which is why §8 will need to change almost nothing.

The Lagrangian (114) is in hand and the question it raises is sharp: L now takes a whole function $\phi(\cdot,t)$ at each instant — it is a *functional*, in exactly the sense of [Mech. §1] — so what replaces $\partial L/\partial q_i$, and what do its Euler–Lagrange equations look like? That is the next section, and the machinery is already built: the Gateaux framework of [Mech. §2.2] was designed for functionals and does not care that this one's argument is a field.

7 The Field Variational Principle

The Lagrangian (114) takes a function and returns a number; its time integral, the action, is therefore a functional of the field's whole history. Extracting equations of motion from such an object is exactly the problem [Mech. §2] solved — the only novelty is that the "curve" being varied is now a function of *two* labels, and the variation will integrate by parts in both. This section runs the machinery once for a general density, supplies the functional derivative the dictionary promised, gives boundary terms the separate treatment they deserve, and closes two threads: the wave equation (a third time) and the fate of the string's accidental Lorentz invariance.

7.1 Hamilton's principle for fields

Let the density depend on the field's value and first derivatives — general enough for everything in this document:

$$S[\phi] = \int_{t_1}^{t_2} \mathrm{d}t \int_0^{\ell} \mathrm{d}x\ \mathcal{L}(\phi,\ \dot{\phi},\ \phi'), \tag{115}$$

(writing ℓ for the spatial extent; the infinite line is the limit $\ell \to \infty$ with decay conditions, §7.3). Hamilton's principle,[1] verbatim from [Mech. §3.1]: *the physical field history makes S stationary* among histories with prescribed field configurations at t_1 and t_2 and prescribed spatial boundary behavior.

Vary in the Gateaux sense of [Mech. §2.2]: $\phi \to \phi + \varepsilon\,\eta(x,t)$ with $\eta(x,t_1) = \eta(x,t_2) = 0$, and

$$\delta S[\phi;\eta] = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\right|_0 S[\phi + \varepsilon\eta] = \iint \mathrm{d}t\,\mathrm{d}x \left[\frac{\partial \mathcal{L}}{\partial \phi}\,\eta + \frac{\partial \mathcal{L}}{\partial \dot{\phi}}\,\dot{\eta} + \frac{\partial \mathcal{L}}{\partial \phi'}\,\eta'\right], \tag{116}$$

by the chain rule, with the slot-derivative shorthand of [Mech. §2.2]. Two integrations by parts free η — one in t per term two, one in x per term three:

$$\delta S = \iint \mathrm{d}t\,\mathrm{d}x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) - \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \phi'}\right)\right]\eta \;+\; \int_{t_1}^{t_2} \mathrm{d}t \left[\frac{\partial \mathcal{L}}{\partial \phi'}\,\eta\right]_{x=0}^{x=\ell}. \tag{117}$$

The temporal boundary term died on the pinned endpoints, as always; the *spatial* boundary term is new, and care keeps it visible — §7.3 is devoted to the ways it vanishes. Granting that it does: η is arbitrary in the interior, the fundamental lemma applies (its proof [Mech. App. B] extends to two integration variables with only notational changes — choose the bump function as a product of two one-variable bumps [stated; the one-line adaptation is left to the reader]), and:

Theorem 7.1 (Field Euler–Lagrange equation)**.** *A field history making* (115) *stationary, with vanishing spatial boundary term, satisfies*

$$\boxed{\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\phi}}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \phi'}\right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0 \qquad \textit{at every } (x,t).} \tag{118}$$

Set beside its ancestor $\mathrm{d}(\partial L/\partial \dot{q})/\mathrm{d}t - \partial L/\partial q = 0$: the only change is that the label x, having joined t in the dictionary of §6.3, now also supplies a derivative term — one term per label, symmetrically, a symmetry §8 will make manifest. And the promised reunion: the spatial integration by parts that produced the $\partial_x(\partial\mathcal{L}/\partial\phi')$ term is the continuum original of the two-neighbor mechanism of Eq. (111) — there, y_j sat in two springs and the two first differences assembled a second difference; here, $\phi(x)$ sits under one derivative and the integration by parts moves it, signs and all, to the same effect.

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[1] A terminological trap worth disarming once: Hamilton's principle, $\delta \int L\,\mathrm{d}t = 0$, is the variational principle *of Lagrangian mechanics* — the name honors its author (Hamilton, 1834–35), not the function varied. Lagrange himself derived his equations from d'Alembert's principle of virtual work [Mech. §3.2], with no action functional in sight; Hamilton's name thus sits on both sides of the Legendre transform — this principle on the Lagrangian side, Hamilton's equations on the other — by biographical accident. The alternative name "principle of least action" is worse: historically it denotes Maupertuis' principle (fixed energy, free transit time, abbreviated action $\int \vec{p} \cdot \mathrm{d}\vec{q}$), a genuinely different variational statement.

7.2 Defining the functional derivative

Section 6.3 promised a meaning for $\delta L/\delta\phi(x)$ — "the sensitivity of L to the single degree of freedom at x." The Gateaux variation supplies it. For the fixed-time functional $L[\phi] = \int \mathrm{d}x\, \mathcal{L}(\phi, \dot\phi, \phi')$, collect the variation (spatial boundary term disposed of as above) into the form

$$\delta L[\phi;\eta] \;=\; \int \mathrm{d}x\, \frac{\delta L}{\delta\phi(x)}\,\eta(x), \qquad \text{which defines} \qquad \boxed{\frac{\delta L}{\delta\phi(x)} = \frac{\partial\mathcal{L}}{\partial\phi} - \frac{\partial}{\partial x}\left(\frac{\partial\mathcal{L}}{\partial\phi'}\right)} \tag{119}$$

(and $\delta L/\delta\dot\phi(x) = \partial\mathcal{L}/\partial\dot\phi$, no derivative to move). In this notation the field equation (118) is, line for line, mechanics:

$$\frac{\partial}{\partial t}\,\frac{\delta L}{\delta\dot\phi(x)} \;=\; \frac{\delta L}{\delta\phi(x)} \qquad\longleftrightarrow\qquad \frac{\mathrm{d}}{\mathrm{d}t}\,\frac{\partial L}{\partial\dot q_i} = \frac{\partial L}{\partial q_i}. \tag{120}$$

The discrete check. The dictionary row $\partial/\partial q_i \leftrightarrow \delta/\delta\phi(x)$ hides a factor worth exposing. On the chain, Eq. (111) gave

$$\frac{\partial L}{\partial y_j} = \frac{T}{a}\big(y_{j-1} - 2y_j + y_{j+1}\big) = a\cdot T\,\frac{y_{j-1} - 2y_j + y_{j+1}}{a^2} \;\longrightarrow\; a\cdot T\,\phi''(x_j), \tag{121}$$

while (119) applied to $\mathcal{L} = \frac{1}{2}\mu\dot\phi^2 - \frac{1}{2}T\phi'^2$ gives $\delta L/\delta\phi(x) = \partial_x(T\phi') = T\phi''$. So

$$\frac{\partial L}{\partial y_j} \;\longrightarrow\; a\,\frac{\delta L}{\delta\phi(x_j)}\;: \tag{122}$$

the discrete partial derivative is the functional derivative times the *cell size* — sensitivity to one bead is sensitivity-density at its site times the length of chain it represents. The factor of a is the same one that converted $\sum_j$ into $\int \mathrm{d}x$, and it must appear here for $\sum_j(\partial L/\partial y_j)\,\eta_j \to \int \mathrm{d}x\,(\delta L/\delta\phi)\,\eta$ to hold.

7.3 Spatial boundary terms

The term (117) left standing,

$$B = \int_{t_1}^{t_2} \mathrm{d}t\, \left[\frac{\partial\mathcal{L}}{\partial\phi'}\,\eta\right]_{x=0}^{x=\ell}, \tag{123}$$

must vanish for Theorem 7.1 to hold, and each way of killing it is a physical situation. For the string, $\partial\mathcal{L}/\partial\phi' = -T\phi'$ is (minus) the transverse force the string exerts across a cross-section, which gives every option below its physical reading.

- **Fixed ends (Dirichlet).** ϕ prescribed at $x = 0, \ell$; admissible variations have $\eta = 0$ there, and B dies on η. The rope between walls of [Osc. §9]. The boundary *condition* is imposed; the variational principle accepts it.

- **Free ends (Neumann).** Nothing prescribed at the boundary, so η is arbitrary there — and then stationarity itself forces

$$\left.\frac{\partial\mathcal{L}}{\partial\phi'}\right|_{x=0,\,\ell} = 0 \qquad \text{(string: } \phi' = 0 \text{ at the free end)}, \tag{124}$$

  a *natural boundary condition*: the variational principle, given no instruction at the edge, issues one — zero transverse force on the massless end ring, which is Newton's law for an end of zero mass. A principle that generates boundary conditions it was not handed is doing more than repackaging the bulk equation.

- **Periodic.** $\phi(0,t) = \phi(\ell,t)$ — a loop: the two evaluations of B cancel each other. The most convenient choice for mode expansions, used in §14.
- **The infinite line.** $\ell \to \infty$ with $\phi \to 0$ (fast enough) at spatial infinity: $B \to 0$ on decay. The default of field theory proper, and this document's standing assumption from §8 on whenever no boundary is mentioned.

The subsection is short but essential: Noether's theorem for fields (§9) converts symmetries into statements about integrals of densities, and every one of those statements holds *up to the same kinds of boundary terms*, disposed of by the same four mechanisms.

7.4 A third route to the wave equation

For the third route to be genuinely a third route, it must not borrow the density from the chain limit of §6.2 — that would smuggle the beads back in. So build $\mathcal{L}$ directly from the continuous rope: mass per length μ, tension T, transverse displacement $\phi(x,t)$, small slopes.

Kinetic density. The rope element over $[x, x + \mathrm{d}x]$ has mass $\mu\,\mathrm{d}x$ and moves transversely at velocity $\dot{\phi}(x,t)$, so

$$\mathrm{d}T_{\mathrm{kin}} = \tfrac{1}{2}\,(\mu\,\mathrm{d}x)\,\dot{\phi}^2 \qquad \Longrightarrow \qquad \text{kinetic density} = \tfrac{1}{2}\,\mu\,\dot{\phi}^2. \tag{125}$$

Potential density. The energy cost of a displaced shape is the work done against the tension by the rope's *excess length* — the continuum original of the spring-by-spring tally in Eq. (110). The displaced element has arc length

$$\mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}\phi^2} = \sqrt{1 + \phi'^2}\,\mathrm{d}x = \Big(1 + \tfrac{1}{2}\,\phi'^2 + O(\phi'^4)\Big)\mathrm{d}x, \tag{126}$$

an excess of $\frac{1}{2}\phi'^2\,\mathrm{d}x$ per element; pulled out against a tension that is itself unchanged at this order (small slopes, [Osc. §9.1]'s standing approximation),

$$\mathrm{d}U = T \cdot \tfrac{1}{2}\,\phi'^2\,\mathrm{d}x \qquad \Longrightarrow \qquad \text{potential density} = \tfrac{1}{2}\,T\,\phi'^2. \tag{127}$$

Assembly. Kinetic minus potential, densely:

$$\mathcal{L} = \tfrac{1}{2}\,\mu\,\dot{\phi}^2 - \tfrac{1}{2}\,T\,\phi'^2, \tag{128}$$

in agreement with the chain limit (114) — as it must be, and the agreement is a real check: two constructions, one through a lattice and its limit, one never leaving the continuum, land on the same density. Feed it to Theorem 7.1:

$$\frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \mu\dot{\phi}, \qquad \frac{\partial \mathcal{L}}{\partial \phi'} = -T\phi', \qquad \frac{\partial \mathcal{L}}{\partial \phi} = 0, \tag{129}$$

so (118) reads $\mu\ddot{\phi} - T\phi'' = 0$:

$$\boxed{\ddot{\phi} = c_s^2\,\phi'', \qquad c_s^2 = \frac{T}{\mu}.} \tag{130}$$

The same physics for the third time, by the third route, each shorter than the last — and now each genuinely independent: Newton on beads, then the limit of the equations [Osc. §9.2]; Lagrange on beads (§6.1), then the same limit; and now two energy densities read off the continuum and one variation, all x at once, no bead ever mentioned. The ranking is not aesthetic. The first two routes need beads — a mechanical substructure to apply Newton or Lagrange *to* — and the fields ahead (Klein–Gordon, Maxwell) have none. The third route needs only a density, and choosing densities is exactly what the covariance principle of §3.6 constrains. (The general solution is $\phi = f(x - c_s t) + g(x + c_s t)$ — d'Alembert's form, derived by passing to characteristic variables in [Osc. §9.5]; its physics is not repeated here.)

Remark 7.2 (The status of the string's Lorentz invariance). Rewrite the string density as

$$\mathcal{L} = \tfrac{\mu}{2}\big(\dot{\phi}^2 - c_s^2\,\phi'^2\big) : \tag{131}$$

up to the factor $\mu/2$, this is the Minkowski norm of the gradient $(\dot{\phi}, \phi')$ *with c_s in the role of c* — so both the density and its wave equation are form-invariant under "Lorentz" transformations of the labels (t, x) built from c_s. Here, at last, is the precise status of the accident flagged in §2.7: the invariance is a true symmetry *of the long-wavelength equations* and a falsehood about the world they describe. It is broken twice over. Physically, the medium has a rest frame — the frame where (114) was derived — detectable by measuring c_s in both directions, so Postulate 2.1 fails for c_s-boosts. And microscopically, the symmetry does not survive below the lattice scale: the chain's true dispersion $\omega = 2\sqrt{T/(ma)}\,\sin(ka/2)$ [Osc. §9.3] is not the c_s-invariant $\omega = c_s|k|$, and the deviation at $ka \sim 1$ is where the lattice becomes visible. An emergent symmetry, valid to $O(k^2a^2)$, absent from the microscopic theory it descends from. The contrast defines the next section's task: write down a field whose Lorentz invariance is *exact* — symmetry of the Lagrangian under the true Lorentz group of Part I, no medium, no lattice, no preferred frame — and see what the covariance principle leaves standing.

8 Relativistic Scalar Fields: Klein–Gordon

Everything is now assembled: a variational principle for fields (§7), a covariance principle that constrains candidate Lagrangians (§3.6), and a sharp question (Remark 7.2): what does a field theory look like when Lorentz invariance is exact? This section repackages the field Euler–Lagrange equation covariantly, then plays the building-block game — and the game has essentially one answer, the Klein–Gordon field, whose dispersion relation turns out to be an old figure wearing new axis labels.

8.1 The covariant Euler–Lagrange equation

Move to 3+1 dimensions and covariant notation in one step: let the density depend on the field and its four-gradient, $\mathcal{L}(\phi, \partial_\mu\phi)$, and write

$$S[\phi] = \int \mathrm{d}^4x\; \mathcal{L}\big(\phi,\, \partial_\mu\phi\big), \qquad \mathrm{d}^4x = \mathrm{d}t\,\mathrm{d}^3x. \tag{132}$$

The measure takes care of itself: under $x' = \Lambda x$ the Jacobian is $|\det\Lambda| = 1$ (§3.8), so d^4x is Lorentz invariant, and *S is a scalar if and only if $\mathcal{L}$ is a scalar field.* That single sentence is the design constraint for everything ahead.

The variation is §7.1 with more labels and better notation. Vary $\phi \to \phi + \varepsilon\eta$, integrate by parts once per spacetime direction — temporal boundary terms die on pinned endpoints, spatial ones by decay at infinity (§7.3, fourth mechanism, henceforth standing) — and apply the fundamental lemma:

$$\boxed{\partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) - \frac{\partial\mathcal{L}}{\partial\phi} \;=\; 0.} \tag{133}$$

Compare with (118): the separate $\partial_t(\partial\mathcal{L}/\partial\dot\phi)$ and $\partial_x(\partial\mathcal{L}/\partial\phi')$ terms — one per label, as §7.1 noted — have fused into a single contracted term, one summed index μ running over all the labels at once. The dictionary of §6.3 put t and x on the same side; Eq. (133) shows what that decision achieves. Note also the index placement: $\partial\mathcal{L}/\partial(\partial_\mu\phi)$ differentiates with respect to a lower-index object and so carries an *upper* μ — it is a four-vector field, contracted by ∂_μ into a scalar equation, every piece certified by §3.6.

8.2 Building scalar Lagrangians

Now the game. $\mathcal{L}$ must be a Lorentz scalar assembled from a scalar field ϕ and its gradient $\partial_\mu\phi$ (§3.6: $\partial_\mu\phi$ is a covariant vector field, and contraction is the only scalar-maker). The full catalogue of blocks:

$$\phi,\ \phi^2,\ \phi^3,\dots;\qquad \partial_\mu\phi\,\partial^\mu\phi,\ \left(\partial_\mu\phi\,\partial^\mu\phi\right)^2,\ \dots;\qquad \phi^n\,\partial_\mu\phi\,\partial^\mu\phi;\qquad \text{higher derivatives } (\Box\phi,\ \dots). \tag{134}$$

Physics thins the catalogue:

- **First derivatives only.** The mechanics of the whole series runs on $L(q,\dot q)$; densities with second derivatives yield higher-order equations of motion whose energies are unbounded below (the Ostrogradsky instability [stated; standard, not proved here]). Discard $\Box\phi$ and friends.
- **At most quadratic.** Quadratic $\mathcal{L}$ means *linear* equations of motion, hence superposition — the defining property of a *free* field, and the natural starting point: every interacting theory in physics is built as a free quadratic core plus higher-power corrections. Discard ϕ^3, $\phi^n(\partial\phi)^2$, … — not as impossible, but as *interactions*, deferred to the outlook.
- **Tidying.** A constant a_0 shifts S without moving its stationary points (though gravity, which weighs $\mathcal{L}$ itself, cares — one sentence of foreshadowing, [stated]). A linear term $a_1\phi$ is removed by the constant shift $\phi\to\phi-a_1/(2a_2)$. The kinetic normalization is fixed to $\frac{1}{2}$ by rescaling ϕ — the field's units are ours to choose.

What survives is a one-parameter family — the most general free scalar field theory:

$$\boxed{\mathcal{L}=\tfrac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi\ -\ \tfrac{1}{2}\,m^2\phi^2,} \tag{135}$$

the parameter written m^2 in anticipation (§8.4 justifies the name). Expanded in 3+1 language,

$$\mathcal{L}=\tfrac{1}{2}\dot\phi^2-\tfrac{1}{2}|\nabla\phi|^2-\tfrac{1}{2}m^2\phi^2, \tag{136}$$

which invites the comparison this Part was built for. The string had $\mathcal{L}=\frac{\mu}{2}(\dot\phi^2-c_s^2\phi'^2)$: same kinetic structure, with $c_s\to c=1$ now *exact* rather than emergent, the overall μ absorbed into the field normalization — and one genuinely new term. Signs are physics, not convention: the relative sign between $\dot\phi^2$ and $|\nabla\phi|^2$ is the metric's signature and is what makes (133) hyperbolic (waves, causal propagation); the sign of the mass term makes $V(\phi)=\frac{1}{2}m^2\phi^2$ a well rather than a hill — the opposite choice, $-m^2\to+\mu^2$, puts every Fourier mode on top of an inverted potential and long-wavelength perturbations grow exponentially (the "tachyonic" instability; when encountered on purpose, it is the engine of spontaneous symmetry breaking — outlook material, [stated]).

Remark 8.1 (What the mass term would mean for a string). The string never had a ϕ^2 term because nothing pushed a displaced bead back toward the axis except its neighbors. Add such a push — mount every bead on its own spring to the equilibrium line, a mattress rather than a rope — and the bead equation gains a term $-\kappa y_j$, whose continuum limit is exactly $-\frac{1}{2}m^2\phi^2$ in the density with $m^2 = \kappa/(\mu a)$ held fixed. The Klein–Gordon field is, mechanically, *a mattress*: a medium whose every element is individually tethered. Each element could oscillate at frequency m even with all its neighbors held still — the physical origin of the mass gap about to appear in the dispersion relation.

8.3 The Klein–Gordon equation

Feed (135) to (133):

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} = \partial^\mu \phi, \qquad \frac{\partial \mathcal{L}}{\partial \phi} = -m^2\phi, \tag{137}$$

(the first equality is worth one private index-expansion: write $\partial_\mu\phi\,\partial^\mu\phi = \eta^{\rho\sigma}\partial_\rho\phi\,\partial_\sigma\phi$ and differentiate — the two slots contribute equally, canceling the $\frac{1}{2}$). Hence

$$\boxed{\left(\Box + m^2\right)\phi = 0} \qquad \text{— the Klein–Gordon equation.} \tag{138}$$

The d'Alembertian now plays the role promised in §3.5: it is the only two-derivative scalar operator there is, so it *had* to be the one. At $m = 0$ the equation is $\Box\phi = 0$ — the wave equation with the invariant speed, form-invariant in every frame by inspection, closing the arc opened at Eq. (6): this is what the light-like wave equation looks like when it is a law of nature rather than a description of a medium.

8.4 Plane waves and the dispersion relation

Hunt for plane-wave solutions with the covariant ansatz

$$\phi(x) = \mathrm{Re}\ \hat{A}\,\mathrm{e}^{-\mathrm{i}\,k\cdot x}, \qquad k^\mu = (\omega, \vec{k}), \qquad k\cdot x = \omega t - \vec{k}\cdot\vec{x}, \tag{139}$$

(complexification per [Osc. Conv. 1.1]; the exponent is a scalar provided k^μ is a four-vector, so the ansatz survives boosts as a plane wave with boosted k^μ). Each derivative pulls down a factor: $\partial_\mu\,\mathrm{e}^{-\mathrm{i}k\cdot x} = -\mathrm{i}k_\mu\,\mathrm{e}^{-\mathrm{i}k\cdot x}$, so $\Box \to (-\mathrm{i}k_\mu)(-\mathrm{i}k^\mu) = -k\cdot k = -(\omega^2 - |\vec{k}|^2)$, and (138) demands

$$\boxed{\omega^2 = |\vec{k}|^2 + m^2.} \tag{140}$$

This is Fig. 6 with its axes relabeled $(E, p) \to (\omega, \vec{k})$, exactly as Remark 4.6 promised: *the dispersion relation of the field is the energy–momentum relation of a particle of mass m* — which is why the coefficient in (135) was named m^2. Classically this is a suggestive pun: m is, as far as (140) knows, an inverse length, $1/m$ the scale (the *Compton wavelength*, once $\hbar$ is restored as $\hbar/mc$) below which the dispersion straightens onto the light line. The pun becomes physics upon quantization, where $E = \hbar\omega$ and $\vec{p} = \hbar\vec{k}$ turn (140) into (83) literally — the outlook (§14) says how.

Two velocities, one check, one warning:

$$v_{\mathrm{ph}} = \frac{\omega}{|\vec{k}|} = \sqrt{1 + \frac{m^2}{|\vec{k}|^2}}\ >\ 1, \qquad v_{\mathrm{g}} = \frac{\partial\omega}{\partial|\vec{k}|} = \frac{|\vec{k}|}{\omega}\ <\ 1, \qquad v_{\mathrm{ph}}\,v_{\mathrm{g}} = 1. \tag{141}$$

The superluminal phase velocity alarms no one who asks what moves: a single infinite plane wave's crests carry no information (nothing distinguishes one crest from the next), and what transports energy and signals is the group velocity of a wave *packet* [Osc. §4.1's beat logic, one wavelength up] — which is not only subluminal but exactly right: $v_{\mathrm{g}} = |\vec{k}|/\omega = |\vec{p}|/E$ is the velocity of the mass-m

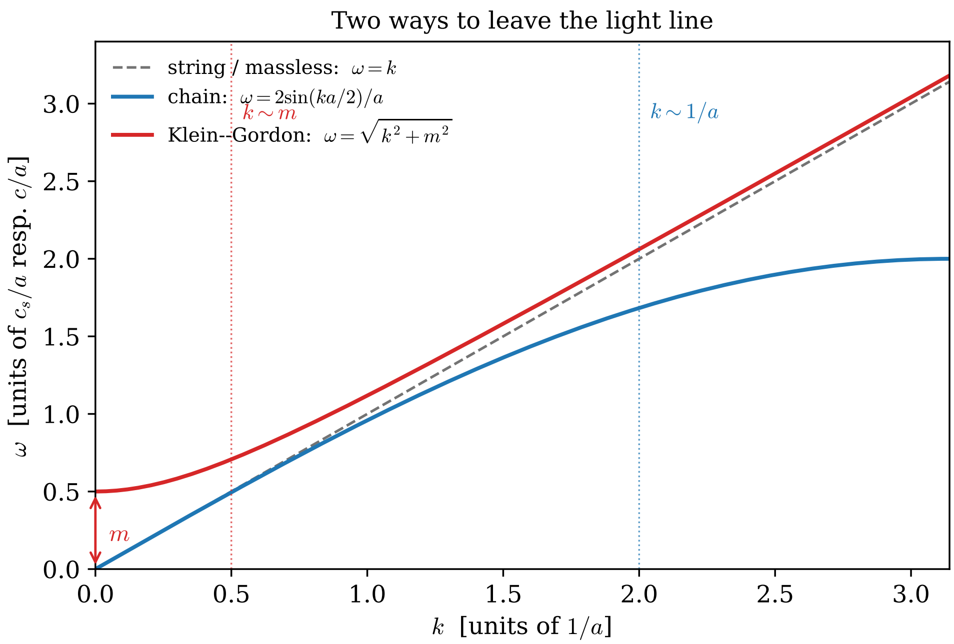

Figure 8: Three dispersion relations, drawn from their exact formulas ($c_s = c = 1$, lattice spacing $a = 1$, $m = 0.5$). The light line $\omega = k$ (gray, dashed) is left in two opposite directions: the chain [Osc. §8.3] bends *below* it at large k, where wavelengths approach the lattice spacing and the medium's discreteness shows; the Klein–Gordon field lifts *above* it at small k, by the mass gap $\omega(0) = m$ — every mattress element oscillates at m even at infinite wavelength (Remark 8.1). Linear, light-like propagation survives only in the window $m \ll k \ll 1/a$: an ultraviolet scale and an infrared scale, bracketing the massless continuum from opposite sides.

particle with that momentum, by Eqs. (81) and (83). The correspondence particle $\leftrightarrow$ packet holds down to the kinematics. [The full story of what can and cannot outrun light in a dispersive medium — front velocity — is genuinely subtle and is flagged, not settled, here.]
Figure 8 assembles the Part's three dispersion relations and reads as a summary of its physics: the lattice scale $1/a$ bounds the wave regime from above (ultraviolet — the chain remembers its beads), the mass m bounds it from below (infrared — the mattress remembers its tethers), and massless light-line propagation is what both cutoffs leave between them. The massive field is *dispersive*: packets built from (140) spread, because their Fourier components travel at different group velocities — in contrast to the massless case, whose general solution $\phi = f(x-t) + g(x+t)$ [Osc. §9.5] carries pulses rigidly forever.

What the field equation cannot yet say is what the field *carries*: energy, momentum, and their currents. That is the subject of Noether's theorem for fields — the next section.

9 Noether's Theorem for Fields

Noether's theorem for mechanics [Mech. §3.3] traded each continuous symmetry for a conserved *number*. For fields the theorem returns with a strictly stronger conclusion: each continuous symmetry yields a conserved *current* — a local accounting of where the conserved stuff sits and how it flows. This section proves the field version, explains why relativity does not merely permit but *demands* the local form, and runs the theorem twice: on an internal symmetry (new in kind — nothing like it exists in Part I) and on spacetime translations, whose four currents assemble into the energy-momentum tensor.

9.1 The theorem

Definition 9.1 (Symmetry of a field theory)**.** A continuous transformation $\phi \to \phi + \varepsilon\,\Delta\phi$ (with $\Delta\phi$ built from ϕ and its derivatives) is a *symmetry* if it changes the Lagrangian density by at most a total divergence,

$$\delta\mathcal{L} = \varepsilon\,\partial_\mu K^\mu \tag{142}$$

for some four-vector K^μ built from the fields — for then δS is a pure boundary term and the equations of motion are unchanged, exactly the logic of §5.2. ($K^\mu = 0$ is the special case of exact invariance.)

Theorem 9.2 (Noether, field version)**.** *If* (142) *holds, then along every solution of the equations of motion* (133) *the current*

$$\boxed{\,j^\mu \;=\; \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\Delta\phi \;-\; K^\mu\,} \tag{143}$$

is conserved:

$$\boxed{\partial_\mu j^\mu = 0.} \tag{144}$$

(For several fields, sum the first term over them.)

Proof. Compute the change of $\mathcal{L}$ under the transformation by the chain rule, in the slot-derivative sense of §7.1:

$$\delta\mathcal{L} = \varepsilon\left[\frac{\partial\mathcal{L}}{\partial\phi}\,\Delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu(\Delta\phi)\right]. \tag{145}$$

On a solution, the equation of motion (133) converts the first coefficient: $\partial\mathcal{L}/\partial\phi = \partial_\mu\big(\partial\mathcal{L}/\partial(\partial_\mu\phi)\big)$. The two terms then assemble into a total divergence by the product rule:

$$\delta\mathcal{L} = \varepsilon\,\partial_\mu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\Delta\phi\right]. \tag{146}$$

Equating with the definition (142) of the symmetry: $\partial_\mu\big[(\partial\mathcal{L}/\partial(\partial_\mu\phi))\Delta\phi - K^\mu\big] = 0$, which is the claim. $\square$

9.2 What "conserved" now means: localization

Unpack $\partial_\mu j^\mu = 0$ in 3+1 language, writing $j^\mu = (\rho, \vec{\jmath})$:

$$\boxed{\frac{\partial\rho}{\partial t} + \nabla\cdot\vec{\jmath} = 0} \qquad \text{— a \textit{continuity equation}:} \tag{147}$$

the density ρ at a point can change only by flux $\vec{\jmath}$ through that point's neighborhood. Integrating over all space and using the divergence theorem,

$$\frac{\mathrm{d}Q}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\int \rho\,\mathrm{d}^3x = -\oint_\partial \vec{\jmath}\cdot\mathrm{d}\vec{A} \;\longrightarrow\; 0 \tag{148}$$

for fields decaying at infinity (§7.3, fourth mechanism): the global charge $Q = \int j^0\,\mathrm{d}^3x$ is constant, and mechanics' conclusion is recovered as a corollary. But the local statement is the physics: the conserved stuff cannot vanish here and reappear there — it must *flow through the intervening space*, and $\vec{\jmath}$ records the flow.

Remark 9.3 (Why relativity demands the local form). In Newtonian physics, local conservation is a refinement of global conservation; in relativity it is the only coherent version. Suppose a charge could teleport: vanish at event P and appear at a distant event P', simultaneous with P in some frame. P and P' are spacelike separated, so their time order is frame-dependent (§3.2) — boosted observers see the charge appear *before* it vanishes, or find slices of simultaneity (§2.6) on which total charge is doubled or absent. "Total Q at time t" presupposes a simultaneity slicing, and Part I spent a section showing that slicings are negotiable; the only conservation statement all observers can share is the slicing-free, pointwise (147). (Conversely, for a conserved current the global Q comes out the *same* on every slicing — Q is a Lorentz scalar. [stated; the proof integrates $\partial_\mu j^\mu = 0$ over the region between two slices.])

9.3 First run: the complex scalar and its charge

The theorem's first application is a symmetry of a kind Part I never met. Package *two* real Klein–Gordon fields of equal mass into one complex field, $\phi = (\phi_1 + \mathrm{i}\phi_2)/\sqrt{2}$, with density

$$\mathcal{L} = \partial_\mu\phi^*\,\partial^\mu\phi - m^2\,\phi^*\phi \tag{149}$$

(the $\frac{1}{2}$'s of (135) absorbed by the $\sqrt{2}$; and one standing trick, justified once: varying the independent real fields ϕ_1, ϕ_2 is equivalent to varying ϕ and ϕ^* *as if independent* — the two variation bases are related by an invertible linear map, so stationarity in one is stationarity in the other). Varying ϕ^* yields $(\Box + m^2)\phi = 0$; varying ϕ, the conjugate.

The point of the packaging: (149) is exactly invariant under the *global phase rotation*

$$\phi \to \mathrm{e}^{-\mathrm{i}\alpha}\phi, \qquad \phi^* \to \mathrm{e}^{+\mathrm{i}\alpha}\phi^*, \qquad \alpha \text{ constant}, \tag{150}$$

since every term pairs a ϕ with a ϕ^*. This is an *internal* symmetry — it turns the field's *value*, touching the spacetime labels not at all — the first symmetry of the series with no spacetime meaning whatsoever, and the origin of everything the word "charge" means in particle physics. Infinitesimally, $\Delta\phi = -\mathrm{i}\phi$, $\Delta\phi^* = +\mathrm{i}\phi^*$, with $K^\mu = 0$ (exact invariance), and Theorem 9.2 yields:

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\Delta\phi + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\,\Delta\phi^* = (\partial^\mu\phi^*)(-\mathrm{i}\phi) + (\partial^\mu\phi)(+\mathrm{i}\phi^*), \tag{151}$$

$$\boxed{j^\mu = \mathrm{i}\left(\phi^*\,\partial^\mu\phi - \phi\,\partial^\mu\phi^*\right), \qquad \partial_\mu j^\mu = 0.} \tag{152}$$

The conservation deserves one direct check — Noether proved it, but the mechanism is instructive:

$$\partial_\mu j^\mu = \mathrm{i}\big(\phi^*\,\Box\phi - \phi\,\Box\phi^*\big) = \mathrm{i}\big(\phi^*(-m^2\phi) - \phi(-m^2\phi^*)\big) = 0, \tag{153}$$

the cross terms $\partial_\mu\phi^*\partial^\mu\phi$ canceling identically and the mass terms canceling *on shell*.

Remark 9.4 (Charge, not probability). The charge density $j^0 = \mathrm{i}(\phi^*\dot\phi - \phi\dot\phi^*)$ is **not** positive definite — complex conjugating a solution flips its sign. Historically this killed the reading of ϕ as a relativistic wavefunction with j^0 a probability density (the Klein–Gordon equation's first, failed interpretation); rightly read, Q is a *charge*, a quantity that legitimately comes in both signs — field and antifield, particle and antiparticle after quantization. And the current will be needed again: (152) is precisely the kind of object that can sit on the right-hand side of a field equation for A_μ — a conserved J^μ is what Part III's Maxwell Lagrangian will demand as its source, and *this* is where such currents come from.

9.4 Second run: translations and the energy-momentum tensor

Now the central application: the four spacetime translations, $x^\mu \to x^\mu + \varepsilon\, a^\mu$. A scalar field is dragged along, $\phi(x) \to \phi(x - \varepsilon a)$, so

$$\Delta\phi = -\,a^\nu\,\partial_\nu\phi. \tag{154}$$

The density, having no explicit x-dependence, is dragged the same way — and *that is already a total divergence*:

$$\delta\mathcal{L} = -\,\varepsilon\, a^\nu\,\partial_\nu\mathcal{L} = \varepsilon\,\partial_\mu\big(-a^\mu\,\mathcal{L}\big), \qquad K^\mu = -\,a^\mu\mathcal{L}\,: \tag{155}$$

a quasi-symmetry in the precise sense of (142), with *four* independent parameters a^ν. Theorem 9.2 gives the current

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\left(-a^\nu\partial_\nu\phi\right) \;+\; a^\mu\,\mathcal{L} \;=\; -\,a^\nu\left[\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\nu\phi - \delta^\mu{}_\nu\,\mathcal{L}\right], \tag{156}$$

and since the four a^ν are independent, the bracket is conserved for each ν separately — four currents, organized as one two-index object:

$$\boxed{T^\mu{}_\nu \;=\; \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\nu\phi \;-\; \delta^\mu{}_\nu\,\mathcal{L}, \qquad \partial_\mu T^\mu{}_\nu = 0 \quad \text{(four equations, one per } \nu\text{)},} \tag{157}$$

the *energy-momentum tensor* (raising the second index, $T^{\mu\nu} = (\partial\mathcal{L}/\partial(\partial_\mu\phi))\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}$; it is a tensor by the algebra of §3.6, being built from tensors by products and contractions). Its sixteen components record the field's entire mechanics:

component	name	role in (147)
T^{00}	energy density	ρ of the $\nu = 0$ current
T^{i0}	energy flux	$\vec{\jmath}$ of the $\nu = 0$ current
T^{0i}	momentum density	ρ of the $\nu = i$ current
T^{ji}	momentum flux (stress)	$\vec{\jmath}$ of the $\nu = i$ current

and the four global charges are the field's four-momentum,

$$P^\nu = \int T^{0\nu}\,\mathrm{d}^3x = \big(E,\ \vec{P}\big), \tag{158}$$

— §4.3 rerun at field level, with the same symmetry yielding the same conserved quantities, now *localized* into densities and fluxes.

The Klein–Gordon case. For (135), $\partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$, so $T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}$, and in particular

$$\boxed{T^{00} = \tfrac{1}{2}\,\dot{\phi}^2 + \tfrac{1}{2}\,|\nabla\phi|^2 + \tfrac{1}{2}\,m^2\phi^2 \ \ \geq\ 0,} \qquad T^{0i} = \dot{\phi}\,\partial^i\phi. \tag{159}$$

The energy density is a sum of three squares — kinetic, gradient (the tension energy of §7.4, one dimension up), and tether (the mattress springs of Remark 8.1) — and its positivity is not accidental: it is the delayed consequence of the sign choices of §8.2. Flip the kinetic sign and T^{00} is unbounded below; flip the mass sign and the tether term goes negative — §8.2's "signs are physics," made quantitative as a bounded energy. As a check against the series' oldest intuition: the same formula (157) applied to the string density gives energy density $\frac{1}{2}\mu\dot{\phi}^2 + \frac{1}{2}T\phi'^2$ and energy flux $T^x{}_0 = -T\phi'\dot{\phi}$ — the transverse force (§7.3) times the transverse velocity, i.e. mechanical power transmitted across the cross-section, exactly as a free-body analysis would say.

Remark 9.5 (The other six, and a symmetry requirement). The remaining Poincaré parameters (three rotations, three boosts; Remark 4.4) yield, by the same theorem, the six currents packaged as $M^{\rho\mu\nu} = x^\mu\,T^{\rho\nu} - x^\nu\,T^{\rho\mu}$. Its conservation is a one-line computation with a condition inside:

$$\partial_\rho M^{\rho\mu\nu} = T^{\mu\nu} - T^{\nu\mu} + x^\mu\,\partial_\rho T^{\rho\nu} - x^\nu\,\partial_\rho T^{\rho\mu} = T^{\mu\nu} - T^{\nu\mu}\ : \tag{160}$$

angular momentum is conserved *if and only if the energy-momentum tensor is symmetric.* For the scalar field it is, automatically — $\partial^\mu\phi\,\partial^\nu\phi$ and $\eta^{\mu\nu}\mathcal{L}$ are symmetric by inspection. For fields that carry indices of their own the canonical (157) comes out *asymmetric*, the mismatch being the field's intrinsic (spin) angular momentum, and a repair — the Belinfante improvement, adding a total-divergence term that symmetrizes T without touching the charges — exists in general. [stated; Part III meets the concrete case for the electromagnetic field in §13, and the general construction is QFT-course material.]

The Lagrangian story of the free scalar field is now complete: equation of motion, symmetries, currents, energy. What remains of the mechanics dictionary is its second column — momenta, Hamiltonian, brackets — and porting it is the last stop before electromagnetism.

10 Hamiltonian Field Theory

One column of the mechanics dictionary remains untranslated: momenta, Hamiltonian, brackets — the phase-space half of [Mech. Part II]. This section ports it, with one genuinely new ingredient (the Dirac delta, introduced as what it is rather than what it is written as) and one promise kept: Remark 4.5 stated that the particle's trade — manifest covariance given up at the Legendre transform — would recur, unchanged, for fields.

10.1 The field momentum and the Hamiltonian

The dictionary of §6.3 translates $p_i = \partial L/\partial \dot q_i$ entry by entry: one momentum per degree of freedom, i.e. one per point,

$$\boxed{\pi(\vec{x}) \;\equiv\; \frac{\delta L}{\delta\dot\phi(\vec{x})} \;=\; \frac{\partial\mathcal{L}}{\partial\dot\phi}} \tag{161}$$

(the two expressions agree because $\mathcal{L}$ contains no spatial derivatives of $\dot\phi$ — nothing for (119)'s integration by parts to move). For Klein–Gordon, $\pi = \dot\phi$; for the complex scalar (149), note the crossing, $\pi = \dot\phi^*$ and $\pi^* = \dot\phi$. The Legendre transform [Mech. §6.1] then runs pointwise — one transform per degree of freedom, aggregated by the dictionary's $\int \mathrm{d}^3x$:

$$\boxed{H = \int \mathrm{d}^3x\, \mathcal{H}, \qquad \mathcal{H} = \pi\,\dot\phi - \mathcal{L} \quad (\dot\phi \text{ eliminated in favor of } \pi).} \tag{162}$$

For Klein–Gordon, $\dot\phi = \pi$ inverts trivially and

$$\mathcal{H} = \tfrac{1}{2}\,\pi^2 + \tfrac{1}{2}\,|\nabla\phi|^2 + \tfrac{1}{2}\,m^2\phi^2 \;=\; T^{00}\big|_{\dot\phi\to\pi} : \tag{163}$$

the Hamiltonian density *is* the energy density (159) — the field-theory instance of [Mech. §6.3]'s "$H = T + V$ for a kinetic term purely quadratic in the velocities and a velocity-independent potential," and positive definite for the reasons established in §9.4.

The discrete check, third installment. On the chain, $p_j = m\dot y_j = (\mu a)\,\dot\phi(x_j) = a\,\pi(x_j)$: the bead's momentum is the momentum *density* times the cell size — the same factor of a as in §7.2, now on the conjugate variable. The two cell factors then combine correctly in the Hamiltonian: $\sum_j p_j^2/(2m) = \sum_j a\,\pi(x_j)^2/(2\mu) \to \int \mathrm{d}x\,\pi^2/(2\mu)$, the kinetic term of (163) (with the string's μ, absorbed to 1 in the Klein–Gordon normalization).

10.2 The Dirac delta as a distribution

Phase-space mechanics needs one more derivative the Lagrangian story never asked for: the sensitivity of a *local* quantity to a local change — $\delta\phi(\vec{y})/\delta\phi(\vec{x})$, the field-theory descendant of $\partial q_i/\partial q_j = \delta_{ij}$. Apply the definition (119) to the evaluation functional $F[\phi] = \phi(\vec{y})$: its variation under $\phi \to \phi + \varepsilon\eta$ is exactly $\varepsilon\,\eta(\vec{y})$, so the sought object is whatever satisfies

$$\int \mathrm{d}^3x\, \frac{\delta\phi(\vec{y})}{\delta\phi(\vec{x})}\,\eta(\vec{x}) = \eta(\vec{y}) \qquad \text{for every test function } \eta. \tag{164}$$

No function does this — a function vanishing away from one point integrates to zero against everything — but (164) itself is a perfectly good definition of a different kind of object: a *distribution*, a linear machine that takes a test function and returns a number, written *as if* it were an integral kernel:

$$\boxed{\frac{\delta\phi(\vec{y})}{\delta\phi(\vec{x})} = \delta^3(\vec{x}-\vec{y}), \qquad \int \mathrm{d}^3x\; \delta^3(\vec{x}-\vec{y})\, f(\vec{x}) = f(\vec{y}).} \tag{165}$$

The integral notation is a fiction with a license: every manipulation performed on δ in this document (integration against smooth functions, integration by parts, change of variables) is the shadow of a legitimate operation on the underlying functional, and Appendix B sets out the small amount of theory that issues the license. The dictionary, meanwhile, closes its last entry with the same cell factor as always:

$$\frac{\partial y_i}{\partial y_j} = \delta_{ij} \qquad \longleftrightarrow \qquad \frac{\delta\phi(x_i)}{\delta\phi(x_j)} = \frac{\delta_{ij}}{a} \;\to\; \delta(x_i - x_j) : \tag{166}$$

the Kronecker delta *per cell volume* is the discrete Dirac delta — check: $\sum_j a\,(\delta_{ij}/a)\,f_j = f_i$ mirrors (165) term for term — and it correctly blows up as $a \to 0$: sensitivity *density* to a single point must be infinite if its integral over a shrinking cell is to stay one.

10.3 Poisson brackets and Hamilton's equations

Port the bracket of [Mech. §8.1] through the dictionary — sum to integral, partials to functional derivatives:

$$\{F, G\} \;=\; \int \mathrm{d}^3x \left[\frac{\delta F}{\delta\phi(\vec{x})}\,\frac{\delta G}{\delta\pi(\vec{x})} - \frac{\delta F}{\delta\pi(\vec{x})}\,\frac{\delta G}{\delta\phi(\vec{x})}\right], \tag{167}$$

inheriting antisymmetry, bilinearity, Leibniz, and Jacobi from the mechanical original (the proofs [Mech. §8.1, App. F] transcribe line by line). The fundamental brackets follow from (165): with $F = \phi(\vec{x})$, $G = \pi(\vec{y})$,

$$\{\phi(\vec{x}),\,\pi(\vec{y})\} = \int \mathrm{d}^3z\;\delta^3(\vec{z}-\vec{x})\,\delta^3(\vec{z}-\vec{y}) = \delta^3(\vec{x}-\vec{y}), \tag{168}$$

$$\boxed{\{\phi(\vec{x}), \pi(\vec{y})\} = \delta^3(\vec{x}-\vec{y}), \qquad \{\phi, \phi\} = \{\pi, \pi\} = 0 \qquad \text{(equal times).}} \tag{169}$$

Dynamics is bracket flow, as in [Mech. §8.2]: $\dot{F} = \{F, H\}$, and in particular

$$\dot{\phi} = \frac{\delta H}{\delta\pi}, \qquad \dot{\pi} = -\,\frac{\delta H}{\delta\phi}. \tag{170}$$

The Klein–Gordon check. From (163): $\delta H/\delta\pi = \pi$, so $\dot{\phi} = \pi$, recovering the definition; and — here the integration by parts built into (119) acts on the gradient term —

$$\dot{\pi} = -\frac{\delta H}{\delta\phi} = \nabla\cdot(\nabla\phi) - m^2\phi = \nabla^2\phi - m^2\phi. \tag{171}$$

Combining, $\ddot{\phi} = \nabla^2\phi - m^2\phi$ — the Klein–Gordon equation (138), re-derived from the phase-space side. And one sentence that the outlook will inflate: promoting the classical bracket (169) to a commutator, $[\hat{\phi}(\vec{x}), \hat{\pi}(\vec{y})] = \mathrm{i}\,\delta^3(\vec{x}-\vec{y})$, is the *first move* of canonical quantization — the entire apparatus of this section is its starting point.

Remark 10.1 (The slicing trade returns for fields). Remark 4.5 described this trade for the particle; here it recurs for fields, in identical form — which is the point. The momentum (161) differentiates with respect to *somebody's* $\dot{\phi}$; the brackets (169) are *equal-time* brackets, defined on a slice; and the Hamiltonian is $H = \int T^{00}\,\mathrm{d}^3x = P^0$ — the 00-component of a tensor, integrated over the chosen slice, the time component of the field's four-momentum. Different observers slice differently and carry different Hamiltonians. Covariance survives as a theorem with the same statement as before: the ten Poincaré charges — P^μ from §9.4, $M^{\mu\nu}$ from Remark 9.5 — close under the bracket (167) into the Poincaré algebra, boost charges explicitly time-dependent, conservation by the same cancellation as in Remark 4.5. [stated; verifying the algebra is a long computation, not run here.] Canonical quantization inherits every word of this, and "the quantum field theory is Lorentz invariant" is a statement about operator charges, proved, not seen — while the path integral, built on the scalar $S[\phi]$ of §8.1, keeps covariance manifest throughout. The two quantization routes part company exactly here, at a fork Part I's free particle already mapped.

Part III

The Electromagnetic Field

11 The Field Tensor

Part I left electromagnetism half-told: the potentials A^μ entered as a background, the variational principle manufactured the antisymmetric combination $F_{\mu\nu}$ (Eq. (100)), and the quotient rule certified it as a tensor. This section opens Part III by taking the tensor seriously as an object: its components are $\vec{E}$ and $\vec{B}$, its transformation law is the long-sought answer to "how do electric and magnetic fields look from a moving frame," and its two invariants are the only field statements all observers share. The section closes by building the object that will source the dynamics: the four-current.

11.1 The six components

From §5.3, with $A^\mu = (\varphi, \vec{A})$, $A_\mu = (\varphi, -\vec{A})$:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad \vec{E} = -\nabla\varphi - \partial_t \vec{A}, \qquad \vec{B} = \nabla \times \vec{A}. \tag{172}$$

Antisymmetry kills the diagonal and pairs the rest: 6 independent components, and §5.3 claimed they are exactly the six numbers above. Verify by direct evaluation. Time-space:

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = \partial_t(-A^i) - \partial_i \varphi = \left(-\nabla\varphi - \partial_t \vec{A}\right)_i = E_i. \tag{173}$$

Space-space:

$$F_{ij} = \partial_i A_j - \partial_j A_i = -\big(\partial_i A^j - \partial_j A^i\big) = -\,\epsilon_{ijk}\, B_k, \tag{174}$$

(the middle sign from lowering two spatial indices; the last equality is the component form of the curl). Assembled, with rows μ and columns ν:

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{pmatrix}, \qquad F^{\mu\nu} = \begin{pmatrix} 0 & -E_x & -E_y & -E_z \\ E_x & 0 & -B_z & B_y \\ E_y & B_z & 0 & -B_x \\ E_z & -B_y & B_x & 0 \end{pmatrix} \tag{175}$$

(raising both indices flips only the mixed time-space entries: $F^{0i} = \eta^{00}\eta^{ij}F_{0j} = -F_{0i}$, while $F^{ij} = F_{ij}$). The moral outranks the matrix: *$\vec{E}$ and $\vec{B}$ are not two vector fields.* They are the time-space and space-space slices of one antisymmetric tensor — a split made by a choice of frame, with no invariant meaning — and every "mystery" of their mixing is the ordinary behavior of tensor components under Eq. (61).

11.2 How $\vec{E}$ and $\vec{B}$ transform

The tensor law now delivers what §3.6 promised: the transformation of $\vec{E}$ and $\vec{B}$ for free. Boost along x with velocity β and apply $F'^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F^{\rho\sigma}$. Two representative components, in full. The (01) entry meets both boost rows:

$$F'^{01} = \Lambda^0{}_0\Lambda^1{}_1\, F^{01} + \Lambda^0{}_1\Lambda^1{}_0\, F^{10} = \gamma^2 F^{01} + \gamma^2\beta^2 F^{10} = \gamma^2(1-\beta^2)\, F^{01} = F^{01}, \tag{176}$$

so $E'_x = E_x$: the component along the boost is untouched (the two γ's and the antisymmetry combine to cancel). The (02) entry meets one boost row and one spectator:

$$F'^{02} = \Lambda^0{}_0\, F^{02} + \Lambda^0{}_1\, F^{12} = \gamma\,(-E_y) + (-\gamma\beta)(-B_z) \implies E'_y = \gamma\big(E_y - \beta B_z\big). \tag{177}$$

The remaining components run identically, and the complete result is:

parallel to the boost	transverse to the boost	
$E'_x = E_x$	$E'_y = \gamma\,(E_y - \beta B_z)$	$E'_z = \gamma\,(E_z + \beta B_y)$
$B'_x = B_x$	$B'_y = \gamma\,(B_y + \beta E_z)$	$B'_z = \gamma\,(B_z - \beta E_y)$

Compactly: $\vec{E}_\parallel$, $\vec{B}_\parallel$ inert; $\vec{E}'_\perp = \gamma(\vec{E} + \vec{\beta} \times \vec{B})_\perp$, $\vec{B}'_\perp = \gamma(\vec{B} - \vec{\beta} \times \vec{E})_\perp$. Electric and magnetic fields feed each other under boosts, in exactly the way t and x do.

Remark 11.1 (Magnetism is boosted electricity)*.* The table contains a physical point usually buried in a third course on the subject. Take a single charge at rest: pure Coulomb $\vec{E}$, no $\vec{B}$ anywhere (elementary electrostatics, imported here as input; §12 derives the field equations, and Example 12.2 the field). View it from a moving frame: the table manufactures $\vec{B}'_\perp = -\gamma(\vec{\beta} \times \vec{E})_\perp \neq 0$ — a magnetic field, from nothing but a boost. Magnetism is not a second force; it is what electrostatics looks like from a frame in which the charges move, and its everyday effects are large not because β is large (drift velocities in wires are mm/s, $\beta \sim 10^{-12}$) but because the electrostatic force it corrects is enormous — wires are neutral to extraordinary precision, and the magnetic force between them is the $O(\beta^2)$ relativistic residue of two large, nearly perfectly canceling electric forces. [The quantitative wire analysis — length contraction unbalancing the two charge densities — is a classic worked example; stated here, not run.] Historically the logic ran backwards: Einstein's 1905 paper is titled "On the Electrodynamics of Moving Bodies," and its opening paragraph is precisely the magnet-and-conductor asymmetry this remark dissolves.

11.3 The two invariants

What do all observers agree on? Full contractions (§3.6). With one tensor and the metric, exactly one is available:

$$F_{\mu\nu}F^{\mu\nu} = 2\,F_{0i}F^{0i} + F_{ij}F^{ij} = -2\,\vec{E}^2 + \epsilon_{ijk}\epsilon_{ijl}B_kB_l \implies \boxed{F_{\mu\nu}F^{\mu\nu} = 2\big(\vec{B}^2 - \vec{E}^2\big),} \tag{178}$$

using $\epsilon_{ijk}\epsilon_{ijl} = 2\delta_{kl}$. A second invariant needs the last invariant tensor, now introduced as §3.6 promised: the totally antisymmetric $\epsilon^{\mu\nu\rho\sigma}$, fixed by $\epsilon^{0123} = +1$. It is invariant under the proper orthochronous transformations only — $\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Lambda^\rho{}_\gamma\Lambda^\sigma{}_\delta\,\epsilon^{\alpha\beta\gamma\delta} = (\det\Lambda)\,\epsilon^{\mu\nu\rho\sigma}$, the definition of the determinant — so objects built with one ϵ flip sign under P or T (*pseudo*-scalars and pseudo-tensors). Define the *dual* tensor and evaluate its slices by the same component patience as (175):

$$\tilde{F}^{\mu\nu} \equiv \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \qquad \tilde{F}^{0i} = -B_i, \quad \tilde{F}^{ij} = \epsilon_{ijk}\,E_k \qquad \text{— i.e. } (\vec{E},\vec{B}) \to (\vec{B}, -\vec{E}): \tag{179}$$

duality trades the two fields. The second invariant:

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = 2F_{0i}\tilde{F}^{0i} + F_{ij}\tilde{F}^{ij} = -2\,\vec{E}\cdot\vec{B} - 2\,\vec{E}\cdot\vec{B} \implies \boxed{F_{\mu\nu}\tilde{F}^{\mu\nu} = -\,4\,\vec{E}\cdot\vec{B}} \tag{180}$$

(a pseudo-scalar: it flips under P, as $\vec{E}\cdot\vec{B}$ visibly does, $\vec{E}$ being a vector and $\vec{B}$ an axial vector). These two numbers are the entire frame-independent content of a field configuration at a point, and they sort configurations into invariant classes: if $\vec{E} \perp \vec{B}$ in one frame, then in all; if $\vec{E} \cdot \vec{B} \neq 0$, both fields are nonzero in *every* frame, and no purely electric or purely magnetic frame exists; if $\vec{E} \cdot \vec{B} = 0$ and $\vec{B}^2 > \vec{E}^2$, some frame sees a pure magnetic field (and symmetrically for $\vec{E}^2 > \vec{B}^2$); and the doubly-null case

$$\vec{E}^2 = \vec{B}^2, \qquad \vec{E} \cdot \vec{B} = 0 \qquad \text{(both invariants zero)} \tag{181}$$

is a class of its own — *null fields*, perpendicular and equal-magnitude in *every* frame, with no rest frame to boost to. Section 13 will find exactly this signature in the electromagnetic wave: light is null, in every sense this document has given the word.

11.4 The four-current

Dynamics needs a source, and §9.3 specified what one must be: a conserved four-vector current. Electromagnetism's version packages charge density and current density,

$$J^\mu = \big(\rho,\ \vec{J}\big), \tag{182}$$

and its four-vector credentials rest on one experimental scalar: *electric charge is Lorentz invariant* (an electron's charge does not depend on its speed — verified to extraordinary precision by the neutrality of atoms, whose electrons move and whose protons' quarks move very differently [experimental input]). For a flow of charges of proper density ρ_0 (density in the local rest frame, a scalar) moving with four-velocity $u^\mu(x)$:

$$J^\mu = \rho_0\, u^\mu = \big(\gamma\rho_0,\ \gamma\rho_0\vec{v}\big) \tag{183}$$

— scalar times four-vector, credentials immediate; the γ in $\rho = \gamma\rho_0$ is length contraction acting on the volume in the denominator of "charge per volume," charge itself inert. A general J^μ is a superposition of such flows. Conservation of charge is then the continuity equation of §9.2,

$$\boxed{\partial_\mu J^\mu = 0 \qquad \Longleftrightarrow \qquad \frac{\partial\rho}{\partial t} + \nabla \cdot \vec{J} = 0,} \tag{184}$$

with Remark 9.3 supplying the reason this *must* be the local statement. For now (184) is an empirical fact about the source; §12 will show the field equations *enforce* it — a source that fails to conserve charge cannot couple to the electromagnetic field at all — and §9.3's Noether current is the standing example of where a microscopic theory gets such a J^μ from.

12 The Maxwell Lagrangian

Now the building-block game of §8.2, replayed for the vector field — with one extra constraint. For the scalar, the covariance principle alone thinned the catalogue; for A_μ, the redundancy met in §5.2 also constrains: *the action may change by at most boundary terms under* $A_\mu \to A_\mu + \partial_\mu\chi$, or the unphysical redundancy would leak into the dynamics. The two constraints turn out to leave essentially one theory standing, and its equations of motion are Maxwell's.

12.1 Building the Lagrangian under two constraints

Candidates quadratic in A_μ (free-field logic, as in §8.2), screened in order:

- **The mass term $A_\mu A^\mu$.** A perfectly good Lorentz scalar — but not gauge invariant: it shifts by $2A^\mu\partial_\mu\chi + \partial_\mu\chi\,\partial^\mu\chi$, no total divergence for generic χ. A photon mass is not merely inelegant; it is *incompatible with the redundancy*, which is why its experimental bound is a test of the framework ($m_\gamma < 10^{-18}\,\mathrm{eV}$ [experimental input]).

- **Generic derivative quadratics** ($\partial_\mu A_\nu\,\partial^\mu A^\nu$, $(\partial_\mu A^\mu)^2$, …). Individually gauge-variant. The *only* first-derivative combination inert under the gauge shift is the one the particle's variational principle manufactured in §5.3: $F_{\mu\nu}$, gauge-invariant because $\partial_\mu\partial_\nu\chi$ is symmetric (Proposition 3.11 in action). Scalars built from it: $F_{\mu\nu}F^{\mu\nu}$ and $F_{\mu\nu}\tilde{F}^{\mu\nu}$.

- **The dual candidate $F\tilde{F}$** drops out: it is a total divergence,

$$F_{\mu\nu}\tilde{F}^{\mu\nu} = 2\,\epsilon^{\mu\nu\rho\sigma}\,\partial_\mu A_\nu\,\partial_\rho A_\sigma = 2\,\partial_\mu\big(\epsilon^{\mu\nu\rho\sigma}A_\nu\,\partial_\rho A_\sigma\big), \tag{185}$$

  the second equality because pulling ∂_μ out costs a term $\epsilon^{\mu\nu\rho\sigma}A_\nu\partial_\mu\partial_\rho A_\sigma$ that dies on Proposition 3.11 ($\partial_\mu\partial_\rho$ symmetric, ϵ antisymmetric). Boundary terms do not affect the equations of motion, so $F\tilde{F}$ contributes nothing classically. (Its quantum role — topological terms, anomalies — is real and deferred. [stated.])

One field-only candidate survives. Fix its normalization by expanding in 3+1 language via (178):

$$-\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}\big(\vec{E}^2 - \vec{B}^2\big), \tag{186}$$

and since $\vec{E}$ carries the time derivatives of the potentials (Eq. (97)), the coefficient $-\frac{1}{4}$ is exactly what gives the kinetic terms the scalar field's $+\frac{1}{2}$ normalization — kinetic minus potential, matching (136).

The source coupling is the linear term $-J^\mu A_\mu$, with J^μ a prescribed external current — and here the gauge constraint becomes decisive. Under the gauge shift,

$$-\,J^\mu\,\partial_\mu\chi = -\,\partial_\mu\big(J^\mu\chi\big) \;+\; \chi\,\partial_\mu J^\mu\,: \tag{187}$$

a total divergence *if and only if* $\partial_\mu J^\mu = 0$. Gauge invariance does not merely tolerate the conservation law of §11.4 — it *demands* it, before a single equation of motion has been derived: a non-conserved current cannot be coupled to the electromagnetic field at all. Assembling:

$$\boxed{\mathcal{L} \;=\; -\,\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu} \;-\; J^\mu A_\mu.} \tag{188}$$

Remark 12.1 (Units in Part III). The normalization of (188) fixes the electromagnetic unit system: these are rationalized Heaviside–Lorentz units (with $c = 1$ as always), in which no factors of 4π appear in the field equations. Restoring SI expressions — as Example 13.2 will do — reinstates ε_0 and μ_0 by dimensional analysis, in the same way Convention 1.3 restores c. [stated; the conversion between unit systems is standard.]

12.2 Variation: the inhomogeneous pair

The field Euler–Lagrange equation (133) runs once per component of A_ν — four fields, one shared index. The derivative of the kinetic term is the section's one full index computation, so it is displayed whole. Write $F_{\rho\sigma}F^{\rho\sigma} = 2\big(\partial_\rho A_\sigma\,\partial^\rho A^\sigma - \partial_\rho A_\sigma\,\partial^\sigma A^\rho\big)$; then, with the renaming discipline of Rule 6 (§3.7),

$$\frac{\partial}{\partial(\partial_\mu A_\nu)}\big(\partial_\rho A_\sigma\,\partial^\rho A^\sigma\big) = 2\,\partial^\mu A^\nu, \qquad \frac{\partial}{\partial(\partial_\mu A_\nu)}\big(\partial_\rho A_\sigma\,\partial^\sigma A^\rho\big) = 2\,\partial^\nu A^\mu, \tag{189}$$

(each factor contributes once, the metric raising what the Kronecker deltas deliver), so

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\tfrac{1}{4}\cdot 2\left(2\,\partial^\mu A^\nu - 2\,\partial^\nu A^\mu\right) = -\,F^{\mu\nu}, \qquad \frac{\partial\mathcal{L}}{\partial A_\nu} = -\,J^\nu, \tag{190}$$

and (133) reads $\partial_\mu(-F^{\mu\nu}) = -J^\nu$:

$$\boxed{\partial_\mu F^{\mu\nu} = J^\nu.} \tag{191}$$

Unpack with the matrix (175) ($F^{i0} = E_i$, $F^{ij} = -\epsilon_{ijk}B_k$). The $\nu = 0$ component:

$$\partial_i F^{i0} = J^0 \qquad \Longrightarrow \qquad \nabla\cdot\vec{E} = \rho \qquad \text{(Gauss)}, \tag{192}$$

and $\nu = j$:

$$\partial_0 F^{0j} + \partial_i F^{ij} = J^j \qquad \Longrightarrow \qquad \nabla\times\vec{B} - \frac{\partial\vec{E}}{\partial t} = \vec{J} \qquad \text{(Ampère–Maxwell)}, \tag{193}$$

using $\partial_i F^{ij} = -\epsilon_{ijk}\partial_i B_k = (\nabla\times\vec{B})_j$. Two of Maxwell's four equations, from one variation.

The field equations enforce charge conservation. Act with ∂_ν on (191):

$$\partial_\nu J^\nu = \partial_\nu\partial_\mu F^{\mu\nu} = 0 \qquad \text{identically} \tag{194}$$

— Proposition 3.11 once more, symmetric $\partial_\nu\partial_\mu$ against antisymmetric $F^{\mu\nu}$. The requirement of §11.4 is now enforced from both ends: gauge invariance demanded a conserved source before the dynamics existed (187), and the dynamics, once derived, is structurally incapable of sourcing itself from anything else. That one three-line proposition from the grammar section is behind gauge invariance of F, the inertness of $F\tilde{F}$, and charge conservation — three results, one identity.

Example 12.2 (The static point charge)**.** The simplest solution of (191). Put a single charge q at rest at the origin: $\rho = q\,\delta^3(\vec{x})$, $\vec{J} = 0$ — the delta of §10.2 placing the whole charge at one point (§12.4 exhibits this as the static case of a worldline current). Seek a static, spherically symmetric field — the source distinguishes no direction and no time — so try $\vec{E} = E(r)\,\hat{r}$ with $\vec{B} = 0$. Integrate Gauss's law over a ball of radius r and apply the divergence theorem, as in §9.2:

$$\oint_{S_r} \vec{E}\cdot \mathrm{d}\vec{A} = 4\pi r^2\,E(r) = \int_{B_r} q\,\delta^3(\vec{x})\;\mathrm{d}^3x = q, \tag{195}$$

so

$$\boxed{\vec{E} = \frac{q}{4\pi r^2}\,\hat{r}} \qquad \left(\text{equivalently } \varphi = \frac{q}{4\pi r},\ \vec{A} = 0:\ \text{check } -\nabla\varphi = \vec{E},\ \vec{B} = \nabla\times\vec{A} = 0\right). \tag{196}$$

The 4π of Heaviside–Lorentz units (Remark 12.1) sits in the field, not in the field equation. This is the pure-electric configuration that Remark 11.1 boosted; that it is the *only* static solution decaying at infinity is a uniqueness statement not proved here. [stated.]

12.3 The homogeneous pair: geometry, not dynamics

Two Maxwell equations remain, and no variation will produce them — because they are not dynamics. They are *identities*, true of any F built from a potential. Substitute $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ into the cyclic sum:

$$\begin{aligned} \partial_\mu F_{\nu\rho} + \partial_\nu F_{\rho\mu} + \partial_\rho F_{\mu\nu} &= \big(\partial_\mu\partial_\nu A_\rho - \partial_\mu\partial_\rho A_\nu\big) + \big(\partial_\nu\partial_\rho A_\mu - \partial_\nu\partial_\mu A_\rho\big) + \big(\partial_\rho\partial_\mu A_\nu - \partial_\rho\partial_\nu A_\mu\big) \\ &= 0, \end{aligned} \tag{197}$$

the six terms canceling in pairs (first with fourth, second with fifth, third with sixth — equality of mixed partials each time). This is the *Bianchi identity*; contracting it with $\frac{1}{2}\epsilon^{\lambda\mu\nu\rho}$ packages the same statement as

$$\boxed{\partial_\mu \tilde{F}^{\mu\nu} = 0,} \tag{198}$$

(ϵ picks out exactly the totally antisymmetric part of ∂F, which is what the cyclic sum is). Unpack with $\tilde{F}^{i0} = B_i$, $\tilde{F}^{ij} = \epsilon_{ijk}E_k$ (§11.3): the $\nu = 0$ component is $\nabla \cdot \vec{B} = 0$ (no magnetic charge), and $\nu = j$ is $\nabla \times \vec{E} + \partial_t \vec{B} = 0$ (Faraday). All four of Maxwell's equations are now in hand, and the division is worth displaying:

pair	origin	content
$\partial_\mu F^{\mu\nu} = J^\nu$	variation of (188)	dynamics; couples to matter
$\partial_\mu \tilde{F}^{\mu\nu} = 0$	identity of $F = \partial A - \partial A$	geometry; no source possible

Half of electromagnetism is a law of motion; the other half is the tautological cost — and reward — of describing the field by a potential. (The converse also holds: any F satisfying the Bianchi identity is, locally, $\partial_\mu A_\nu - \partial_\nu A_\mu$ for some A — the Poincaré lemma [stated; a real theorem, with global exceptions on topologically nontrivial regions that become physics in their own right — magnetic monopoles, Aharonov–Bohm].)

12.4 One coupling, two faces

Part I coupled a *particle* to the field through $-q\int A_\mu\,\mathrm{d}x^\mu$; this Part couples the *field* to a current through $-\int \mathrm{d}^4x\, J^\mu A_\mu$. These are the same term. Write the current of a single point charge on a worldline $x^\mu(\lambda)$ — the delta apparatus of §10.2 finally building something:

$$J^\mu(x) = q\int \mathrm{d}\lambda\, \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\, \delta^4\big(x - x(\lambda)\big) \tag{199}$$

(reparametrization invariant, since $\mathrm{d}\lambda$ cancels; and the $\mu = 0$ component integrates over the δ in time to give $J^0 = q\,\delta^3(\vec{x} - \vec{x}(t))$ — a point charge's density, correctly). Insert into the field-side coupling; the δ^4 collapses the spacetime integral:

$$-\int \mathrm{d}^4x\; J^\mu A_\mu = -\,q\int \mathrm{d}\lambda\, \frac{\mathrm{d}x^\mu}{\mathrm{d}\lambda}\, A_\mu\big(x(\lambda)\big) = -\,q\int A_\mu\, \mathrm{d}x^\mu \tag{200}$$

— the particle-side coupling of Eq. (89), recovered exactly. One interaction term, two Euler–Lagrange addressees: vary the *worldline* through it and the Lorentz force emerges (§5.3); vary the *potential* through it and the source of Maxwell's equations emerges (§12.2). A single term in S supplies both action and reaction. And when the source is itself a field rather than a particle, §9.3 has already built the J^μ that supplies the coupling's other side — the conserved $U(1)$ current, whose conservation is exactly what (187) requires of any source.

What the Lagrangian has not yet delivered is the thing electromagnetism is famous for: waves. They are hiding inside (191), behind the gauge redundancy — and extracting them is the next section's business.

13 Gauge Freedom and Electromagnetic Waves

Maxwell's equations are in hand; light is not yet. This section extracts it, and the extraction is an exercise in managing the redundancy: the wave equation for A^μ appears only after the gauge freedom is used well, the physical content of the wave — two transverse polarizations — appears only after the redundancy is counted carefully, and the section closes with the field's energy-momentum tensor, which settles two outstanding items at once (Remark 9.5's concrete Belinfante case, and §11.3's claim that light is null).

13.1 Fixing a gauge: the Lorenz condition

Write the dynamical Maxwell pair (191) in terms of the potential:

$$\partial_\mu F^{\mu\nu} = \partial_\mu\big(\partial^\mu A^\nu - \partial^\nu A^\mu\big) = \Box A^\nu - \partial^\nu\big(\partial_\mu A^\mu\big) = J^\nu. \tag{201}$$

The first term is four Klein–Gordon operators at $m = 0$; the second term couples the components and blocks the wave reading. But $\partial_\mu A^\mu$ is exactly the kind of thing gauge freedom can reach: under $A_\mu \to A_\mu + \partial_\mu\chi$,

$$\partial_\mu A^\mu \;\longrightarrow\; \partial_\mu A^\mu + \Box\chi, \tag{202}$$

so choosing χ to solve the sourced wave equation $\Box\chi = -\,\partial_\mu A^\mu$ (solvable; standard [stated]) lands the potential in the *Lorenz gauge*,[2]

$$\partial_\mu A^\mu = 0, \tag{203}$$

where (201) collapses to

$$\boxed{\Box A^\nu = J^\nu\,:} \tag{204}$$

four decoupled, sourced wave equations, one per component — the massless Klein–Gordon field of §8, four copies, plus a source. The arc that began with Eq. (6) closes here: *this* is the wave equation whose Galilean non-invariance started the document, now derived from a Lorentz-scalar action, invariant by construction, no medium anywhere in sight. (What this section does *not* do is solve the sourced equation: the retarded potentials — the field propagating outward from its source at speed c — are the standard next step and are deferred [stated]; from §13.2 on we set $J^\nu = 0$ and study the waves themselves.)

Two housekeeping notes. First, the gauge freedom is not exhausted: (203) survives any further shift with $\Box\chi = 0$ — *residual* freedom, about to be needed. Second, an alternative choice exists:

Remark 13.1 (The Coulomb gauge). One may instead impose $\nabla\cdot\vec{A} = 0$ (Coulomb gauge), whereupon Gauss's law becomes $\nabla^2\varphi = -\rho$ — the scalar potential is determined *instantaneously* by the charge distribution, with no wave equation and no retardation. Nothing acausal follows (φ is not measurable; the gauge-invariant $\vec{E}$, $\vec{B}$ propagate causally [stated; this follows from the retarded solution just deferred]), but manifest covariance is gone: the condition $\nabla\cdot\vec{A} = 0$ names a frame. The trade is the same as in Remark 4.5 — Coulomb gauge gains explicit physical degrees of freedom at the cost of manifest symmetry, Lorenz gauge the reverse — and quantum theory uses both, for exactly those reasons. [stated; this document stays in Lorenz gauge.]

[2] Lorenz, not Lorentz: the condition is due to Ludvig Lorenz (Danish, 1867), not Hendrik Lorentz (Dutch) — and since (203) is built from a four-divergence, the Lorenz condition is Lorentz-invariant, a coincidence that has corrupted spellings for a century.

13.2 Counting the degrees of freedom: two polarizations

Vacuum ($J^\nu = 0$), plane-wave ansatz as in §8.4:

$$A^\mu(x) = \mathrm{Re}\,\varepsilon^\mu\,\mathrm{e}^{-\mathrm{i}k\cdot x}, \tag{205}$$

with a constant *polarization vector* ε^μ. Three conditions now stack, and the count they leave is the physics:

- **Dynamics.** $\Box A^\mu = 0$ demands $k \cdot k = 0$: the wavevector is *null*, $\omega = |\vec{k}|$ — the $m = 0$ dispersion of Fig. 8, light on the light line, as the name always promised. Four amplitudes ε^μ remain.
- **The gauge condition.** $\partial_\mu A^\mu = 0$ demands $k \cdot \varepsilon = 0$: one linear constraint, $4 \to 3$.
- **The residual freedom.** The leftover shifts, $\chi = \mathrm{Re}\,(\mathrm{i}\lambda\,\mathrm{e}^{-\mathrm{i}k\cdot x})$ (which obey $\Box\chi = 0$ precisely because $k^2 = 0$), move the polarization by

  $$\varepsilon^\mu \;\longrightarrow\; \varepsilon^\mu + \lambda\,k^\mu\,: \tag{206}$$

  the component of ε *along* k^μ is pure gauge — present in the description, absent from the physics — $3 \to 2$.

$$\boxed{4\text{ (components)} \;-\; 1\text{ (gauge condition)} \;-\; 1\text{ (residual freedom)} \;=\; 2\;\text{ physical polarizations.}} \tag{207}$$

Concretely, for $\vec{k} = \omega\hat{z}$: $k \cdot \varepsilon = 0$ kills one combination of $\varepsilon^0, \varepsilon^z$, the residual shift kills the other, and what remains is $\varepsilon^\mu = (0, \varepsilon_x, \varepsilon_y, 0)$ — *transverse* light, two independent directions of it. This is the pattern of Remark 4.2 and §5.2 brought to its conclusion: a redundant description carries components that the equations of motion decline to govern, and a correct count must subtract them *twice* — once for the condition that can be imposed, once for the freedom that survives it.

The fields of the wave close a promise. For the transverse ε above, Eqs. (97) give $\vec{E} \parallel \vec{\varepsilon}_\perp$, $\vec{B} = \hat{k} \times \vec{E}$: mutually perpendicular, perpendicular to the propagation, equal in magnitude. Hence both invariants vanish,

$$\vec{B}^2 - \vec{E}^2 = 0, \qquad \vec{E} \cdot \vec{B} = 0 \tag{208}$$

— the electromagnetic wave is a *null field* in the precise sense of §11.3, with a null wavevector to match: light is null in every sense this document has given the word, in every frame, with no rest frame to boost to — which is where the document came in (Postulate 2.2), now as output rather than input.

13.3 Improving the energy-momentum tensor

Run the canonical recipe (157), summed over the four fields A_ρ as Theorem 9.2 prescribes:

$$T^{\mu\nu}_{\text{can}} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu A_\rho)}\,\partial^\nu A_\rho - \eta^{\mu\nu}\mathcal{L} = -\,F^{\mu\rho}\,\partial^\nu A_\rho + \tfrac{1}{4}\,\eta^{\mu\nu}F_{\rho\sigma}F^{\rho\sigma}, \tag{209}$$

and meet the defect Remark 9.5 predicted for a field with indices of its own: the first term is neither symmetric in $\mu\nu$ nor gauge invariant — a bare $\partial^\nu A_\rho$, not an F. The repair is one total divergence. Add

$$\Delta^{\mu\nu} = \partial_\rho\big(F^{\mu\rho}A^\nu\big) = F^{\mu\rho}\,\partial_\rho A^\nu + \big(\partial_\rho F^{\mu\rho}\big)A^\nu = F^{\mu\rho}\,\partial_\rho A^\nu \quad \text{(free field)}, \tag{210}$$

which changes no charge ($\Delta^{0\nu}$ is a spatial divergence, integrating to zero) and no conservation law ($\partial_\mu \Delta^{\mu\nu} = \partial_\mu \partial_\rho (F^{\mu\rho} A^\nu) = 0$: symmetric derivatives on a $\mu\rho$-antisymmetric object, Proposition 3.11, fourth appearance). The first term of T_{can} then completes to an F:

$$\boxed{T^{\mu\nu} = -\,F^{\mu\rho}\,F^{\nu}{}_{\rho} + \tfrac{1}{4}\,\eta^{\mu\nu}\,F_{\rho\sigma}F^{\rho\sigma}} \tag{211}$$

— symmetric, gauge invariant, conserved, all by inspection: the Belinfante improvement in its simplest working instance. Its physical slices, evaluated with the matrix (175):

$$\boxed{T^{00} = \tfrac{1}{2}\big(\vec{E}^2 + \vec{B}^2\big), \qquad T^{0i} = \big(\vec{E} \times \vec{B}\big)_i} \tag{212}$$

— the classical energy density of the electromagnetic field (positive definite, confirming the sign choice of §12.1's $-\frac{1}{4}$), and the *Poynting vector*, the energy flux, obtained here without a single line of the traditional $\vec{E} \times \vec{B}$ manipulation: it is a component of a tensor that a symmetry built. One structural remark is worth adding, and foreshadows much: the trace vanishes, $T^{\mu}{}_{\mu} = -F^{\mu\rho}F_{\mu\rho} + \frac{1}{4} \cdot 4 \cdot F_{\rho\sigma}F^{\rho\sigma} = 0$ — the fingerprint of a theory with no mass and hence no intrinsic scale. [stated; tracelessness and scale invariance is a QFT-course thread.]

For the plane wave of §13.2, with $|\vec{E}| = |\vec{B}|$ and $\vec{E} \perp \vec{B} \perp \hat{k}$:

$$T^{00} = \vec{E}^2, \qquad T^{0i} = \vec{E}^2\,\hat{k}_i \qquad \Longrightarrow \qquad |\text{flux}| = (\text{density}) \times c\,: \tag{213}$$

the wave's energy travels at exactly the speed of light — as it must, there being nothing else in a null field's kinematics for it to do.

Example 13.2 (Sunlight: the worked number)**.** Restore SI units and point (212) at the sky. The solar constant — the time-averaged Poynting flux at Earth's orbit — is $\langle S\rangle = 1.36\,\text{kW/m}^2$. For a wave of amplitude E_0, the SI average is $\langle S\rangle = \frac{1}{2}\varepsilon_0 c\,E_0^2$, so

$$E_0 = \sqrt{\frac{2\langle S\rangle}{\varepsilon_0 c}} \approx 1.0 \times 10^3\,\text{V/m}, \qquad B_0 = \frac{E_0}{c} \approx 3.4\,\mu\text{T} \tag{214}$$

— a kilovolt per meter of oscillating field in ordinary daylight, and a magnetic partner a tenth of the Earth's field. The momentum flux of the wave is a pressure, $\langle S\rangle/c \approx 4.5\,\mu\text{Pa}$: minute on skin, decisive on comet tails and solar sails, and a direct classical reading of the momentum density that (211) localizes in empty space.

Remark 13.3 (Conservation with sources)*.* The tensor (211) was conserved for the *free* field; with a source, the field alone no longer conserves energy-momentum, and the failure is exactly the exchange with matter. Differentiate (211):

$$\partial_\mu T^{\mu\nu} = -\big(\partial_\mu F^{\mu\rho}\big)F^{\nu}{}_{\rho} \;-\; F^{\mu\rho}\,\partial_\mu F^{\nu}{}_{\rho} \;+\; \tfrac{1}{2}\,F_{\rho\sigma}\,\partial^\nu F^{\rho\sigma}. \tag{215}$$

The first term is $-F^{\nu\rho}J_\rho$ by the inhomogeneous pair (191). The last two cancel by the Bianchi identity: substituting the cyclic sum in the form $\partial^\nu F^{\rho\sigma} = -\partial^\rho F^{\sigma\nu} - \partial^\sigma F^{\nu\rho}$ into the third term turns it, after renaming dummies (Rule 1), into two copies of $+\frac{1}{2}\,F^{\mu\rho}\partial_\mu F^{\nu}{}_{\rho}$, which cancel the middle term. Hence

$$\boxed{\partial_\mu T^{\mu\nu} = -\,F^{\nu\rho}\,J_\rho\,.} \tag{216}$$

The $\nu = 0$ component, in 3+1 language:

$$\frac{\partial}{\partial t}\,\tfrac{1}{2}\big(\vec{E}^2 + \vec{B}^2\big) + \nabla \cdot \big(\vec{E} \times \vec{B}\big) = -\,\vec{E} \cdot \vec{J}\,: \tag{217}$$

the field's energy budget loses, per unit volume, exactly the power the Lorentz force (98) delivers to the charges — §5's force law, seen from the field's side of the one shared coupling (§12.4). Only the sum of field and matter energy-momentum is conserved; the transfer between them is what (98) describes from the matter side.

The classical story is complete: one scalar action per object — the free scalar (135), the Maxwell field (188), the charged particle (89) stitching them together — equations of motion, symmetries, currents, energy, waves. What the classical theory cannot say is why the light arrives in lumps. The outlook locates that boundary.

14 Outlook

This document has run out of classical physics to derive, and the outlook's first job is not a preview but a closing of the series' longest arc. The second job is to mark the three directions this apparatus opens, each with one paragraph and no pretense of following them.

14.1 Every field is a family of oscillators

Put the free Klein–Gordon field in a periodic box of volume V (§7.3, third mechanism) and expand in the Fourier modes the box allows:

$$\phi(\vec{x},t) = \frac{1}{\sqrt{V}} \sum_{\vec{k}} q_{\vec{k}}(t)\, \mathrm{e}^{\mathrm{i}\vec{k}\cdot\vec{x}}, \qquad q_{-\vec{k}} = q_{\vec{k}}^{*} \tag{218}$$

(the reality condition ties the modes in $\pm\vec{k}$ pairs, each pair carrying two real degrees of freedom — the amplitude's real and imaginary parts; the steps are standard and not belabored). Insert into the Lagrangian $L = \int \mathrm{d}^3x \left[\frac{1}{2}\dot{\phi}^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{1}{2}m^2\phi^2\right]$ and let the orthogonality $\int \mathrm{d}^3x\, \mathrm{e}^{\mathrm{i}(\vec{k}-\vec{k}')\cdot\vec{x}} = V\delta_{\vec{k}\vec{k}'}$ collapse the integrals:

$$\boxed{L = \sum_{\vec{k}} \left[\tfrac{1}{2}\left|\dot{q}_{\vec{k}}\right|^2 - \tfrac{1}{2}\,\omega_{\vec{k}}^2 \left|q_{\vec{k}}\right|^2\right], \qquad \omega_{\vec{k}}^2 = |\vec{k}|^2 + m^2.} \tag{219}$$

This is the statement the whole series was built to reach: *a free field is a countable family of independent simple harmonic oscillators*, one per wavevector, each with the frequency the dispersion relation (140) assigns. The object that opened the first document [Osc. §2] — one mass, one spring — is the *atom of field theory*: every result proved there applies here, mode by mode, verbatim. The chain's normal modes becoming Fourier modes (§6.3, last dictionary row) was this fact in embryo; the Hamiltonian

$$H = \sum_{\vec{k}} \left[\tfrac{1}{2}\left|p_{\vec{k}}\right|^2 + \tfrac{1}{2}\,\omega_{\vec{k}}^2 \left|q_{\vec{k}}\right|^2\right] \tag{220}$$

is [Osc.]'s energy, summed over infinitely many springs; and the Maxwell field, in Lorenz gauge four massless copies of the same (§13.1) minus the gauge redundancy, is two such families — one per polarization (§13.2).

14.2 Toward the quantum theory

Everything canonical quantization needs is already on the table. The bracket $\{\phi(\vec{x}), \pi(\vec{y})\} = \delta^3(\vec{x}-\vec{y})$ of §10.3 becomes, in mode variables, one canonical pair per oscillator; promote brackets to commutators (§10.3's one-sentence hook, now inflated to three). The quantum harmonic oscillator has the spectrum $E_n = \omega\,(n + \frac{1}{2})$ [stated; the only quantum-mechanics input used in this outlook], so a state of the quantum *field* is an assignment of an excitation number $n_{\vec{k}}$ to every mode, with

$$E = \sum_{\vec{k}} \omega_{\vec{k}}\, n_{\vec{k}} \;+\; E_0 : \tag{221}$$

energy arrives in lumps of $\omega_{\vec{k}}$, momentum in lumps of $\vec{k}$, and each lump sits on the mass shell $\omega^2 = |\vec{k}|^2 + m^2$ — Fig. 6, at last, *literally*: the lumps are **particles**, and the field's excitations of the Maxwell family are *photons*, two polarizations per wavevector, which is the answer to why the light of §13 arrives in pieces. (The constant $E_0 = \sum \frac{1}{2}\omega_{\vec{k}}$ diverges — the theory's first ultraviolet infinity, cousin of Appendix B's warning about products of distributions, and the problem named *renormalization*. [stated; a QFT course's first month.]) From here the handoff is clean: this document's Lagrangians, currents, and brackets are the exact classical inputs of the canonical formalism of Schwartz Chs. 2–3, and Remark 10.1 already mapped the fork in the road there.

14.3 Two directions not taken

Curved spacetime. Every structure in this document stood on one fixed object: $\eta_{\mu\nu}$. Promote it to a *field*, $g_{\mu\nu}(x)$, and the template of §3.1 survives pointwise while the physics transforms: freely falling particles still extremize $\int \mathrm{d}\tau$, but the proper time is now measured by g, and the equation of motion picks up terms encoding the metric's variation — gravity as geometry. The source of the metric's own dynamics is an object this document built: the energy-momentum tensor of §9, sitting on the right-hand side of Einstein's equations exactly where J^μ sits in (191) — and the once-harmless additive constant in the Lagrangian (§8.2) stops being harmless, because gravity weighs $\mathcal{L}$ itself: the cosmological constant. [stated; a general relativity course.]

The gauge principle. This document met gauge freedom as a property electromagnetism *has* (§5.2); modern field theory runs the logic backwards, as a property interactions *come from*. Take the global $U(1)$ of §9.3 and demand it hold *locally*, $\alpha \to \alpha(x)$: the gradient $\partial_\mu \phi$ no longer transforms covariantly, and repairing it forces the introduction of a field A_μ with exactly the transformation law (91) and exactly the minimal coupling (107) — the electromagnetic field is *what localizing a phase demands*. Replace the phase circle by larger internal symmetry groups and the same demand manufactures the weak and strong interactions: the Standard Model is three runs of this argument. [stated; the actual run is Schwartz Ch. 8 and beyond.]

14.4 Closing the series

One mass on one spring; two, coupled; N, chained; the chain, continued to a rope; the rope's equation caught transforming badly; spacetime rebuilt so that transforming well means something; mechanics re-founded on a variational principle and Noether's dictionary between symmetry and conservation; the principle handed a field, then the simplest relativistic field, then the field that carries light — and at the end, expanded in modes, it is one mass on one spring again, infinitely many times. The recycled ideas were few: quadratic Lagrangians and their normal modes, coordinates adapted to a preserved structure, symmetry converted to conservation, redundancy counted correctly. Physics ahead — quantum fields, curved spacetime, gauge theory — will add machinery, but it will replace remarkably little of what these documents built; mostly it will run the same few ideas again, on stranger stages.

A Linearity of the Transformation Between Inertial Frames

This appendix proves Lemma 2.3 and disposes of the transverse coordinates. Write the transformation as a map $f : \mathbb{R}^4 \to \mathbb{R}^4$, $x \mapsto x' = f(x)$, between the coordinates that two inertial frames assign to the same events. We assume f is a bijection (each frame labels every event exactly once) and continuous (nearby events get nearby labels).

A.1 Homogeneity forces affinity

The physical input is the *homogeneity of space and time*: no event is distinguishable from any other by the laws of physics, so the transformation law for coordinate *differences* cannot depend on where in spacetime the difference sits. Formally: for every displacement $a \in \mathbb{R}^4$ there is a displacement $g(a)$ such that

$$f(x+a) - f(x) = g(a) \qquad \text{for all } x. \tag{222}$$

(If the left side depended on x, two identically prepared experiments at different locations, separated by the same a, would transform differently — a violation of homogeneity.)

Lemma A.1. *A continuous f satisfying (222) is affine: $f(x) = \mathsf{L}x + b$ with L linear and b constant.*

Proof. Evaluate (222) at two displacements. First, $f(x + a_1 + a_2) - f(x) = g(a_1 + a_2)$ directly. Second, insert the intermediate point:

$$f(x+a_1+a_2) - f(x) = \big[f((x+a_1)+a_2) - f(x+a_1)\big] + \big[f(x+a_1) - f(x)\big] = g(a_2) + g(a_1). \tag{223}$$

Hence g satisfies the Cauchy functional equation $g(a_1 + a_2) = g(a_1) + g(a_2)$. Additivity gives $g(qa) = q\,g(a)$ for all rational q (integers by iteration, reciprocals by substitution), and continuity of g — inherited from f via $g(a) = f(a) - f(0)$ — extends this to all real scalars. So g is linear; call it L. Setting $x = 0$ in (222): $f(a) = \mathsf{L}a + f(0)$, which is the claim with $b = f(0)$. ☐

In the standard configuration the origins coincide, $f(0) = 0$, so $b = 0$ and the transformation is linear, proving Lemma 2.3.

Remark A.2 (An alternative route). A different argument, requiring no appeal to homogeneity as a separate principle, starts from the definition of inertial frames: free particles move on straight worldlines in both, so f maps lines to lines, and a continuous line-preserving bijection of $\mathbb{R}^4$ is affine by the fundamental theorem of affine geometry. The conclusion is the same; the theorem invoked is substantially harder than Lemma A.1, so the main text leans on homogeneity.

A.2 The transverse coordinates are spectators

Let the relative motion lie along x. Isotropy about the motion axis (no direction transverse to the boost is preferred) and homogeneity force $y' = \kappa(v)\,y$, $z' = \kappa(v)\,z$ with a single scale $\kappa(v) > 0$: the transverse map can involve no shear into t, x, or the other transverse direction, since any such term would single out a transverse direction.

Now run the reciprocity–isotropy argument of Theorem 2.4 Step 3 on κ: composing the boost with its inverse gives $\kappa(v)\kappa(-v) = 1$, while reflection symmetry of the transverse plane under $x \to -x$ gives $\kappa(-v) = \kappa(v)$. Hence $\kappa(v)^2 = 1$, and positivity (continuity from $\kappa(0) = 1$) yields

$$\boxed{y' = y, \qquad z' = z.} \tag{224}$$

Lengths transverse to the motion are absolute — which is why the entire derivation of §2 could be run in the (ct, x) plane.

B Distributions, and the License for the Delta Function

This appendix issues the license claimed in §10.2: what kind of object δ is, and why the integral notation for it is safe. Everything is stated for one dimension; three dimensions is the same theory with products of test functions.

Test functions and distributions. Let $\mathcal{D}$ be the space of *test functions*: infinitely differentiable functions $f : \mathbb{R} \to \mathbb{R}$ vanishing outside some bounded set. A *distribution* is a linear functional $T : \mathcal{D} \to \mathbb{R}$, continuous in the appropriate sense (if $f_n \to f$ with all derivatives, uniformly, supports in a common bounded set, then $T[f_n] \to T[f]$). Distributions are exactly the objects that *know how to be integrated against test functions* — and nothing more.

Functions are distributions; δ is not a function. Every locally integrable function g defines a distribution by ordinary integration, $T_g[f] = \int gf\,\mathrm{d}x$; such distributions are called *regular*, and the map $g \mapsto T_g$ loses no information. The evaluation functional

$$\delta_y[f] \;=\; f(y) \tag{225}$$

is manifestly linear and continuous — a distribution — but it is not regular: a candidate g would have to satisfy $\int gf = f(y)$ for all f, forcing $g = 0$ almost everywhere away from y, whence $\int gf = 0 \neq f(y)$ in general. The notation

$$\delta_y[f] \;=\; \int \delta(x-y)\,f(x)\,\mathrm{d}x \tag{226}$$

is therefore a *notation*: the right side is defined by the left. Its utility is that the rules of integral manipulation, applied formally to the fiction, compute correctly — because each rule is secretly a definition:

- **Derivatives.** Define $T'[f] \equiv -T[f']$ — which is what integration by parts *would* give if T were a decaying function, boundary terms and all. Every distribution is thereby infinitely differentiable, and manipulations like "integrate the delta by parts" are applications of this definition, not of calculus.
- **Multiplication by smooth functions.** Define $(gT)[f] \equiv T[gf]$ for smooth g; e.g. $g(x)\,\delta(x-y) = g(y)\,\delta(x-y)$ as distributions.
- **Change of variables, scaling.** Defined by transporting the test function; e.g. $\delta(ax) = \delta(x)/|a|$ is the statement $\int \delta(ax)f(x)\mathrm{d}x = f(0)/|a|$, i.e. substitution performed on the test-function side.

What the fiction does *not* license is multiplying two distributions: δ^2 is not defined by any of the above, and no consistent general definition exists. (Classical field theory never needs such products; quantum field theory does — products of field operators at a point — and a substantial part of its machinery, under the name *renormalization*, is the repair. One sentence of foreshadowing, no more.)

Smeared fields. The careful reading of a classical field mirrors that of δ: the physically meaningful pairing is the field *smeared* against a test function, $\phi[f] = \int \phi(x)f(x)\,\mathrm{d}x$ — an average over the resolution profile f, which is all any finite-resolution measurement returns. "The field at a point" is the idealized $f \to \delta$ limit, harmless classically for continuous ϕ and adopted without further comment in the main text; the smeared object is the one that survives, unaltered, into the quantum theory.

Part IV
Quantum Mechanics

From Brackets to Amplitudes: Quantum Mechanics

Quantizing the canonical structure

This document is the quantum layer of the series: it takes the canonical structures the previous four documents built — conjugate pairs, Poisson brackets, symmetries traded for conserved charges, the harmonic oscillator, coordinates adapted to preserved structure — and quantizes them. The route is structural rather than historical. Part I begins where the mechanics notes ended: the Poisson bracket's algebra admits a second realization, by commutators of operators, and Dirac's rule $\{\,,\,\} \to [\,,\,]/\mathrm{i}\hbar$ is the replacement of realization with the algebra kept; the canonical commutator forces an infinite-dimensional stage (a three-line trace argument), the stage is Hilbert space, and the measurement postulates are installed on it, with the uncertainty relations derived from them. Position and momentum representations are constructed, with $\hat{p} = -\mathrm{i}\hbar\,\partial_x$ derived from the translation group. Part II is dynamics: the evolution operator, the Heisenberg picture *first* — quantum bracket flow, the verbatim mirror of the mechanics notes — with the Schrödinger picture as a change of bookkeeping; the harmonic oscillator solved algebraically, the series' opening object now with a discrete spectrum; and the propagator computed twice, canonically and by path integral, making explicit at the one level where both routes are fully tractable the fork that the field-theory notes could only mark. Part III is symmetry: rotations represented unitarily, angular momentum as their Noether charges, the $SU(2)$ double cover with its measurable $2\pi \neq \mathbf{1}$, the angular momentum spectrum from pure algebra, and the hydrogen atom as the capstone worked example. The outlook quantizes the field modes of the companion notes — every mode one quantum oscillator, occupation numbers, particles — closing the series' longest arc for the second and final time. Functional-analytic fine print (domains, self-adjointness, the Stone–von Neumann theorem) is flagged where it lives and deferred to Hall; every statement used is either proved here or explicitly tagged.

1 Conventions and Notation

Operators wear hats ($\hat{q}$, $\hat{p}$, $\hat{H}$) until §3 declares them safe to remove in unambiguous contexts; classical quantities are bare. Dots are time derivatives. Three-vectors carry arrows. Cross-references to the companion documents: [Osc. §n] for the oscillator reference, [Mech. §n] for the variational mechanics course, and [Fld. §n] for the spacetime-and-fields notes.

Convention 1.1 ($\hbar$ stays visible)**.** This document keeps $\hbar = 1.055 \times 10^{-34}\,\mathrm{J\,s}$ explicit throughout — a deliberate reversal of the companion notes' $c = 1$ policy [Fld. §3.9], and the contrast is the reason. There, c was a conversion factor between units of length and time, carrying no dynamics; hiding it lost nothing. Here, $\hbar$ is the *subject*: it is the deformation parameter by which quantum mechanics differs from classical mechanics, the knob whose $\hbar \to 0$ limit must recover everything the series built, and the scale that decides when quantum effects matter. Setting it to 1 would hide the very parameter under study. Note its dimensions: $[\hbar] = [q][p] = \textit{action}$ — the very quantity the series has been extremizing since [Mech. §3].

Convention 1.2 (Nonrelativistic throughout)**.** The speed of light does not appear: this document's mechanics is Galilean, its Hamiltonians are $\hat{p}^2/2m + V$, and the marriage of quantum mechanics with special relativity is exactly the handoff of the outlook (§17) — quantum *field* theory, for the reasons [Fld. §4.4] previewed.

Recurring symbols, each re-introduced where it first appears:

Symbol	Definition	First appears	Meaning
$[\hat{A}, \hat{B}]$	$\hat{A}\hat{B} - \hat{B}\hat{A}$	§2	commutator
$\hbar$	$h/2\pi$	§2	quantum of action
$\mathcal{H}$	—	§3	Hilbert space
$\lvert\alpha\rangle$, $\langle\alpha\rvert$	—	§3	ket, bra
$\hat{U}(t)$	$\mathrm{e}^{-\mathrm{i}\hat{H}t/\hbar}$	§6	evolution operator
$\hat{a}$, $\hat{a}^\dagger$	—	§9	ladder operators
$K(x_b, t_b; x_a, t_a)$	—	§10	propagator
$\hat{J}_i$	—	§11	angular momentum

(The $\lvert\,\rangle/\langle\,\rvert$ notation is defined in §3; the table lists it for reference only. Mathematical fine print — operator domains, self-adjointness versus Hermiticity, spectral theory of unbounded operators — is flagged with [stated; Hall] where it arises, pointing to Hall, *Quantum Theory for Mathematicians*; this document works at the rigor of a careful physicist, with the potholes marked rather than paved.)

Part I
Kinematics

2 From Brackets to Commutators

The mechanics notes ended with dynamics as bracket flow: observables are functions on phase space, the Poisson bracket organizes them into an algebra, and $\dot{F} = \{F, H\}$ runs the world [Mech. §8.2]. This section makes the single structural observation on which the whole document stands: *that algebra admits a second realization*, by operators, and nature uses it. Everything else — Hilbert space, wavefunctions, spectra — is the working-out of where the second realization is forced to live.

2.1 What classical mechanics could not do

Three experimental facts, each fatal on its own, name the problem. [All three: experimental input.]

- **Atoms are stable and discrete.** A classical electron orbiting a nucleus is an accelerating charge, and accelerating charges radiate (Larmor's formula — one classical result this series did *not* derive [stated; Griffiths, *Introduction to Electrodynamics*, Ch. 11]); losing energy to radiation, the electron must spiral in, in about 10^{-11} s. It does not — and when atoms do radiate, they emit *lines*, not continua: something restricts the energies available to the electron to a discrete set, and no classical Hamiltonian $H(q, p)$ with a continuum of orbits does that.

- **The oscillator's energy comes in lumps.** Blackbody radiation and low-temperature specific heats both fail classically in the same way, and both are repaired by one assumption: an oscillator of frequency ω can hold energy only in units of $\hbar\omega$. The failing system is the series' own mascot [Osc. §2].

- **Measurements interfere with each other.** A Stern–Gerlach apparatus sorts silver atoms by a two-valued quantity ($S_z = \pm\hbar/2$). Sort by S_z, keep the + beam, sort it by S_x, keep the + beam, and sort *that* by S_z again: both S_z values reappear. Measuring S_x *destroyed* the previously sharp value of S_z. Classical observables — mere functions on phase space — can all be read simultaneously; these two demonstrably cannot. The order of observations matters: whatever mathematical objects S_z and S_x are, *they do not commute.*

The third fact is the structural one, and it points directly at the bracket.

2.2 The algebra the bracket built

Recall from [Mech. §8.1] what makes the Poisson bracket more than a formula — its axioms. For observables A, B, C and numbers λ:

$$\begin{aligned}
&\text{antisymmetry:} && \{A, B\} = -\{B, A\}, \\
&\text{linearity:} && \{A, \lambda B + C\} = \lambda\{A, B\} + \{A, C\}, \\
&\text{Leibniz:} && \{A, BC\} = \{A, B\}\,C + B\,\{A, C\}, \\
&\text{Jacobi:} && \{A, \{B, C\}\} + \{B, \{C, A\}\} + \{C, \{A, B\}\} = 0.
\end{aligned} \tag{1}$$

Everything the mechanics notes did with brackets — conserved quantities, canonical transformations, the flow $\dot{F} = \{F, H\}$ — used only (1) plus the fundamental bracket $\{q, p\} = 1$; the specific realization as $\sum(\partial_q A\, \partial_p B - \partial_p A\, \partial_q B)$ mattered only for computing.

Now the observation. Let observables instead be *operators* — linear maps on some vector space, composed by application, so that $\hat{A}\hat{B}$ and $\hat{B}\hat{A}$ need not agree — and define

$$[\hat{A}, \hat{B}] \;\equiv\; \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{2}$$

Antisymmetry and linearity are immediate. Leibniz is one line of adding and subtracting:

$$[\hat{A}, \hat{B}\hat{C}] = \hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = \big(\hat{A}\hat{B} - \hat{B}\hat{A}\big)\hat{C} + \hat{B}\big(\hat{A}\hat{C} - \hat{C}\hat{A}\big) = [\hat{A}, \hat{B}]\,\hat{C} + \hat{B}\,[\hat{A}, \hat{C}], \tag{3}$$

and Jacobi is a patient expansion — twelve terms, canceling in pairs (write it out once; the cancellation needs no cleverness). So the commutator satisfies *every axiom* in (1): **the bracket algebra has a second realization**, in which observables are operators, "bracket" means commutator, and — this is the license the Stern–Gerlach fact demanded — observables need not commute.

Definition 2.1 (Dirac's quantization rule)**.** Quantum mechanics keeps classical mechanics' algebra and replaces its realization:

$$\boxed{\{A, B\} \quad\longrightarrow\quad \frac{1}{\mathrm{i}\hbar}\,[\hat{A}, \hat{B}],} \tag{4}$$

and in particular the fundamental bracket $\{q, p\} = 1$ becomes the *canonical commutation relation*,

$$\boxed{[\hat{q}, \hat{p}] = \mathrm{i}\hbar.} \tag{5}$$

Three notes on the rule before it is put to work. *The* i: observables should be represented by operators whose measured values are real — Hermitian operators, §3 — and the commutator of two Hermitian operators is *anti*-Hermitian ($[\hat{A}, \hat{B}]^\dagger = [\hat{B}^\dagger, \hat{A}^\dagger] = -[\hat{A}, \hat{B}]$); the i repairs the mismatch, making $[\hat{A}, \hat{B}]/\mathrm{i}\hbar$ Hermitian again, eligible to be an observable as $\{A, B\}$ was. *The* $\hbar$: each term of the Poisson bracket carries one ∂_q and one ∂_p, so the bracket operation costs a factor $1/([q][p]) = 1/[\text{action}]$ relative to the plain product — $\{A, B\}$ has dimensions $[A][B]/[\text{action}]$, and $\{q, p\} = 1$ is dimensionless. The commutator involves no derivatives: $[\hat{q}, \hat{p}]$ has dimensions $[q][p] = [\text{action}]$, plain. The two realizations differ by exactly one unit of action, so the dictionary between them needs a constant with those dimensions — and nature supplies exactly one. And it is small in a precisely quantifiable sense: a desk pendulum ($m = 0.1\,\mathrm{kg}$, energy $\sim 10^{-2}\,\mathrm{J}$, period $\sim 1\,\mathrm{s}$) carries an action per cycle of order $10^{-2}\,\mathrm{J\,s} \sim 10^{32}\,\hbar$, while one vibrational quantum of a molecule carries, per cycle, exactly $ET = \hbar\omega \cdot 2\pi/\omega = h \approx 6\hbar$. The dimensionless ratio $S/\hbar$ is the switch: where it is astronomical, the deformation (4) is invisible and the mechanics notes rule; where it is of order one, this document takes over. (§10 converts this slogan into a theorem — the stationary-phase limit of the path integral.) *The status*: (4) is a *construction principle*, not a theorem — for polynomials in q, p beyond quadratic it suffers ordering ambiguities (which operator does q^2p^2 become?), and the precise statement is that (5) plus the algebra is the postulate, with (4) its guiding reading. [The obstruction to a perfect $\{\,,\,\} \to [\,,\,]$ dictionary is a real theorem — Groenewold's no-go theorem — stated here, proved nowhere near here; Hall §13.4.]

2.3 First consequences from the algebra alone

Remarkably much follows from (5) before any stage for the operators is chosen — three results, in increasing order of consequence.

Derivative identities from the algebra. By induction with the Leibniz rule (3): $[\hat{q}, \hat{p}^n] = [\hat{q}, \hat{p}]\hat{p}^{n-1} + \hat{p}\,[\hat{q}, \hat{p}^{n-1}]$, and unwinding,

$$[\hat{q}, \hat{p}^n] = \mathrm{i}\hbar\, n\, \hat{p}^{n-1} \qquad \Longrightarrow \qquad \boxed{[\hat{q}, F(\hat{p})] = \mathrm{i}\hbar\, F'(\hat{p}), \qquad [\hat{p}, G(\hat{q})] = -\mathrm{i}\hbar\, G'(\hat{q})} \tag{6}$$

for functions defined by power series — the commutator with $\hat{q}$ *differentiates in* $\hat{p}$, exactly as the classical $\{q, F(p)\} = F'(p)$ does (one line from the bracket's definition [Mech. §8.1]: $\{q, F\} = \partial F/\partial p$): the new realization reproduces the old one's derivative structure operator-by-operator, which is the algebra-keeping promise of (4) kept.

No common eigenvectors. Suppose $|\psi\rangle$ had sharp values of both: $\hat{q}|\psi\rangle = q_0|\psi\rangle$ and $\hat{p}|\psi\rangle = p_0|\psi\rangle$ (notation borrowed from §3, content clear already). Then $[\hat{q}, \hat{p}]|\psi\rangle = (q_0p_0 - p_0q_0)|\psi\rangle = 0$, contradicting $\mathrm{i}\hbar|\psi\rangle \neq 0$. *No state has sharp position and sharp momentum* — the uncertainty principle's skeleton, three lines from the algebra; §4 adds the flesh (the quantitative bound $\Delta q\, \Delta p \geq \hbar/2$).

The stage must be infinite-dimensional. Here is the theorem that decides where the subject must live.

Theorem 2.2 (No finite-dimensional realization)**.** *No pair of finite-dimensional matrices satisfies* $[\hat{q}, \hat{p}] = \mathrm{i}\hbar\,\mathbf{1}$.

Proof. The trace is cyclic, $\mathrm{tr}(\hat{A}\hat{B}) = \mathrm{tr}(\hat{B}\hat{A})$, so the trace of any commutator vanishes: $\mathrm{tr}[\hat{q}, \hat{p}] = 0$. But $\mathrm{tr}(\mathrm{i}\hbar\,\mathbf{1}) = \mathrm{i}\hbar\, d \neq 0$ in dimension d. No $d < \infty$ survives. □

So the canonical commutation relation cannot live where the mechanics of the series lived — on finite lists of coordinates — nor even on finite-dimensional vector spaces: it *forces* an infinite-dimensional stage, on which the trace argument dissolves (the trace of $\mathbf{1}$ is no longer finite, and indeed $\hat{q}, \hat{p}$ turn out to be unbounded operators, for which the cyclic property fails — the first pothole, marked: [stated; Hall Ch. 9, esp. §9.8 for these two operators]). The spin observables of Stern–Gerlach, by contrast, close their commutators without any $\mathbf{1}$ on the right (§11: $[\hat{S}_x, \hat{S}_y] = \mathrm{i}\hbar\, \hat{S}_z$), which is why *they* fit in two dimensions — for both algebras here, the size of the smallest stage is visible in the right-hand sides.

2.4 The representation question

An algebra is not yet a physical theory; it needs a stage — a vector space for the operators to act on, an inner product to make predictions with, and a dictionary between vectors and physical states. Building that stage is §3's job, and one theorem, quoted now and leaned on throughout, guarantees the job has essentially one answer:

Remark 2.3 (Stone–von Neumann and why the founders agreed)*.* Up to unitary equivalence, the canonical commutation relation (5) has exactly *one* irreducible representation (in the technically careful, exponentiated form). [stated; Hall Ch. 14 — the theorem needs the Weyl form of the CCR (Hall §14.2) precisely because of the unboundedness pothole above.] The physical content is uniqueness of quantum mechanics itself: Heisenberg's matrix mechanics and Schrödinger's wave mechanics, born separately in 1925–26, are the same representation in different coordinates — the position representation of §5 is not *a* choice among physically distinct options but a convenient basis in the unique theory, exactly as normal coordinates were a basis in the unique chain [Osc. §8]. When §7 finds two descriptions of dynamics and §10 finds two routes to the propagator, the same moral will hold: one theory, many coordinates.

3 The Stage: Hilbert Space and Dirac Notation

Theorem 2.2 ordered an infinite-dimensional complex vector space with an inner product; this section builds it, installs Dirac's notation on it, and stocks it with the operator vocabulary the rest of the document speaks. One deliberate exception to the infinite-dimensionality runs through the section and the document: the *two-state system*, spin-$\frac{1}{2}$, small enough to compute everything in and rich enough to exhibit almost every quantum phenomenon — the standing laboratory, introduced at the end and never dismissed.

3.1 States: kets, superposition, and rays

Postulate 3.1 (States)**.** The states of a quantum system are the nonzero vectors of a complex Hilbert space $\mathcal{H}$ — a complex vector space with an inner product, complete in the induced norm [the completeness clause is the first pothole: it matters only in infinite dimensions, and only for the analysis this document defers; stated here, leaned on never — Hall App. A.4 does it right] — with two vectors describing the same physical state when they differ by a nonzero complex factor.

A state is written as a *ket*, $|\alpha\rangle$, the label α naming whatever physics identifies it. Two pieces of physics hide in "vector space" and one in "complex":

- **Superposition.** If $|\alpha\rangle$ and $|\beta\rangle$ are states, so is $c_1|\alpha\rangle + c_2|\beta\rangle$ — not a mixture, not an either-or, but a genuinely new state. This is the vector-space axiom promoted to physics, and it is the structural fact behind interference: amplitudes add before probabilities are taken.
- **Rays.** Since $|\alpha\rangle$ and $c|\alpha\rangle$ are the same physics, states are really *rays*; in practice one normalizes, $\langle\alpha|\alpha\rangle = 1$, leaving a physically empty overall phase $\mathrm{e}^{\mathrm{i}\gamma}$. (Empty for a lone state — *relative* phases inside superpositions are physical, and §3.5 measures one.)
- **Complex.** The i in $[\hat{q},\hat{p}] = \mathrm{i}\hbar$ already forced complex scalars: a real vector space has no operator algebra in which (5) can hold with Hermitian $\hat{q},\hat{p}$ (§3.2's Hermiticity computation shows the commutator of Hermitians is anti-Hermitian — on a real space, where anti-Hermitian means antisymmetric, $\mathrm{i}\hbar\,\mathbf{1}$ has no home).

Bras. Every ket $|\beta\rangle$ defines a linear machine that eats kets and returns numbers — "take the inner product with $|\beta\rangle$" — written as the *bra* $\langle\beta|$, so that the inner product is the bracket

$$\langle\beta|\alpha\rangle \;=\; \big(\langle\beta|\big)\big(|\alpha\rangle\big), \qquad \langle\beta|\alpha\rangle = \langle\alpha|\beta\rangle^*, \qquad \langle\alpha|\alpha\rangle \geq 0, \tag{7}$$

conjugate-linear in the bra's label and linear in the ket's. (That every bounded linear machine is of this form — that the bras are exactly the duals — is the Riesz representation theorem [stated; Hall App. A.4.3]; in finite dimensions it is row-vectors-meet-column-vectors, and that mental model is safe throughout.) The notation's design intent will pay off in §3.3: bars and brackets are engineered so that the move made on nearly every page below — inserting a complete set of states — is typographically frictionless.

3.2 Operators: Hermitian and unitary

Observables need operators whose measurement outcomes are real; time evolution and symmetries need operators that preserve probability. Two definitions cover both needs.

The *adjoint* $\hat{A}^\dagger$ of an operator $\hat{A}$ is defined by

$$\langle\beta|\big(\hat{A}^\dagger|\alpha\rangle\big) = \big(\langle\alpha|\hat{A}|\beta\rangle\big)^* \qquad \text{for all } |\alpha\rangle, |\beta\rangle, \tag{8}$$

i.e. $\hat{A}^\dagger$ acting on kets matches $\hat{A}$ acting inside bras; in a basis it is the conjugate transpose. An operator is *Hermitian* if $\hat{A}^\dagger = \hat{A}$, and *unitary* if $\hat{U}^\dagger\hat{U} = \hat{U}\hat{U}^\dagger = \mathbf{1}$ — the latter exactly the condition $\langle \hat{U}\beta\,|\hat{U}\alpha\rangle = \langle\beta|\alpha\rangle$: unitaries are the inner product's isometries, the Hilbert-space entry in the series' table of structure-preserving maps (rotations for δ_{ij}, Lorentz for $\eta_{\mu\nu}$ [Fld. §3.1] — and now unitaries for $\langle\cdot\,|\cdot\rangle$; same template, third preserved pairing).

Two properties of Hermitian operators carry the probabilistic interpretation, and both proofs take two lines:

Proposition 3.2. *A Hermitian operator's eigenvalues are real, and eigenvectors with distinct eigenvalues are orthogonal.*

Proof. Let $\hat{A}|a\rangle = a|a\rangle$ and $\hat{A}|a'\rangle = a'|a'\rangle$, eigenkets normalized. Sandwich once: $\langle a|\hat{A}|a\rangle = a\langle a|a\rangle = a$, while Hermiticity says the same number equals its own conjugate, $a = a^*$. Sandwich twice, crosswise: $\langle a'|\hat{A}|a\rangle$ evaluated right-to-left gives $a\langle a'|a\rangle$, and left-to-right (using $\hat{A}^\dagger = \hat{A}$ and a' real) gives $a'\langle a'|a\rangle$; subtracting, $(a-a')\langle a'|a\rangle = 0$, so $a \neq a'$ forces $\langle a'|a\rangle = 0$. □

Remark 3.3 (The pothole made concrete). In infinite dimensions, "Hermitian" as used above (symmetric on its domain) is strictly weaker than *self-adjoint* — equality of $\hat{A}$ and $\hat{A}^\dagger$ *including their domains* — and the spectral machinery of the next subsection belongs to the self-adjoint. The gap is not pedantic. Take the momentum-type operator $\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ on $[0,1]$ with hard-wall conditions $\psi(0) = \psi(1) = 0$: integration by parts kills the boundary terms, so it is symmetric — and it has *no eigenvectors whatsoever* (the candidates $\mathrm{e}^{-\mathrm{i}\lambda x/\hbar}$ refuse to vanish at the endpoints), no spectral decomposition, no basis of its own. Self-adjointness is restored only by *choosing* boundary conditions of the periodic type $\psi(1) = \mathrm{e}^{\mathrm{i}\theta}\psi(0)$, one legitimate operator per choice of θ. [stated, with the example verified against Hall §9.6; the systematic theory of such choices — deficiency indices — is Hall Ch. 9.] The working policy of this document: "Hermitian" everywhere, potholes flagged where a domain actually bites (it bites once more in §4).

Hats, henceforth. With the operator vocabulary in place, the promise of §1 is kept: hats are dropped whenever context distinguishes operator from number (H, p, a), and retained where confusion threatens ($\hat{q}$ versus an eigenvalue q).

3.3 Eigenbases and the resolution of the identity

Theorem 3.4 (Spectral theorem, working version)**.** *A Hermitian operator on a finite-dimensional Hilbert space possesses an orthonormal basis of eigenvectors, $A|a\rangle = a|a\rangle$, $\langle a'|a\rangle = \delta_{a'a}$. [For genuinely self-adjoint operators in infinite dimensions the statement survives with sums traded for integrals where the spectrum is continuous — §5 uses exactly that trade — and its full formulation and proof are Hall Ch. 10; stated.]*

The basis being complete, every ket expands in it, and the expansion is worth writing as an operator identity:

$$\boxed{\mathbf{1} \;=\; \sum_a |a\rangle\langle a|} \qquad \text{(the \textit{resolution of the identity})}, \tag{9}$$

for then every expansion, every change of basis, and every matrix element is the *same* typographical move — insert (9) where needed:

$$|\psi\rangle = \sum_a |a\rangle \underbrace{\langle a|\psi\rangle}_{\text{coefficients}}, \qquad \langle\beta|B|\alpha\rangle = \sum_{a,a'} \langle\beta|a\rangle \underbrace{\langle a|B|a'\rangle}_{\text{matrix of } B} \langle a'|\alpha\rangle. \tag{10}$$

An operator, in its own eigenbasis, is diagonal: $A = \sum_a a\,|a\rangle\langle a|$ — the spectral theorem restated as a formula.

Remark 3.5 (Adapted coordinates: third appearance in the series). This is the series' standing pattern. Normal coordinates diagonalized the coupled pair and the chain [Osc. §§6, 8]; null coordinates diagonalized the boost [Fld. §2.4, Remark 2.7]; and here, the eigenbasis of whatever operator matters diagonalizes it — energy eigenstates for dynamics (§6), momentum eigenstates for free motion, angular momentum eigenstates for rotations. What is new is not the strategy but its *cost*: on the chain, finding normal coordinates was work; here, Dirac notation makes changing to adapted coordinates a one-symbol operation, (9) inserted at will. The notation is not a convenience layered on the theory — it is a systematic, zero-cost form of the series' oldest strategy.

3.4 What the notation computes

The notation so far is grammar; here is its semantics. Bras, kets, and operators combine into exactly four composite objects, and each carries a physical reading. (The readings involving probability lean on the measurement postulate, stated properly in §4; they are previewed here because the notation is unintelligible without them — and because Example 3.7 below already needs them.)

(a) The inner product is an amplitude. The number $\langle\beta|\alpha\rangle$ is the *overlap* of the two states — operationally, the *probability amplitude* that a system prepared in $|\alpha\rangle$ answers "yes" when asked "are you $|\beta\rangle$?", with

$$P(\alpha \to \beta) = \left|\langle\beta|\alpha\rangle\right|^2 \tag{11}$$

the probability itself (Born's rule, installed as a postulate in §4). The two extremes calibrate the reading: $\langle\alpha|\alpha\rangle = 1$ is certainty of self, and orthogonality $\langle\beta|\alpha\rangle = 0$ is *perfect distinguishability* — a measurement can tell the two states apart without error, which is why eigenstates of an observable (orthogonal, by Proposition 3.2) are the states a measurement of it can cleanly report.

(b) The outer product is an operator. The object $|\alpha\rangle\langle\beta|$ — ket first, bra second — eats a ket and returns a ket:

$$\big(|\alpha\rangle\langle\beta|\big)|\gamma\rangle = |\alpha\rangle\,\langle\beta|\gamma\rangle, \tag{12}$$

"detect β, output α, weighted by the overlap." Its adjoint swaps the factors, $(|\alpha\rangle\langle\beta|)^\dagger = |\beta\rangle\langle\alpha|$ (one line from the adjoint's definition: take matrix elements of both sides). The special case $\beta = \alpha$ is the workhorse: the *projector*

$$P_\alpha = |\alpha\rangle\langle\alpha|, \qquad P_\alpha^2 = |\alpha\rangle\,\underbrace{\langle\alpha|\alpha\rangle}_{1}\langle\alpha| = P_\alpha, \qquad P_\alpha^\dagger = P_\alpha : \tag{13}$$

Hermitian and idempotent — asking the question twice is the same as asking it once. The resolution of the identity (9) now reads as physics: the projectors of a complete eigenbasis exhaust the space, $\mathbf{1} = \sum_a P_a$ — "the answer is *always* one of the a's" — and they are the mathematical atoms out of which §4 builds measurement.

(c) The closed sandwich is an average. For a normalized state, define the *expectation value* $\langle A\rangle_\psi \equiv \langle\psi|A|\psi\rangle$. Its statistical meaning is two insertions of (9) away:

$$\langle\psi|A|\psi\rangle = \sum_a \langle\psi|A|a\rangle\langle a|\psi\rangle = \sum_a a\,\langle\psi|a\rangle\langle a|\psi\rangle = \boxed{\sum_a a\,\left|\langle a|\psi\rangle\right|^2} \tag{14}$$

— the eigenvalues, weighted by exactly the Born probabilities (11): $\langle A\rangle$ is the *mean of the measurement statistics*, not the outcome of any single measurement. The distinction has teeth: a spin-$\frac{1}{2}$ state can have $\langle S_z\rangle = 0.3\,\hbar$ while every individual measurement returns $\pm\hbar/2$ and nothing else. The spread about the mean,

$$(\Delta A)^2 \equiv \big\langle (A - \langle A\rangle)^2 \big\rangle = \langle A^2\rangle - \langle A\rangle^2, \tag{15}$$

is the quantity the uncertainty relations of §4 will bound.

(d) The open sandwich is a transition amplitude. The general matrix element $\langle\beta|A|\alpha\rangle$ is the amplitude for α to be found as β *with the process A in between* — the object dynamics computes: §6's central quantity is $\langle x|\,U(t)|\psi\rangle$, "the amplitude to be found at x after evolving for time t," and §10 is one long evaluation of a single such sandwich. In its own eigenbasis an operator's sandwich structure is laid bare by the spectral formula $A = \sum_a a\,P_a$: every operator this document meets is a weighted sum of "are you a?"-questions.

In summary — the four moves, which the laboratory below runs in miniature:

object	type	reading
$\langle\beta\|\alpha\rangle$	number	amplitude to find α as β
$\|\alpha\rangle\langle\beta\|$	operator	detect β, output α; $P_\alpha = \|\alpha\rangle\langle\alpha\|$ asks "are you α?"
$\langle\psi\|A\|\psi\rangle$	number	mean of A's measurement statistics
$\langle\beta\|A\|\alpha\rangle$	number	amplitude for $\alpha\to\beta$ through the process A

3.5 The standing laboratory: spin-$\frac{1}{2}$

Theorem 2.2 exempted observables whose commutators close without an $\mathbf{1}$, and the exemption's smallest instance is the two-dimensional Hilbert space of a spin-$\frac{1}{2}$ system — the silver atom of §2.1, finally given its mathematics. Take the S_z eigenbasis as computational home,

$$S_z|\pm\rangle = \pm\tfrac{\hbar}{2}|\pm\rangle, \qquad \langle +|-\rangle = 0, \quad \langle +|+\rangle = \langle -|-\rangle = 1, \tag{16}$$

and write the three spin components in it. In matrix form $S_i = \frac{\hbar}{2}\,\sigma_i$ with the *Pauli matrices*

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{17}$$

Hermitian, traceless, and obeying the identity that packages their entire algebra (verify once by direct multiplication — nine products, each a one-liner):

$$\boxed{\sigma_i\,\sigma_j = \delta_{ij}\,\mathbf{1} + \mathrm{i}\,\epsilon_{ijk}\,\sigma_k,} \tag{18}$$

whose antisymmetric part is the commutation relation $[S_i, S_j] = \mathrm{i}\hbar\,\epsilon_{ijk}S_k$ — a promise of §2.3 that this verifies and that §11 will derive from rotations — and whose symmetric part, $\{\sigma_i,\sigma_j\} = 2\delta_{ij}\mathbf{1}$, is a structure the Klein–Gordon-to-Dirac story will one day need [stated; used in the QFT sequel].

The general normalized state, with the empty overall phase fixed to make the $|+\rangle$ coefficient real:

$$|\theta,\varphi\rangle = \cos\tfrac{\theta}{2}\,|+\rangle \;+\; \mathrm{e}^{\mathrm{i}\varphi}\sin\tfrac{\theta}{2}\,|-\rangle, \tag{19}$$

two real parameters — and they are exactly the polar coordinates of a unit vector: a direct computation of the closed sandwich (14) with the matrices (17) gives

$$\langle\theta,\varphi|\vec{S}|\theta,\varphi\rangle = \tfrac{\hbar}{2}\big(\sin\theta\cos\varphi,\ \sin\theta\sin\varphi,\ \cos\theta\big) : \tag{20}$$

the state space of a spin-$\frac{1}{2}$ is a sphere (Fig. 1), each state "pointing" somewhere, with $|\pm\rangle$ at the poles and the relative phase φ — physical, as promised — as the longitude.

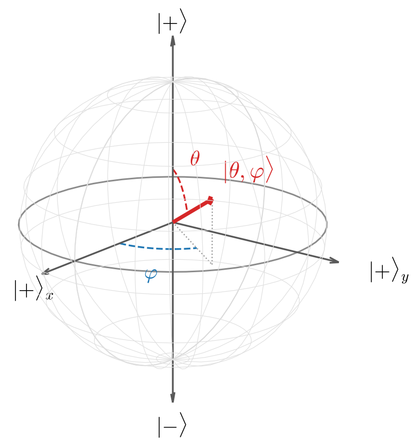

Figure 1: The *pure-state* Bloch sphere: the states (19) of a spin-$\frac{1}{2}$ system, mapped to unit vectors by the exact correspondence $\langle\vec{S}\rangle = \frac{\hbar}{2}(\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta)$. Poles: the S_z eigenstates; equator at $\varphi = 0$: the S_x eigenstate $|+\rangle_x$. Antipodal points — and only antipodal points — are orthogonal: the sphere's geometry encodes the Hilbert space's. (This surface is the boundary of a ball; Remark 3.6.)

Remark 3.6 (The sphere is the boundary of a ball)*.* The name "Bloch sphere" refers to a larger object than Postulate 3.1 describes. The general state concept of quantum mechanics — covering ensembles and subsystems, not just maximally-known preparations — is the *density matrix* ρ, and for two levels every such object is

$$\rho = \tfrac{1}{2}\big(\mathbf{1} + \vec{r}\cdot\vec{\sigma}\big), \qquad |\vec{r}\,| \le 1, \tag{21}$$

one point of the solid *Bloch ball* per state. Purity is the boundary condition: $\rho^2 = \rho \iff |\vec{r}\,| = 1$, and there $\vec{r} = \langle\vec{\sigma}\rangle$ reproduces exactly the map of Fig. 1 — what this document calls a state is the ball's surface, with the interior (mixed states, the center $\frac{1}{2}\mathbf{1}$ the "knows-nothing" state) describing situations where the preparation itself is uncertain. This document works with pure states throughout and does not develop ρ; the machinery is stated here so the figure is not mistaken for the whole story. [stated; see Sakurai's treatment of density operators and ensembles — density matrices are where quantum mechanics meets statistical mechanics and decoherence.]

Example 3.7 (Stern–Gerlach made quantitative)**.** The sequence of §2.1 can now be computed. The S_x eigenstates sit on the equator, at $(\theta, \varphi) = (\frac{\pi}{2}, 0)$ and $(\frac{\pi}{2}, \pi)$ in (19):

$$|\pm\rangle_x = \tfrac{1}{\sqrt{2}}\big(|+\rangle \pm |-\rangle\big), \qquad \big|{}_x\langle+|+\rangle\big|^2 = \Big|\tfrac{1}{\sqrt{2}}\Big|^2 = \tfrac{1}{2}. \tag{22}$$

So: the beam filtered to $|+\rangle$ splits 50:50 at the S_x magnet; the $|+\rangle_x$ beam, expressed in the z basis, is an equal superposition again, and the final S_z magnet splits it 50:50 — one quarter of the twice-filtered atoms emerge with $S_z = -\hbar/2$, the value the first filter had *removed*. The numbers say what §2.1 could only gesture at: measuring S_x prepared an S_x eigenstate, which is not an S_z eigenstate but a superposition of both — filtering is *preparation*, and incompatible preparations overwrite each other. (What "measurement prepares eigenstates" means as a postulate is §4's opening business.)

3.6 Composite systems: the tensor product

One construction completes the kit. Two systems with spaces $\mathcal{H}_1, \mathcal{H}_2$ compose into one with the *tensor product* $\mathcal{H}_1 \otimes \mathcal{H}_2$: basis kets are pairs $|a\rangle \otimes |b\rangle \equiv |a, b\rangle$, dimensions *multiply*, operators of system 1 act as $A \otimes \mathbf{1}$ (each subsystem's operators ignore the other), and inner products factorize on product kets. The physically loaded fact is what the construction contains beyond products: a general vector of $\mathcal{H}_1 \otimes \mathcal{H}_2$ is a superposition of product kets that need not *be* a product ket — for two spins,

$$\tfrac{1}{\sqrt{2}}\big(|+,-\rangle - |-,+\rangle\big) \tag{23}$$

assigns neither spin a state of its own. Such states are called *entangled*, they are the generic case (product states are a measure-zero subset), and their consequences — correlations no classical joint distribution reproduces — are flagged here and delivered as a physical specimen in §15 (Remark 15.2); the outlook's identical particles are where nature makes the choice of symmetric versus antisymmetric combinations for us. [stated at flag level; the Bell-inequality story is real, quantitative, and outside this document's arc.]

The stage is built. What has not been said is what happens on it when someone *looks* — the postulates connecting the formalism to laboratory statistics, and the inequality that §2.3 promised to make quantitative.

4 Measurement and Uncertainty

The stage of §3 connects to laboratory statistics through two postulates — what the outcomes are, and what remains afterward. Both were used, unproved, in Example 3.7; this section supplies them, extracts the algebraic criterion for when two observables tolerate each other, and converts §2.3's qualitative "no common eigenvectors" into the quantitative uncertainty relation — derived, then *stress-tested*: the section ends by exhibiting, with Hall, a state that seems to violate it, and locating the loophole.

4.1 The measurement postulates

Postulate 4.1 (Born rule)**.** Measuring an observable $A = \sum_a a\, P_a$ (spectral form, §3.4) on a normalized state $|\psi\rangle$ returns one of the eigenvalues a, with probability

$$\boxed{P(a) = \langle\psi|P_a|\psi\rangle} \qquad \big(= |\langle a|\psi\rangle|^2 \text{ when } a \text{ is nondegenerate}\big). \tag{24}$$

Postulate 4.2 (Projection)**.** If the outcome is a, the state immediately after the measurement is

$$|\psi\rangle \;\longrightarrow\; \frac{P_a|\psi\rangle}{\sqrt{P(a)}}. \tag{25}$$

Three consistency checks, each one line, each delivering a piece of §3.4:

- **Probabilities sum to one** because the projectors do: $\sum_a P(a) = \langle\psi|\big(\sum_a P_a\big)|\psi\rangle = \langle\psi|\psi\rangle = 1$ — the resolution of the identity (9) *is* the conservation of probability, which is the physical reason completeness sat in the postulates at all.

- **Measurement is repeatable**: measuring A again on the post-measurement state (25) returns a with probability $\langle\psi|P_a^3|\psi\rangle/P(a) = 1$ by idempotence — "asking twice equals asking once," now a law rather than a slogan, and exactly the "filtering is preparation" that Example 3.7 read off the beams.

- **Statistics match**: $\sum_a a\,P(a) = \langle\psi|A|\psi\rangle$ — the expectation value (14) is the mean of the Born distribution, as §3.4 promised.

Remark 4.3 (What the postulates do not say)*.* Equation (25) is a *rule*, not a mechanism: it does not say what physical process implements the jump, when exactly "a measurement" has occurred, or how the linear, deterministic dynamics of Part II coexists with a nonlinear, stochastic update. These questions — the measurement problem, decoherence, the classical limit of measuring devices — are genuine, quantitative, and open in parts; this document uses the postulates as every working calculation does, and flags the foundations rather than litigating them. [stated; decoherence theory is where the modern discussion lives.]

4.2 Compatible and incompatible observables

When can two observables have sharp values together? Exactly when their operators share an eigenbasis — and the criterion is the commutator, closing the loop opened by Stern–Gerlach in §2.1.

Proposition 4.4 (Compatibility criterion)**.** *Two Hermitian operators with* $[A, B] = 0$ *possess a common orthonormal eigenbasis, and conversely.*

Proof (nondegenerate case, displayed; general case stated). Let $A|a\rangle = a|a\rangle$ with a nondegenerate. Then

$$A\,\big(B|a\rangle\big) = B\,A|a\rangle = a\,\big(B|a\rangle\big)\;: \tag{26}$$

$B|a\rangle$ is again an a-eigenvector of A, hence (nondegeneracy) proportional to $|a\rangle$ — which says $B|a\rangle = b|a\rangle$: the A-eigenbasis already diagonalizes B. The converse is immediate (diagonal matrices commute). [Degenerate case: diagonalize B within each A-eigenspace; Sakurai's compatible-observables discussion runs the induction.] □

Compatible observables can be measured in either order with the same statistics, and their eigenvalues can label states jointly — the practice, formalized as a *complete set of commuting observables* (CSCO), of naming a state by the full list of sharp quantum numbers it carries. The hydrogen atom's $|n, \ell, m\rangle$ (§14) is the series' destination example. Incompatible observables are the interesting ones: $[S_z, S_x] = \mathrm{i}\hbar\, S_y \neq 0$, and the Stern–Gerlach overwriting of §2.1 now has its one-line algebraic diagnosis. How *much* incompatibility costs is the next subsection's theorem.

4.3 Derivation of the uncertainty relation

Theorem 4.5 (Robertson)**.** *For Hermitian A, B and a normalized state $|\psi\rangle$ (in the domains of the operators involved — a clause that Remark 4.8 will show is load-bearing),*

$$\boxed{\Delta A\,\Delta B \;\geq\; \tfrac{1}{2}\big|\langle\psi|\,[A,B]\,|\psi\rangle\big|.} \tag{27}$$

Proof. Center the operators: $\bar{A} \equiv A - \langle A\rangle$, $\bar{B} \equiv B - \langle B\rangle$, so that $(\Delta A)^2 = \|\bar{A}|\psi\rangle\|^2$ by (15), and note the shifts cancel inside the commutator, $[\bar{A}, \bar{B}] = [A, B]$. Cauchy–Schwarz on the two vectors $\bar{A}|\psi\rangle$, $\bar{B}|\psi\rangle$:

$$(\Delta A)^2(\Delta B)^2 = \|\bar{A}|\psi\rangle\|^2\,\|\bar{B}|\psi\rangle\|^2 \;\geq\; \big|\langle\psi|\bar{A}\,\bar{B}|\psi\rangle\big|^2. \tag{28}$$

Split the product into its Hermitian and anti-Hermitian parts,

$$\bar{A}\bar{B} = \tfrac{1}{2}\{\bar{A}, \bar{B}\} + \tfrac{1}{2}[\bar{A}, \bar{B}], \tag{29}$$

whose expectation values are respectively real and purely imaginary (the anticommutator of Hermitians is Hermitian; the commutator, anti-Hermitian — the same computation as in §2's "the i" note). A complex number's modulus dominates its imaginary part:

$$\big|\langle\psi|\bar{A}\bar{B}|\psi\rangle\big|^2 = \tfrac{1}{4}\big\langle\{\bar{A}, \bar{B}\}\big\rangle^2 + \tfrac{1}{4}\big|\big\langle[A,B]\big\rangle\big|^2 \;\geq\; \tfrac{1}{4}\big|\big\langle[A,B]\big\rangle\big|^2, \tag{30}$$

and chaining (28)–(30) and taking square roots gives (27). (Keeping the anticommutator term instead of discarding it strengthens the bound to the *Schrödinger* inequality; the discarded term measures the A–B correlations in the state.) □

Feed the canonical pair to the theorem: $[\hat{q}, \hat{p}] = \mathrm{i}\hbar\,\mathbf{1}$ has a *state-independent* expectation, so

$$\boxed{\Delta q\;\Delta p \;\geq\; \frac{\hbar}{2}} \tag{31}$$

for every state in the operators' domains (the theorem's parenthetical clause rides along) — the quantitative flesh on §2.3's skeleton, with the classical limit visible in the formula: the floor under the statistics is set by exactly the deformation parameter of Convention 1.1. For non-canonical pairs the right-hand side is an expectation value and therefore *state-dependent*:

$$\Delta S_x\,\Delta S_y \;\geq\; \tfrac{\hbar}{2}\,\big|\langle S_z\rangle\big|, \tag{32}$$

a weaker-looking statement with a physical point — there exist states ($\langle S_z\rangle = 0$, anywhere on the Bloch equator) where nothing prevents sharp S_x *and* S_y statistics from coexisting in the bound, and the bound correctly goes slack.

Example 4.6 (The pole saturates)**.** Run (32) on the laboratory. In the state $|+\rangle$ (north pole of Fig. 1): the Pauli identity (18) gives $S_x^2 = S_y^2 = \frac{\hbar^2}{4}\mathbf{1}$, while $\langle S_x\rangle = \langle S_y\rangle = 0$ (the matrices (17) are purely off-diagonal). So

$$\Delta S_x = \Delta S_y = \frac{\hbar}{2}, \qquad \Delta S_x\,\Delta S_y = \frac{\hbar^2}{4} = \frac{\hbar}{2}\,\big|\langle S_z\rangle\big| : \tag{33}$$

the inequality is *saturated* — the spin-up state is a minimum-uncertainty state of the transverse pair, its unavoidable transverse spread exactly the floor the algebra charges for a sharp S_z. (Geometrically: a Bloch vector pinned to the pole has its transverse components maximally undetermined — the sphere picture and the algebra agree.)

What saturation requires in general. Equality in (27) needs equality at both inequalities of the proof: Cauchy–Schwarz collapses only when the two vectors are parallel, $\bar{B}|\psi\rangle = c\,\bar{A}|\psi\rangle$, and the discarded anticommutator term vanishes only when $\operatorname{Re} c = 0$ (substitute the parallelism into $\langle\{\bar{A}, \bar{B}\}\rangle = 2\operatorname{Re}(c)\,(\Delta A)^2$). Hence the *minimum-uncertainty condition*:

$$\big(\bar{B} - \mathrm{i}\lambda\,\bar{A}\big)|\psi\rangle = 0, \qquad \lambda \in \mathbb{R}. \tag{34}$$

For the canonical pair this is a first-order equation whose solution requires a representation to solve in — an IOU: §5 builds the position representation, solves (34) there in two lines, and finds the Gaussian wave packet: *the* state, unique up to translations, that sits exactly on the floor (31).

Remark 4.7 (Interpretive care: preparation statistics versus disturbance). Everything in (27) refers to *preparation statistics*: ΔA and ΔB are standard deviations of outcome distributions over many runs on identically prepared systems, each run measuring *one* observable. The theorem says incompatible statistics cannot both be sharp *in the same preparation* — it does not, by itself, quantify how measuring A disturbs a subsequent measurement of B on the same system (Heisenberg's original microscope heuristic). The disturbance question is real physics with its own modern formulations, and conflating the two readings is a standard misstatement of the uncertainty principle. [stated; error–disturbance relations are a living literature.]

Remark 4.8 (The counterexample and where the proof leans). Now the stress test, following Hall §12.2. Work on $L^2[-1,1]$ with the bounded "position" $A\psi(x) = x\,\psi(x)$ and the "momentum" $B = -\mathrm{i}\hbar\,\mathrm{d}/\mathrm{d}x$ on continuously differentiable functions with *periodic* boundary values $\psi(-1) = \psi(1)$ — a legitimate, essentially self-adjoint operator (its eigenvectors $\psi_n = \mathrm{e}^{\mathrm{i}\pi n x}/\sqrt{2}$, $n \in \mathbb{Z}$, form an orthonormal basis). On a dense domain one checks $AB - BA = \mathrm{i}\hbar$, exactly the canonical value. Yet take $|\psi\rangle = \psi_n$: it is an eigenvector of B, so $\Delta B = 0$, while ΔA is finite — and

$$\Delta A\,\Delta B = 0\ <\ \frac{\hbar}{2}\,, \tag{35}$$

an apparent violation. The resolution is the parenthetical clause of Theorem 4.5: the proof applies B after A, and $A\psi_n = x\,\psi_n$ *violates the periodic boundary condition* ($x\,\psi_n$ takes different values at ± 1), so $\psi_n \notin \operatorname{Dom}(BA)$ — the state sits exactly outside the hypotheses, and the inequality never applied to it. Two morals, one local, one global. Local: the physical system here is a particle on a *ring*, A is (essentially) an angle and B an angular momentum, and the lesson is that a bounded, periodic coordinate admits no naive uncertainty relation with its conjugate momentum — a fact that returns, constructively, when §13 builds angular momentum. Global: this is the promised second bite of Remark 3.3 — domains are not decoration, and the hypotheses of a theorem are part of its physics. [Example verified against Hall §12.2.]

Kinematics still lacks its most-used representation: the one in which states are *functions*, $\psi(x)$, and (34) becomes a differential equation. Building it — and deriving $\hat{p} = -\mathrm{i}\hbar\,\partial_x$ within it — is the next section's business.

5 Position, Momentum, and Translation

Kinematics closes with its most-used representation: states as functions. Building it means confronting an operator whose spectrum is a continuum — which imports the delta apparatus of the field-theory notes — and the payoff is structural: the canonical commutator, *postulated* in §2 as a deformation of the classical bracket, will here be *derived* a second, independent way, from the geometry of translations. Two roads, one commutator; their convergence is the first entry of the quantum Noether dictionary.

5.1 The position basis and its fine print

A particle on a line has a position operator $\hat{q}$ whose possible sharp values fill $\mathbb{R}$: a continuous spectrum. Write formal eigenkets $\hat{q}|x\rangle = x|x\rangle$ and trade Theorem 3.4's sums for integrals, replacing the Kronecker delta of orthonormality by its continuum original:

$$\langle x'|x\rangle = \delta(x - x'), \qquad \boxed{\mathbf{1} = \int \mathrm{d}x\, |x\rangle\langle x|.} \tag{36}$$

The delta function arrives with the license already issued in [Fld. §10.2, App. B]: not a function but a distribution, every manipulation below the shadow of a legitimate operation on smeared objects — the identical fiction, imported wholesale. The fine print: $|x\rangle$ itself is *not* a vector of the Hilbert space (its "norm" is $\delta(0)$); it is a computational scaffold, and the physically meaningful objects are the coefficients it produces,

$$\psi(x) \equiv \langle x|\psi\rangle, \qquad \langle\psi|\psi\rangle = \int \mathrm{d}x\, \left|\psi(x)\right|^2 = 1, \tag{37}$$

— the *wavefunction*: the state's components in the position basis, exactly as $\langle a|\psi\rangle$ were its components in a discrete one. Born's rule (24) reads through the trade: with no normalizable eigenvector, $|\psi(x)|^2$ is a probability *density*, its integral over a region the probability of finding the particle there. [The mathematically clean route is the spectral theorem without eigenvectors, Hall Ch. 10; rigged Hilbert spaces legitimizing $|x\rangle$ directly are a second route, outside Hall; stated, and the working notation is safe for everything this document does.]

5.2 Translation and the operator it generates

Now the geometry. Define the *translation operator* by its action on the position basis,

$$T(a)\,|x\rangle = |x + a\rangle, \tag{38}$$

extended by linearity. It is unitary (it maps the orthonormal continuum (36) onto itself), composes as the line's additive group, $T(a)\,T(b) = T(a + b)$, $T(0) = \mathbf{1}$, and carries one commutator on its face: on any basis ket,

$$\hat{q}\,T(a)|x\rangle = (x + a)|x + a\rangle, \qquad T(a)\,\hat{q}|x\rangle = x|x + a\rangle \qquad \Longrightarrow \qquad [\hat{q}, T(a)] = a\,T(a). \tag{39}$$

Look at an infinitesimal translation. Unitarity and smoothness force the form

$$T(\mathrm{d}a) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\,\hat{p}\,\mathrm{d}a + O(\mathrm{d}a^2), \tag{40}$$

which *defines* an operator $\hat{p}$ — the *generator* of translations — Hermitian because T is unitary (expand $T^\dagger T = \mathbf{1}$ to first order: $\hat{p}^\dagger = \hat{p}$), with the $\hbar$ inserted so that $[\hat{p}] = [\text{action}]/[\text{length}] = [\text{momentum}]$: the name is a promise, about to be kept. Insert (40) into (39) and keep first order:

$$-\frac{\mathrm{i}}{\hbar}\,[\hat{q}, \hat{p}]\,\mathrm{d}a = \mathrm{d}a\,\mathbf{1} \qquad \Longrightarrow \qquad \boxed{[\hat{q}, \hat{p}] = \mathrm{i}\hbar\,\mathbf{1}.} \tag{41}$$

Remark 5.1 (Two roads to one commutator: the quantum Noether dictionary). Equation (41) is the canonical commutation relation (5) again — reached by a completely different road. Section 2 *deformed* the classical bracket ($\{q, p\} = 1$, Dirac's rule); this section never mentioned a bracket, only the statement *"$\hat{p}$ is what translates."* That the two roads meet is not a coincidence to admire but a dictionary to use: in mechanics, Noether traded each symmetry for a conserved quantity, and momentum *was* the translation entry [Mech. §3.3], as it was again for fields [Fld. §9.4]. Quantum mechanics keeps the trade and sharpens it: *the conserved quantity is the symmetry's generator*, as an operator —

symmetry	generator	where
spatial translation	$\hat{p}$	here
time translation	$\hat{H}$	§6
rotation	$\hat{J}_i$	§11

— and the commutators among generators are the transcription of the classical bracket algebra, one theorem per row ahead.

5.3 $\hat{p}$ in the position representation

What does $\hat{p}$ do to wavefunctions? Compute $\langle x|T(\mathrm{d}a)|\psi\rangle$ twice. Geometrically: acting leftward, $\langle x|T(\mathrm{d}a) = \langle x - \mathrm{d}a|$ (the adjoint of $T(\mathrm{d}a)|x\rangle = |x + \mathrm{d}a\rangle$, since $T^\dagger(\mathrm{d}a) = T(-\mathrm{d}a)$), so

$$\langle x|T(\mathrm{d}a)|\psi\rangle = \psi(x - \mathrm{d}a) = \psi(x) - \mathrm{d}a\,\frac{\partial\psi}{\partial x} + O(\mathrm{d}a^2). \tag{42}$$

Algebraically, via (40): $\langle x|T(\mathrm{d}a)|\psi\rangle = \psi(x) - \frac{\mathrm{i}}{\hbar}\,\mathrm{d}a\,\langle x|\hat{p}|\psi\rangle$. Equate the first orders:

$$\boxed{\langle x|\,\hat{p}\,|\psi\rangle = -\,\mathrm{i}\hbar\,\frac{\partial}{\partial x}\,\psi(x)} \tag{43}$$

— the textbook formula for momentum: here not a postulate but the unique answer to "what must the translation generator look like on functions that get dragged." Hermiticity in this representation is one integration by parts, $\int \psi_1^*(-\mathrm{i}\hbar\,\partial_x\psi_2)\,\mathrm{d}x = \int(-\mathrm{i}\hbar\,\partial_x\psi_1)^*\psi_2\,\mathrm{d}x$ + boundary terms — and the boundary terms are precisely where the potholes of Remarks 3.3 and 4.8 lived: on the full line with decay they vanish and all is well; on intervals, they were the whole story.

The eigenvalue problem of (43) is immediate: $-\mathrm{i}\hbar\,\partial_x\langle x|p\rangle = p\langle x|p\rangle$ gives

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}}\,\mathrm{e}^{\mathrm{i}px/\hbar}, \qquad \langle p'|p\rangle = \frac{1}{2\pi\hbar}\int \mathrm{d}x\,\mathrm{e}^{\mathrm{i}(p-p')x/\hbar} = \delta(p - p') \tag{44}$$

(the normalization check is the Fourier representation of the delta: the $\sigma \to \infty$ limit of the Gaussian pair (178), read distributionally under the license of [Fld. App. B]). A momentum eigenstate is a plane wave of wavelength

$$\lambda = \frac{2\pi\hbar}{p} = \frac{h}{p} \tag{45}$$

— de Broglie's relation, here neither postulate nor analogy but the shape of the translation generator's eigenfunctions.

5.4 The momentum representation and the Fourier bridge

The momentum eigenkets are delta-normalized and complete in their own right, so a state has momentum-basis components $\varphi(p) \equiv \langle p|\psi\rangle$, and the bridge between the two descriptions is one insertion of (36):

$$\varphi(p) = \int \mathrm{d}x\,\langle p|x\rangle\,\langle x|\psi\rangle = \int \frac{\mathrm{d}x}{\sqrt{2\pi\hbar}}\,\mathrm{e}^{-\mathrm{i}px/\hbar}\,\psi(x) \tag{46}$$

— the Fourier transform, with its inverse running the other way. This is Remark 3.5 delivered in full: position and momentum representations are the two adapted coordinate systems ($\hat{q}$ diagonal versus $\hat{p}$ diagonal), the change between them is a basis change like any other, and the celebrated transform is nothing but $\langle p|x\rangle$ used as a transition matrix. In the momentum representation the roles swap symmetrically: $\hat{p}$ multiplies and $\langle p|\hat{q}|\psi\rangle = +\mathrm{i}\hbar\,\partial_p\varphi(p)$ (same derivation, sign from conjugating (44)).

5.5 The Gaussian minimum-uncertainty packet

Section 4.3 left one item open: the minimum-uncertainty condition (34), waiting for a representation to be solved in. In the position representation, with $\bar{p} = \hat{p} - p_0$ and $\bar{q} = \hat{q} - x_0$ (expectations p_0, x_0), the condition $(\bar{p} - \mathrm{i}\lambda\bar{q})|\psi\rangle = 0$ reads

$$\Big(-\mathrm{i}\hbar\,\frac{\partial}{\partial x} - p_0 - \mathrm{i}\lambda\,(x - x_0)\Big)\,\psi(x) = 0 \qquad\Longrightarrow\qquad \frac{\partial\psi}{\partial x} = \Big[\frac{\mathrm{i}p_0}{\hbar} - \frac{\lambda}{\hbar}(x - x_0)\Big]\psi, \tag{47}$$

a first-order linear equation with a two-line solution:

$$\boxed{\psi(x) = N\,\exp\Big[\frac{\mathrm{i}p_0 x}{\hbar} - \frac{(x - x_0)^2}{4(\Delta x)^2}\Big], \qquad (\Delta x)^2 = \frac{\hbar}{2\lambda},} \tag{48}$$

normalizable precisely when $\lambda > 0$, with the variance read off the profile $|\psi|^2 \propto \exp[-(x - x_0)^2/2(\Delta x)^2]$ (integrals: Appendix A). No further computation is needed for the momentum spread: the state was *constructed* to saturate, so $\Delta p = \hbar/(2\,\Delta x)$ by Theorem 4.5's equality case — and the direct check confirms it: the Fourier bridge (46) of a Gaussian is again a Gaussian (Appendix A), with exactly the reciprocal width. *The Gaussian wave packet is the unique family of states sitting on the floor* $\Delta x\,\Delta p = \hbar/2$ (Fig. 2): squeeze the left panel and the right panel broadens in exact compensation, the deformation parameter $\hbar$ fixing the product.

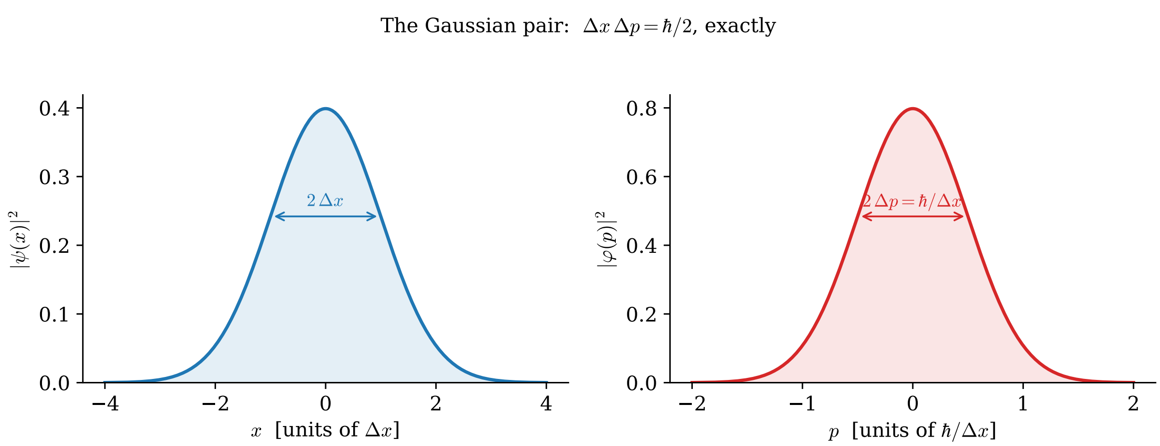

Figure 2: The minimum-uncertainty state (48) in both adapted coordinate systems, drawn from the exact formulas ($\hbar = 1$, $\Delta x = 1$, $x_0 = p_0 = 0$). The widths are reciprocal, $\Delta p = \hbar/(2\Delta x)$: narrowing the position distribution forces a wider momentum distribution, the product fixed at $\Delta x\,\Delta p = \hbar/2$.

Example 5.2 (Why atoms are the size they are)**.** The uncertainty floor is not philosophy; it is a pressure, and it holds atoms up — answering the first puzzle of §2.1. Confine an electron to an atomic length, $\Delta x \approx 0.5\,\text{Å} = 5 \times 10^{-11}$ m. The floor (31) forces

$$p \gtrsim \frac{\hbar}{2\,\Delta x} \approx 1.1 \times 10^{-24}\,\mathrm{kg\,m/s} \qquad \Longrightarrow \qquad E_{\mathrm{kin}} \sim \frac{p^2}{2m_e} \approx 6.1 \times 10^{-19}\,\mathrm{J} \approx 3.8\,\mathrm{eV}: \tag{49}$$

electron-volt-scale kinetic energy, forced by localization alone — the atomic energy scale, from nothing but $\hbar$, m_e, and a length. Squeezing the electron further toward the nucleus raises this cost as $1/(\Delta x)^2$, faster than the Coulomb energy $\sim -1/\Delta x$ can compensate; the balance point *is* the atom, and the classical spiral-in of §2.1 is arrested by the same inequality drawn in Fig. 2. (The balance is made exact, with the numerical constant earned, at the hydrogen capstone of §14.)

Part I is complete: an algebra (§2), a stage (§3), measurement rules (§4), and coordinates (§5). Nothing yet *moves*. Part II turns on time — and the first thing it will do is run the quantum Noether dictionary's second row.

Part II
Dynamics

6 Time Evolution

Nothing in Part I moved. This section turns on time, and the structure is deliberately a rerun: exactly as §5.2 derived the momentum operator from "something must implement spatial shifts," here the Hamiltonian is derived from "something must implement waiting" — which fills the second row of the quantum Noether dictionary (Remark 5.1). The section ends where quantum dynamics always ends: in the energy eigenbasis, where the whole of time evolution is a set of rotating phases — and with the series' laboratory precessing on the Bloch sphere — the working principle of MRI.

6.1 The evolution operator and where H comes from

Let $|\psi(t)\rangle$ be the state at time t of a system prepared as $|\psi(0)\rangle$. Two physical requirements pin down the map between them:

- **Probability is conserved**: $\langle\psi(t)|\psi(t)\rangle = 1$ at all times, for every initial state — the map $U(t)$ with $|\psi(t)\rangle = U(t)|\psi(0)\rangle$ must be *unitary* (§3.2: unitaries are exactly the inner product's isometries).
- **Waiting composes**: evolving for t_1 and then t_2 is evolving for t_1+t_2, and no moment is special (a closed system; explicit time dependence would encode an outside agent): $U(t_2)\,U(t_1) = U(t_1+t_2)$, $U(0) = \mathbf{1}$.

These are verbatim the properties that §5.2 used for spatial shifts — a one-parameter unitary group, now in time — so the same move applies: expand the infinitesimal element,

$$U(\mathrm{d}t) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\,\hat{H}\,\mathrm{d}t + O(\mathrm{d}t^2), \tag{50}$$

which *defines* a Hermitian operator $\hat{H}$ — the generator of time translations — with $\hbar$ inserted so that $[\hat{H}] = [\text{action}]/[\text{time}] = [\text{energy}]$. The name *Hamiltonian* is the dictionary's second row: in mechanics, the conserved quantity guaranteed by time-translation invariance *was* the Hamiltonian [Mech. §§3.3, 6.2], and quantum mechanics promotes that conserved quantity to the generator itself. [That every strongly continuous one-parameter unitary group has such a self-adjoint generator is Stone's theorem — Hall §10.2; stated, and it is the precise justification for (50).]

Differentiate the composition law at finite t — or simply apply (50) to $U(t+\mathrm{d}t) = U(\mathrm{d}t)\,U(t)$ — to get the operator equation and its state-level shadow:

$$\boxed{\mathrm{i}\hbar\,\frac{\partial}{\partial t}\,U(t) = \hat{H}\,U(t), \qquad \mathrm{i}\hbar\,\frac{\partial}{\partial t}\,|\psi(t)\rangle = \hat{H}\,|\psi(t)\rangle} \tag{51}$$

— the *Schrödinger equation*, obtained here as the differential form of "waiting is unitary and composes," with all its famous features inherited rather than postulated: linear (because U is), probability-conserving (because U is), and first order in time (because groups have one-parameter tangents). For an $\hat{H}$ with no explicit time dependence the solution closes:

$$U(t) = \exp\!\Big[-\frac{\mathrm{i}\,\hat{H}\,t}{\hbar}\Big]. \tag{52}$$

(Sandwiching (51) in the position basis, with the Hamiltonian $\hat{p}^2/2m + V(\hat{q})$ and the representation (43), produces the wave equation $\mathrm{i}\hbar\,\partial_t\psi = -\frac{\hbar^2}{2m}\partial_x^2\psi + V\psi$ — the form §8 will live in.)

6.2 The energy eigenbasis: dynamics is phases

Now the adapted-coordinates discipline (Remark 3.5), fourth appearance — and its most consequential. Diagonalize the operator that generates the dynamics: $\hat{H}|n\rangle = E_n|n\rangle$. In this basis (52) is diagonal, and every initial state evolves by

$$\boxed{|\psi(t)\rangle = \sum_n c_n\, \mathrm{e}^{-\mathrm{i}E_n t/\hbar}\,|n\rangle, \qquad c_n = \langle n|\psi(0)\rangle\,:} \tag{53}$$

the entire dynamics of quantum mechanics is a set of rotating phases, one per energy level, at rates set by the spectrum. Solving a quantum system *means* finding the spectrum; everything after is (53). Two immediate consequences, one per extreme:

- **One term: stationary states.** An energy eigenstate evolves as $\mathrm{e}^{-\mathrm{i}E_n t/\hbar}|n\rangle$ — an overall phase, physically empty (§3.1): every probability, every expectation value is frozen. Here, at last, is the full answer to §2.1's stability crisis: an atom in its ground state is not an electron *orbiting* — not a charge accelerating, not a system doing anything — it is a stationary state, and stationary states have no dynamics to radiate away.

- **Two terms: beats at the Bohr frequency.** A superposition $c_1|1\rangle + c_2|2\rangle$ has one *relative* phase — physical (§3.1) — rotating at

$$\omega_{21} = \frac{E_2 - E_1}{\hbar}, \tag{54}$$

  so every observable connecting the two levels oscillates at ω_{21}: the quantum system *beats*, exactly as two coupled classical modes did [Osc. §6.6], with energy differences playing the role of frequency differences. This is why atoms emit *lines* and why the lines sit at differences of a fixed set of terms (the Rydberg–Ritz combination principle, mysterious in 1908, one line in (54)): this answers the second experimental puzzle of §2.1 in structure; the numbers arrive at the hydrogen capstone. [One gap, flagged: an atom *prepared* in an excited stationary state also radiates — spontaneous emission — which no isolated-atom stationary state can explain; the mechanism is the coupling to the quantized field of the outlook, and the beat picture above is its semiclassical face. Stated.]

6.3 The laboratory precesses: spin in a magnetic field

Run the machinery end to end on the two-state laboratory. A spin-$\frac{1}{2}$ magnetic moment in a uniform field $\vec{B} = B\hat{z}$ has

$$\hat{H} = -\gamma B\,\hat{S}_z, \qquad E_\pm = \mp\frac{\gamma\hbar B}{2}, \qquad \omega \equiv \frac{E_- - E_+}{\hbar} = \gamma B \tag{55}$$

(γ the gyromagnetic ratio; the S_z eigenstates are the stationary states, and ω is the Bohr frequency of the only pair there is). Prepare the state at the equator, $|\psi(0)\rangle = |+\rangle_x = (|+\rangle + |-\rangle)/\sqrt{2}$, and apply (53):

$$|\psi(t)\rangle = \frac{1}{\sqrt{2}}\Big(\mathrm{e}^{\mathrm{i}\omega t/2}\,|+\rangle + \mathrm{e}^{-\mathrm{i}\omega t/2}\,|-\rangle\Big), \tag{56}$$

which is the Bloch state (19) with $\theta = \frac{\pi}{2}$, $\varphi = -\omega t$: the expectation vector,

$$\langle S_x \rangle = \tfrac{\hbar}{2} \cos \omega t, \qquad \langle S_y \rangle = -\tfrac{\hbar}{2} \sin \omega t, \qquad \langle S_z \rangle = 0, \tag{57}$$

precesses around the field at the *Larmor frequency* $\omega = \gamma B$ — uniform circular motion of the Bloch vector on the equator of Fig. 1: the figure of §3.5, now a movie.

Example 6.1 (Larmor precession in the clinic: MRI)**.** For a proton, $\gamma_p/2\pi = 42.58\,\mathrm{MHz/T}$ [experimental input; CODATA]. In the 3 T field of a clinical MRI magnet, the hydrogen nuclei of the body's water precess at

$$f = \frac{\gamma_p B}{2\pi} = 42.58\,\mathrm{MHz/T} \times 3\,\mathrm{T} \approx 128\,\mathrm{MHz} \tag{58}$$

— a radio frequency. An MRI machine is a Larmor-precession transceiver: it tips the protons' Bloch vectors off axis, listens for the 128 MHz signal of their collective precession, and maps tissue by how the precession decays. A clinical scan is Eq. (53) run on $\sim 10^{27}$ copies of the two-state laboratory.

Remark 6.2 (The 2π seed)*.* Watch the state, not just its expectation vector, through one full turn. At $t = 2\pi/\omega$ the Bloch vector has returned — but the ket has not:

$$|\psi(2\pi/\omega)\rangle = \frac{1}{\sqrt{2}}\big(\mathrm{e}^{\mathrm{i}\pi}|+\rangle + \mathrm{e}^{-\mathrm{i}\pi}|-\rangle\big) = -\,|\psi(0)\rangle\ : \tag{59}$$

a full 2π of precession multiplies the state by -1, and only 4π restores it. For an isolated spin the sign is an overall phase, invisible (§3.1) — but a phase becomes *relative*, and hence measurable, the moment the spin is one arm of an interferometer. That experiment exists, with neutrons, and §12 owes the reader its number; the present remark is the seed: the -1 is not an artifact of this Hamiltonian but the signature of what kind of representation of the rotation group a spin-$\frac{1}{2}$ carries.

Remark 6.3 (Energy–time is not Robertson)*.* A caution the two-level beat invites. There is no time operator to feed Theorem 4.5 — in this formalism t is a parameter, not an observable [stated; the obstruction to a self-adjoint time conjugate to a bounded-below H is Pauli's observation], so "$\Delta E\,\Delta t \geq \hbar/2$" cannot mean what $\Delta q\,\Delta p$ means. The precise version is one line away. Differentiate an expectation value with (51):

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle A \rangle = \frac{\mathrm{i}}{\hbar}\big\langle\, [\hat{H}, \hat{A}]\,\big\rangle \qquad (A \text{ with no explicit } t\text{-dependence}), \tag{60}$$

then run Robertson on the pair $(\hat{A}, \hat{H})$ and substitute: $\Delta A\,\Delta E \geq \frac{\hbar}{2}\,|\mathrm{d}\langle A \rangle/\mathrm{d}t|$, i.e.

$$\Delta E\ \tau_A\ \geq\ \frac{\hbar}{2}, \qquad \tau_A \equiv \frac{\Delta A}{|\mathrm{d}\langle A \rangle/\mathrm{d}t|} \tag{61}$$

(Mandelstam–Tamm): τ_A is the time for the statistics of A to shift by one standard deviation, and the theorem says *states with sharp energy change slowly* — the stationary state ($\Delta E = 0$, nothing moves) is its extreme case, and a spectral line's width–lifetime relation its laboratory face. Equation (60) is more than this remark's tool: it is the quantum mirror of $\dot{F} = \{F, H\}$ [Mech. §8.2], about to be promoted from expectation values to the operators themselves — which is precisely the next section.

7 Two Pictures of One Physics

Remark 6.3 ended mid-promotion: the flow equation (60) moved *expectation values*, and the promise was to move the operators themselves. This section keeps it — and the payoff, for a reader of the mechanics notes: the quantum equation of motion *is* the classical bracket flow, transcribed by Dirac's rule with nothing lost in translation. The section then locates the classical limit (Ehrenfest), and closes by dissolving a historical rivalry into a choice of coordinates.

7.1 The Heisenberg picture: operators flow

Everything measurable is a sandwich, and a sandwich has two slices:

$$\langle\psi(t)|\,\hat{A}\,|\psi(t)\rangle = \langle\psi(0)|\,\underbrace{U^\dagger(t)\,\hat{A}\,U(t)}_{\equiv\ \hat{A}_H(t)}\,|\psi(0)\rangle. \tag{62}$$

Section 6 attached the evolution to the state (the *Schrödinger picture*); attach it instead to the operator — states frozen at $|\psi(0)\rangle$, observables carrying the time — and nothing measurable changes, because (62) is an identity. This is the *Heisenberg picture*, and its equation of motion is one derivative away. With $\hat{H}$ time-independent, $U(t) = \mathrm{e}^{-\mathrm{i}\hat{H}t/\hbar}$ commutes with $\hat{H}$, and differentiating $\hat{A}_H = U^\dagger \hat{A}\,U$ by the product rule (using (51) and its adjoint):

$$\frac{\mathrm{d}\hat{A}_H}{\mathrm{d}t} = \frac{\mathrm{i}}{\hbar}\,U^\dagger\,[\hat{H},\hat{A}]\,U = \frac{\mathrm{i}}{\hbar}\,[\hat{H},\hat{A}_H], \tag{63}$$

i.e., written to make the point unmissable:

$$\boxed{\frac{\mathrm{d}\hat{A}_H}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\,[\hat{A}_H,\,\hat{H}]} \qquad \text{versus} \qquad \frac{\mathrm{d}F}{\mathrm{d}t} = \{F,H\} \quad \text{[Mech. §8.2]}. \tag{64}$$

Dirac's rule (4) was installed in §2 as a statement about *kinematics* — replace the bracket's realization, keep its algebra. Equation (64) says the replacement extends to *dynamics* verbatim: the law of motion is the same sentence in the new realization. Three corollaries fall out at the speed of substitution:

- **Conservation is commutation.** $[\hat{A},\hat{H}] = 0$ freezes $\hat{A}_H$ for all time — the quantum Noether loop closes at the operator level: the generators of Remark 5.1's table are conserved exactly when their symmetry is a symmetry *of the dynamics*, and by §4.2 the conserved quantities are precisely the observables sharable with energy. A CSCO containing $\hat{H}$ is therefore the natural naming scheme for stationary states — the (n,ℓ,m) of the hydrogen capstone are exactly such a list.

- **The canonical structure rides along.** Unitary conjugation preserves commutators, so $[\hat{q}_H(t),\hat{p}_H(t)] = \mathrm{i}\hbar$ at every (equal) time — the quantum echo of the classical fact that time evolution preserves the fundamental brackets [time evolution is the flow generated by H, Mech. §8.3; that Hamiltonian flows preserve the fundamental brackets is part of the canonical-transformation theory the mechanics notes' outlook only points to — stated there, proved neither there nor here].

- **The algebra of §2.3 was built for this.** The derivative formulas (6) turn (64) into operator-valued classical mechanics, next.

7.2 Operator Newton: two flows run explicitly

Free particle, $\hat{H} = \hat{p}^2/2m$: the flow equations

$$\frac{\mathrm{d}\hat{p}_H}{\mathrm{d}t} = 0, \qquad \frac{\mathrm{d}\hat{q}_H}{\mathrm{d}t} = \frac{1}{\mathrm{i}\hbar}\Big[\hat{q}_H, \frac{\hat{p}_H^2}{2m}\Big] = \frac{\hat{p}_H}{m} \tag{65}$$

(the second by (6)) integrate to

$$\hat{q}_H(t) = \hat{q}(0) + \frac{\hat{p}(0)}{m}\, t \tag{66}$$

— the classical trajectory, *as an operator identity.* The quantum content hides in what the classical formula cannot say:

$$\big[\hat{q}_H(t),\, \hat{q}_H(0)\big] = \frac{t}{m}\,[\hat{p}(0), \hat{q}(0)] = -\,\frac{\mathrm{i}\hbar\, t}{m} \;\neq\; 0: \tag{67}$$

the position now and the position later are incompatible observables. By Theorem 4.5, a sharp position at $t = 0$ forces growing position spread later — $\Delta q(t)\,\Delta q(0) \geq \hbar t/2m$ — which is wave-packet spreading, obtained with no wavefunction in sight; the wave-mechanical version, with the full time profile, is §8's opening act.

General potential, $\hat{H} = \hat{p}^2/2m + V(\hat{q})$: the same two lines give

$$\frac{\mathrm{d}\hat{q}_H}{\mathrm{d}t} = \frac{\hat{p}_H}{m}, \qquad \frac{\mathrm{d}\hat{p}_H}{\mathrm{d}t} = -\,V'(\hat{q}_H) \qquad \Longrightarrow \qquad m\,\frac{\mathrm{d}^2\hat{q}_H}{\mathrm{d}t^2} = -\,V'(\hat{q}_H) \tag{68}$$

— Newton's second law, wearing hats.

7.3 Ehrenfest and where the classical world sits

Sandwich (68) in the (frozen) state:

$$m\,\frac{\mathrm{d}^2}{\mathrm{d}t^2}\,\langle\hat{q}\rangle = -\,\big\langle V'(\hat{q})\big\rangle \qquad \text{(Ehrenfest)}, \tag{69}$$

and resist the tempting misreading: the right side is $-\langle V'(\hat{q})\rangle$, *not* $-V'(\langle\hat{q}\rangle)$, and the two differ whenever the packet is wide enough to sample the potential's curvature — Taylor-expand about the center,

$$\big\langle V'(\hat{q})\big\rangle = V'(\langle q\rangle) + \tfrac{1}{2}\,V'''(\langle q\rangle)\,(\Delta q)^2 + \cdots, \tag{70}$$

so the center of a quantum state obeys classical mechanics exactly when (i) the potential is at most quadratic — $V''' = 0$: the free particle, the uniform field, and, decisively for §9, *the harmonic oscillator*, whose packet centers are perfectly classical — or (ii) the spread is negligible on the scale over which the force varies: the $S \gg \hbar$ regime of §2, now with its precise small parameter, $(\Delta q)^2 V'''/V'$. Classical mechanics is the dynamics of centers, valid where centers are all one can resolve.

Remark 7.1 (Matrix mechanics recovered). In the energy eigenbasis, the Heisenberg operator's matrix elements evolve by (53) applied twice:

$$\big(\hat{A}_H(t)\big)_{mn} = A_{mn}\,\mathrm{e}^{\mathrm{i}\omega_{mn} t}, \qquad \omega_{mn} = \frac{E_m - E_n}{\hbar}: \tag{71}$$

every observable is a table of numbers oscillating at the Bohr frequencies (54) — which is literally what Heisenberg wrote down in 1925, abstracted from the frequencies atoms were observed to emit, before wavefunctions existed. Matrix mechanics *is* the Heisenberg picture in the energy basis; the document's structural route has arrived at the historical starting point, from the other side.

7.4 One physics, a choice of bookkeeping

The formal content of the section fits in a table — where the time dependence is attached, everything else being forced by (62):

	Schrödinger picture	Heisenberg picture
states	$\lvert\psi(t)\rangle = U(t)\lvert\psi(0)\rangle$	frozen
operators	frozen	$\hat{A}_H(t) = U^\dagger \hat{A}\, U$
equation of motion	(51) on kets	(64) on operators
natural kin	wave mechanics (§5)	matrix mechanics (Remark 7.1)

Remark 7.2 (One theory, many coordinates — the dynamical instance). Remark 2.3 promised this moment: Heisenberg's matrix mechanics (1925) and Schrödinger's wave mechanics (1926) were briefly rival theories, with partisans; they are the two columns of the table, related by the unitary bookkeeping shift (62) — one theory, coordinatized twice, the rivalry dissolved by a change of variables. The pattern is the series' oldest (Remark 3.5), and it has one more consequence worth planting: when quantum mechanics meets relativity, it is the *Heisenberg* picture that survives gracefully — operators carrying the spacetime label, states carrying none, is exactly the form the quantization program of [Fld. §14.2] leads to once run in full, and the outlook carries this out. (A third attachment — splitting $\hat{H}$ and letting each piece carry part of the time — is the *interaction picture*, the natural home of time-dependent perturbation theory; it belongs to the sequel course and is flagged, not built. [stated.])

8 Stationary States in One Dimension

The energy eigenbasis is where dynamics lives (§6.2); this section actually finds it, for the potentials a line can hold — and it opens with a conservation law the document has met before, in relativistic dress. Everything here runs in the position representation: the Schrödinger equation of §6.1,

$$\mathrm{i}\hbar\,\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\,\frac{\partial^2\psi}{\partial x^2} + V(x)\,\psi, \tag{72}$$

with stationary states $\psi(x,t) = \mathrm{e}^{-\mathrm{i}Et/\hbar}\,u(x)$ and the time-independent problem $-\frac{\hbar^2}{2m}u'' + Vu = Eu$.

8.1 The probability current

Track the probability density $\rho = |\psi|^2$ in time, using (72) and its conjugate — the potential terms cancel (real V), the kinetic terms assemble into a total derivative:

$$\frac{\partial\rho}{\partial t} = \frac{\mathrm{i}\hbar}{2m}\Big(\psi^*\,\partial_x^2\psi - \psi\,\partial_x^2\psi^*\Big) = -\,\partial_x\underbrace{\Big[\frac{\hbar}{m}\,\mathrm{Im}\big(\psi^*\partial_x\psi\big)\Big]}_{\equiv\; j(x,t)}, \tag{73}$$

$$\boxed{\frac{\partial\rho}{\partial t} + \frac{\partial j}{\partial x} = 0, \qquad j = \frac{\hbar}{m}\,\mathrm{Im}\big(\psi^*\partial_x\psi\big).} \tag{74}$$

A continuity equation — and the reader of the field-theory notes has seen this exact machine: (74) is [Fld. §9.2]'s local conservation law, and its pedigree is the same too. The Schrödinger equation inherits the global phase symmetry $\psi \to \mathrm{e}^{\mathrm{i}\alpha}\psi$ of the complex scalar, the conserved current of that symmetry is exactly j, and probability conservation is *Noether's theorem for the phase* — the nonrelativistic shadow of the $U(1)$ current of [Fld. §9.3], with one instructive difference: there, j^0 was not positive definite and had to be read as charge; here $\rho = |\psi|^2 \geq 0$, and the probability reading Born installed is consistent — the very consistency that failed relativistically and forced the reinterpretation. (Global conservation, $\int \rho \,\mathrm{d}x = 1$ forever, was already guaranteed by unitarity in §6.1; (74) is the *local* refinement — probability does not teleport, it flows.)

Two immediate readings calibrate j. A plane wave $\mathrm{e}^{\mathrm{i}(px-Et)/\hbar}$ carries $j = \rho\, p/m = \rho v$ — density times velocity, as a current should. And in a *stationary* state $\partial_t \rho = 0$, so j is constant in x: for a bound state (u real up to a global phase, via the nondegeneracy of one-dimensional bound states[1]) that constant is zero, while for scattering states it is the nonzero number out of which the transmission coefficients of §8.5 are built.

8.2 The spreading of the free packet

Section 7.2 proved positions at different times incompatible, (67); here is the full time profile, and the Heisenberg picture delivers it without a single integral. From the operator trajectory (66), the variance at time t is

$$\big(\Delta q(t)\big)^2 = \Big\langle \big(\bar{q} + \tfrac{t}{m}\,\bar{p}\big)^2 \Big\rangle = (\Delta q_0)^2 + \frac{t^2}{m^2}(\Delta p_0)^2 + \frac{t}{m}\,\big\langle \{\bar{q}, \bar{p}\} \big\rangle, \tag{75}$$

all expectations in the frozen initial state. For the Gaussian packet (48) the cross term vanishes — elegantly: the minimum-uncertainty condition (34) makes $\langle \bar{q}\,\bar{p} \rangle = \mathrm{i}\hbar/2$ *purely imaginary*, so the anticommutator's (real) expectation is zero — and saturation fixes $\Delta p_0 = \hbar/2\Delta q_0$:

$$\boxed{\Delta q(t) = \Delta q_0 \sqrt{1 + \Big(\frac{\hbar\, t}{2m\,(\Delta q_0)^2}\Big)^2}} \tag{76}$$

— ballistic spreading at late times, at speed $\Delta p_0/m$: the packet's own momentum spread, in motion. The timescale $\tau = 2m(\Delta q_0)^2/\hbar$ is the $S/\hbar$ switch of §2 wearing dynamics: an electron localized to $1\,\text{Å}$ doubles its width in $\tau\sqrt{3} \approx 3 \times 10^{-16}\,\mathrm{s}$ — quantum delocalization is, for electrons, effectively instantaneous — while a microgram dust grain localized to a micron needs $\sim 10^6$ *years*: the same formula, thirty orders of magnitude apart, and the reason tables do not diffuse.

8.3 The infinite well: an old set of modes

Hard walls at $x = 0, L$ impose $u(0) = u(L) = 0$ (the boundary-condition discipline of Remark 3.3, here doing constructive work), and inside, $u'' = -k^2 u$ with $k = \sqrt{2mE}/\hbar$. Requiring the sine to fit both walls quantizes it:

$$\boxed{u_n(x) = \sqrt{\frac{2}{L}}\,\sin\frac{n\pi x}{L}, \qquad E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots} \tag{77}$$

(Fig. 3, left). Three readings, in ascending order of reach:

- **The floor is the uncertainty floor.** $E_1 > 0$: a confined particle cannot sit still, and the scaling is Example 5.2's — $\Delta x \lesssim L$ forces $E \gtrsim \hbar^2/8mL^2$, and the exact $\pi^2\hbar^2/2mL^2$ is that estimate with its constant earned.

[1] Three lines, so [verified here]: two solutions u_1, u_2 of $-\frac{\hbar^2}{2m}u'' + Vu = Eu$ at the same E have Wronskian $W = u_1 u_2' - u_2 u_1'$ with $W' = 0$, since both satisfy the same equation; a bound state and its derivative vanish at infinity, so $W \to 0$, hence $W \equiv 0$ — which integrates to $u_2 \propto u_1$.

- **Discreteness comes from confinement.** The continuum of free momenta became a ladder because the walls admit only fitting waves — the same mechanism, and in fact the same *modes*, as the pinned string: $\sin(n\pi x/L)$ is verbatim [Osc. §9.3]'s pinned-endpoint mode set. One eigenvalue problem, two theories sharing it: the string attaches a frequency $\omega_n \propto n$ (linear dispersion), the box an energy $E_n \propto n^2$ (quadratic) — the difference between the two columns of physics built on one column of mathematics.

- **Nodes count.** u_n has $n-1$ interior nodes — the ground state none, each excitation adding one — an instance of the general one-dimensional oscillation theorem [stated; Sturm–Liouville theory], useful as a free consistency check on any computed spectrum.

8.4 The finite well: counting bound states

Remark 8.1 (Parity). For a symmetric potential, $V(-x) = V(x)$, define the *parity operator* $(\hat{P}\psi)(x) \equiv \psi(-x)$: Hermitian, $\hat{P}^2 = \mathbf{1}$ (so eigenvalues ± 1), and $[\hat{H}, \hat{P}] = 0$ (the kinetic term is even in ∂_x, and V is even by hypothesis). A *nondegenerate* bound state u is therefore an eigenstate of $\hat{P}$ — two lines: $\hat{P}u$ solves the same eigenvalue problem at the same E, nondegeneracy forces $\hat{P}u = c\,u$, and $\hat{P}^2 = \mathbf{1}$ forces $c = \pm 1$. Every bound state of the well below is *even or odd*; this is why its catalogue splits into two families, and the same operator, promoted to $L^2(\mathbb{R}^3)$, returns as the $(-1)^\ell$ of the dipole selection rules (§16.5).

Soften the walls: $V = -V_0$ on $|x| < a/2$, zero outside. Bound states ($-V_0 < E < 0$) oscillate inside and decay outside, $u \sim \mathrm{e}^{-\kappa|x|}$ with $\kappa = \sqrt{2m|E|}/\hbar$, and matching u and u' at the edges turns the spectrum into a transcendental intersection problem (the even and odd families of Remark 8.1, $\kappa = k\tan(ka/2)$ and $\kappa = -k\cot(ka/2)$). Two facts survive the details and are worth owning:

- **A one-dimensional well always binds.** However shallow, the symmetric well holds at least one (even) bound state — the even matching curve always intersects once [stated; the graphical argument is two lines, and the fact fails in three dimensions, where shallow wells can be empty — a genuine dimension-dependence with physical consequences, e.g. for the deuteron].

- **The count is the well's "size in quanta."** The number of bound states is $N = 1 + \lfloor z_0/\pi \rfloor$ with $z_0 = \sqrt{2mV_0}\,a/\hbar$ [stated; read off the intersection graph]: the dimensionless depth z_0 — an action in units of $\hbar$, once again — measures how many fitting waves the well can hold.

8.5 Tunneling and the microscope built on it

Now the inverted problem: a barrier, $V = V_0 > 0$ on $[0, a]$, and a wave incident with $0 < E < V_0$ — classically, certain reflection. Quantum mechanically the interior solution does not vanish; it decays, $u \sim \mathrm{e}^{\pm\kappa x}$ with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, and a decaying tail that reaches the far edge leaks. Matching u, u' at both edges (the algebra is mechanical and not displayed) and defining the transmission coefficient as the ratio of §8.1's currents, $T = j_{\text{trans}}/j_{\text{inc}}$:

$$\boxed{T = \left[1 + \frac{V_0^2 \sinh^2(\kappa a)}{4E\,(V_0 - E)}\right]^{-1} \xrightarrow[\kappa a \gg 1]{} \frac{16\,E(V_0 - E)}{V_0^2}\,\mathrm{e}^{-2\kappa a}:} \tag{78}$$

through a thick barrier, transmission is *exponential* in the width, with decay rate 2κ set by the deficit $V_0 - E$ (Fig. 3, right — which also shows the above-barrier resonances, $T = 1$ whenever a whole number of half-wavelengths fits the barrier: the same fitting-waves mechanism as the box, working in transmission). Lowest checks: $a \to 0$ gives $T \to 1$ — no barrier, no reflection — and $E \to V_0$ gives the finite $T = [1 + z_0^2/4]^{-1}$ from both branches, the sinh and sin forms meeting continuously at the top.

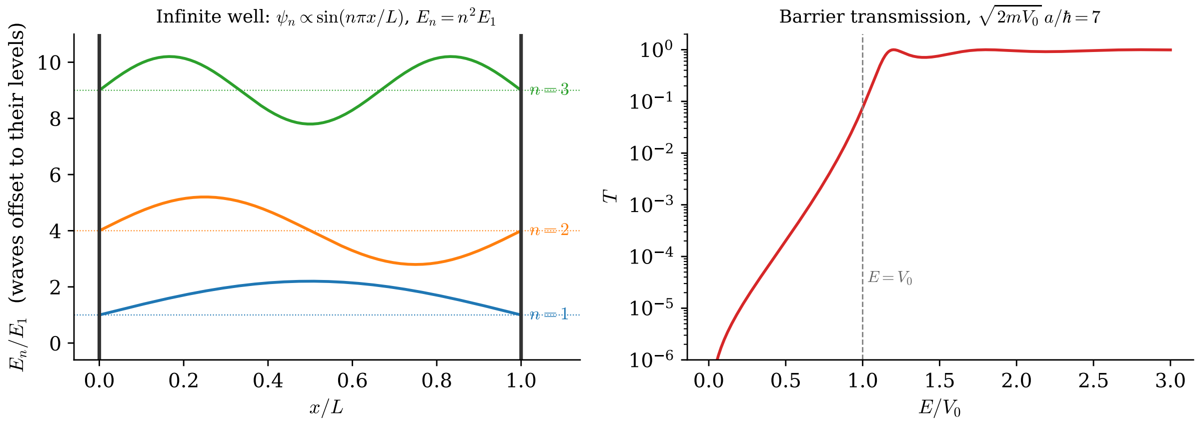

Figure 3: One dimension, both regimes, exact formulas. *Left*: the infinite well's first three stationary states (77), each drawn at its energy — the pinned string's mode shapes, now carrying quantum energies ($E_n \propto n^2$). *Right*: barrier transmission (78) (log scale) for $z_0 = \sqrt{2mV_0}\,a/\hbar = 7$: exponential tunneling below V_0, unit-transmission resonances above — fitting waves again.

Example 8.2 (The scanning tunneling microscope)**.** The exponential in (78) is an instrument. Between an STM tip and a metal surface, the vacuum gap is a barrier of height set by the work function, $\phi \approx 4.5\,\mathrm{eV}$ for typical metals, so

$$\kappa = \frac{\sqrt{2m_e\phi}}{\hbar} \approx 1.09\ \text{Å}^{-1}, \qquad \frac{T(a)}{T(a+1\,\text{Å})} = \mathrm{e}^{2\kappa\cdot 1\,\text{Å}} \approx 9: \tag{79}$$

the tunneling current changes by nearly an *order of magnitude per ångström* of tip height. That extreme sensitivity is the whole instrument: a feedback loop holding the current constant traces the surface to sub-atomic precision, and every STM image of individual atoms is Eq. (78)'s exponential read out as topography.

Remark 8.3 (Slow potentials: WKB in brief)*.* Between the exactly solvable and the numerical lies the *slowly-varying* potential, and the ansatz that owns it is worth one display even unproved:

$$u(x) \;\approx\; \frac{C}{\sqrt{p(x)}}\,\exp\!\Big[\pm\frac{\mathrm{i}}{\hbar}\int^{x} p(x')\,\mathrm{d}x'\Big], \qquad p(x) = \sqrt{2m\big(E - V(x)\big)} \tag{80}$$

— the phase is the *classical action* accumulating along the path, the amplitude $p^{-1/2}$ is flux conservation ((74): constant $j = \rho\,p/m$ forces $\rho \propto 1/p$), and the patching across turning points (where $p \to 0$ and the ansatz fails; the exact linear-potential solution, an Airy function, supplies the splice [stated; Hall Ch. 15]) yields the Bohr–Sommerfeld quantization $\oint p\,\mathrm{d}x = (n + \frac{1}{2})\,h$ of the old quantum theory [stated; Hall Ch. 15 gives the full treatment]. Why the action sits in the phase is not explicable at this level — but it is two sections ahead: §10 will show every amplitude is a sum of $\mathrm{e}^{\mathrm{i}S/\hbar}$ over paths, and (80) is what that sum looks like when one path dominates. The remark is a promise; §10 keeps it.

9 Quantizing the Harmonic Oscillator

The object that opened the series — one mass, one spring [Osc. §2] — makes its fifth appearance, and its most consequential: quantized here, it becomes in the outlook the atom of every quantum field. The solution method is pure algebra — no differential equation is solved until the wavefunctions are wanted — and the algebraic engine, a ladder built from the commutator, will be reused, unchanged, for angular momentum in §13.

9.1 Factorizing the Hamiltonian

The Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}\, m\omega^2 \hat{q}^2 \tag{81}$$

calls for factoring as a sum of squares, $u^2 + v^2 = (u - \mathrm{i}v)(u + \mathrm{i}v)$ — and the obstruction to doing so, $[\hat{q}, \hat{p}] \neq 0$, turns out to carry all of the problem's quantum content, as the derivation below makes explicit. Define the dimensionless combination (and its adjoint)

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}}\Big(\hat{q} + \frac{\mathrm{i}\,\hat{p}}{m\omega}\Big), \qquad \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\Big(\hat{q} - \frac{\mathrm{i}\,\hat{p}}{m\omega}\Big), \tag{82}$$

and multiply out, letting the cross terms manufacture the commutator:

$$\hat{a}^\dagger \hat{a} = \frac{m\omega}{2\hbar}\Big(\hat{q}^2 + \frac{\hat{p}^2}{m^2\omega^2}\Big) + \frac{\mathrm{i}}{2\hbar}\,[\hat{q}, \hat{p}] = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}, \qquad [\hat{a}, \hat{a}^\dagger] = 1 \tag{83}$$

(the second relation by the same two lines with the sign flipped). Hence

$$\boxed{\hat{H} = \hbar\omega\left(\hat{N} + \tfrac{1}{2}\right), \qquad \hat{N} \equiv \hat{a}^\dagger \hat{a}, \qquad [\hat{a}, \hat{a}^\dagger] = 1.} \tag{84}$$

Remark 9.1 (The classical amplitude promoted to an operator). Equation (82) is not a clever guess; it is the series' oldest move. From the classical oscillator's amplitude–phase solution $q = A\cos(\omega t - \phi)$ [Osc. §2.3], form the combination $z \equiv q + \mathrm{i}p/m\omega = q + \mathrm{i}\dot{q}/\omega$: one line gives $z = A\,\mathrm{e}^{-\mathrm{i}(\omega t - \phi)}$ [verified here] — a complex amplitude that rotates rigidly, $z(t) = z(0)\,\mathrm{e}^{-\mathrm{i}\omega t}$. The operator $\hat{a}$ *is* the promotion of that combination — and the Heisenberg flow confirms it inherits the rigid rotation: $\mathrm{d}\hat{a}_H/\mathrm{d}t = [\hat{a}_H, \hat{H}]/\mathrm{i}\hbar = -\mathrm{i}\omega\,\hat{a}_H$, so $\hat{a}_H(t) = \hat{a}\,\mathrm{e}^{-\mathrm{i}\omega t}$, verbatim the classical solution with a hat.

9.2 The spectrum from the algebra alone

Everything now follows from (84) — three steps, no analysis.

Step 1: the ladder. From $[\hat{N}, \hat{a}] = [\hat{a}^\dagger \hat{a}, \hat{a}] = [\hat{a}^\dagger, \hat{a}]\,\hat{a} = -\,\hat{a}$ (Leibniz, (3)), and its adjoint $[\hat{N}, \hat{a}^\dagger] = +\,\hat{a}^\dagger$: if $\hat{N}|\nu\rangle = \nu|\nu\rangle$, then

$$\hat{N}\left(\hat{a}|\nu\rangle\right) = (\nu - 1)\left(\hat{a}|\nu\rangle\right), \qquad \hat{N}\left(\hat{a}^\dagger|\nu\rangle\right) = (\nu + 1)\left(\hat{a}^\dagger|\nu\rangle\right): \tag{85}$$

$\hat{a}$ steps down the spectrum by one, $\hat{a}^\dagger$ up.

Step 2: the floor. Norms bound the ladder:

$$\big\|\hat{a}|\nu\rangle\big\|^2 = \langle\nu|\hat{a}^\dagger\hat{a}|\nu\rangle = \nu \;\geq\; 0. \tag{86}$$

Descending from any ν produces eigenvalues $\nu-1, \nu-2, \dots$, which (86) forbids from passing below zero — so the descent must *terminate*: some rung is annihilated, $\hat{a}|\nu_{\min}\rangle = 0$, and then (86) says $\nu_{\min} = 0$. Termination is only possible if the starting ν was a whole number of steps above zero:

$$\nu = n \in \{0, 1, 2, \dots\}. \tag{87}$$

Step 3: the bookkeeping. Fixing phases,

$$\boxed{E_n = \hbar\omega\Big(n + \tfrac{1}{2}\Big), \qquad \hat{a}|n\rangle = \sqrt{n}\,|n{-}1\rangle, \qquad \hat{a}^\dagger|n\rangle = \sqrt{n{+}1}\,|n{+}1\rangle, \qquad |n\rangle = \frac{(\hat{a}^\dagger)^n}{\sqrt{n!}}\,|0\rangle.} \tag{88}$$

The spectrum §2.1 demanded — energy in uniform lumps of $\hbar\omega$ — is now a theorem: the ladder has constant rungs because $[\hat{a}, \hat{a}^\dagger]$ is a *number*. [Completeness of the $|n\rangle$ — that the ladder built from $|0\rangle$ exhausts the Hilbert space — is true and quietly nontrivial; Hall §11.4. Stated.]

Remark 9.2 (The floor is the Gaussian again). The termination condition $\hat{a}|0\rangle = 0$ is, written out via (82), $(\hat{p} - \mathrm{i}\, m\omega\, \hat{q})|0\rangle = 0$ — which is *verbatim* the minimum-uncertainty condition (34) with $\lambda = m\omega$ and centers at the origin. The oscillator's ground state *is* the Gaussian of §5.5, width $(\Delta q)^2 = \hbar/2m\omega$, sitting exactly on the floor $\Delta q\, \Delta p = \hbar/2$; and the zero-point energy is that floor expressed as an energy — minimizing $\langle H\rangle = (\Delta p)^2/2m + \frac{1}{2}m\omega^2(\Delta q)^2$ subject to the uncertainty bound gives exactly $\hbar\omega/2$, the algebra's $\frac{1}{2}$ re-derived as a variational fact. The particle cannot sit at the bottom of the well for the same reason the atom of Example 5.2 cannot collapse: stillness is not an available state.

Example 9.3 (One molecule's quantum and frozen heat)**.** Carbon monoxide's vibrational frequency, spectroscopically $\tilde{\nu} = 2143\,\mathrm{cm}^{-1}$, puts its quantum at

$$\hbar\omega = hc\tilde{\nu} \approx 4.26 \times 10^{-20}\,\mathrm{J} \approx 0.266\,\mathrm{eV}, \qquad \left.\frac{\hbar\omega}{k_B T}\right|_{300\,\mathrm{K}} \approx 10.3: \tag{89}$$

ten times the thermal energy, so the Boltzmann weight of even the first excited rung is $\mathrm{e}^{-10.3} \approx 3 \times 10^{-5}$ — at room temperature the vibration is *frozen out*, the molecule locked to $n = 0$, and the vibrational degree of freedom silently withdraws from the specific heat. Classical equipartition's room-temperature failure — half of §2.1's second bullet — is this number; the first rung sits an order of magnitude above the thermal energy.

9.3 Wavefunctions from the ladder

When shapes are wanted, the algebra dictates them. The ground state solves $\hat{a}\, u_0 = 0$ in the position representation — a first-order equation, $(m\omega x + \hbar\, \partial_x)\, u_0 = 0$, whose solution is the Gaussian Remark 9.2 promised — and every excited state is $(\hat{a}^\dagger)^n$ applied to it: each application of $\hat{a}^\dagger \propto (m\omega x - \hbar\partial_x)$ multiplies by a polynomial of one higher degree, generating the Hermite family,

$$u_n(x) = \Big(\frac{m\omega}{\pi\hbar}\Big)^{1/4} \frac{1}{\sqrt{2^n n!}}\, H_n(\xi)\, \mathrm{e}^{-\xi^2/2}, \qquad \xi = x\sqrt{\frac{m\omega}{\hbar}} \tag{90}$$

[normalizations stated; the generation from $\hat{a}^\dagger$ is the proof strategy, and it also delivers n nodes for u_n — the oscillation theorem of §8.3 passing its check]. Figure 4 shows the family at its rungs, tails penetrating the classically forbidden region — tunneling's kinship (§8.5) — and, at $n = 20$, the correspondence principle in the flesh: the quantum density's ripples average to the classical dwell-time distribution $1/\pi\sqrt{A^2 - x^2}$, largest at the turning points where the classical particle lingers.

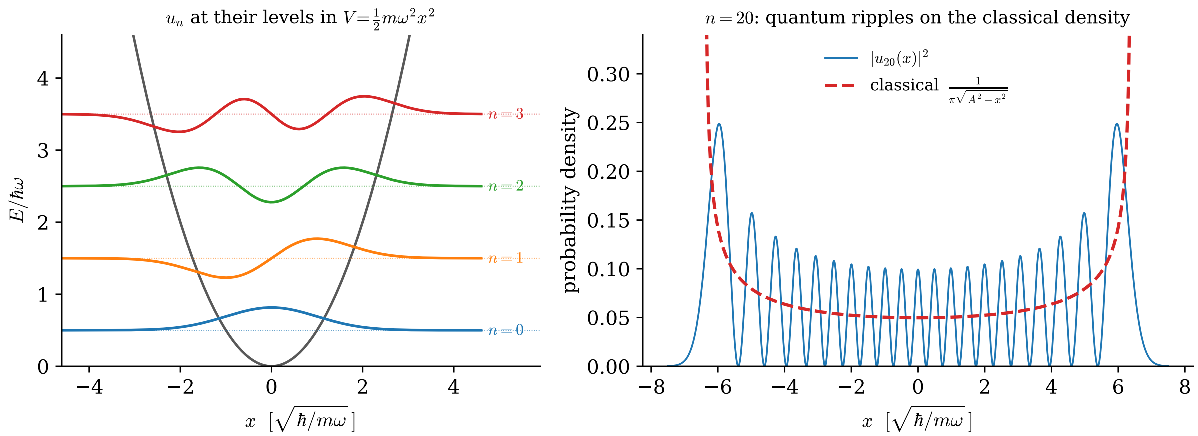

Figure 4: The quantized oscillator, exact formulas ($\hbar = m = \omega = 1$). *Left*: eigenfunctions (90), $n = 0$–3, at their levels $E_n = n + \frac{1}{2}$ in the potential — equal rungs, n nodes, Gaussian floor, forbidden-region tails. *Right*: at $n = 20$ the quantum density ripples about the classical dwell-time distribution (dashed), which diverges at the turning points $\pm A$, $A = \sqrt{2E_n}$ — the correspondence principle, drawn.

9.4 Coherent states: where the classical oscillator lives

Ehrenfest (§7.3) promised that quadratic potentials evolve packet centers exactly classically; here are the packets that take fullest advantage. Define *coherent states* as eigenstates of the (non-Hermitian — eigenkets are still legal) lowering operator,

$$\hat{a}\,|\alpha\rangle = \alpha\,|\alpha\rangle, \qquad \alpha \in \mathbb{C}, \qquad |\alpha\rangle = \mathrm{e}^{-|\alpha|^2/2}\sum_{n=0}^{\infty}\frac{\alpha^n}{\sqrt{n!}}\,|n\rangle \tag{91}$$

(the expansion follows by feeding $\sum c_n|n\rangle$ to the eigenvalue equation, whence $c_n = \alpha\,c_{n-1}/\sqrt{n}$ by (88)). Three properties make them the classical limit's residents:

- **They stay coherent, and α rotates classically.** Apply (53) to (91): the phases $\mathrm{e}^{-\mathrm{i}E_nt/\hbar}$ reassemble the same sum with $\alpha \to \alpha\,\mathrm{e}^{-\mathrm{i}\omega t}$ (times a global phase) — the label traces exactly the classical complex amplitude of Remark 9.1, so $\langle q\rangle(t)$ and $\langle p\rangle(t)$ run the classical trajectory *exactly*, at any amplitude.

- **They ride the floor.** A coherent state is the ground -state Gaussian rigidly displaced (in q and p by Re, Im of α), so it saturates $\Delta q\,\Delta p = \hbar/2$ at every instant [one-line check: $\hat{a}|\alpha\rangle = \alpha|\alpha\rangle$ *is* the minimum-uncertainty condition with shifted centers] — a wave packet that oscillates forever *without spreading*, the loophole to §8.2 that only the quadratic potential's rigidity permits.

- **They are what "classical amplitude A" means.** The excitation statistics are Poisson, $P(n) = \mathrm{e}^{-|\alpha|^2}|\alpha|^{2n}/n!$, with mean $\langle N\rangle = |\alpha|^2$: a macroscopic oscillation is a coherent state with $|\alpha|^2 \sim S_{\mathrm{cl}}/\hbar \gg 1$ quanta — the $S/\hbar$ switch of §2 counting rungs — and laser light is, to excellent approximation, this state of the electromagnetic field's modes. [stated; quantum optics' founding fact.]

One bookkeeping note closes the section, aimed at the outlook: the pair $(\hat{a}, \hat{a}^\dagger)$ with $[\hat{a}, \hat{a}^\dagger] = 1$ is the entire quantum content of *one* oscillator — and [Fld. §14.1] showed a free field is a countable family of them, one per wavevector. Quantizing the field will mean writing one ladder per mode, $[\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = \delta_{\vec{k}\vec{k}'}$, and reading $\hat{a}^\dagger_{\vec{k}}|0\rangle$ as *a particle.* Every technical prerequisite for that sentence is now on the table.

10 The Propagator by Two Routes

The field-theory notes could only *mark* the fork between the two quantizations — canonical, with its time slicing, and the path integral, built on the scalar action [Fld. §10.3]. This section walks both roads in full, at the one level where both are completely tractable, and verifies they arrive at the same place. Along the way it keeps the promise of Remark 8.3, answers the series' oldest open question — *why* nature extremizes actions — and returns the central object of the mechanics notes, the action, as a phase.

10.1 The object: the propagator and its composition law

Define the *propagator* as the open sandwich of §3.4(d) with the evolution inside:

$$\boxed{K(x_b, T;\, x_a, 0) \;\equiv\; \langle x_b|\, U(T)\, |x_a\rangle} \tag{92}$$

— the amplitude to be found at x_b after time T, having started at x_a. It carries the whole of dynamics: inserting $\mathbf{1} = \int \mathrm{d}x_a |x_a\rangle\langle x_a|$ into $|\psi(T)\rangle = U(T)|\psi(0)\rangle$,

$$\psi(x_b, T) = \int \mathrm{d}x_a\; K(x_b, T; x_a, 0)\; \psi(x_a, 0), \tag{93}$$

so solving for K solves every initial-value problem at once. And it obeys one structural identity, obtained by splitting $U(T) = U(T-t_c)\,U(t_c)$ and inserting a position resolution between the factors:

$$K(x_b, T; x_a, 0) = \int \mathrm{d}x_c\; K(x_b, T; x_c, t_c)\; K(x_c, t_c; x_a, 0) : \tag{94}$$

amplitudes for successive legs multiply; amplitudes for alternative waypoints add. Equation (94) looks like bookkeeping. Iterated, it is a theory (§10.3).

10.2 Route 1: the spectral representation

Canonically, K is computed by diagonalizing: insert the energy eigenbasis into (92),

$$K(x_b, T; x_a, 0) = \sum_n u_n(x_b)\, u_n^*(x_a)\; \mathrm{e}^{-\mathrm{i}E_nT/\hbar} \tag{95}$$

— the whole kernel, from the spectrum. For the free particle the sum is an integral over momentum eigenstates (44), and it is a Gaussian (Appendix A, in its Fresnel form (179)):

$$\begin{aligned} K_{\text{free}} &= \int \frac{\mathrm{d}p}{2\pi\hbar}\; \exp\Big[\frac{\mathrm{i}}{\hbar}\Big(p\,(x_b - x_a) - \frac{p^2T}{2m}\Big)\Big] \\ &= \sqrt{\frac{m}{2\pi\mathrm{i}\hbar T}}\; \exp\Big[\frac{\mathrm{i}}{\hbar}\,\frac{m\,(x_b - x_a)^2}{2T}\Big] \end{aligned} \tag{96}$$

(complete the square in p; the shifted Gaussian is (177)–(179)). Now look hard at the exponent. The quantity $m(x_b - x_a)^2/2T$ is $\frac{1}{2}mv^2 T$ for the uniform motion connecting the endpoints — it is *the classical action of the classical path*,

$$K_{\text{free}} = \sqrt{\frac{m}{2\pi \mathrm{i}\hbar T}}\; \mathrm{e}^{\,\mathrm{i}S_{\text{cl}}/\hbar}. \tag{97}$$

Nothing in Route 1 asked for the action; no Lagrangian was mentioned; and yet the action of [Mech.] stands in the exponent. Route 1 cannot explain this. Route 2 is the explanation.

10.3 Route 2: time slicing and the sum over paths

Iterate the composition law (94) $N-1$ times with uniform slices $\varepsilon = T/N$:

$$K(x_b, T; x_a, 0) = \int \mathrm{d}x_1 \cdots \mathrm{d}x_{N-1} \prod_{j=0}^{N-1} \langle x_{j+1}|\, U(\varepsilon)\, |x_j\rangle, \qquad x_0 = x_a,\ x_N = x_b. \tag{98}$$

Only the *short-time* amplitude is needed, and for $\hat{H} = \hat{p}^2/2m + V(\hat{q})$ the two terms can be exponentiated separately at cost $O(\varepsilon^2)$ per slice [stated; the Trotter product formula is the precise justification]. The kinetic factor is computed by inserting momentum states — the identical Gaussian as (96) with $T \to \varepsilon$ — and the potential factor is diagonal:

$$\langle x_{j+1}|U(\varepsilon)|x_j\rangle \approx \sqrt{\frac{m}{2\pi \mathrm{i}\hbar\varepsilon}}\; \exp\left[\frac{\mathrm{i}\varepsilon}{\hbar}\left(\frac{m}{2}\Big(\frac{x_{j+1}-x_j}{\varepsilon}\Big)^2 - V(x_j)\right)\right]. \tag{99}$$

Multiply the slices: the exponents *add*, and what they add up to is a Riemann sum —

$$\sum_j \varepsilon\Big[\frac{m}{2}\Big(\frac{\Delta x_j}{\varepsilon}\Big)^2 - V(x_j)\Big] \;\xrightarrow[N\to\infty]{}\; \int_0^T \mathrm{d}t\,\Big[\frac{m}{2}\,\dot{x}^2 - V(x)\Big] \;=\; S[x], \tag{100}$$

the *action* of [Mech. §3], kinetic minus potential, evaluated along the broken-line path through the waypoints. In the limit:

$$\boxed{K(x_b, T; x_a, 0) \;=\; \int \mathcal{D}x(t)\;\; \mathrm{e}^{\,\mathrm{i}S[x]/\hbar}} \tag{101}$$

— the *path integral*, with $\mathcal{D}x$ shorthand for the sliced measure and its prefactors [the continuum measure's existence is a genuine mathematical subtlety; stated, with the sliced definition as the operational meaning]. The reading (Fig. 5): *every* path from $(x_a, 0)$ to (x_b, T) contributes — jagged, backtracking, absurd ones included — all with the *same magnitude*, distinguished only by a phase, and the phase is the classical action in units of $\hbar$. Democracy of paths; the classical trajectory holds no privilege in the integrand.

10.4 Stationary phase: the origin of Hamilton's principle

Where does the classical world sit inside (101)? In the interference. When $S \gg \hbar$ — the switch of §2, one last time — the phase $S[x]/\hbar$ spins wildly under any deformation of the path, and contributions cancel in pairs *except* where the phase is stationary:

$$\delta S[x] = 0 \qquad \text{— Hamilton's principle.} \tag{102}$$

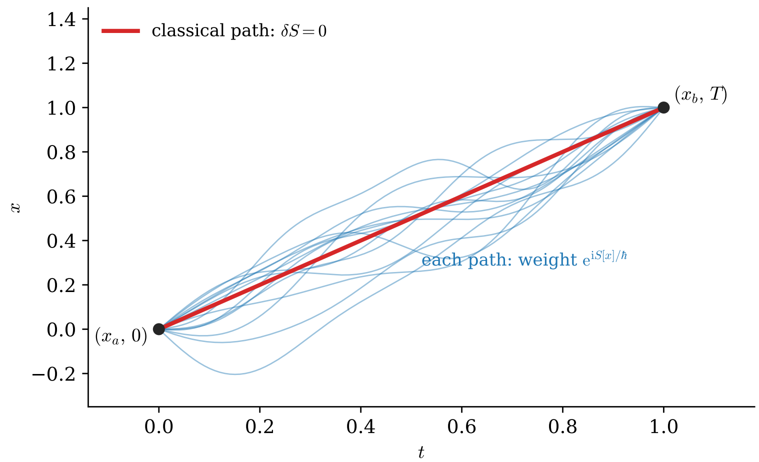

Figure 5: The sum over paths (101): every history from $(x_a, 0)$ to (x_b, T) carries the same weight in magnitude, phase $S[x]/\hbar$. The classical path (bold) is distinguished not by the integrand but by *interference*: it is where the phase is stationary, so neighboring paths add coherently instead of canceling.

The variational principle that the mechanics notes installed as their axiom [Mech. §3] is hereby demoted and thereby explained: *it is the stationary-phase approximation to the sum over histories* — nature does not prefer the extremal path; it takes all of them, and only near the extremum do the amplitudes survive their own interference. This answers the question the mechanics notes could only postpone — why extremizing S was ever the right principle — and it keeps Remark 8.3's promise: the WKB wavefunction (80), action in the phase and flux in the amplitude, is exactly what (101) looks like when one stationary path dominates and Gaussian fluctuations around it supply the prefactor.

For *quadratic* Lagrangians the story sharpens from approximation to identity: expanding $x(t) = x_{\mathrm{cl}}(t) + \eta(t)$, the action splits exactly — $S[x] = S_{\mathrm{cl}} + S_2[\eta]$ with no higher terms, the linear term absent by $\delta S = 0$ — and the fluctuation integral factors off as an endpoint-independent function of time:

$$K = F(T)\,\mathrm{e}^{\mathrm{i}S_{\mathrm{cl}}/\hbar} \qquad \text{(exactly, for quadratic } L\text{).} \tag{103}$$

That is why (97)'s exponent was *exactly* the classical action: the free particle is quadratic. So is the oscillator — the payoff:

Route 2 applied to the oscillator, with Route 1 as the check. The classical action of the harmonic path between fixed endpoints is a classical-mechanics exercise in [Mech.]'s toolkit:

$$S_{\mathrm{cl}} = \frac{m\omega}{2\sin\omega T}\Big[(x_a^2 + x_b^2)\cos\omega T - 2\,x_a x_b\Big], \tag{104}$$

and the fluctuation factor [computable as a product over the modes of S_2; stated — or fixed, more cheaply, by matching the $\omega \to 0$ free limit (96)] completes the kernel:

$$K_{\mathrm{SHO}} = \sqrt{\frac{m\omega}{2\pi\mathrm{i}\hbar\,\sin\omega T}}\;\exp\Big[\frac{\mathrm{i}}{\hbar}\,S_{\mathrm{cl}}\Big]. \tag{105}$$

Route 1 now checks Route 2. Set $x_a = x_b = x$ and integrate — the *trace* of the propagator, which by the spectral representation (95) must equal $\sum_n \mathrm{e}^{-\mathrm{i}E_nT/\hbar}$. The integral is one more Gaussian (exponent $\mathrm{i}m\omega\, x^2(\cos\omega T - 1)/\hbar\sin\omega T$, Appendix A), and with $1-\cos\omega T = 2\sin^2(\omega T/2)$:

$$\int \mathrm{d}x\; K_{\mathrm{SHO}}(x,T;x,0) = \frac{1}{2\mathrm{i}\,\sin(\omega T/2)} = \frac{\mathrm{e}^{-\mathrm{i}\omega T/2}}{1-\mathrm{e}^{-\mathrm{i}\omega T}} = \sum_{n=0}^{\infty} \mathrm{e}^{-\mathrm{i}\omega\left(n+\frac{1}{2}\right)T} \tag{106}$$

[the geometric series converges under the standard $T \to T(1-\mathrm{i}\epsilon)$ prescription; stated] — a geometric series whose exponents read off $E_n = \hbar\omega(n+\frac{1}{2})$: *the path integral knows the ladder's spectrum* (88), rung for rung, zero-point included. Two routes, one kernel, one spectrum; the fork's two roads, walked to the same destination and checked against each other.

Remark 10.1 (The fork revisited). [Fld. §10.3] could only gesture at what has just been done in full. The canonical route (Routes 1 here, Heisenberg/Schrödinger machinery throughout) slices time: unitarity is manifest, but a frame has been chosen — the same trade-off, at the same Legendre step, as in the mechanics and field notes. The path-integral route weights histories by the *scalar* $S[x]$: no slicing in the weight, no picture, no operator ordering — but unitarity is now a theorem to prove rather than a structure to see. At this document's level the choice is convenience; one level up, where S is the Lorentz scalar of [Fld. §8.1], the path integral is what keeps covariance visible while the canonical formalism keeps probability visible — and quantum field theory, needing both, keeps both formalisms and translates freely. The translation's proof of concept is this section: (95) and (101) are the same object, as (106) verifies.

Part II is complete: evolution, pictures, spectra, and the two grand formalisms reconciled on the series' own mascot. Part III turns from time to *space* — rotations, and everything the word "angular momentum" is about to mean.

Part III
Symmetry

11 Rotations and Angular Momentum

Part III is the symmetry part, and it opens by finishing a table. The quantum Noether dictionary (Remark 5.1) has run twice — translations yielded $\hat{p}$, waiting yielded $\hat{H}$ — and its third row, rotations, is richer than both: three generators instead of one, and, for the first time, generators that *do not commute with each other*. That non-commutativity is inherited straight from the geometry of turning, it reproduces the classical bracket algebra of the mechanics notes on cue, and it is loose enough — no **1** on its right-hand side — to admit the finite-dimensional representations that Theorem 2.2 exempted. The standing laboratory is about to receive its derivation.

11.1 Symmetries act unitarily

What should "the system, rotated" mean on Hilbert space? A map $|\alpha\rangle \to |\alpha'\rangle$ that preserves all physical predictions — all transition probabilities, $|\langle\beta|\alpha\rangle|^2$. A foundational theorem pins down what such maps can be:

Theorem 11.1 (Wigner)**.** *Any map of rays preserving all transition probabilities is implemented by an operator that is either unitary or antiunitary. [stated; the antiunitary option is realized in nature by time reversal, and is deferred with it.]*

Continuous symmetries connect to the identity, which is unitary, so each rotation R acts by a unitary $U(R)$, and composition respects the group — up to a phase, since states are rays (§3.1): $U(R_2)U(R_1) = \mathrm{e}^{\mathrm{i}\gamma}\, U(R_2R_1)$. That innocent-looking phase has consequences: for most groups it can be removed by redefinitions; for the rotation group the irreducible ambiguity is exactly a *sign* — the sign Remark 6.2 watched a precessing spin acquire — and §12 is where it surfaces. [stated at this level; the classification of the phases is the theory of projective representations.]

For an infinitesimal rotation by $\mathrm{d}\varphi$ about the axis $\hat{n}$, the third run of the generator move (§5.2, §6.1):

$$U(\hat{n}, \mathrm{d}\varphi) = \mathbf{1} - \frac{\mathrm{i}}{\hbar}\,\mathrm{d}\varphi\;\hat{n}\cdot\hat{\vec{J}} + O(\mathrm{d}\varphi^2), \tag{107}$$

defining three Hermitian generators $\hat{J}_x, \hat{J}_y, \hat{J}_z$ — the *angular momentum* operators, $[\hat{J}] = [\text{action}]$ this time, no $\hbar$-repair needed — and completing the dictionary's table. What remains is their algebra.

11.2 The algebra by both roads

The document's standing pattern — one commutator, two independent derivations — applies verbatim.

Road A: deform the classical bracket. The mechanics notes recorded the closed bracket of the classical angular momenta, $\{L_x, L_y\} = L_z$ [Mech. §8.2]; the full family $\{L_i, L_j\} = \epsilon_{ijk}L_k$ is the same two-line computation from the fundamental brackets, run for each pair [verify once]. Dirac's rule (4) transcribes in one line:

$$\boxed{[\hat{J}_i, \hat{J}_j] = \mathrm{i}\hbar\,\epsilon_{ijk}\,\hat{J}_k.} \tag{108}$$

Road B: read the geometry of turning. Rotations about different axes do not commute, and the failure is quantitative. For the 3×3 rotation matrices, a direct expansion to second order gives the classic identity

$$R_x(\varepsilon)\, R_y(\varepsilon)\, R_x(-\varepsilon)\, R_y(-\varepsilon) = R_z(\varepsilon^2) + O(\varepsilon^3) \tag{109}$$

— go around the little x–y commutator loop and you have turned, slightly, about z (verify once with the matrices: the $O(\varepsilon)$ terms cancel by construction, and the surviving ε^2 entries sit exactly in the xy block). Demand that the unitaries represent this, insert (107) on both sides, and match at order ε^2: the left side's surviving second-order term is $-[\hat{J}_x, \hat{J}_y]\, \varepsilon^2/\hbar^2$ (the $\hat{J}_x^2$ and $\hat{J}_y^2$ terms cancel pairwise between the $+\varepsilon$ and $-\varepsilon$ factors), the right side's is $-\mathrm{i}\hat{J}_z\, \varepsilon^2/\hbar$, whence $[\hat{J}_x, \hat{J}_y] = \mathrm{i}\hbar\, \hat{J}_z$ — (108) again, cyclically. The two roads' agreement is Remark 5.1's convergence, third instance: the bracket algebra of mechanics *was* the geometry of the symmetry group all along, and quantization preserves both because they are one.

Remark 11.2 (Why the laboratory was allowed). Look at the right-hand side of (108): operators, not $\mathrm{i}\hbar\,\mathbf{1}$. The trace obstruction of Theorem 2.2 evaporates — $\mathrm{tr}\ \epsilon_{ijk}\hat{J}_k = 0$ is perfectly satisfiable by traceless matrices — and finite-dimensional representations are *allowed*: the algebra itself does not fix the size of its stage, and §13 will find one stage per half-integer. The two-dimensional laboratory of §3.5 has operated on exactly this exemption since §2.3 first noted it; the exemption is now explained.

11.3 Orbital angular momentum: the classical entry

The oldest realization of (108) is the quantized classical angular momentum,

$$\hat{L}_i = \epsilon_{ijk}\, \hat{q}_j\, \hat{p}_k \tag{110}$$

(no ordering ambiguity: ϵ_{ijk} only pairs $j \neq k$, and $[\hat{q}_j, \hat{p}_k] = 0$ for $j \neq k$ — Dirac's rule applies without ambiguity). Verify the family with a single commutator:

$$\begin{aligned}[\hat{L}_x, \hat{L}_y] &= [\hat{q}_y\hat{p}_z - \hat{q}_z\hat{p}_y,\ \hat{q}_z\hat{p}_x - \hat{q}_x\hat{p}_z] \\ &= \hat{q}_y\hat{p}_x\,[\hat{p}_z, \hat{q}_z] + \hat{q}_x\hat{p}_y\,[\hat{q}_z, \hat{p}_z] = \mathrm{i}\hbar\left(\hat{q}_x\hat{p}_y - \hat{q}_y\hat{p}_x\right) = \mathrm{i}\hbar\,\hat{L}_z\end{aligned} \tag{111}$$

(only the terms sharing a z-pair survive; Leibniz (3) does the bookkeeping). And $\hat{L}$ is a generator in the concrete sense: in polar coordinates on the plane, the same two-line computation as §5.3 — drag the wavefunction, $\psi(\varphi - \mathrm{d}\varphi)$, match against the generator — gives

$$\langle \vec{x}|\, \hat{L}_z\, |\psi\rangle = -\,\mathrm{i}\hbar\, \frac{\partial}{\partial\varphi}\, \psi(\vec{x}) : \tag{112}$$

$\hat{L}_z$ is to the angle what $\hat{p}$ was to position — with the one crucial difference that the angle is *periodic*, which is about to quantize $\hat{L}_z$'s spectrum (§13) by the same fitting-waves mechanism as §8.3, and which is the constructive face of the bounded-coordinate warning in Remark 4.8.

11.4 Spin: internal rotation and the sign

Now the laboratory itself. Spin is angular momentum with no $\hat{q} \times \hat{p}$ inside — generators acting on an *internal* index of the state, touching the spatial argument not at all: the precise analogue of the internal–spacetime split that [Fld. §9.3] drew for field symmetries. For spin-$\frac{1}{2}$, $\hat{S}_i = \frac{\hbar}{2}\sigma_i$, and the Pauli identity (18)'s antisymmetric part *is* (108) — verified back in §3.5, now recognized as the rotation algebra in its smallest allowed representation (Remark 11.2).

The finite rotations come cheap, and they carry the section's punchline. Exponentiate the generator using the Pauli identity's symmetric part, $(\hat{n}\cdot\vec{\sigma})^2 = \mathbf{1}$: the power series splits into even and odd,

$$\boxed{U(\hat{n},\varphi) = \exp\Big[-\frac{\mathrm{i}\varphi}{2}\,\hat{n}\cdot\vec{\sigma}\Big] = \cos\frac{\varphi}{2}\,\mathbf{1}\; - \;\mathrm{i}\sin\frac{\varphi}{2}\,\hat{n}\cdot\vec{\sigma},} \tag{113}$$

— and set $\varphi = 2\pi$:

$$U(\hat{n},2\pi) = -\,\mathbf{1}. \tag{114}$$

A full turn is *minus* the identity — for any axis, any spin-$\frac{1}{2}$ state. Remark 6.2 watched one Hamiltonian produce this sign dynamically; (114) says the Hamiltonian was innocent: the sign belongs to the *representation*, hard-wired into (113)'s half-angles, which trace back through the exponential to the $\frac{\hbar}{2}$ in the generators — to what spin-$\frac{1}{2}$ *is*. It is also exactly the projective phase Theorem 11.1's fine print permitted: $U(2\pi)$ represents the same rotation as $U(0)$, differing by a sign that rays cannot see but interferometers can. What kind of mathematical object carries such a representation — and what experiment measures the sign — is the next section entire.

12 $SU(2)$, $SO(3)$, and the Double Cover

Section 11.4 left a sign in the room: $U(2\pi) = -\mathbf{1}$, hard-wired into the half-angles of (113). This section identifies the sign's origin — the states of a spin-$\frac{1}{2}$ do not carry the rotation group at all, but its *double cover* — explains why no redefinition can remove it, and then does what a physics document must: reports the experiment in which the sign was measured, interference fringes with a period of 720°.

12.1 Two groups for one geometry

The rotation group $SO(3)$ is an old acquaintance: the $\mathsf{G} = \mathbf{1}$, det $= +1$ member of the one-template family $\mathsf{M}^{\mathsf{T}}\mathsf{G}\,\mathsf{M} = \mathsf{G}$ [Fld. §3.1] — this subsection is that remark's quantum sequel. The new group is the one §11.4 was unknowingly computing in: $SU(2)$, the 2×2 unitaries of unit determinant. Its general element is exactly (113),

$$U = a\,\mathbf{1} + \mathrm{i}\,\vec{b}\cdot\vec{\sigma}, \qquad a,\vec{b}\in\mathbb{R}, \qquad a^2 + |\vec{b}|^2 = 1 \tag{115}$$

(unitarity plus $\det U = 1$ force real $(a,\vec{b})$ on the unit sphere) — so as a space, $SU(2)$ *is the three-sphere* S^3. The bridge to rotations is conjugation. Package a vector as a matrix, $X = \vec{x}\cdot\vec{\sigma}$, and let U act by

$$X \;\longrightarrow\; U\,X\,U^\dagger \;\equiv\; \vec{x}'\cdot\vec{\sigma}. \tag{116}$$

The map $\vec{x}\to\vec{x}'$ is linear, and it is a *rotation* by a two-line argument: $\det X = -|\vec{x}|^2$ (write it out from (17)), conjugation preserves determinants, so lengths are preserved; and $SU(2)$ being connected, the map connects to the identity — orientation preserved, $R(U)\in SO(3)$. Composition is automatic, $R(U_2U_1) = R(U_2)R(U_1)$. But the correspondence is not one-to-one:

$$\boxed{R(-U) = R(U)} \qquad \text{— two elements of } SU(2) \text{ per rotation,} \tag{117}$$

since the sign cancels between U and $U^\dagger$ in (116) (and $\pm U$ are the *only* such pairs: the kernel is the center $\{\pm\mathbf{1}\}$). $SU(2)$ is a *double cover* of $SO(3)$: the sphere S^3, wrapped twice around the space of rotations, antipodal points $U, -U$ landing on the same R. The spin-$\frac{1}{2}$ sign is now identified rather than mysterious: the state transforms under $SU(2)$; "rotate by 2π" is a path in $SO(3)$ that returns to the identity, but its *lift* to $SU(2)$ walks from $\mathbf{1}$ to the other preimage, $-\mathbf{1}$ — which re-derives Eq. (114) as a statement of geography.

Remark 12.1 (Why the sign cannot be redefined away). Could a cleverer assignment of phases make $U(2\pi) = +\mathbf{1}$? No, and the obstruction is topological. In $SO(3)$ the 2π-rotation loop *cannot be continuously contracted to a point* — the group is not simply connected, $\pi_1(SO(3)) = \mathbb{Z}_2$ [stated; the "belt trick" is the tabletop proof, and the 4π loop *does* contract] — so a representation is free to assign the loop a sign, and no continuous redefinition can change a discrete assignment. $SU(2) = S^3$ *is* simply connected: it is the universal cover, its true (non-projective) representations are exactly the projective representations Theorem 11.1's fine print allowed for $SO(3)$, and the classification splits cleanly in two — representations blind to the sign (the true $SO(3)$ representations: integer spin) and representations that carry it (half-integer spin), a dichotomy the algebra of §13 is about to reproduce from the other end. [The general theory — covering groups, projective representations — is stated at exactly this depth and no further.]

12.2 Measuring the sign: neutrons and $720°$

An overall sign on an isolated state is invisible (§3.1); the measurement therefore requires making the sign *relative*, as Remark 6.2's closing sentence anticipated. Split a beam in two, rotate the spin on *one* arm, recombine:

$$I \;\propto\; \big\||\psi\rangle + U(\hat{n},\varphi)|\psi\rangle\big\|^2 = 2 + 2\,\mathrm{Re}\langle\psi|\,U(\hat{n},\varphi)\,|\psi\rangle \;\overset{(113)}{=}\; 2\Big(1 + \cos\frac{\varphi}{2}\Big) \tag{118}$$

(for $|\psi\rangle$ an eigenstate of $\hat{n}\cdot\vec{\sigma}$; the general case only adds contrast factors). The fringe pattern is periodic in the rotation angle with period

$$\varphi_{\text{period}} = 4\pi = 720°, \tag{119}$$

and at $\varphi = 2\pi$ — one full rotation — the two arms interfere *destructively*: the state rotated by a full turn is minus itself, measurably.

Example 12.2 (The experiment and its numbers)**.** The experiment was run in 1975 (Rauch and collaborators; Werner and collaborators, independently) with thermal neutrons in a single-crystal silicon interferometer — the beam split and recombined by Bragg diffraction, one arm threading a magnet. The rotation is Larmor precession (§6.3): a neutron of thermal speed $v \approx 2200\,\mathrm{m/s}$ crossing a field region of length $\ell \approx 2\,\mathrm{cm}$ precesses by $\varphi = \gamma_n B\,\ell/v$, and with $|\gamma_n| = 1.83\times10^8\,\mathrm{rad/(s\,T)}$ [experimental input] the field needed for one *fringe* period ($\varphi = 4\pi$) is

$$B_{4\pi} = \frac{4\pi\,v}{|\gamma_n|\,\ell} \approx 7.5\,\mathrm{mT} \tag{120}$$

— laboratory-bench scale, and sweepable. Sweeping it, the measured intensity oscillated with a period of $\approx 716° \pm 4°$ [experimental input; consistent with $720°$, the small offset understood from field inhomogeneity], and emphatically not $360°$: the neutron count itself, plotted against the precession angle, is a graph of Eq. (118) — a direct measurement of the minus sign in (114). A rotation by 2π is not nothing; it is -1, and a silicon crystal can tell.

12.3 Finite rotations tabulated: the $\mathscr{D}$-matrices

For bookkeeping that §16.3 will reference — the finite form of the tensor-operator definition — name the matrix elements of a finite rotation in a given representation. Parametrize rotations by Euler angles, $U(\alpha,\beta,\gamma) = \mathrm{e}^{-\mathrm{i}\alpha J_z/\hbar}\,\mathrm{e}^{-\mathrm{i}\beta J_y/\hbar}\,\mathrm{e}^{-\mathrm{i}\gamma J_z/\hbar}$, and define

$$\mathscr{D}^{(j)}_{m'm}(\alpha,\beta,\gamma) \;\equiv\; \langle j,m'|\,U(\alpha,\beta,\gamma)\,|j,m\rangle \tag{121}$$

— the *Wigner $\mathscr{D}$-functions*, one unitary matrix per rotation per representation (the labels (j, m) are §13's to construct; the definition is parked here with the group, where it belongs). For the one representation already in hand, everything is explicit: with $J_z = \frac{\hbar}{2}\sigma_z$, $J_y = \frac{\hbar}{2}\sigma_y$ and the half-angle formula (113),

$$\mathscr{D}^{(1/2)}(\alpha, \beta, \gamma) = \begin{pmatrix} \mathrm{e}^{-\mathrm{i}(\alpha+\gamma)/2} \cos\frac{\beta}{2} & -\,\mathrm{e}^{-\mathrm{i}(\alpha-\gamma)/2} \sin\frac{\beta}{2} \\ \mathrm{e}^{\mathrm{i}(\alpha-\gamma)/2} \sin\frac{\beta}{2} & \mathrm{e}^{\mathrm{i}(\alpha+\gamma)/2} \cos\frac{\beta}{2} \end{pmatrix}, \tag{122}$$

half-angles throughout — the double cover legible in every entry ($\alpha \to \alpha + 2\pi$ multiplies the whole matrix by -1, Eq. (114) again). The higher-j families follow the same definition once §13 builds their stages, and they are the rotation group's complete answer to "how does anything transform"; §16.3 consumes them as the finite form of the tensor-operator definition.

The group theory is settled: two groups for one geometry, and the sign has been identified, measured, and tabulated. What has *not* been done is find the representations themselves — which pairs (j, m) exist, and why j comes in halves. That is pure ladder algebra, the oscillator's argument reused, and it is next.

13 The Angular Momentum Spectrum from the Algebra

Sections 11 and 12 settled which group acts and what its topology permits; this section finds the stages — every Hilbert space the algebra (108) can act on irreducibly — using nothing but the algebra itself. The engine is a ladder bounded by norms, the oscillator's argument (§9.2) reused; the output is the celebrated (j, m) catalogue, half-integers included; and the closing subsection explains why *orbital* angular momentum, for all its seniority, realizes only half the catalogue.

13.1 The commuting set and the ladder

The three generators do not commute with each other, so pick the largest compatible set (§4.2). The rotation-invariant combination $\hat{\vec{J}}^2 = \hat{J}_x^2 + \hat{J}_y^2 + \hat{J}_z^2$ commutes with all three components —

$$[\hat{\vec{J}}^2, \hat{J}_z] = \sum_i \big(\hat{J}_i\,[\hat{J}_i, \hat{J}_z] + [\hat{J}_i, \hat{J}_z]\,\hat{J}_i\big) = \mathrm{i}\hbar \sum_{i,k} \epsilon_{izk}\big(\hat{J}_i\hat{J}_k + \hat{J}_k\hat{J}_i\big) = 0 \tag{123}$$

(Leibniz, then the sym-against-antisym lemma of [Fld. §3.7]: the antisymmetric ϵ_{izk} contracts a product symmetrized in $i \leftrightarrow k$ to zero — a one-line lemma the series uses repeatedly) — so $\{\hat{\vec{J}}^2, \hat{J}_z\}$ is the working pair. Label the joint eigenstates

$$\hat{\vec{J}}^2|\lambda, m\rangle = \hbar^2\lambda\,|\lambda, m\rangle, \qquad \hat{J}_z|\lambda, m\rangle = \hbar m\,|\lambda, m\rangle, \tag{124}$$

and build the ladder from the transverse pair:

$$\hat{J}_\pm \equiv \hat{J}_x \pm \mathrm{i}\,\hat{J}_y, \qquad [\hat{J}_z, \hat{J}_\pm] = \pm\hbar\,\hat{J}_\pm, \qquad [\hat{J}_+, \hat{J}_-] = 2\hbar\,\hat{J}_z, \qquad [\hat{\vec{J}}^2, \hat{J}_\pm] = 0 \tag{125}$$

(each a one-line consequence of (108)). The first relation is the oscillator's $[\hat{N}, \hat{a}^\dagger] = \hat{a}^\dagger$ wearing a z: $\hat{J}_\pm$ shifts m by ±1; the last says it does so *within* a fixed λ — the ladder climbs a finite tower, if the norms allow.

13.2 The norm constraint and the catalogue

They allow only so much. Multiply out (using (125)):

$$\hat{J}_{\mp}\hat{J}_{\pm} = \hat{J}_x^2 + \hat{J}_y^2 \pm \mathrm{i}\,[\hat{J}_x, \hat{J}_y] = \hat{\vec{J}}^2 - \hat{J}_z^2 \mp \hbar\,\hat{J}_z, \tag{126}$$

so the squared norms of the rungs above and below are

$$\big\|\hat{J}_{\pm}|\lambda, m\rangle\big\|^2 = \hbar^2\left[\lambda - m(m \pm 1)\right] \;\geq\; 0. \tag{127}$$

Unlike the oscillator's one-sided constraint (86), this one squeezes from *both* ends: large $|m|$ of either sign violates it. So the ladder must terminate both ways — there is a top rung killed by $\hat{J}_+$ and a bottom rung killed by $\hat{J}_-$ — and (127) fixes them:

$$\lambda = m_{\max}(m_{\max}+1) \quad \text{and} \quad \lambda = m_{\min}(m_{\min}-1) \qquad \Longrightarrow \qquad m_{\min} = -\,m_{\max} \equiv -j \tag{128}$$

(the quadratic's other root, $m_{\min} = m_{\max} + 1$, contradicts $m_{\min} \leq m_{\max}$). The ladder runs from $-j$ to $+j$ in unit steps, so $2j$ is a nonnegative *integer* — and that is the entire derivation:

$$\boxed{\begin{aligned} &\hat{\vec{J}}^2 :\; \hbar^2\, j(j+1), \qquad j \in \left\{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, 2, \dots\right\}, \\ &\hat{J}_z :\; \hbar m, \qquad m \in \{-j, -j{+}1, \dots, +j\} \quad \text{— } 2j+1 \text{ states}, \\ &\hat{J}_{\pm}|j,m\rangle = \hbar\sqrt{j(j+1) - m(m\pm 1)}\;|j, m{\pm}1\rangle. \end{aligned}} \tag{129}$$

[The derivation fixes the coefficient's modulus; taking it real and positive is a phase convention — Condon–Shortley, the one App. B uses.]

Three readings of the box:

- **The halves are half the answer.** Nothing was assumed but the commutators, and the algebra delivered integer *and* half-integer j on equal footing — the half-integers are not an exotic appendix but fifty percent of the catalogue, and §12 has already identified them: they are precisely the representations that ride the double cover and carry the sign (114). Setting $j = \frac{1}{2}$ in (129) reproduces the standing laboratory from nothing — two states, $\hat{J}_z$ eigenvalues $\pm\hbar/2$, and the $\hat{J}_{\pm}$ matrix elements assemble exactly $\frac{\hbar}{2}\sigma_{x,y}$ — a derivation of the Pauli matrices that §3.5 could only posit.

- **Same engine, different termination.** The oscillator's norm constraint (86) bounded the ladder below only — one termination, infinite tower, one representation. Here (127) bounds both ends — two terminations that must be an integer number of rungs apart, *finite* towers, one per j: the finite-dimensionality that Remark 11.2 allowed is here explicitly constructed. One lemma of bookkeeping distinguishes a particle count from a spin.

- **The catalogue is complete.** Every irreducible representation of the algebra is one of (129)'s towers [stated; the standard induction], so the catalogue is exhaustive: this *is* what angular momentum can be.

13.3 Orbital angular momentum realizes only the integers

Now the realization of §11.3. In the position representation $\hat{L}_z = -\mathrm{i}\hbar\,\partial_\varphi$ (112), so its eigenfunctions are $\mathrm{e}^{\mathrm{i}m\varphi}$ — and the coordinate φ is an *angle*: the harvest of the periodicity planted in §11.3 and flagged in Remark 4.8. A wavefunction on physical space takes one value per point, $\psi(\varphi+2\pi) = \psi(\varphi)$, and the fitting-waves mechanism of §8.3 — walls there, closure here — quantizes

$$m \in \mathbb{Z}: \tag{130}$$

half-integer m would make $\mathrm{e}^{\mathrm{i}m\varphi}$ double-valued, not a function on the circle at all. Orbital angular momentum therefore realizes only the integer-j towers of (129). [Caveat: as stated this is the quick argument, and its edge — *why* must ψ be single-valued when phases are cheap? — deserves acknowledgment. The airtight version: $\hat{\vec{L}} = \hat{\vec{q}} \times \hat{\vec{p}}$ acts on $L^2(\mathbb{R}^3)$, and that space decomposes into integer representations only, the spherical harmonics below being a complete angular basis; the half-integer towers simply do not occur in it. Half-integer angular momentum is not a funny wavefunction — it requires an *internal* index, which is what spin is. Stated, with the decomposition as the citation-level fact.]

The integer towers, realized as functions on the sphere, are the *spherical harmonics* $Y_{\ell m}(\theta,\varphi) = \langle\theta,\varphi|\ell,m\rangle$: joint eigenfunctions of $\hat{\vec{L}}^2$ (eigenvalue $\hbar^2\ell(\ell+1)$) and $\hat{L}_z$ ($\hbar m$), constructible by the ladder itself — kill the top, $\hat{L}_+ Y_{\ell\ell} = 0$, a first-order equation giving $Y_{\ell\ell} \propto \sin^\ell\theta\,\mathrm{e}^{\mathrm{i}\ell\varphi}$, and descend with $\hat{L}_-$ — the oscillator's $(\hat{a}^\dagger)^n|0\rangle$ construction, sideways. The low family explicitly:

$$Y_{00} = \frac{1}{\sqrt{4\pi}}, \qquad Y_{10} = \sqrt{\frac{3}{4\pi}}\cos\theta, \qquad Y_{1,\pm1} = \mp\sqrt{\frac{3}{8\pi}}\sin\theta\,\mathrm{e}^{\pm\mathrm{i}\varphi} \tag{131}$$

[normalizations stated; completeness on the sphere stated] — $Y_{10} \propto \cos\theta = z/r$ is chemistry's p_z orbital, and Fig. 6 draws the angular shapes through $\ell = 2$: ℓ counts total angular nodes, $|m|$ tilts them between polar and equatorial, and the φ-independence of every $|Y_{\ell m}|^2$ is $\hat{L}_z$-eigenstate stationarity made visible (§6.2: sharp m, uniformly distributed conjugate angle).

The kinematics of rotation is now complete: the group (§11), its cover (§12), and every stage it can play on (129), with the spherical harmonics as the integer towers' concrete realization. What is missing is a *dynamics* to put on the stage — a Hamiltonian that respects the rotations — and the payoff for choosing one is the hydrogen atom, the document's capstone.

14 Central Potentials and Hydrogen

The document's threads converge here. The stage is §13's; the boundary-condition discipline and the one-dimensional intuition are §8's; the naming scheme is §7.1's CSCO; and the open puzzle is §2.1's — an atom that should not exist, emitting discrete lines with no classical origin. This section builds the hydrogen atom from the Coulomb potential and answers that puzzle in nanometers.

14.1 Central potentials: three dimensions become one

Let $V = V(r)$: rotationally invariant, so $[\hat{H}, \hat{\vec{L}}] = 0$ — angular momentum is conserved (§7.1), and $\{\hat{H}, \hat{\vec{L}}^2, \hat{L}_z\}$ is a commuting set: the states can carry the labels (E,ℓ,m) jointly. Separate accordingly. In the position representation the kinetic operator splits into radial and angular parts,

$$-\hbar^2\nabla^2 = -\hbar^2\,\frac{1}{r}\,\frac{\partial^2}{\partial r^2}\,r \;+\; \frac{\hat{\vec{L}}^2}{r^2} \tag{132}$$

[the identity is a coordinate computation; stated, with the angular part recognizable as $\hat{\vec{L}}^2$ from (112) and its siblings], so the ansatz $\psi = R(r)\,Y_{\ell m}(\theta,\varphi)$ absorbs the angular dependence entirely

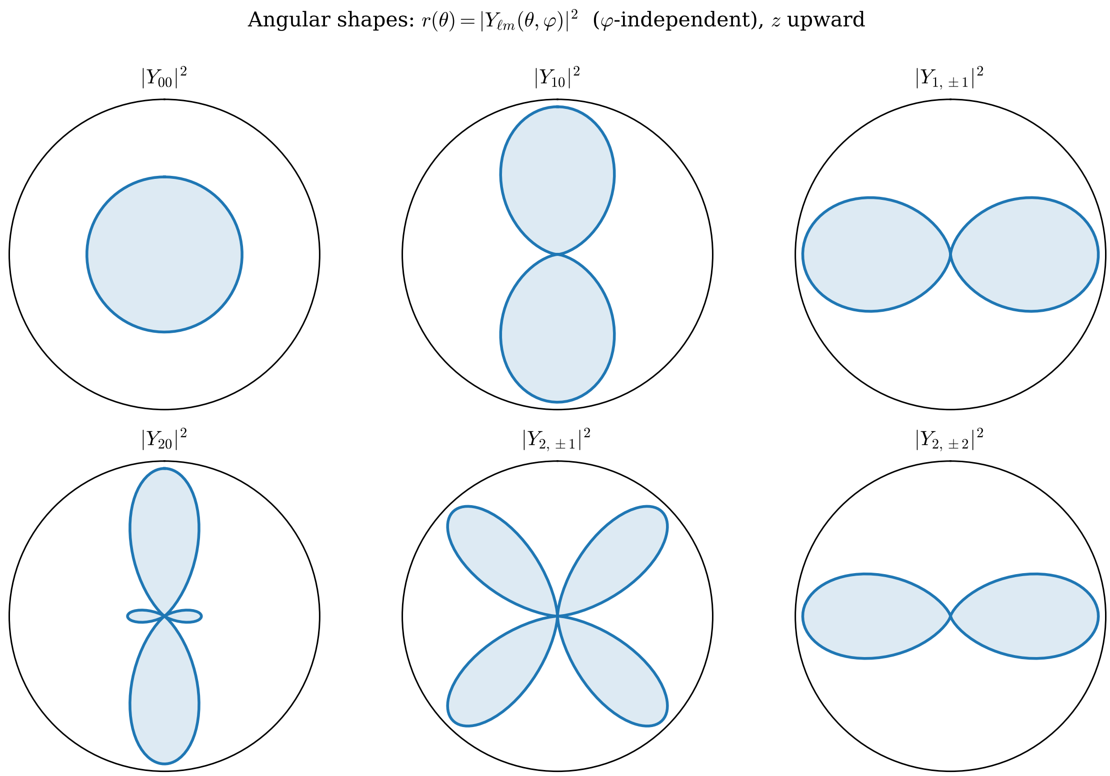

Figure 6: The integer catalogue on the sphere: angular probability profiles $|Y_{\ell m}(\theta)|^2$ through $\ell = 2$, exact closed forms, z upward ($|Y_{\ell m}|^2$ is φ-independent, so the planar cut is the whole story). ℓ sets the angular structure; m distributes it between pole and equator — $|m| = \ell$ hugs the equator, $m = 0$ the axis.

into §13's eigenvalue $\hbar^2\ell(\ell+1)$, and the substitution $u(r) = r\,R(r)$ lands on

$$\boxed{-\frac{\hbar^2}{2m}\,u'' + \Big[V(r) + \frac{\hbar^2\,\ell(\ell+1)}{2m\,r^2}\Big]u = E\,u, \qquad u(0) = 0} \tag{133}$$

— a *one-dimensional* Schrödinger problem on the half-line: the whole of §8 applies wholesale, with two new fixtures. The boundary condition $u(0) = 0$ is a hard wall at the origin ($R = u/r$ must stay finite — the quick reason; the airtight one: $u(0) \neq 0$ gives $R \sim 1/r$, and $\nabla^2(1/r) = -4\pi\,\delta^3(\vec{x})$ plants a delta source the equation cannot balance [stated]), so the origin is §8.3's wall relocated; and the potential has acquired the *centrifugal barrier* — the quantum version of the classical $\ell^2/2mr^2$ of the central-force reduction [Mech. §3.3], with $\ell(\ell+1)$ where ℓ^2 might have been guessed (the ladder's $j(j+1)$, (129), insists). States of higher ℓ are pushed off the origin; only $\ell = 0$ states touch the nucleus.

14.2 Hydrogen: scales first, then the spectrum

The Coulomb problem is $V(r) = -e^2/r$, with $e^2 \equiv q_e^2/4\pi\epsilon_0$ as shorthand. Before solving anything, do what the house style demands: ask what the parameters $(\hbar, m, e^2)$ can build. Exactly one length and one energy can be built,

$$a_0 = \frac{\hbar^2}{me^2} \approx 0.529\,\text{Å}, \qquad \text{Ry} = \frac{me^4}{2\hbar^2} = \frac{e^2}{2a_0} \approx 13.61\,\text{eV} \tag{134}$$

— and these are Example 5.2's balance point made exact: the radius where localization cost ($\sim \hbar^2/2ma_0^2$) and Coulomb gain ($\sim e^2/a_0$) compete on equal terms. The atom's size was fixed by dimensional analysis before a single equation; the equation's job is the numerical coefficients and the *pattern* of levels.

Solve (133) in units $\rho = r/a_0$, $\epsilon = E/\text{Ry}$. The two ends dictate asymptotics — decay $u \sim \mathrm{e}^{-\sqrt{-\epsilon}\,\rho}$ at infinity (bound state), $u \sim \rho^{\ell+1}$ at the origin (imposed by the centrifugal wall) — so peel both off, $u = \rho^{\ell+1}\,\mathrm{e}^{-\sqrt{-\epsilon}\,\rho}\,w(\rho)$, and feed w a power series. The recursion that falls out relates consecutive coefficients by

$$\frac{c_{k+1}}{c_k} = \frac{2\big[\sqrt{-\epsilon}\,(k+\ell+1) - 1\big]}{(k+1)(k+2\ell+2)} \xrightarrow[k\to\infty]{} \frac{2\sqrt{-\epsilon}}{k}, \tag{135}$$

and the large-k tail is the series of $\mathrm{e}^{+2\sqrt{-\epsilon}\,\rho}$: a non-terminating series overwhelms the decay and the state fails to normalize — the same normalizability constraint that bounded §9.2's ladder. Survival requires *termination*: some numerator must vanish, $\sqrt{-\epsilon}\,(n_r + \ell + 1) = 1$ for a nonnegative integer n_r, i.e., with $n \equiv n_r + \ell + 1$:

$$\boxed{E_n = -\frac{\text{Ry}}{n^2}, \qquad n = 1, 2, 3, \ldots, \qquad \ell \in \{0, 1, \ldots, n-1\}, \qquad \text{degeneracy } \sum_{\ell=0}^{n-1}(2\ell+1) = n^2.} \tag{136}$$

Example 14.1 (Hydrogen's spectral lines in nanometers)**.** Section 2.1 opened with spectral lines the classical atom could not have; §6.2 showed lines live at Bohr frequencies between levels; here are the levels, so here are the lines. Between $n = 2$ and $n = 1$:

$$\Delta E = \text{Ry}\Big(1 - \tfrac{1}{4}\Big) = 10.20\,\text{eV} \qquad \Longrightarrow \qquad \lambda = \frac{hc}{\Delta E} = 121.5\,\text{nm} \tag{137}$$

— Lyman-α, the ultraviolet workhorse of astronomy. Between 3 and 2: $\Delta E = 1.89\,\mathrm{eV}$, $\lambda = 656\,\mathrm{nm}$ — Hα, *the* red line of emission nebulae, now a two-line computation. The whole zoo compresses to the Rydberg formula $1/\lambda = R_\infty(1/n_1^2 - 1/n_2^2)$ with $R_\infty = \mathrm{Ry}/hc = 1.097 \times 10^7\,\mathrm{m}^{-1}$ — which derives the empirical rule of 1888, instantiates the Rydberg–Ritz combination principle (§6.2), and answers the document's opening puzzle in full, in the observed decimal places. [Scope: nonrelativistic, spinless electron, infinitely heavy proton — the reduced-mass correction is a factor $(1 + m_e/m_p)^{-1} \approx 1 - 1/1836$, already visible in precision data, and fine structure needs the tools of §§15–16 plus relativity; stated.]

14.3 The radial catalogue

Termination at $n_r = n - \ell - 1$ makes w a polynomial of that degree (an associated Laguerre polynomial [normalization stated]), so the radial function $R_{n\ell}$ has exactly $n - \ell - 1$ nodes — the oscillation theorem of §8.3 passing its check in the radial column. The low family, explicitly ($\rho = r/a_0$, in units $a_0^{-3/2}$):

$$R_{10} = 2\,\mathrm{e}^{-\rho}, \qquad R_{20} = \frac{2-\rho}{2\sqrt{2}}\,\mathrm{e}^{-\rho/2}, \qquad R_{21} = \frac{\rho}{2\sqrt{6}}\,\mathrm{e}^{-\rho/2}, \tag{138}$$

and Fig. 7 draws the radial probability densities $P_{n\ell} = r^2|R_{n\ell}|^2$ through $n = 3$. Three features to read off: the ground state peaks at exactly $r = a_0$ (differentiate $\rho^2\mathrm{e}^{-2\rho}$); size grows quadratically, $\langle r\rangle \sim n^2 a_0$ — Rydberg atoms are *large*; and at fixed n, higher ℓ means fewer radial nodes but a stronger centrifugal push off the origin — the $(3,0)$ state touches the nucleus, the $(3,2)$ state cannot, a distinction that controls everything from hyperfine structure to which orbitals chemistry builds bonds with.

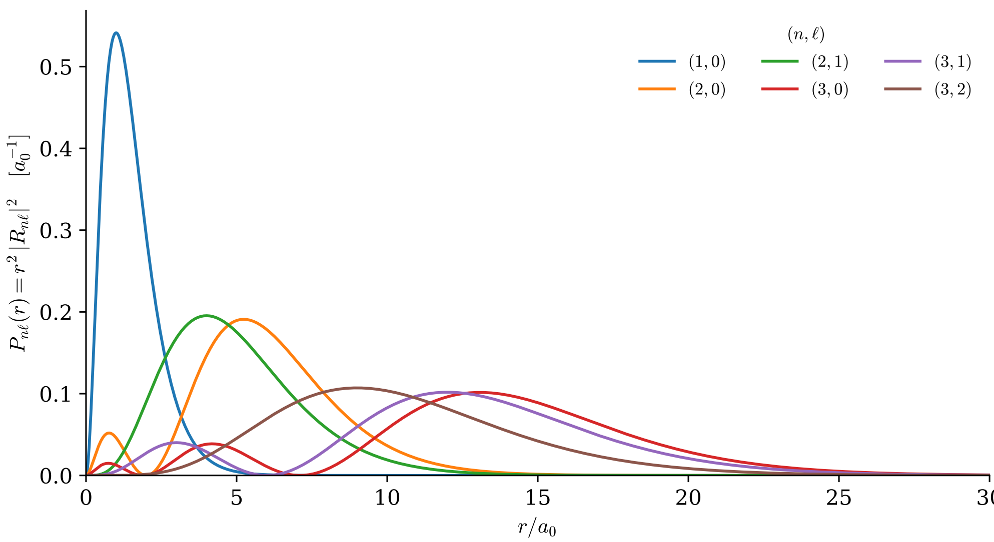

Figure 7: Hydrogen's radial probability densities $P_{n\ell}(r) = r^2|R_{n\ell}|^2$ through $n = 3$, exact closed forms; each curve numerically verified to unit norm by the figure script. Node count $n - \ell - 1$; ground-state peak at exactly a_0; the n^2 growth of size; the centrifugal exclusion of high-ℓ states from the origin.

Remark 14.2 (The degeneracy is not an accident — it is a bigger symmetry). Rotational symmetry explains the $(2\ell + 1)$-fold degeneracy in m — states of one ℓ-tower are rotations of each other —

but it does *not* explain why $\ell = 0, \ldots, n{-}1$ all land on the same E_n: a generic central potential splits them, and (136)'s n^2 is special to $1/r$. The classical fingerprint of the same specialness is that Coulomb orbits *close* (ellipses that do not precess — a property $1/r$ shares only with r^2 [stated; Bertrand's theorem]), and closed orbits mean an extra conserved quantity: the Runge–Lenz vector, pointing along the ellipse's fixed axis. Quantum mechanically the corresponding operator commutes with $\hat{H}$, and together with $\hat{\vec{L}}$ it closes a *larger* algebra — $SO(4)$, rotations of a four-dimensional space — whose representation theory delivers exactly the n^2-dimensional multiplets [stated; one of the subject's classic algebraic solutions, Pauli 1926]. The moral is the document's in miniature: where there is an unexplained degeneracy, look for the symmetry that explains it.

One atom is built. But the electron has spin the capstone ignored, and real atoms have more than one angular momentum to their name — orbital and spin, electron and nucleus. Composing them is not optional bookkeeping; it is where the tensor product of §3.6 finally does physics, and it is next.

15 Addition of Angular Momenta

The hydrogen capstone ended on a confession: the electron carries a spin the calculation ignored, so even one hydrogen atom holds *two* angular momenta — $\hat{\vec{L}}$ of the orbit, $\hat{\vec{S}}$ of the electron — and nature's Hamiltonians couple them. This section works out what "the total angular momentum of a composite" means and what values it can take. The material has a reputation for abstraction, so the treatment is deliberately slow: the *problem* is stated before the machinery, the commutator computation that creates the two-bases problem is displayed in full rather than left at [stated] level, and the simplest nontrivial case is then worked out completely.

15.1 A careful statement of the problem

Two subsystems with angular momenta j_1 and j_2 live, by §3.6, on the tensor product

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2, \qquad \dim \mathcal{H} = (2j_1+1)(2j_2+1), \tag{139}$$

with the obvious product basis $|j_1 m_1\rangle \otimes |j_2 m_2\rangle \equiv |m_1, m_2\rangle$ (the j's fixed throughout, suppressed when unambiguous). Each factor carries its own generators, acting only on its own slot — $\hat{\vec{J}}_1 \otimes \mathbf{1}$ and $\mathbf{1} \otimes \hat{\vec{J}}_2$, written $\hat{\vec{J}}_1$, $\hat{\vec{J}}_2$ for short — and because they touch different slots they commute with each other: $[\hat{J}_{1i}, \hat{J}_{2k}] = 0$, always.

Now rotate the *whole* system. A rotation turns both parts at once, $U(R) = U_1(R) \otimes U_2(R)$, and expanding each factor to first order ((107), twice) shows the composite's generator is the sum:

$$\hat{\vec{J}} \;\equiv\; \hat{\vec{J}}_1 + \hat{\vec{J}}_2 \qquad (\text{meaning } \hat{\vec{J}}_1 \otimes \mathbf{1} + \mathbf{1} \otimes \hat{\vec{J}}_2). \tag{140}$$

This is what "adding angular momenta" means — not a numerical addition but an operator sum on a tensor product — and the first sanity check is that the sum still *is* an angular momentum:

$$[\hat{J}_i, \hat{J}_k] = [\hat{J}_{1i}, \hat{J}_{1k}] + [\hat{J}_{2i}, \hat{J}_{2k}] = \mathrm{i}\hbar\, \epsilon_{ikl}\big(\hat{J}_{1l} + \hat{J}_{2l}\big) = \mathrm{i}\hbar\, \epsilon_{ikl}\, \hat{J}_l \tag{141}$$

(the cross terms vanish — different slots), so the entire catalogue (129) applies to $\hat{\vec{J}}$: its representation content is some collection of (j, m) towers. The question of the section: *which* towers, and how they sit inside the product basis.

Here is why the question has content. There are now two natural complete sets of commuting observables:

uncoupled set	*coupled* set
$\hat{\vec{J}}_1^2,\ \hat{\vec{J}}_2^2,\ \hat{J}_{1z},\ \hat{J}_{2z}$	$\hat{\vec{J}}_1^2,\ \hat{\vec{J}}_2^2,\ \hat{\vec{J}}^2,\ \hat{J}_z$
basis $\lvert m_1, m_2\rangle$	basis $\lvert j, m\rangle$
answers: each part's z-component	answers: the total's magnitude and z-component

Both are legitimate; they answer different experimental questions; and they are *different bases*, because the coupled set's $\hat{\vec{J}}^2$ refuses to commute with the uncoupled set's $\hat{J}_{1z}$. That refusal is the crux of the whole subject, so it deserves its computation. Expand

$$\hat{\vec{J}}^2 = \hat{\vec{J}}_1^2 + \hat{\vec{J}}_2^2 + 2\,\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2, \tag{142}$$

note the first two terms commute with everything in sight, and hit the cross term:

$$\big[\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2,\ \hat{J}_{1z}\big] = [\hat{J}_{1x},\hat{J}_{1z}]\,\hat{J}_{2x} + [\hat{J}_{1y},\hat{J}_{1z}]\,\hat{J}_{2y} = \mathrm{i}\hbar\big(\hat{J}_{1x}\hat{J}_{2y} - \hat{J}_{1y}\hat{J}_{2x}\big)\ \neq\ 0 \tag{143}$$

(only slot-1 operators feel the commutator; the result is the z-component of $\hat{\vec{J}}_1\times\hat{\vec{J}}_2$). Physically: the total magnitude involves the *relative orientation* of the two spins, and pinning down one spin's z-component disturbs exactly that. So sharp (m_1, m_2) and sharp (j, m) are incompatible bookkeeping — two adapted coordinate systems (Remark 3.5, once more) for one space, and the change of basis between them is this section's deliverable.

Two members are shared: $\hat{J}_z = \hat{J}_{1z} + \hat{J}_{2z}$ commutes with both sets, and acting on a product state,

$$\hat{J}_z\lvert m_1, m_2\rangle = \hbar\,(m_1+m_2)\lvert m_1, m_2\rangle \qquad\Longrightarrow\qquad m = m_1 + m_2\,: \tag{144}$$

the z-components *do* add numerically. This one additive quantum number is the crack through which the whole structure can be pried open.

15.2 Which totals occur: counting by m

Since every j-tower contributes exactly one state at each $m \in \{-j,\dots,j\}$, the multiplicities $N(m)$ — how many product states carry a given total m — determine the tower content completely: the number of towers whose top rung is at m is $N(m) - N(m+1)$. And $N(m)$ is elementary counting on (144): the number of ways to write $m = m_1 + m_2$ with $|m_1| \le j_1$, $|m_2| \le j_2$.

Count it for a concrete case first, $j_1 = 1$, $j_2 = \frac{1}{2}$ (six product states):

m	product states (m_1, m_2)	$N(m)$
$+\frac{3}{2}$	$(1, +\frac{1}{2})$	1
$+\frac{1}{2}$	$(1, -\frac{1}{2}),\ (0, +\frac{1}{2})$	2
$-\frac{1}{2}$	$(0, -\frac{1}{2}),\ (-1, +\frac{1}{2})$	2
$-\frac{3}{2}$	$(-1, -\frac{1}{2})$	1

Read the staircase top-down: one tower tops out at $m = \frac{3}{2}$ (so $j = \frac{3}{2}$ occurs, once), one more appears at $m = \frac{1}{2}$ (N jumped $1 \to 2$: a $j = \frac{1}{2}$ tower), and then nothing new — N never exceeds 2. Conclusion: $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$, dimensions $6 = 4 + 2$. The general case is the same staircase: $N(m)$ starts at 1 for $m = j_1 + j_2$, climbs by one per step down, and plateaus once the shorter ladder is exhausted — at $m = |j_1 - j_2|$, after which no new towers appear. Hence the *triangle rule*:

$$\boxed{\,j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j, \qquad \text{each } j \text{ exactly once,}\,} \tag{145}$$

the possible totals being exactly the triangle-compatible values — the quantum shadow of the classical fact that two vectors of lengths j_1, j_2 compose to anything between $|j_1 - j_2|$ and $j_1 + j_2$, discretized in unit steps. The dimension check is a telescoping sum worth displaying once:

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (j_1+j_2+1)^2 - (j_1-j_2)^2 = (2j_1+1)(2j_2+1) \checkmark \tag{146}$$

(each $2j+1$ is a difference of consecutive squares, $(j+1)^2 - j^2$; the sum collapses to its endpoints).

15.3 Clebsch–Gordan coefficients: the change of adapted coordinates

The two bases are related by a unitary matrix whose entries have a name:

$$|j, m\rangle = \sum_{m_1+m_2=m} \underbrace{\langle m_1, m_2 | j, m\rangle}_{\text{Clebsch–Gordan coefficient}} \quad |m_1, m_2\rangle, \tag{147}$$

the sum restricted by (144) — the one selection rule every coefficient obeys for free. Nothing about them is new machinery: (147) is a change of basis exactly as in §3.3, computable in every case by the two tools already in hand — *ladders* ($\hat{J}_\pm = \hat{J}_{1\pm} + \hat{J}_{2\pm}$ walks down a tower once its top is known, and the top of the top tower is unmissable: $|j_1 + j_2,\, j_1 + j_2\rangle = |j_1, j_2\rangle$, the unique maximal state) — and *orthogonality* (each new tower's top is fixed, up to phase, by being orthogonal to everything already built at its m). Conventional phases (Condon–Shortley: all coefficients real, and the highest-m_1 coefficient in each tower positive) pin the remaining signs [stated]; beyond that there is nothing to memorize, only an algorithm — run explicitly in the next subsection — and a lookup habit: Appendix B tabulates the small products, every entry machine-generated and verified.

15.4 The complete worked case: $\frac{1}{2} \otimes \frac{1}{2}$

Two spin-$\frac{1}{2}$'s: four product states $|{+}{+}\rangle, |{+}{-}\rangle, |{-}{+}\rangle, |{-}{-}\rangle$, and the triangle rule (145) promises $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$: a *triplet* and a *singlet*, $4 = 3 + 1$.

Step 1: the top. The maximal state is the unique $m = 1$ state:

$$|1, 1\rangle = |{+}{+}\rangle. \tag{148}$$

Step 2: ladder down, factors and all. Apply $\hat{J}_- = \hat{J}_{1-} + \hat{J}_{2-}$. On the left, the catalogue's matrix element (129) with $j = 1$, $m = 1$: $\hat{J}_-|1,1\rangle = \hbar\sqrt{1 \cdot 2 - 1 \cdot 0}\,|1,0\rangle = \hbar\sqrt{2}\,|1,0\rangle$. On the right, each one-particle lowering carries $\hbar\sqrt{\frac{1}{2} \cdot \frac{3}{2} - \frac{1}{2} \cdot (-\frac{1}{2})} = \hbar$:

$$\big(\hat{J}_{1-} + \hat{J}_{2-}\big)|{+}{+}\rangle = \hbar\,|{-}{+}\rangle + \hbar\,|{+}{-}\rangle. \tag{149}$$

Equate and divide by $\hbar\sqrt{2}$:

$$|1, 0\rangle = \frac{1}{\sqrt{2}}\big(|{+}{-}\rangle + |{-}{+}\rangle\big). \tag{150}$$

Once more: $\hat{J}_-|1,0\rangle = \hbar\sqrt{2}\,|1,-1\rangle$, while the right side gives $\hbar(|{-}{-}\rangle + |{-}{-}\rangle)/\sqrt{2} = \hbar\sqrt{2}\,|{-}{-}\rangle$, so $|1,-1\rangle = |{-}{-}\rangle$ — the ladder closes exactly where the catalogue said it must.

Step 3: the leftover. Four states, three used. The $m = 0$ subspace is two-dimensional and the triplet took one direction; the orthogonal complement, normalized and with the conventional phase, is

$$|0, 0\rangle = \frac{1}{\sqrt{2}}\big(|{+}{-}\rangle - |{-}{+}\rangle\big) \qquad \text{— the } \textit{singlet}. \tag{151}$$

Step 4: verify the labels. That $|0,0\rangle$ really has $j=0$ should be *checked*, not trusted, and the check exercises the machinery. First rewrite the cross term of (142) in ladder form — one line,

$$\hat{J}_{1+}\hat{J}_{2-}+\hat{J}_{1-}\hat{J}_{2+}=2\big(\hat{J}_{1x}\hat{J}_{2x}+\hat{J}_{1y}\hat{J}_{2y}\big)\implies\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2=\hat{J}_{1z}\hat{J}_{2z}+\tfrac{1}{2}\big(\hat{J}_{1+}\hat{J}_{2-}+\hat{J}_{1-}\hat{J}_{2+}\big),\tag{152}$$

then evaluate on the singlet, term by term. The zz term gives $(\frac{\hbar}{2})(-\frac{\hbar}{2})=-\frac{\hbar^2}{4}$ on both $|{+}{-}\rangle$ and $|{-}{+}\rangle$. The ladder terms swap the spins with a factor $\hbar^2$ each ($\hat{J}_{1-}\hat{J}_{2+}|{+}{-}\rangle=\hbar^2|{-}{+}\rangle$, $\hat{J}_{1+}\hat{J}_{2-}|{-}{+}\rangle=\hbar^2|{+}{-}\rangle$, the other orderings annihilate), so on the antisymmetric combination the swap costs a sign:

$$\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2\,|0,0\rangle=\Big(-\frac{\hbar^2}{4}-\frac{\hbar^2}{2}\Big)|0,0\rangle=-\frac{3\hbar^2}{4}\,|0,0\rangle,\tag{153}$$

and assembling (142) with $\hat{\vec{J}}^2_{1,2}=\frac{3}{4}\hbar^2$:

$$\hat{\vec{J}}^2|0,0\rangle=\Big(\tfrac{3}{4}+\tfrac{3}{4}-\tfrac{3}{2}\Big)\hbar^2\,|0,0\rangle=0\ \checkmark\tag{154}$$

— and the same evaluation on $|1,0\rangle$ (swap costs no sign) gives $+\frac{\hbar^2}{4}+\frac{\hbar^2}{2}$, hence $\hat{\vec{J}}^2=2\hbar^2=1(1{+}1)\hbar^2$ ✓. The summary table, with every entry now derived:

state	(j,m)	$\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2/\hbar^2$
$\lvert{+}{+}\rangle$	$(1,+1)$	$+\frac{1}{4}$
$\frac{1}{\sqrt{2}}(\lvert{+}{-}\rangle+\lvert{-}{+}\rangle)$	$(1,0)$	$+\frac{1}{4}$
$\lvert{-}{-}\rangle$	$(1,-1)$	$+\frac{1}{4}$
$\frac{1}{\sqrt{2}}(\lvert{+}{-}\rangle-\lvert{-}{+}\rangle)$	$(0,0)$	$-\frac{3}{4}$

The last column is the section's most-used practical output: any Hamiltonian containing $\hat{\vec{J}}_1\cdot\hat{\vec{J}}_2$ — and nature's spin–spin couplings do — is *diagonal* in the coupled basis, constant on each multiplet, splitting singlet from triplet by $\hbar^2$.

Example 15.1 (The algorithm scales: $\frac{3}{2}\otimes\frac{1}{2}$)**.** Re-run the machine, compactly, on the next case up — eight product states $|m_1,m_2\rangle$, $m_1\in\{\pm\frac{3}{2},\pm\frac{1}{2}\}$, $m_2=\pm\frac{1}{2}$. First the staircase:

m	product states	$N(m)$
$+2$	$(\frac{3}{2},+\frac{1}{2})$	1
$+1$	$(\frac{3}{2},-\frac{1}{2})$, $(\frac{1}{2},+\frac{1}{2})$	2
0	$(\frac{1}{2},-\frac{1}{2})$, $(-\frac{1}{2},+\frac{1}{2})$	2
-1	$(-\frac{1}{2},-\frac{1}{2})$, $(-\frac{3}{2},+\frac{1}{2})$	2
-2	$(-\frac{3}{2},-\frac{1}{2})$	1

One tower tops at $m=2$, one new one at $m=1$, none after: $\frac{3}{2}\otimes\frac{1}{2}=2\oplus 1$, dimensions $8=5+3$, as (145) promised. Now two rows of coefficients, by the same two tools. *Ladder*: $|2,2\rangle=|\frac{3}{2},+\frac{1}{2}\rangle$, and applying $\hat{J}_-$ — left side $\hbar\sqrt{2\cdot 3-2\cdot 1}\,|2,1\rangle=2\hbar\,|2,1\rangle$, right side $\hbar\sqrt{3}\,|\frac{1}{2},+\frac{1}{2}\rangle+\hbar\,|\frac{3}{2},-\frac{1}{2}\rangle$ (the $\sqrt{3}$ from (129) at $j_1=\frac{3}{2}$, $m_1=\frac{3}{2}$; the $\hbar$ from the spin-$\frac{1}{2}$ slot) — gives

$$|2,1\rangle=\frac{\sqrt{3}}{2}\,|\tfrac{1}{2},+\tfrac{1}{2}\rangle+\frac{1}{2}\,|\tfrac{3}{2},-\tfrac{1}{2}\rangle.\tag{155}$$

Orthogonality: the $m = 1$ subspace is two-dimensional and the $j = 2$ tower has claimed one direction; the $j = 1$ tower's top is the complement, with the overall sign fixed by the Condon–Shortley convention — *the coefficient of the highest-m_1 state is positive*:

$$|1, 1\rangle = \frac{\sqrt{3}}{2}\,|\tfrac{3}{2}, -\tfrac{1}{2}\rangle - \frac{1}{2}\,|\tfrac{1}{2}, +\tfrac{1}{2}\rangle, \tag{156}$$

and continued lowering fills in the rest — nothing new happens, only more square roots. The case is not academic: a nucleus of spin $\frac{3}{2}$ coupled to an electron spin is exactly this decomposition, and the two hyperfine levels $F = 2, 1$ of ^{87}Rb's ground state — the pair between which laser cooling and atomic-clock physics runs — are these two towers [stated].

Remark 15.2 (The entanglement flag delivered). Section 3.6 flagged that most states of a composite are entangled and promised a physical specimen. The singlet is *the* specimen: $|0, 0\rangle$ is manifestly not a product (try $(a|+\rangle + b|-\rangle) \otimes (c|+\rangle + d|-\rangle)$: matching coefficients forces $ac = bd = 0$ against $ad = -bc \neq 0$), and its correlations are total — measure either spin along *any* axis and the other is fixed opposite, since $j = 0$ makes the state invariant under the simultaneous rotation $U \otimes U$ (no direction is special: a rotationally invariant state has no Bloch vector to point). This is the two-particle state of the Einstein–Podolsky–Rosen discussion and of Bell's inequalities [stated; the flag planted in §3.6 is delivered to exactly this depth and the literature does the rest], and it was built here by nothing more exotic than a ladder and an orthogonality.

Example 15.3 (The 21-centimeter line)**.** Hydrogen's ground state, $(n, \ell) = (1, 0)$, carries two idle spins: the electron's and the proton's. Their magnetic coupling is proportional to $\hat{\vec{S}}_e \cdot \hat{\vec{S}}_p$ — so the table above *is* the level diagram: a triplet and a singlet, split by the hyperfine interaction, singlet lower. The measured splitting is tiny,

$$\Delta E = 5.87\,\mu\text{eV}, \qquad f = \Delta E/h = 1420.4\,\text{MHz}, \qquad \lambda = 21.1\,\text{cm} \tag{157}$$

[experimental input; predicting ΔE's magnitude needs the magnetic moments and the contact interaction — note it involves $|\psi(0)|^2$, nonzero only for the $\ell = 0$ states of Fig. 7 — stated], corresponding to $\Delta E/k_B \approx 68\,\text{mK}$: any astrophysical environment populates the triplet, and the slow magnetic-dipole decay back to the singlet emits the *21-centimeter line* — the radio signature by which neutral hydrogen is mapped across galaxies, including the first maps of the Milky Way's spiral arms. The structure of that line — why there are exactly two levels with statistical weights $3 : 1$ — is precisely the four-row table derived above; radio astronomy runs on a Clebsch–Gordan decomposition.

The composition of *states* is settled: totals by the triangle rule, bases by ladders and orthogonality. What remains for the symmetry story is the composition of states with *operators* — how observables themselves transform under rotation, and the theorem that squeezes every rotational matrix element through one reduced number. That theorem is next.

16 Tensor Operators and the Wigner–Eckart Theorem

One asymmetry remains in the symmetry story. States have been sorted into rotation multiplets — the (j, m) towers — but *operators* have not, and physical predictions end in matrix elements $\langle\beta|\hat{A}|\alpha\rangle$ with an operator in the middle. This closing technical section repairs the asymmetry, carefully: it defines what it means for an operator to transform, discovers that operators fall into the *same* towers as states, and proves — to the depth of one displayed recursion — the theorem that then squeezes every rotational matrix element in atomic physics through a single number. The abstraction is front-loaded; the payoff, at the end, is as concrete as which spectral lines exist.

16.1 How operators transform

Rotate the system: states go $|\psi\rangle \to U|\psi\rangle$. For an operator, "transform" means whatever keeps the bookkeeping consistent, and expectation values dictate it: the rotated experiment measures $\langle\psi|U^\dagger \hat{A}\, U|\psi\rangle$, so the operator's transform is $\hat{A} \to U^\dagger \hat{A}\, U$. Now impose physics on specific operators. The position operator is a *vector*: rotating the apparatus must rotate its readings,

$$U^\dagger(R)\,\hat{q}_i\,U(R) = R_{ik}\,\hat{q}_k. \tag{158}$$

Feed in an infinitesimal rotation (107) about $\hat{n}$: the left side is $\hat{q}_i + \frac{\mathrm{i}}{\hbar}\,\mathrm{d}\varphi\,[\hat{n}\cdot\hat{\vec{J}}, \hat{q}_i]$, the right side is $\hat{q}_i + \mathrm{d}\varphi\,(\hat{n}\times\hat{\vec{q}})_i$, and matching — for every axis — gives the commutator form of "vectorhood":

$$\boxed{[\hat{J}_i, \hat{V}_k] = \mathrm{i}\hbar\,\epsilon_{ikl}\,\hat{V}_l} \qquad \text{(vector operator)}, \qquad [\hat{J}_i, \hat{S}\,] = 0 \quad \text{(scalar operator)}. \tag{159}$$

This is a *definition by commutators*, and it already has instances: $\hat{\vec{q}}$, $\hat{\vec{p}}$, $\hat{\vec{L}}$, $\hat{\vec{S}}$ all satisfy it (for $\hat{\vec{L}}$ it is (108) itself), while $\hat{\vec{q}}\cdot\hat{\vec{p}}$, $\hat{\vec{q}}^{\,2}$, $\hat{H}_{\text{central}}$ are scalars. Notice what (159) formally *is*: the statement that the components $\hat{V}_k$, under the adjoint action of $\hat{\vec{J}}$ (action by commutator), shuffle among themselves exactly the way a $j = 1$ triple of states would. That observation — operators organize into the same multiplets as states — is the entire section in embryo, and making it precise is next.

16.2 Cartesian tensors: the obvious generalization and its redundancy

Given (159), the natural next move is more indices. Define a *Cartesian tensor* of rank n as an object with n vector indices, transforming with one rotation matrix per index — constructively, take products of vectors: the *dyadic*

$$\hat{T}_{ik} = \hat{U}_i\hat{V}_k, \qquad U^\dagger\,\hat{T}_{ik}\,U = R_{ii'}\,R_{kk'}\,\hat{T}_{i'k'}, \tag{160}$$

nine components, and the pattern continues to any rank. This is the tensor grammar of the field-theory notes [Fld. §3.7], imported to operator components, and it is perfectly *correct*. It is also, as an organizing scheme, *redundant* — and watching the redundancy is the fastest route to the right definition. Split the dyadic identically into three pieces:

$$\hat{U}_i\hat{V}_k = \underbrace{\frac{\hat{\vec{U}}\cdot\hat{\vec{V}}}{3}\,\delta_{ik}}_{\text{trace: 1}} + \underbrace{\frac{\hat{U}_i\hat{V}_k - \hat{U}_k\hat{V}_i}{2}}_{\text{antisymmetric: 3}} + \underbrace{\left(\frac{\hat{U}_i\hat{V}_k + \hat{U}_k\hat{V}_i}{2} - \frac{\hat{\vec{U}}\cdot\hat{\vec{V}}}{3}\,\delta_{ik}\right)}_{\text{symmetric traceless: 5}} \tag{161}$$

(an identity — add the three terms and cancel). Now rotate. Each block stays strictly within itself, for a transparent reason each: the trace is a scalar because $R\,\delta\,R^{\mathsf{T}} = \delta$ — the *defining equation* of the rotation group, [Fld. §3.1]'s template once more; the antisymmetric part packages as a vector, $\frac{1}{2}\epsilon_{ikl}(\hat{U}_k\hat{V}_l - \hat{U}_l\hat{V}_k) = (\hat{\vec{U}}\times\hat{\vec{V}})_i$, and vectors rotate among themselves; and symmetry-plus-tracelessness survives conjugation by any R. So the nine components never mix as nine: rotations act on (161) block-diagonally, in pieces of size

$$3\times 3 = 1 + 3 + 5. \tag{162}$$

The Cartesian packaging is *reducible*: it lumps together subspaces that never speak to one another, like insisting on one 9×9 matrix where three small blocks act. The counts $1, 3, 5$ are a broad hint — they are $2k+1$ for $k = 0, 1, 2$, the dimensions of the state catalogue (129) — and they pose the section's real question: what are the *minimal* rotation-closed operator families, the irreducible pieces the Cartesian language keeps bundling?

16.3 Spherical tensors: the irreducible organization

The answer's shape is already in hand. Irreducible multiplets of the rotation algebra are the $(2k{+}1)$-component towers of §13, laddered by $\hat{J}_\pm$ and graded by $\hat{J}_z$; demand operator families organized the same way. Define a *spherical tensor of rank* k as any $2k{+}1$ operators $\hat{T}_q^{(k)}$, $q = -k, \ldots, k$, obeying

$$\boxed{[\hat{J}_z, \hat{T}_q^{(k)}] = \hbar\, q\, \hat{T}_q^{(k)}, \qquad [\hat{J}_\pm, \hat{T}_q^{(k)}] = \hbar\sqrt{k(k+1) - q(q\pm 1)}\, \hat{T}_{q\pm1}^{(k)}} \tag{163}$$

— the catalogue's matrix elements (129), transplanted verbatim from action-on-states to action-by-commutator: *an operator multiplet is a tower that commutators ladder.* (Equivalently, in finite form, the components mix by the $\mathscr{D}^{(k)}$-matrices of §12.3; the infinitesimal version (163) is taken as the definition because it is convention-proof and computation-ready. That the adjoint action is a legitimate "action" at all — that acting by nested commutators obeys the same algebra — is the Jacobi identity of §2, invoked one last time.)

A scalar is the $k = 0$ case. A vector fits with the spherical repackaging

$$\hat{V}_0^{(1)} = \hat{V}_z, \qquad \hat{V}_{\pm1}^{(1)} = \mp\,\frac{\hat{V}_x \pm \mathrm{i}\,\hat{V}_y}{\sqrt{2}}, \tag{164}$$

and one check shows the definition engaging: using (159),

$$[\hat{J}_z,\ \hat{V}_{+1}^{(1)}] = -\frac{[\hat{J}_z, \hat{V}_x] + \mathrm{i}\,[\hat{J}_z, \hat{V}_y]}{\sqrt{2}} = -\frac{\mathrm{i}\hbar\,\hat{V}_y + \hbar\,\hat{V}_x}{\sqrt{2}} = \hbar\,\hat{V}_{+1}^{(1)} \quad \checkmark \tag{165}$$

— eigenvalue $q = +1$, as (163) demands (the $\hat{J}_\pm$ relations check the same way). The paradigm rank-k example: the spherical harmonics themselves, read as multiplication operators, $Y_{km}(\hat{q})$ — they were *built* as (k, m) towers in §13.3.

Now return to the dyadic and name its blocks in the new language. The counts (162) identify the three pieces of (161) as spherical tensors of ranks $0, 1, 2$ — dot product, cross product, and *quadrupole* — so the decomposition every linear-algebra student knows (trace, antisymmetric, symmetric-traceless) is a Clebsch–Gordan statement about operators,

$$1 \otimes 1 = 0 \oplus 1 \oplus 2, \tag{166}$$

with the triangle rule (145) presiding, exactly as it did for composite states. Two morals. First, *Cartesian tensors are reducible; spherical tensors are their irreducible pieces* — the spherical language is not a rival convention but the finer sieve, and the route of §16.2 was deliberate: the redundancy was found first, the irreducible organization built to remove it. Second, a warning about the overloaded word "rank": a Cartesian tensor of n indices contains spherical ranks up to $k = n$ *and below* (the dyadic holds $k = 0, 1, 2$ at once); when a physicist says "the quadrupole operator," only the five-component $k = 2$ piece is meant, the trace and antisymmetric pieces already stripped. [Sakurai's tensor-operator discussion is the standard reference for this decomposition.]

16.4 The key observation and the theorem

The section converges on one line. Act with a generator on the state $\hat{T}_q^{(k)}|j, m\rangle$ and let Leibniz split the product:

$$\hat{J}_i\left(\hat{T}_q^{(k)}|j, m\rangle\right) = \underbrace{[\hat{J}_i,\, \hat{T}_q^{(k)}]}_{\text{rotates the operator: rep } k}\, |j, m\rangle \;+\; \hat{T}_q^{(k)}\, \underbrace{\hat{J}_i|j, m\rangle}_{\text{rotates the state: rep } j}. \tag{167}$$

Compare with §15.1: this *is* the two-slot structure of $\hat{\vec{J}}_1 + \hat{\vec{J}}_2$, with the adjoint action in slot one and the state in slot two — Leibniz has reproduced the addition of angular momenta, verbatim. Therefore $\hat{T}^{(k)}_q|j,m\rangle$ transforms exactly like the product state $|k,q\rangle \otimes |j,m\rangle$, and all of §15 applies for free: its total-j' content obeys the triangle rule, its $\hat{J}_z$-eigenvalue is $\hbar(q+m)$, and its overlap with $\langle j',m'|$ must be proportional to the corresponding Clebsch–Gordan coefficient. That is already the theorem, seen from above; here is its statement and proof mechanism.

Theorem 16.1 (Wigner–Eckart)**.**

$$\boxed{\langle j',m'|\,\hat{T}^{(k)}_q\,|j,m\rangle \;=\; \langle k,q;\,j,m|j',m'\rangle \;\times\; \langle j'\|\,\hat{T}^{(k)}\|\,j\rangle} \tag{168}$$

— *the Clebsch–Gordan coefficient carries* all *dependence on* (m',q,m)*, and the second factor, the* reduced matrix element*, is one single number for the whole* (j',k,j) *family, independent of every orientation label.*

Proof mechanism, one recursion displayed. Sandwich the defining relation $[\hat{J}_\pm, \hat{T}^{(k)}_q] = \hbar\sqrt{k(k+1)-q(q\pm1)}\,\hat{T}^{(k)}_{q\pm1}$ between $\langle j',m'|$ and $|j,m\rangle$, letting $\hat{J}_\pm$ act left and right by the catalogue (129):

$$\begin{aligned}\sqrt{j'(j'{+}1)-m'(m'{\mp}1)}\;\langle j',m'{\mp}1|\hat{T}^{(k)}_q|j,m\rangle \;-\; \sqrt{j(j{+}1)-m(m{\pm}1)}\;\langle j',m'|\hat{T}^{(k)}_q|j,m{\pm}1\rangle \\ = \sqrt{k(k{+}1)-q(q{\pm}1)}\;\langle j',m'|\hat{T}^{(k)}_{q\pm1}|j,m\rangle.\end{aligned} \tag{169}$$

Regard the numbers $\langle j',m'|\hat{T}^{(k)}_q|j,m\rangle$, at fixed (j',k,j), as one unknown per label triple (m',q,m): (169) is a homogeneous linear recursion linking them — and it is, coefficient for coefficient, *the same recursion the Clebsch–Gordan coefficients satisfy* (obtained by the identical sandwich of $\hat{J}_\pm = \hat{J}_{1\pm} + \hat{J}_{2\pm}$ on product states in §15.3). A linear system does not care what its unknowns mean: its solution space here is one-dimensional [the induction walking the recursion through all labels from the maximal one is stated], so any solution is a multiple of the CG solution — and the multiple is (168)'s reduced matrix element. Geometry determines everything but one number; the one number is where the physics lives. ☐

Two immediate harvests, one structural, one practical:

- **Selection rules for free.** The matrix element vanishes unless the CG coefficient survives:

$$m' = q + m, \qquad |j-k| \;\le\; j' \;\le\; j+k \tag{170}$$

  — with no dynamical computation required.

- **Ratios are universal.** Within a multiplet family, ratios of matrix elements are ratios of CG coefficients — pure geometry, identical for every rank-k operator and every physical system. Dynamics enters only through the one reduced element: the theorem *factorizes* every rotational matrix element into (universal geometry) × (one number of physics), the same division of labor the document has met as kinematics-versus-dynamics throughout, here made into an equation.

16.5 The payoff in lines: dipole selection rules

Aim the theorem at Fig. 7. Atoms radiate, in leading approximation, through the *dipole* operator $\hat{\vec{q}}$ — a vector, hence via (164) a rank-1 spherical tensor. The selection rules (170) with $k=1$, applied to the orbital labels of hydrogen:

$$\Delta m \in \{-1,0,+1\}, \qquad \Delta\ell \in \{-1,0,+1\}, \tag{171}$$

and one further cut: under the parity operator of Remark 8.1, now on $L^2(\mathbb{R}^3)$ ($\vec{x} \to -\vec{x}$), the spherical harmonics carry $(-1)^\ell$ [stated; visible in (131)], while the dipole operator is odd — so the matrix element survives only if the parities differ, killing $\Delta\ell = 0$:

$$\boxed{\Delta\ell = \pm 1.} \tag{172}$$

Now read hydrogen's low levels through this rule. The $2p \to 1s$ transition is allowed — that *is* Lyman-α (Example 14.1), lifetime 1.6 ns [experimental]. But $2s \to 1s$ has $\Delta\ell = 0$: dipole-*forbidden*, and with no lower level to reach otherwise, the $2s$ state is stranded — *metastable*, decaying only by simultaneous two-photon emission with a lifetime of about an eighth of a second [experimental; a factor $\sim 10^8$ beyond its neighbor]. A hundred-million-fold lifetime ratio between two states one figure apart, traced to a single vanishing Clebsch–Gordan coefficient: that is the Wigner–Eckart theorem doing what it was built for — and it is why astrophysicists can find hydrogen's $2s$ population glowing in a continuum while $2p$ blinks in a line.

The symmetry story is complete, and it closes symmetric: states in (j, m) towers, operators in (k, q) towers, matrix elements factored through the geometry that both share. What remains is not another theorem but the outlook, and the handoff the series has been building toward.

17 Outlook

The document closes as the others in this series did: a list of what was left out, each item with its pointer; one last piece of physics that the tools in hand can almost reach, and the handoff. The last piece is the largest — the series' second closing of a circle — and it is why the whole five-document arc was walked in this order.

17.1 Deferred material, with pointers

What a first graduate course contains and this document deferred, each with its pointer: *time-dependent perturbation theory* lives in the interaction picture flagged at §7.4 — transition rates, the golden rule, and the actual strength of the spectral lines whose existence §16.5 settled; *scattering theory* continues §8.5's transmission problem into three dimensions, with the propagator of §10 as its natural engine; *density matrices* (Remark 3.6) carry the statistical and open-system story — decoherence, the modern descendant of the gap Remark 4.3 flagged; *time reversal*, with its antiunitary implementation, is the deferral Theorem 11.1 announced; *stationary perturbation theory* and the *variational method as an approximation scheme* are the standard machinery for Hamiltonians near ones solved here (the exact variational computations of §9 and Example 5.2 are its seeds); *charged particles in electromagnetic fields* — minimal coupling, $\hat{H} = (\hat{\vec{p}} - q\vec{A})^2/2m + q\phi$, and the gauge structure it carries — is deliberately left untouched here: it belongs with the field theory; *relativistic* quantum mechanics is not a patch but a rebuild, and it is where this outlook is headed. One omission, however, is too structural to defer without a paragraph, because without it quantum mechanics does not even predict that matter is stable in bulk.

17.2 Identical particles

Two electrons are not two similar objects; they are *indistinguishable in principle* — no measurement labels which is which. On the tensor product $\mathcal{H} \otimes \mathcal{H}$ of §3.6, swapping the slots is an operator P_{12} with $P_{12}^2 = \mathbf{1}$, so its eigenvalues are ± 1 — and indistinguishability demands physical states be eigenstates (anything else would let the swap, which changes nothing physical, change the state by more than a phase). Nature's rule, an empirical input at this level: *every species commits globally to one sign* —

$$\psi(2,1) = +\,\psi(1,2) \quad \text{(bosons)} \qquad \text{or} \qquad \psi(2,1) = -\,\psi(1,2) \quad \text{(fermions)}, \tag{173}$$

and the assignment is by spin: integer-spin species are bosons, half-integer fermions [stated — the *spin–statistics theorem*, unprovable inside nonrelativistic quantum mechanics and a celebrated theorem of relativistic QFT: the sign of §12, which a 2π rotation writes on a spinor, grows up to become the minus sign of exchange]. The fermionic minus sign has an immediate, world-building corollary: put two fermions in the same single-particle state and antisymmetry forces $\psi = -\psi = 0$ —

$$\boxed{\text{at most one fermion per state}} \tag{174}$$

— the *Pauli exclusion principle*. Electrons therefore cannot all collapse into hydrogen-like ground states: they stack, filling the (n, ℓ, m, m_s) catalogue of §14 floor by floor — which is the periodic table — the organizing structure of chemistry — read off the labels this document constructed. [The many-electron problem in earnest — Slater determinants, exchange energies — is deferred with the rest; the principle is the point.]

17.3 The second closing: quantizing the field

Now the closing the series was built for. Recall where the field-theory notes ended [Fld. §14.1–14.2]: a free field, expanded in its modes, *is* a countable family of independent harmonic oscillators — one complex amplitude $a_{\vec{k}}$ per wavevector, each obeying the classical oscillator equations — and the notes could only *state* what quantization would mean. This document's §9 closed with the missing sentence's grammar: the pair $(\hat{a}, \hat{a}^\dagger)$, $[\hat{a}, \hat{a}^\dagger] = 1$, is the entire quantum content of one oscillator. Compose the two documents:

$$\boxed{a_{\vec{k}} \;\longrightarrow\; \hat{a}_{\vec{k}}, \qquad [\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = \delta_{\vec{k}\vec{k}'}, \qquad \hat{H} = \sum_{\vec{k}} \hbar\omega_{\vec{k}}\Big(\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} + \tfrac{1}{2}\Big)} \tag{175}$$

— one ladder per mode, nothing else. And now read the spectrum with this document's eyes. The ground state $|0\rangle$ (every mode on its floor) is the *vacuum.* The state $\hat{a}^\dagger_{\vec{k}}|0\rangle$ costs energy $\hbar\omega_{\vec{k}}$ and carries momentum $\hbar\vec{k}$ (the mode's Noether charges, [Fld. §9.4], now operator eigenvalues) — a discrete, countable lump of field excitation with definite energy and momentum. There is a word for that:

a particle.

$\hat{a}^\dagger_{\vec{k}}$ *creates* one; $\hat{N}_{\vec{k}}$ counts them; multi-particle states are multi-rung states; and the identical-particle symmetrization of §17.2 is automatic — the ladder operators carry no particle labels to antisymmetrize, only mode labels, so "which quantum is which" is not even a question the formalism can pose. The photoelectric lumps of §2.1, the first experimental puzzle on the first page, are the rungs of (175) applied to the electromagnetic field: *photons*, each $E = \hbar\omega$ — Einstein's 1905 heuristic, now a two-line consequence of composing two documents. The coherent states of §9.4 become laser light; the zero-point halves, summed over infinitely many modes, diverge — the subject's first renormalization, flagged and left for the sequel [stated].

This is the series' second closed circle. The first [Fld. §14.4] ran *mechanics* → *chains* → *fields*: the continuum limit. This one runs *oscillator* → *quantum oscillator* → *quantum field*: the quantization. The two arcs compose into the diagram the five documents have been drawing —

$$\begin{array}{c} \text{classical mechanics} \xrightarrow{\text{[Mech.], [Osc.]}} \text{classical fields} \xrightarrow{\text{[Fld.]}} \,? \\ \text{classical mechanics} \xrightarrow{\text{this document}} \text{quantum mechanics} \xrightarrow{?} \,? \end{array}$$

— and both question marks are the same subject: quantum field theory, where the Heisenberg-picture operators carry spacetime labels (Remark 7.2's closing prediction), the action of [Mech.] weighs histories of *fields* in the path integral of §10, and the spin–statistics sign of §17.2 finally gets its proof. The reader holding a QFT course — Schwartz, or the lecture notes this series was built to accompany — now has every prerequisite for its first three chapters.

17.4 Coda

The document promised, in §1, to keep $\hbar$ visible so its arrival and departure could be watched; here is the departure. Every classical limit met on the way was the same limit: $S \gg \hbar$ collapsed the path sum onto Hamilton's principle (§10.4); $\hbar\omega \ll k_B T$ unfroze the ladders into equipartition (Example 9.3); $|\alpha|^2 \gg 1$ turned rungs into amplitudes (§9.4); $n \gg 1$ laid quantum ripples on the classical dwell time (Fig. 4). One deformation parameter, dialed down, and the whole classical series reassembles — which is the final sense in which these five documents are one: the mechanics notes are this document's $\hbar \to 0$ shadow, and the field notes are its $N \to \infty$ stage. The brackets became commutators; the commutators built amplitudes; the amplitudes, summed over histories, explained why the brackets' own variational principle was ever true. The circle does not just close; it explains why it was a circle.

A Gaussian Integrals

This appendix keeps the Gaussian reference material where reference material belongs: out of the narrative, available on demand. Everything is elementary; the point is to have the formulas stated once, with their conditions, so the text can cite rather than rederive.

The basic integral. For $a > 0$,

$$I(a) = \int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-ax^2} = \sqrt{\frac{\pi}{a}}, \tag{176}$$

by the classic squaring trick: I^2 is a two-dimensional integral, polar coordinates make it elementary, $I^2 = \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} \mathrm{d}r\, r\, \mathrm{e}^{-ar^2} = \pi/a$.

Linear term: complete the square. For $a > 0$ and any complex b,

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-ax^2+bx} = \sqrt{\frac{\pi}{a}}\, \mathrm{e}^{b^2/4a} \tag{177}$$

— write $-ax^2 + bx = -a(x - b/2a)^2 + b^2/4a$ and shift the contour (legitimate for entire integrands with Gaussian decay; [stated] at this document's level of rigor). The imaginary-b case is the Fourier transform of the Gaussian:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{-x^2/4\sigma^2}\, \mathrm{e}^{-\mathrm{i}kx} = 2\sigma\sqrt{\pi}\, \mathrm{e}^{-\sigma^2 k^2} \qquad \text{— a Gaussian again, width } \tfrac{1}{2\sigma} \text{ in } k, \tag{178}$$

the reciprocal-width fact that §5.5 turns into physics.

Complex Gaussian. Equation (176) extends to complex a with $\mathrm{Re}\, a > 0$ (same formula, principal branch of the root), and — with more care — to purely imaginary a as an improper (Fresnel) limit:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \mathrm{e}^{\mathrm{i}cx^2} = \sqrt{\frac{\pi}{|c|}}\, \mathrm{e}^{\mathrm{i}\frac{\pi}{4}\operatorname{sgn} c}, \qquad c \in \mathbb{R} \setminus \{0\} \tag{179}$$

[stated; the contour rotation is standard]. This is the version the path integral of §10 consumes in bulk: the weights there are $\mathrm{e}^{\mathrm{i}S/\hbar}$, pure phases, and (179) is how pure phases nonetheless integrate to finite answers.

Moments. Differentiating (176) under the integral sign with respect to a:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, x^2\, \mathrm{e}^{-ax^2} = \frac{1}{2a}\sqrt{\frac{\pi}{a}}, \qquad \text{hence} \qquad \langle x^2 \rangle = \frac{\int x^2\, \mathrm{e}^{-x^2/2\sigma^2} \mathrm{d}x}{\int \mathrm{e}^{-x^2/2\sigma^2} \mathrm{d}x} = \sigma^2 : \tag{180}$$

the parameter in $\mathrm{e}^{-x^2/2\sigma^2}$ *is* the variance — the reading-off used at Eq. (48). Odd moments vanish by symmetry; higher even moments follow by further differentiation, $\langle x^{2n} \rangle = (2n-1)!!\, \sigma^{2n}$.

B Computed Clebsch–Gordan Tables

This appendix tabulates the Clebsch–Gordan coefficients $\langle m_1, m_2|J, M\rangle$ for the smallest products, in the Condon–Shortley convention of §15.3 (all coefficients real; the highest-m_1 coefficient in each tower positive). Every entry below was *machine-generated* by symbolic computation from the ladder-and-orthogonality algorithm of §15.4 and checked: each M-block of each table was verified to form an orthonormal matrix, and spot values were checked against the standard particle-data-group table. Signs sit outside the roots. Rows are grouped by $M = m_1 + m_2$ (Eq. (144)); the columns are the towers J that the triangle rule (145) admits; a dot marks a combination the m selection rule forbids, while an explicit 0 marks an allowed combination whose coefficient the symmetries force to vanish.

Two symmetry relations extend the tables beyond what is printed [stated; both follow from the ladder construction and the Condon–Shortley phases]:

$$\langle j_1\, j_2\, m_1\, m_2|JM\rangle = (-1)^{J-j_1-j_2}\,\langle j_2\, j_1\, m_2\, m_1|JM\rangle, \tag{181}$$

so a product in the opposite order ($\frac{1}{2}\otimes 1$, say) is read from the table printed here ($1\otimes\frac{1}{2}$) at the cost of the sign $(-1)^{J-j_1-j_2}$; and

$$\langle j_1\, j_2\, m_1\, m_2|JM\rangle = (-1)^{J-j_1-j_2}\,\langle j_1\, j_2, \,-m_1, \,-m_2|J, \,-M\rangle, \tag{182}$$

which is why each table's lower half mirrors its upper half up to that same sign. Equation (182) also explains the explicit zeros below: at $m_1 = m_2 = M = 0$ it reads $\langle 0\,0|J\,0\rangle = (-1)^{J-j_1-j_2}\langle 0\,0|J\,0\rangle$, forcing the coefficient to vanish whenever $J - j_1 - j_2$ is odd — the $J = 1$ column of $1\otimes 1$ and the $J = 2$ column of $2\otimes 1$. Larger products ($\frac{3}{2}\otimes\frac{3}{2}$, $2\otimes\frac{3}{2}$, $2\otimes 2$) are in the particle-data-group table.

$\frac{1}{2}\otimes\frac{1}{2}$

m_1	m_2	$J=1$	$J=0$
$+\frac{1}{2}$	$+\frac{1}{2}$	1	$\cdot$
$+\frac{1}{2}$	$-\frac{1}{2}$	$\sqrt{1/2}$	$\sqrt{1/2}$
$-\frac{1}{2}$	$+\frac{1}{2}$	$\sqrt{1/2}$	$-\sqrt{1/2}$
$-\frac{1}{2}$	$-\frac{1}{2}$	1	$\cdot$

$1\otimes\frac{1}{2}$

m_1	m_2	$J=\frac{3}{2}$	$J=\frac{1}{2}$
$+1$	$+\frac{1}{2}$	1	$\cdot$
$+1$	$-\frac{1}{2}$	$\sqrt{1/3}$	$\sqrt{2/3}$
$+0$	$+\frac{1}{2}$	$\sqrt{2/3}$	$-\sqrt{1/3}$
$+0$	$-\frac{1}{2}$	$\sqrt{2/3}$	$\sqrt{1/3}$
-1	$+\frac{1}{2}$	$\sqrt{1/3}$	$-\sqrt{2/3}$
-1	$-\frac{1}{2}$	1	$\cdot$

$\frac{3}{2} \otimes \frac{1}{2}$

m_1	m_2	$J=2$	$J=1$
$+\frac{3}{2}$	$+\frac{1}{2}$	1	$\cdot$
$+\frac{3}{2}$	$-\frac{1}{2}$	$\sqrt{1/4}$	$\sqrt{3/4}$
$+\frac{1}{2}$	$+\frac{1}{2}$	$\sqrt{3/4}$	$-\sqrt{1/4}$
$+\frac{1}{2}$	$-\frac{1}{2}$	$\sqrt{1/2}$	$\sqrt{1/2}$
$-\frac{1}{2}$	$+\frac{1}{2}$	$\sqrt{1/2}$	$-\sqrt{1/2}$
$-\frac{1}{2}$	$-\frac{1}{2}$	$\sqrt{3/4}$	$\sqrt{1/4}$
$-\frac{3}{2}$	$+\frac{1}{2}$	$\sqrt{1/4}$	$-\sqrt{3/4}$
$-\frac{3}{2}$	$-\frac{1}{2}$	1	$\cdot$

$1 \otimes 1$

m_1	m_2	$J=2$	$J=1$	$J=0$
$+1$	$+1$	1	$\cdot$	$\cdot$
$+1$	$+0$	$\sqrt{1/2}$	$\sqrt{1/2}$	$\cdot$
$+0$	$+1$	$\sqrt{1/2}$	$-\sqrt{1/2}$	$\cdot$
$+1$	-1	$\sqrt{1/6}$	$\sqrt{1/2}$	$\sqrt{1/3}$
$+0$	$+0$	$\sqrt{2/3}$	0	$-\sqrt{1/3}$
-1	$+1$	$\sqrt{1/6}$	$-\sqrt{1/2}$	$\sqrt{1/3}$
$+0$	-1	$\sqrt{1/2}$	$\sqrt{1/2}$	$\cdot$
-1	$+0$	$\sqrt{1/2}$	$-\sqrt{1/2}$	$\cdot$
-1	-1	1	$\cdot$	$\cdot$

$2 \otimes \frac{1}{2}$

m_1	m_2	$J=\frac{5}{2}$	$J=\frac{3}{2}$
$+2$	$+\frac{1}{2}$	1	$\cdot$
$+2$	$-\frac{1}{2}$	$\sqrt{1/5}$	$\sqrt{4/5}$
$+1$	$+\frac{1}{2}$	$\sqrt{4/5}$	$-\sqrt{1/5}$
$+1$	$-\frac{1}{2}$	$\sqrt{2/5}$	$\sqrt{3/5}$
$+0$	$+\frac{1}{2}$	$\sqrt{3/5}$	$-\sqrt{2/5}$
$+0$	$-\frac{1}{2}$	$\sqrt{3/5}$	$\sqrt{2/5}$
-1	$+\frac{1}{2}$	$\sqrt{2/5}$	$-\sqrt{3/5}$
-1	$-\frac{1}{2}$	$\sqrt{4/5}$	$\sqrt{1/5}$
-2	$+\frac{1}{2}$	$\sqrt{1/5}$	$-\sqrt{4/5}$
-2	$-\frac{1}{2}$	1	$\cdot$

$\frac{3}{2} \otimes 1$

m_1	m_2	$J = \frac{5}{2}$	$J = \frac{3}{2}$	$J = \frac{1}{2}$
$+\frac{3}{2}$	$+1$	1	$\cdot$	$\cdot$
$+\frac{3}{2}$	$+0$	$\sqrt{2/5}$	$\sqrt{3/5}$	$\cdot$
$+\frac{1}{2}$	$+1$	$\sqrt{3/5}$	$-\sqrt{2/5}$	$\cdot$
$+\frac{3}{2}$	-1	$\sqrt{1/10}$	$\sqrt{2/5}$	$\sqrt{1/2}$
$+\frac{1}{2}$	$+0$	$\sqrt{3/5}$	$\sqrt{1/15}$	$-\sqrt{1/3}$
$-\frac{1}{2}$	$+1$	$\sqrt{3/10}$	$-\sqrt{8/15}$	$\sqrt{1/6}$
$+\frac{1}{2}$	-1	$\sqrt{3/10}$	$\sqrt{8/15}$	$\sqrt{1/6}$
$-\frac{1}{2}$	$+0$	$\sqrt{3/5}$	$-\sqrt{1/15}$	$-\sqrt{1/3}$
$-\frac{3}{2}$	$+1$	$\sqrt{1/10}$	$-\sqrt{2/5}$	$\sqrt{1/2}$
$-\frac{1}{2}$	-1	$\sqrt{3/5}$	$\sqrt{2/5}$	$\cdot$
$-\frac{3}{2}$	$+0$	$\sqrt{2/5}$	$-\sqrt{3/5}$	$\cdot$
$-\frac{3}{2}$	-1	1	$\cdot$	$\cdot$

$2 \otimes 1$

m_1	m_2	$J = 3$	$J = 2$	$J = 1$
$+2$	$+1$	1	$\cdot$	$\cdot$
$+2$	$+0$	$\sqrt{1/3}$	$\sqrt{2/3}$	$\cdot$
$+1$	$+1$	$\sqrt{2/3}$	$-\sqrt{1/3}$	$\cdot$
$+2$	-1	$\sqrt{1/15}$	$\sqrt{1/3}$	$\sqrt{3/5}$
$+1$	$+0$	$\sqrt{8/15}$	$\sqrt{1/6}$	$-\sqrt{3/10}$
$+0$	$+1$	$\sqrt{2/5}$	$-\sqrt{1/2}$	$\sqrt{1/10}$
$+1$	-1	$\sqrt{1/5}$	$\sqrt{1/2}$	$\sqrt{3/10}$
$+0$	$+0$	$\sqrt{3/5}$	0	$-\sqrt{2/5}$
-1	$+1$	$\sqrt{1/5}$	$-\sqrt{1/2}$	$\sqrt{3/10}$
$+0$	-1	$\sqrt{2/5}$	$\sqrt{1/2}$	$\sqrt{1/10}$
-1	$+0$	$\sqrt{8/15}$	$-\sqrt{1/6}$	$-\sqrt{3/10}$
-2	$+1$	$\sqrt{1/15}$	$-\sqrt{1/3}$	$\sqrt{3/5}$
-1	-1	$\sqrt{2/3}$	$\sqrt{1/3}$	$\cdot$
-2	$+0$	$\sqrt{1/3}$	$-\sqrt{2/3}$	$\cdot$
-2	-1	1	$\cdot$	$\cdot$